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ENHANCED DECISION SUPPORT SYSTEM FOR WILDFIRE DISASTER
RESPONSE AND MANAGEMENT (DISASTER-AWARE)

A THESIS SUBMITTED TO
THE GRADUATE SCHOOL OF NATURAL AND APPLIED SCIENCES
OF
MIDDLE EAST TECHNICAL UNIVERSITY

BY
ÇAĞLAR AKMAN

IN PARTIAL FULFILLMENT OF THE REQUIREMENTS
FOR
THE DEGREE OF DOCTOR OF PHILOSOPHY
IN
ELECTRICAL AND ELECTRONIC ENGINEERING

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**ENHANCED DECISION SUPPORT SYSTEM FOR WILDFIRE DISASTER
RESPONSE AND MANAGEMENT (DISASTER-AWARE)**

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1 **ABSTRACT**

2
3 **ENHANCED DECISION SUPPORT SYSTEM FOR WILDFIRE DISASTER**
4 **RESPONSE AND MANAGEMENT (DISASTER-AWARE)**
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8 Akman, Çağlar
9 Doctor of Philosophy, Electrical and Electronic Engineering
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16 Existing wildfire decision support systems are constrained by closed decision spaces,
17 centralized reasoning, tight simulation–decision coupling, and weak representation
18 of epistemic uncertainty. This thesis proposes DisasterAware, a hierarchically
19 distributed, explainable, and uncertainty-aware framework in which six bounded
20 features are fuzzified under partition of unity into a four-dimensional situational
21 concept vector (Fire Threat, Suppression Feasibility, Asset Exposure, Intervention
22 Urgency) that drives a Mamdani-type inference pipeline distributed across regional
23 agents. Concept abstraction yields a proven ninefold rule complexity reduction (729
24 → 81 rules per intervention type; 351 vs. 2,187 total), enabling improved scalability.
25 Formal guarantees of output boundedness, Lipschitz continuity, and structural
26 partial-observability compliance are established, together with confidence-gated
27 concept modulation that encodes epistemic uncertainty without probabilistic
28 assumptions. A non-inferential global coordination operator assembles resource-
29 normalized regional fields into an intervention input strictly separated from the
30 simulation state, while an Explainability and AI Governance Layer provides audit
31 logging, and operator override. Comparative simulation against a centralized

32 baseline validates adaptability, interpretability, and operational feasibility,
33 positioning DisasterAware as a reproducible alternative to optimization-based and
34 black-box AI wildfire decision support.

35

36 Keywords: Wildfire decision support, Hierarchically distributed multi-agent
37 systems, Concept-based fuzzy reasoning, Explainable AI, Partial observability

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ÖZ

ORMAN YANGINI AFET MÜDAHALE VE YÖNETİMİ İÇİN GELİŞTİRİLMİŞ KARAR DESTEK SİSTEMİ (DISASTER-AWARE)

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Mevcut orman yangını karar destek sistemleri; kapalı karar uzayları, merkezi akıl yürütme, simülasyon–karar arasındaki sıkı bağlantı ve epistemik belirsizliğin yetersiz temsili nedeniyle kısıtlanmaktadır. Bu tez; hiyerarşik olarkdağıtık, açıklanabilir ve belirsizliğe duyarlı bir çerçeve olan DisasterAware'i önermektedir. Çerçeve, altı sınırlı öznitelik birim bölüşümü altında bulanıklaştırılarak dört boyutlu bir durumsal kavram vektörüne (Yangın Tehdidi, Bastırma Uygunluğu, Varlık Maruziyeti, Müdahale Aciliyeti) dönüştürülmekte ve bu vektör, bölgesel ajanlar arasında dağıtılmış bir Mamdani tipi çıkarım hattını beslemektedir. Kavram soyutlaması, kanıtlanmış dokuz kat kural karmaşıklığı azaltımı sağlar (müdahale tipi başına 729 → 81 kural; toplamda 351'e karşı 2.187 kural); bu da ölçeklenebilirlikte belirgin iyileşme sağlar. Çıktı sınırlılığı, Lipschitz sürekliliği ve yapısal kısmi gözlemlenebilirlik uyumluluğu biçimsel olarak garanti altına alınmış; ayrıca, olasılıksal varsayımlara gerek duymadan epistemik belirsizliği kodlayan güven kapılı kavram modülasyonu tanımlanmıştır. Çıkarımsal olmayan küresel bir koordinasyon işleci, kaynak-normalize bölgesel alanları simülasyon durumundan kesin biçimde ayrılmış bir müdahale girdisi olarak birleştirirken, Açıklanabilirlik ve Yapay Zekâ (YZ) Yönetişim Katmanı denetim kaydı ve operatör geçersiz kılma

70 işlevlerini sağlar. Merkezi bir temel çizgiye karşı yapılan karşılaştırmalı simülasyon;
71 uyarlanabilirliği, yorumlanabilirliği ve operasyonel uygulanabilirliği doğrulayarak,
72 DisasterAware'i optimizasyon tabanlı ve kara kutu yapay zekâ yaklaşımlarına
73 yeniden üretilebilir bir alternatif olarak konumlandırır.
74
75 Anahtar Kelimeler: Orman yangını karar desteği, Hiyerarşik dağıtık çok-ajanlı
76 sistemler, Kavram tabanlı bulanık mantık, Açıklanabilir yapay zekâ, Kısmi
77 gözlemlenebilirlik
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To my beloved wife Emine and our sons Kuzey, Rüzgar, and Atlas for their endless love, patience, and encouragement. You are my calm after every storm.

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To my sons, Kuzey, Rüzgar, and Atlas, I hope this dissertation serves as an example that perseverance and discipline can turn even the most challenging dreams into reality. You are my greatest inspiration and the true meaning behind this effort.

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28

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LIST OF ABBREVIATIONS

AHP	: Analytic Hierarchy Process
AI	: Artificial Intelligence
COP	: Common Operational Picture
DM	: Decision Maker
DSS	: Decision Support System
ELECTRE	: ELimination Et Choix Traduisant la REalité
FIS	: Fuzzy Inference System
GIS	: Geographic Information System
HIL	: Human-in-the-loop
LIME	: Local Interpretable Model-agnostic Explanations
MAGDM	: Multi-Attribute Group Decision Making
MCDM	: Multi-Criteria Decision Making
POMDP	: Partially Observable Markov Decision Process
SHAP	: SHapley Additive exPlanations
TOPSIS	: Technique for Order of Preference by Similarity to Ideal Solution
UI	: User Interface
VIKOR	: Visekriterijumska Optimizacija i Kompromisno Resenje
XAI	: Explainable Artificial Intelligence
WFDSS	: Wildfire Decision Support System

276 The following list defines all symbols used in this thesis, organized by functional
 277 category. Four notational conventions apply throughout: subscript k denotes the
 278 discrete time step; arguments (x, y) denote the grid cell; superscript i identifies the
 279 operational region; a hat accent ($\hat{}$) indicates an observed estimate; and an overbar ($\bar{}$)
 280 indicates a confidence-gated or spatially averaged quantity. See Remark 2.2 for the
 281 complete convention specification.

1. Fire State and Propagation Variables

$B_k(x, y)$	Burning status indicator
$B_{k+1}^{\text{pers}}(x, y)$	Burning persistence term
$B_{k+1}^{\text{prop}}(x, y)$	Burning propagation term
$\hat{B}_k^i(x, y)$	Observed estimate of burning status in region i
$\bar{B}_k^\delta(x, y)$	Mean burning status over δ -extended neighborhood of (x, y)
$F_{\text{burn},k}(x, y)$	Fractional burn rate
$I_k(x, y)$	Local fire intensity proxy
$\hat{I}_k^i(x, y)$	Observed estimate of fire intensity in region i
$I_{\text{ign},k}(x, y)$	External ignition injection indicator
$R_{\text{spread},k}(x, y)$	Rate of spread at time step k
$\tau_k(x, y)$	Time since ignition
$\hat{\tau}_k^i(x, y)$	Observed estimate of ignition time in region i
$\Psi_k(x, y)$	Propagation influence
Ψ_{ref}	Reference normalization constant for spread potential feature z_3
θ_{ign}	Ignition threshold
Φ	State transition operator

2. Fuel Variables

$F_{\text{load},k}(x, y)$	Available combustible fuel mass
$\hat{F}_{\text{load},k}^i(x, y)$	Observed estimate of fuel load in region i
$F_{\text{moist},k}(x, y)$	Surface fuel moisture

$F_{red,k}(x, y)$	Suppression-induced fuel reduction
$F_{type}(x, y)$	Fuel type classification
$U_{Fuel,k}(x, y)$	Fuel-related input
ε_{fuel_load}	Fuel extinction threshold

3. Terrain / Geographic Variables

$G_{access}(x, y)$	Terrain accessibility index
$G_{aspect}(x, y)$	Terrain orientation (aspect)
$G_{elev}(x, y)$	Terrain elevation
$G_{slope}(x, y)$	Terrain slope
$U_{Geo}(x, y)$	Terrain-related input

4. Meteorological Variables

$W_{gust,k}(x, y)$	Wind gust field
$W_{prec,k}$	Precipitation (domain-uniform scalar, no spatial variation)
$W_{rh,k}(x, y)$	Relative humidity field
$W_{temp,k}(x, y)$	Air temperature field
$W_{wd,k}(x, y)$	Wind direction field
$W_{ws,k}(x, y)$	Wind speed field
$U_{Meteo,k}(x, y)$	Meteorological input
δ_{Meteo}	Meteorological perturbation applied by operator

5. Value / Asset Variables

$V_{bld}(x, y)$	Building footprint indicator
$V_{crit}(x, y)$	Critical facility index
$V_{evac}^{norm}(x, y)$	Accessibility to evacuation routes
$V_{pop}^{norm}(x, y)$	Population density at grid cell (person/km ²)
$V_{prio}(x, y)$	Asset priority index
$U_{Val,k}(x, y)$	Value-related contextual input
σ_{Val}	Value priority scaling factor in θ_{UI}

6. Resource / Suppression Variables

C_k	Total available resource capacity
-------	-----------------------------------

$C_{k,m}$	Type-specific available capacity for type m
D_k	Total integrated intervention demand over the domain at step k
$D_{k,m}$	Type-specific integrated intervention demand for type m
M	Number of intervention types
$R_{avail,k}(x, y)$	Operational availability of suppression resources
$R_{cap,k}(x, y)$	Available suppression capacity
$R_{eff,k}(x, y)$	Nominal suppression effectiveness
$R_{time,k}(x, y)$	Estimated travel time for suppression resources
$U_{Res,k}(x, y)$	Resource-related contextual input
$u_{k,m}^i(x, y)$	Candidate intervention degree for type m in region i
α_m	Priority weight for intervention type m
$\bar{\alpha}_m$	Normalized priority weight for intervention type m
$\delta_{Res}(x, y)$	Resource availability perturbation applied by operator
$\mathcal{N}_{Res,k}(\bullet)$	Resource normalization operator
$\chi_{k,m}$	Conflict ratio for intervention type m
$\hat{u}_{k,m}(x, y)$	Conflict-aware per-type intervention field

7. Observation / Sensing Variables

$O_k^i(x, y)$	Region-specific observable information at time step k
$O_k(x, y)$	Observable information at time step k
$h(\bullet)$	Observation process
$h^i(\bullet)$	Region-specific observation function for Ω_i
$conf_k^i(x, y)$	Observation confidence coefficient
$\rho_k^i(x, y)$	Normalized sensor coverage density at (x, y)
γ_k^i	Communication reliability factor for region i
$\varepsilon_k^i(x, y)$	Observation uncertainty
$\Delta t_{rep,k}^i(x, y)$	Elapsed time since last observation report at (x, y)
$\theta_{j,k}^i(x, y)$	Observability weight for state component j , region i , step k
λ_{conf}	Confidence decay rate with reporting delay

8. Spatial Domain / Grid Variables

$\mathcal{G}$	Spatial grid domain
$\mathcal{G}_k^+$	Active domain at step k: burning or fire-threatened cells
N	Total number of operational regions / local agents
N_x	Number of grid cells along the horizontal axis
N_y	Number of grid cells along the vertical axis
$\mathcal{N}^8(x, y)$	8-connected neighborhood
$\mathcal{N}^\delta(x, y)$	Set of cells within radius δ of cell (x, y)
Ω	Full spatial domain
Ω_i	Operational region of agent i
$S_k(x, y)$	Wildfire system state over spatial grid at time step k

9. Decision Support System (DSS) Variables

$\mathcal{A}_{DSS}$	Global coordination and spatial field assembly operator
$\mathcal{A}_{DSS}^{sp}$	Spatial assembly operator for regional intervention fields
$\bar{c}_{sev,k}$	Mean operational severity
$\mathcal{F}_{DSS,k}$	Contextual input set
$\mathcal{F}_{DSS,k}^*$	UI-modified decision-context set at step k
$\mathcal{F}_{in,k}$	External input set
$\mathcal{F}_{in,k}^*$	UI-modified external input set at step k
$\mathcal{L}_k(x, y)$	Audit record tuple at time step k and cell (x, y)
$R_m^*(x, y)$	Dominant explanation rule for intervention type m
$U_{DSS,k}(x, y)$	DSS-generated intervention input parameters
$U_{DSS,k}^i(x, y)$	Regional intervention output from agent i
$U_{DSS,k}^m(x, y)$	Per-intervention-type global intervention field
$U_{DSS,k}^{final}(x, y)$	Final intervention field after decision acceptance or fail-safe attenuation
$U_{safe,k}(x, y)$	Conservative baseline intervention under fail-safe policy
$\mathcal{U}_{DSS}$	Feasible intervention space
$U_{ign,k}(x, y)$	Ignition input

$Q_{DSS,k}$	Multi-criteria decision quality score
π_{DSS}^i	Regional decision-support mapping
$\pi_{i,m}$	Local decision mapping for intervention type m
$\bar{\rho}_k$	Mean observation confidence over active domain
$\bar{s}_k^*$	Mean resource sufficiency ratio
$\delta_{DSS}(x, y)$	Resource parameter perturbation applied by operator
$\delta_{Ign}(x, y)$	Ignition perturbation applied by operator
θ_{UI}	User interaction operator
η_k	Decision satisfaction threshold
η_{min}	Lower bound of adaptive satisfaction threshold
η_{max}	Upper bound of adaptive satisfaction threshold
ζ_k	Decision attenuation coefficient (fail-safe blending weight)

10. Fuzzy Logic Variables

$\mathcal{N}_{concept}$	Number of situational concepts
$\mathcal{N}_{feat}$	Number of input features
$\mathcal{N}_\mu$	Number of linguistic terms per fuzzy variable
$\mathcal{R}^{concept}$	Max rule base size with concept abstraction
$\mathcal{R}^{direct}$	Max rule base size without concept abstraction
κ	Rule complexity reduction ratio
$\mathbf{c}_k^i(x, y)$	Concept vector at time step k
$\bar{\mathbf{c}}_k^i(x, y)$	Confidence-gated concept vector
$c_{k,h}^i(x, y)$	Concept vector component (h-th concept at time step k)
$\bar{c}_{k,h}^i(x, y)$	Confidence-gated concept value for concept h , region i
$c_{prior,h}^i(x, y)$	Concept prior: persistence-based estimate from previous step
μ_k^i	Membership matrix for region i at step k
$\mu_{k,n,l}^i$	Membership degree
$\mathcal{S}_h$	Index set of (n, l) membership pairs contributing to concept h
v_R	Singleton consequent value for rule R

α_R	Activation strength of rule R = min over antecedent membership degrees
$w_{h,n,l}$	Weight assigned to (n,l) membership pair contributing to concept h
w_j	Criterion weight for f_{eval}

11. Functions / Mappings

f_{agg}	Action aggregation function
$f_{concept,h}$	Concept construction mapping
f_{eval}	Multi-criteria decision evaluation function
f_{feat}	Feature extraction mapping
f_{fuzzy}	Fuzzification mapping
$f_{intensity}$	Fire intensity approximation function
f_{ros}	Rate-of-spread mapping function
f_{supp}	Suppression-to-mitigation mapping
$g_{dir,k}$	Directional spread weighting factor
$q_{j,k}$	Partial criterion score for criterion j at step k

12. Feature Variables and Parameters

$z_{k,l}^i(x,y)$	l-th scalar feature at (x,y), region i, step k; $z \in [0,1]$, $l = 1 \dots 6$
β_t	Travel-time decay coefficient
Δt	Simulation time step
δ	Neighborhood radius for asset exposure feature z_4 (cells)
$\varepsilon_{j,k}^i(x,y)$	Epistemic disturbance for component j, region i
$\bar{\varepsilon}_j^i$	Maximum epistemic disturbance bound for component j, region i
λ	Saturation shape parameter for temporal urgency feature z_6
τ_{ref}	Reference time scale for temporal urgency feature z_6 (simulation steps)

283

CHAPTER 1

284

INTRODUCTION

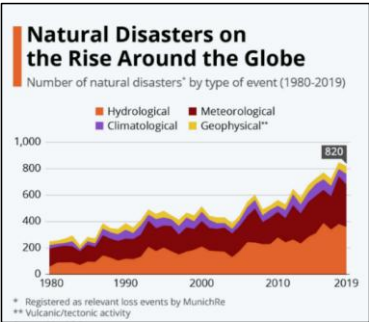
285 Decisions in domains such as disaster response, healthcare, and critical infrastructure
286 are made under uncertainty, time pressure, and resource constraints, where errors
287 carry severe and often irreversible consequences. Decision makers must synthesize
288 heterogeneous streams of information while operating under uncertainty, time
289 pressure, and resource constraints, conditions that intensify in domains such as
290 disaster management, healthcare, transportation, and critical infrastructure, where
291 poor decisions can carry severe and irreversible consequences. Decision Support
292 Systems (DSS) address this gap by providing analytical capabilities that extend the
293 information an operator can integrate and evaluate under pressure (Sprague &
294 Carlson, 1982; Power, 2002; Turban, 2011; Sharda, Delen, & Turban, 2018).

295 Over the past decades, DSS have evolved alongside advances in data analytics,
296 computational intelligence, and cyber-physical sensing infrastructures (Power &
297 Heavin, 2017; Sharda, Delen, & Turban, 2018; Arnott & Pervan, 2005). Despite this
298 progress, a fundamental structural mismatch remains unresolved between the
299 assumptions embedded in classical DSS models and the conditions that define real-
300 world operational environments. Analytical frameworks based on optimization and
301 formal decision analysis typically presuppose full observability, stationary system
302 dynamics, and well-defined objective functions (Keeney & Raiffa, 1976). However,
303 decision makers operate under incomplete, evolving, and often conflicting
304 information, leading to significant epistemic uncertainty that such models cannot
305 adequately represent (French, Maule, & Papamichail, 2019; Walker, et al., 2003).

306 In response, DSS research has shifted toward hybrid architectures that integrate
307 simulation, expert knowledge representation, and uncertainty-aware reasoning
308 (Marakas & O'Brien, 2013; Arnott & Pervan, 2005). Concurrently, recent advances

309 in artificial intelligence (AI) and machine learning have improved predictive
310 capabilities in complex operational domains.

311 However, many of these approaches, particularly those based on deep learning and
312 ensemble methods, operates as opaque systems whose internal reasoning is difficult
313 to expose at the decision level. Even when predictive performance is high, the
314 absence of transparent, traceable reasoning pathways limits trust and operational
315 adoption in high-risk environments (Ribeiro, Singh, & Guestrin, 2016; Abdul,
316 Vermeulen, Wang, Lim, & Kankanhalli, 2018; Sanneman & Shah, 2020). In domains
317 such as disaster management, decision support outputs must be interpretable,
318 traceable, and aligned with human reasoning to support effective human-machine
319 collaboration (McNeese, Schelble, Canonico, & Demir, 2021). Consequently, next-
320 generation DSS must simultaneously address epistemic uncertainty, preserve
321 interpretability, and operate adaptively under dynamic and partially observable
322 conditions (French, Maule, & Papamichail, 2019; Thompson & Calkin, 2011;
323 Comfort, Boin, & Demchak, 2010; Walker, et al., 2003).



324
325 Figure 1.1 Global trend in reported natural disasters, 1900–2022

326 Figure 1.1 shows the number of reported natural disaster events globally from 1900
327 to 2022. The record reflects a substantial rise through the twentieth century followed
328 by a partial plateau in the most recent two decades, with weather- and climate-related
329 events, including wildfires, accounting for a growing share of the reported incidents
330 (CRED/UCLouvain, 2023; UNEP, 2022). Wildfires are among the most

331 consequential environmental hazards of the twenty-first century. Their frequency,
332 spatial extent, and destructive intensity have grown under climate change, land-use
333 transformation, and the expansion of wildland–urban interfaces (Bowman, et al.,
334 2020; UNEP, 2022), producing ecological damage, economic losses, and threats to
335 human settlements that outpace conventional response capabilities (Thompson &
336 Calkin, 2011; Pacheco, 2015; Finney, 2006).

337 Wildfire management represents a particularly demanding decision domain because
338 fire behavior evolves dynamically across both space and time under the simultaneous
339 influence of meteorological forcing, terrain characteristics, fuel availability, and
340 suppression interventions. Wind speed and humidity govern ignition probability and
341 spread rate; slope and aspect introduce directional propagation effects; fuel structure
342 determines combustion intensity and burn duration; and suppression actions alter
343 local conditions in ways that feed back into fire dynamics (Rothermel, 1972; Finney,
344 2006; Sullivan, 2009a; Sullivan, 2009b; Sullivan, 2009c).

345 Operational wildfire response therefore relies not only on numerical indicators, but
346 also on incomplete and linguistically expressed situational assessments such as “high
347 spread potential,” “limited suppression feasibility,” or “uncertain fire boundary.”
348 This characteristic makes wildfire management especially suitable for fuzzy
349 reasoning, which enables approximate inference over partial and imprecise evidence
350 without requiring probability distributions that are rarely available in real-time
351 operational settings (Thompson & Calkin, 2011; Pacheco, 2015).

352 Numerous DSS approaches have been proposed for wildfire management, including
353 optimization-based resource allocation, simulation-based systems, geospatial risk
354 assessment, and fuzzy logic reasoning (Thompson & Calkin, 2011; Pacheco, 2015;
355 Finney, 2006; Vásquez, 2021; Figueiredo, Macedo, Alves, & Figueiredo, 2024).
356 Mature deployed systems such as WFDSS, FlamMap, and recent ML-based hybrids
357 deliver real operational value on subsets of these requirements. For instance, WFDSS
358 provides robust spatio-temporal fire modeling and resource documentation. The
359 remaining challenge is therefore architectural: no single framework simultaneously
360 addresses adaptability under dynamic conditions, representation of epistemic

361 uncertainty, explainability of decision outputs, and support for distributed multi-
362 level coordination within one coherent design (Thompson & Calkin, 2011; Vásquez,
363 2021; McNeese, Schelble, Canonico, & Demir, 2021).

364 This thesis addresses these limitations through the development of DisasterAware,
365 an explainable, uncertainty-aware DSS framework for wildfire disaster response and
366 management. DisasterAware integrates a spatio-temporal wildfire simulation
367 environment with concept-driven fuzzy reasoning in a layered, distributed
368 architecture designed to generate interpretable and adaptive intervention guidance
369 under partial observability.

370 The proposed framework is motivated not by a single methodological deficiency, but
371 by a combination of unresolved structural limitations in existing DSS, particularly in
372 decision-space openness, distributed coordination, uncertainty representation,
373 explainability, and simulation-decision separation. These limitations are examined
374 in the Gap Analysis, and the thesis contributions are then mapped to them explicitly.

375 1.1 Gap Analysis

376 Despite substantial advances in DSS research, the literature still exhibits several
377 unresolved limitations that constrain operational effectiveness in the real world.
378 Existing systems have contributed important capabilities in fire spread prediction,
379 geospatial risk analysis, resource planning, and situation monitoring. However,
380 many of these contributions remain bounded by assumptions that are difficult to
381 sustain in dynamic conditions, including predefined decisions, centralized reasoning,
382 limited treatment of epistemic uncertainty, weak explainability, and insufficient
383 support for multi-level coordination (Arnott & Pervan, 2005; Turban, 2011).

384 Wildfire response is a dynamic and partially observable decision environment in
385 which fire behavior, operational feasibility, asset exposure, and decision urgency
386 evolve simultaneously across space and time. Under such conditions, DSS cannot be
387 reduced to static prediction, isolated optimization, or centralized ranking of

predefined options. The central challenge is therefore architectural rather than algorithmic: wildfire DSS requires a framework that is simultaneously open-ended, distributed, uncertainty-aware, explainable, modular, constraint-sensitive, and spatio-temporally coordinated, properties that no existing system satisfies in combination. The research gaps identified in this section, summarized in Table 1.1, are each a manifestation of this overarching architectural shortfall.

Table 1.1 DSS gaps identified in wildfire decision support literature

Gaps of Existing Wildfire DSS Architectures

G1	Existing DSS frameworks rely on predefined decision configurations, limiting adaptability under dynamic and unforeseen conditions.
G2	Centralized DSS architectures reduce scalability and local responsiveness in spatially distributed wildfire environments.
G3	Current systems inadequately represent epistemic uncertainty arising from incomplete, delayed, or conflicting observations.
G4	Many approaches rely on opaque AI or optimization models, limiting explainability and reducing operator trust.
G5	Existing frameworks lack a clear separation between simulation dynamics and decision reasoning, reducing traceability and modularity.
G6	Operational constraints such as resource availability, travel time, and suppression feasibility are insufficiently integrated into decision reasoning.
G7	Spatial and temporal aspects of wildfire dynamics are often treated separately, limiting coherent spatio-temporal decision support.
G8	Lack of hierarchical coordination between local and global decision levels.

G1 Closed Decision Spaces in Traditional DSS

Many classical DSS frameworks assume that decision alternatives can be explicitly defined before the decision process begins. Early DSS models focused on structuring decision problems and evaluating predefined alternatives through analytical models and decision rules (Power, Decision support systems: Concepts and resources for

managers, 2002; Pacheco, 2015; Sprague & Carlson, 1982). Similarly, many multi-criteria decision-making frameworks require a predefined set of alternatives that can be ranked according to decision criteria (Malczewski, 1999; Keeney & Raiffa, 1976). However, wildfire management rarely conforms to this assumption. Fire behavior evolves continuously due to environmental factors such as wind shifts, fuel variability, and terrain interactions. As a result, operational conditions may generate situations that cannot be represented by a fixed set of predefined intervention strategies. Studies reviewing DSS implementations confirm that many existing systems primarily evaluate suppression strategies or mitigation actions that have been defined beforehand rather than generating new decision options dynamically (Pacheco, 2015; Vásquez, 2021; Calkin, O'Connor, Thompson, & Stratton, 2021). This limitation restricts the adaptability of DSS frameworks in emergency environments where novel operational scenarios frequently emerge. Field studies of operational wildfire DSS use further confirm that practitioners tend to rely on existing tools to validate decisions already made on the basis of experience, rather than to generate new response alternatives dynamically (Fillmore & Paveglio, 2023; Rapp, Rabung, Wilson, & Toman, 2020). This behavioral pattern reflects a structural mismatch between the closed decision spaces embedded in current DSS architectures and the open-ended operational demands of dynamic environments.

G2 Dependence on Centralized Decision Architectures

Many DSS systems rely on centralized reasoning architectures in which observation aggregation, decision evaluation, and response generation are concentrated within a single analytical core. While such architectures simplify global coordination and system management, they often exhibit reduced responsiveness and limited scalability in spatially distributed, rapidly evolving, and time-critical environments. DSS research has long recognized that centralized decision structures become vulnerable to computational bottlenecks when the operational state changes rapidly or when the supported domain expands spatially (Arnott & Pervan, 2005; Shim, et al., 2002). In wildfire response, this limitation is especially pronounced because

decision-relevant conditions evolve simultaneously across multiple spatial regions, each with distinct fire behavior, constraints, resource states, and local priorities.

Accordingly, the central architectural issue is not merely whether a system can compute a global recommendation, but whether it can sustain timely and region-sensitive decision generation under dynamic and spatially heterogeneous conditions. Recent reviews of wildfire DSS research increasingly emphasize the need for distributed DSS mechanisms capable of preserving local responsiveness while maintaining higher-level coordination (Vásquez, 2021). Without such architectures, centralized systems risk delayed decision cycles, reduced adaptability, and degraded operational scalability in large wildfire incidents.

Operational evidence supports this concern. Post-incident analyses of large-scale wildfire events in the United States indicate that centralized coordination structures frequently delay localized action because regional observations and requests must be routed through central decision nodes before actionable guidance can be issued (Calkin, O'Connor, Thompson, & Stratton, 2021). Resource allocation systems that depend on single-point centralized optimization have likewise been shown to lose responsiveness as incident scale and spatial dispersion increase (Orphanoudakis, Betzelos, & Leligou, 2025). This reveals a structural gap between centralized DSS assumptions and the distributed character of real wildfire response operations.

G3 Insufficient Representation of Epistemic Uncertainty

Uncertainty plays a central role in wildfire decision environments. While fire behavior models capture stochastic variability associated with combustion dynamics, operational decisions are often affected more strongly by epistemic uncertainty arising from incomplete or unreliable information. Studies in wildfire risk management emphasize that decision-makers frequently rely on uncertain estimates of fire spread, weather conditions, and resource availability. More broadly, uncertainty management research distinguishes between aleatory uncertainty related to stochastic variability and epistemic uncertainty arising from incomplete knowledge (Morgan & Henrion, 1990; Walker, et al., 2003). Despite this distinction,

459 many wildfire DSS implementations rely on deterministic thresholds or simplified
460 probabilistic models that do not adequately represent epistemic uncertainty.

461 Thompson and Calkin (2011) identify epistemic uncertainty as the primary challenge
462 in wildfire risk assessment, noting that incomplete field observations, sensor
463 limitations, and communication delays systematically undermine the reliability of
464 information available to decision-makers. Empirical studies of DSS use in practice
465 further reveal that operators frequently distrust model outputs precisely because
466 existing tools do not expose or communicate the degree of observational uncertainty
467 underlying their recommendations (Rapp, Rabung, Wilson, & Toman, 2020). Fuzzy
468 logic has been proposed as a principled mechanism for encoding such epistemic
469 uncertainty through graded membership representations rather than crisp thresholds
470 (Duarte, Teodoro, Gonçalves, Soares, & Cunha, 2022), yet its systematic integration
471 into operational wildfire DSS architectures remains limited.

472 **G4 Limited Explainability in Decision-Support Systems**

473 Recent advances in AI have led to the development of predictive models for wildfire
474 behavior and risk assessment. While such models may improve predictive accuracy,
475 they often operate as black-box systems whose internal reasoning is difficult for
476 human operators to interpret. This limitation is not merely technical but operational,
477 because opaque recommendations are difficult to justify, validate, and trust during
478 time-critical response. Research in XAI shows that transparency and interpretability
479 are critical for building trust in DSS (Ribeiro, Singh, & Guestrin, 2016; Guidotti, et
480 al., 2018). In emergency response contexts, decision recommendations must be
481 understandable and traceable in order to support human decision-making and
482 accountability. Human–AI collaboration research further demonstrates that
483 explainability significantly improves situational awareness and decision quality in
484 high-risk environments (McNeese, Schelble, Canonico, & Demir, 2021).

485 This challenge is particularly acute in wildfire management, where time-critical
486 decisions must be justified to multiple stakeholders at different organizational levels.
487 Empirical studies indicate that wildfire managers frequently override or ignore DSS

recommendations when the underlying reasoning is not transparent, reducing the operational value of otherwise capable analytical systems (Fillmore & Paveglio, 2023; Rapp, Rabung, Wilson, & Toman, 2020). Recent hybrid frameworks that combine high-accuracy predictive models with fuzzy rule-based systems illustrate that interpretability and predictive performance need not be mutually exclusive (Mohammadi, Alesheikh, & Alimohammadi, 2025). Such explainable architectures remain absent from mainstream operational wildfire DSS implementations.

G5 Lack of Separation Between Simulation and Decision Logic

Many wildfire DSS implementations tightly integrate fire spread simulation models with decision reasoning mechanisms. Although such integration may simplify system design, it often reduces system transparency and complicates validation. In complex DSS, modular architectures are essential for maintaining traceability between system components (Sugumaran & Degroote, 2010). Without a clear separation between dynamics and decision reasoning, it is difficult to determine whether system outputs are driven by physical fire models or by decision algorithms. In the context of wildfire management, this tight coupling is particularly problematic because fire spread models operate on physical timescales that differ from decision cycles. Combining both within a single computational unit makes it difficult to update or validate either component independently (Pacheco, 2015; Vásquez, 2021). The Wildland Fire Decision Support System (WFDSS) illustrates this coupling in practice: its fire behavior modeling subsystems and decision documentation components are architecturally integrated within a single platform, which constrains independent validation and limits the substitutability of individual components (Noonan-Wright, et al., 2011). As wildfire simulation science continues to advance independently of DSS research, this tight coupling increasingly impedes the incorporation of new modeling capabilities into operational decision frameworks.

514 **G6 Limited Integration of Operational Constraints**

515 Operational wildfire response decisions depend strongly on practical constraints
516 such as resource availability, travel time, and suppression effectiveness. Classical
517 decision analysis frameworks often represent these constraints through optimization
518 models or cost functions (Keeney & Raiffa, 1976).

519 However, acknowledging the importance of these constraints is not equivalent to
520 integrating them into real-time decision reasoning. Wildfire suppression operations
521 involve distributed resources whose availability changes continuously during an
522 incident; existing frameworks that formally represent constraints through
523 optimization models require complete and accurate resource state information that is
524 rarely available in practice (Pais, Carrasco, Moudio, & Shen, 2021). The gap,
525 therefore, is not a lack of awareness (i.e., the operational significance of these
526 constraints is well-documented) but a persistent failure to model them with sufficient
527 fidelity within DSS architectures that must operate under incomplete and evolving
528 conditions (Martell, 2007; Finney, 2006).

529 More recent work demonstrates that resource accessibility, travel time penalties, and
530 suppression effectiveness under high fire intensity represent particularly critical
531 operational constraints that are rarely modeled with sufficient fidelity in existing
532 DSS frameworks (Oliveira, Barros, Ager, & Fernandes, 2022).

533 **G7 Weak Integration of Spatio-Temporal Reasoning**

534 Wildfire dynamics evolve across both spatial and temporal dimensions. However,
535 many DSS frameworks treat spatial risk analysis and temporal fire evolution as
536 separate analytical processes. Spatial DSS research emphasizes the importance of
537 integrating spatial context with decision analysis (Malczewski, 1999). For instance,
538 a system that identifies high-risk zones based on current fire extent without
539 accounting for projected spread trajectories over the next time steps may recommend
540 interventions that are spatially correct but temporally too late to be effective
541 (Thompson & Calkin, 2011; Finney, 2006).

Landscape-level fuel treatment studies confirm that decisions based on static spatial risk assessments, without accounting for projected fire spread trajectories over the decision horizon, systematically misallocate resources toward areas that are at risk in the current time step but are no longer reachable or relevant by the time interventions can be deployed (Ager, Vaillant, & Finney, 2010). More recent approaches that integrate spatiotemporal predictive models with fuzzy decision layers demonstrate improvements in intervention timeliness, but their architectural coupling of prediction and decision logic introduces the traceability limitations described in G5 (Mohammadi, Alesheikh, & Alimohammadi, 2025).

G8 Lack of Hierarchical Decision Coordination

Wildfire response operations do not occur at a single decision level. They involve local intervention decisions, regional operational coordination, and broader strategic prioritization. However, many DSS implementations still treat decision support as a single-layer reasoning process, without explicitly structuring interactions between these distinct levels of decision authority. This limitation differs from centralized architectural dependence. The issue here is not where computation is performed, but whether the DSS provides formal mechanisms for reconciling decisions made at different organizational and operational levels. Disaster management research consistently emphasizes that effective response depends on hierarchical coordination between tactical execution, operational management, and strategic oversight (Turoff, 2004; Comfort, Boin, & Demchak, 2010). Without such mechanisms, local decisions may be operationally rational in isolation while remaining globally inconsistent with higher-priority containment, protection, or resource-allocation objectives.

In wildfire response, this coordination challenge is particularly important because priorities can differ across scales. A locally feasible suppression action may conflict with regional resource preservation needs; a tactical containment preference may be suboptimal relative to broader asset-protection strategy; and delayed feedback between levels may produce inconsistent or fragmented intervention patterns.

570 Field studies of wildfire response organizations in the United States show that
571 incident-level commanders often make locally appropriate decisions that become
572 globally suboptimal in the absence of structured coordination and feedback across
573 decision levels (Schultz, Miller, Greiner, & Kooistra, 2021). Dunn, Calkin, and
574 Thompson (2017) likewise demonstrate that the absence of formal hierarchical
575 coordination contributes to delayed escalation, inefficient deployment, and reduced
576 strategic coherence. These findings indicate that DSS design requires not only local
577 reasoning capacity, but also explicit mechanisms for integrating local, regional, and
578 strategic decision layers into a coherent response architecture.

579 1.2 Research Question and Hypothesis

580 Formally, this thesis addresses the following research question:

581 *How can a hierarchically distributed and explainable DSS architecture effectively*
582 *generate adaptive and operationally feasible intervention guidance under partial*
583 *observability in dynamic wildfire environments?*

584 This question gives rise to the following main hypothesis:

585 *The DisasterAware framework, based on hierarchically distributed concept-driven*
586 *fuzzy reasoning with non-inferential global coordination, achieves superior*
587 *performance compared to functionally equivalent centralized DSS baselines in*
588 *terms of adaptability, interpretability, and operational feasibility under identical*
589 *simulation conditions.*

590 This hypothesis is decomposed into three testable sub-claims:

591 **H1 (Adaptability):** The hierarchically distributed generation of decisions across
592 spatially decomposed regions leads to faster response times and improved resource
593 allocation accuracy under evolving wildfire conditions compared to centralized
594 decision mechanisms.

H2 (Interpretability): The concept-driven fuzzy reasoning structure enables traceable and interpretable decision outputs at the local level, improving transparency and human–machine collaboration compared to centralized approaches with aggregated representations.

H3 (Operational Feasibility): The integration of resource-aware and constraint-sensitive fuzzy evaluation mechanisms produces intervention strategies that remain operationally feasible across a wider range of dynamic wildfire scenarios.

Figure 1.2 summarizes the central research question, the main hypothesis, and the three testable sub-claims.

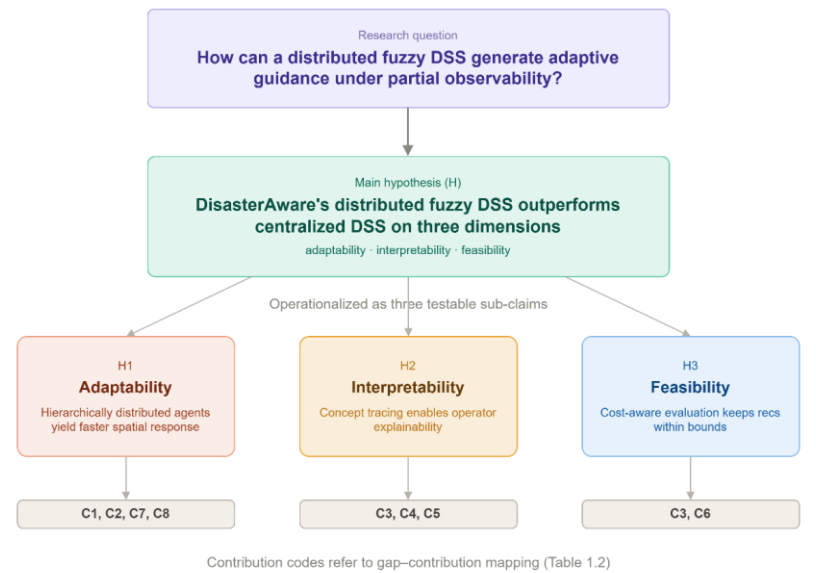


Figure 1.2 Research question, main hypothesis, and associated sub-claims

Each sub-claim targets one or more structural limitations identified in the gap analysis and maps directly to the architectural contributions detailed in Table 1.2: H1 is addressed by contributions C1, C2, C7, and C8, thereby targeting gaps G1 (closed decision spaces), G2 (centralized architecture), G7 (weak spatio-temporal

610 integration), and G8 (lack of hierarchical coordination). H2 is addressed by
 611 contributions C3, C4, and C5, targeting gaps G3 (insufficient epistemic uncertainty
 612 representation), G4 (limited explainability), and G5 (tight coupling between
 613 simulation and decision logic). H3 is addressed by contributions C3 and C6, targeting
 614 gaps G3 (epistemic uncertainty) and G6 (limited integration of operational
 615 constraints). This mapping ensures that every structural gap identified in Table 1.1
 616 is covered by at least one testable hypothesis, and that each hypothesis targets a
 617 distinct validation objective. These hypotheses are evaluated through simulation-
 618 based comparative analysis as detailed in Chapter 7.

619 **1.3 Main Contributions of the Thesis**

620 The contributions of this thesis introduce a structural shift from static, centralized,
 621 and predefined decision paradigms toward a dynamic, hierarchically distributed,
 622 uncertainty-aware, and explainable DSS framework. Unlike conventional DSS
 623 approaches that operate on fixed decision alternatives and assume complete state
 624 information, the DisasterAware framework generates adaptive intervention
 625 strategies directly from partially observable and evolving system conditions. In this
 626 framework, DSS is reframed as a continuous reasoning process over an open
 627 decision space rather than an optimization problem over predefined choices.

628 Each identified gap listed in Table 1.1 is explicitly mapped to a contribution of the
 629 proposed framework. This mapping establishes a traceable link between the
 630 limitations of existing systems and the architectural design choices introduced in this
 631 thesis, as summarized in Table 1.2.

632 Table 1.2 Mapping of thesis contributions to identified gaps

	<i>Limitation in Existing Systems</i>	<i>Contribution of this Thesis</i>
CI	G1: Predefined decision alternatives limit adaptability	Introduction of an open decision space using concept-driven fuzzy reasoning

	<i>Limitation in Existing Systems</i>	<i>Contribution of this Thesis</i>
C2	G2: Centralized DSS reduces responsiveness and scalability	Development of a hierarchically distributed decision architecture with non-inferential global coordination
C3	G3: Limited representation of epistemic uncertainty	Integration of FIS for epistemic uncertainty modeling
C4	G4: Lack of explainability in DSS and AI-based systems	Design of explainable fuzzy reasoning with AI governance
C5	G5: Tight coupling between simulation and decision logic	Architectural separation of Simulation Core and DSS Core
C6	G6: Limited representation of operational constraints in decision logic	Integration of cost-aware fuzzy decision evaluation including resources and travel time
C7	G7: Weak integration of spatial and temporal reasoning	Introduction of spatio-temporal fuzzy reasoning mechanisms
C8	G8: Lack of hierarchical coordination between decision levels	Development of a multi-level decision architecture combining local and global decision processes

C1 Open Decision Space Through Concept-Based Reasoning

To address G1, the DisasterAware framework introduces an open decision space in which intervention guidance emerges dynamically from concept-driven fuzzy reasoning rather than from a fixed set of predefined alternatives. Instead of selecting among fixed strategies, it generates context-sensitive decision guidance based on evolving operational context, enabling adaptive responses to unforeseen conditions.

C2 Hierarchically Distributed Decision Architecture

To address G2, DisasterAware introduces a hierarchically distributed DSS in which local fuzzy agents execute independent region-level inference in parallel, eliminating

643 the inferential bottleneck of centralized architectures. A non-inferential global
644 coordination operator subsequently assembles the resulting regional intervention
645 fields through spatial assignment and resource normalization, without performing
646 fuzzy inference itself. This separation preserves local responsiveness and scalability
647 while ensuring global spatial consistency and resource feasibility.

648 **C3 Epistemic Uncertainty Representation**

649 To address G3, the proposed framework incorporates fuzzy inference mechanisms
650 to represent epistemic uncertainty through linguistic variables and membership
651 functions. This allows the system to reason under incomplete or ambiguous
652 information conditions without requiring precise probability distributions.

653 **C4 Explainable Decision Support**

654 To address G4, DisasterAware integrates explainable fuzzy reasoning with an AI-
655 based governance layer. Fuzzy rules provide transparent and traceable reasoning
656 mechanisms, while the governance layer delivers traceability, post-hoc auditability,
657 and operator override mechanisms without replacing human decision authority.

658 **C5 Separation Between Simulation and Decision Logic**

659 To address G5, DisasterAware enforces a strict architectural separation between the
660 Simulation and DSS cores, ensuring traceability, modularity, and independent
661 validation of each component. The simulation layer models wildfire dynamics
662 through structured state transitions, while the DSS observes simulation outputs and
663 generates intervention recommendations without modifying internal state variables.

664 **C6 Integration of Operational Constraints**

665 To address G6, DisasterAware incorporates cost-aware fuzzy decision evaluation
666 that integrates operational constraints including resource availability, travel time,
667 and suppression feasibility directly into the reasoning process. This enables
668 distributed evaluation of intervention feasibility at the local agent level, ensuring that
669 interventions remain operationally executable under dynamic resource constraints.

670	C7 Spatio-Temporal Decision Reasoning
671	To address G7, the DisasterAware framework introduces spatio-temporal fuzzy
672	reasoning mechanisms that combine spatial context awareness with temporal
673	decision evolution. This enables the system to reason about how wildfire risk
674	propagates across regions over time, supporting intervention decisions that are both
675	spatially targeted and temporally appropriate.
676	C8 Multi-Level Decision Coordination
677	To address G8, DisasterAware instantiates a level-differentiated architecture in
678	which each decision level is realized by a distinct mechanism: parallel evaluation of
679	a shared local fuzzy policy at the operational level, non-inferential global assembly
680	at the tactical level, and AI-governed oversight at the strategic level. Feedback across
681	these levels keeps decisions locally responsive, globally consistent, and strategically
682	supervised.
683	Collectively, these contributions define DisasterAware not as a single decision
684	algorithm, but as an integrated DSS paradigm in which intervention guidance
685	emerges from the coordinated interaction of concept-based reasoning, distributed
686	local inference, non-inferential global coordination, and physically separated
687	simulation dynamics.
688	The eight structural gaps G1–G8 identified above are not uniformly distributed
689	across the wildfire DSS literature. Individual frameworks address some gaps while
690	remaining constrained by others, and no single existing system resolves all eight in
691	combination. Table 1.3 positions six representative frameworks against the identified
692	gaps, selected to span the principal methodological families currently in operational
693	use or active academic development: an operationally deployed incident support
694	platform (WFDSS), a widely used fire-behavior simulation system (FlamMap), a
695	recent multi-criteria optimization toolkit (MORA), a real-time operational DSS
696	(Figueiredo, Macedo, Alves, & Figueiredo, 2024), a deep-learning–fuzzy hybrid

697 prediction framework (Mohammadi, Alesheikh, & Alimohammadi, 2025), and the
698 proposed DisasterAware framework.

699 The comparison in Table 1.3 confirms that no existing framework simultaneously
700 addresses all eight structural gaps. Operationally deployed systems such as WFDSS
701 (Noonan-Wright, et al., 2011) offer mature incident-support capabilities but remain
702 constrained by closed decision spaces, centralized reasoning, and tight coupling
703 between fire-behavior modeling and decision documentation. Pure fire-behavior
704 simulators such as FlamMap (Finney, 2006) provide high-fidelity physical modeling
705 but do not constitute decision support systems in the operational sense, as they lack
706 structured decision-generation mechanisms and are therefore marked as non-
707 applicable across most gaps. Recent optimization-based toolkits such as MORA
708 (Orphanoudakis, Betzelos, & Leligou, 2025) advance resource-allocation reasoning
709 but retain a centralized, closed-alternative paradigm with limited explainability.
710 Real-time DSS implementations (Figueiredo, Macedo, Alves, & Figueiredo, 2024)
711 improve responsiveness under dynamic conditions but do not integrate concept-level
712 abstraction or distributed coordination. The hybrid fuzzy–transformer framework of
713 Mohammadi et al. (2025) advances explainable AI for wildfire management, but its
714 focus remains on prediction rather than on the generation of structured, operationally
715 feasible intervention fields under partial observability. Against this landscape,
716 DisasterAware is positioned not as superior along any single dimension, but as the
717 first framework to resolve all eight structural gaps within a single coherent
718 architecture, as summarized in the bottom row of Table 1.3. This combined coverage
719 directly motivates the research question and hypotheses formulated in Section 1.2.

720 Table 1.3 Structural comparison of representative wildfire DSS frameworks

Framework	G1	G2	G3	G4	G5	G6	G7	G8
WFDSS	○	○	○	●	○	●	●	●
FlamMap	—	—	○	○	—	○	●	—
MORA	○	○	●	○	●	●	●	○
Figueiredo et al.	○	●	●	●	●	●	●	○

Mohammadi et al.	○	○	●	◐	○	○	●	○
DisasterAware	●	●	◐	●	●	●	●	◐

Legend: ● Addressed · ◐ Partial · ○ Not targeted · – Not applicable

The assessment in Table 1.3 reflects design intent and architectural coverage rather than operational validation maturity. DisasterAware receives partial support on G3 because its non-probabilistic treatment of epistemic uncertainty via confidence-gated fuzzy modulation represents one of several principled paradigms; Bayesian, Dempster-Shafer, and type-2 fuzzy approaches remain complementary rather than dominated. Partial support on G8 reflects the two-level local–global coordination structure, which does not yet instantiate the strategic–tactical–operational triad canonical in hierarchical decision theory (Anthony, 1965). WFDSS, although constrained by closed decision spaces, retains mature spatio-temporal and resource-integration capabilities, which is reflected in its G6 and G7 ratings.

1.4 Organization of the Thesis

Chapter 2 presents the formal problem definition of wildfire DSS. It describes the operational challenges of wildfire response and identifies key decision requirements arising from dynamic environmental conditions, uncertain observations, and limited operational resources. The chapter formalizes the system state, observation process, decision variables, and feasibility conditions, and concludes with a unified spatio-temporal problem statement that bounds the scope of the proposed framework.

Chapter 3 provides the theoretical background for the proposed methodology. It introduces the core principles of DSS and presents fuzzy logic as a reasoning paradigm for reasoning under uncertainty. It discusses the limitations of classical multi-criteria decision-making approaches, examines the temporal and spatial structure of wildfire decision problems, and reviews centralized versus distributed DSS architectures and their implications for the proposed design.

Chapter 4 introduces the DisasterAware Simulation Framework, which forms the computational foundation of the system. It presents the overall system architecture, the spatial grid-based representation of wildfire dynamics, and the mathematical formulation of state evolution, including the hybrid fire spread model, external data layers, and the UI dashboard for visualization and interaction.

Chapter 5 establishes the formal design specification of DisasterAware. It derives the six-feature input space from Chapter 4 state variables, specifies the fuzzy variable design and membership functions, defines the four situational concepts and their construction mappings, and provides a formal proof of the ninefold rule complexity reduction. It specifies the intervention decision rule base across intervention types and the action aggregation function that composes them into a final decision.

Chapter 6 presents the DisasterAware architecture. It describes the layered decision structure comprising the Observation, Concept Mapping and Abstraction, Local DSS Model, and Global Coordination layers. The chapter details the execution of the epistemic encoding of partial observability, concept mapping pipeline, and its formal properties namely, output boundedness, partition of unity, and Lipschitz continuity. It also introduces the Explainability and AI Governance Layer as a supervisory overlay providing traceability, auditability, and operator override mechanisms.

Chapter 7 presents the empirical evaluation of DisasterAware against a functionally equivalent centralized DSS baseline. Three test scenarios of increasing complexity evaluate the stated hypotheses across decision quality, adaptability, interpretability, and operational feasibility, followed by a sensitivity analysis over the feature space, membership functions, and priority weights.

Finally, Chapter 8 summarizes the main contributions of the study and outlines directions for future research.

Together, these chapters form a coherent argument: from problem formulation to theoretical grounding, from system design to formal specification, and from empirical evaluation to synthesis, each layer building on the one before it.

773

CHAPTER 2

774

PROBLEM DEFINITION

775 This chapter formally defines the wildfire DSS problem addressed in this thesis. The
776 formulation captures wildfire dynamics, partial observability, and operational
777 intervention constraints within a unified spatio-temporal framework. The chapter
778 introduces the system state, observation process, decision variables, and feasibility
779 conditions, and concludes with a formal problem statement. The objective of this
780 chapter is to establish a mathematically consistent formulation of wildfire decision
781 support under partial observability while explicitly separating physical system
782 dynamics from decision-generation mechanisms.

783 Section 2.1 frames wildfire response as a dynamic decision problem shaped by
784 interacting environmental processes, operational interventions, and changing system
785 conditions.

786 Section 2.2 defines the wildfire system state representation and the corresponding
787 observation model under partial observability. It distinguishes between the true
788 system state and the observable information available to the DSS, and formally
789 introduces the observation mapping, explicitly incorporating epistemic uncertainty
790 arising from incomplete, delayed, and noisy measurements.

791 Section 2.3 formalizes the decision variables and the feasible intervention space,
792 establishing how operational actions are represented as structured intervention
793 inputs. It further characterizes the role of epistemic uncertainty in shaping admissible
794 decisions and constraining the feasible intervention space, emphasizing that
795 decision-making is performed over incomplete and evolving information rather than
796 fully known system states.

797 Section 2.4 presents the multi-level and spatially distributed decision structure, in
798 which region-specific local decisions are generated based on localized observations
799 and contextual inputs, and subsequently integrated through a global coordination
800 mechanism. This section establishes the formal distinction between local decision
801 generation and global decision consistency, ensuring both spatial responsiveness and
802 system-level coherence.

803 Section 2.5 provides the unified formal problem statement, integrating the state
804 transition model, observation process, and DSS mapping into a consistent spatio-
805 temporal framework. It defines wildfire decision support as a distributed, partially
806 observable, and constraint-driven intervention-generation process, explicitly
807 distinguishing it from classical optimization-based or control-theoretic formulations.

808 Section 2.6 concludes by defining the modeling scope, assumptions, and abstraction
809 boundaries of the proposed framework. It specifies the structural separation between
810 physical system dynamics and decision-generation, clarifies assumptions related to
811 spatial discretization, temporal evolution, uncertainty representation, and operational
812 constraints, and establishes the formal conditions that enable the simulation and DSS
813 architectures developed in subsequent chapters.

814 2.1 Wildfire Response as a Dynamic Decision Problem

815 Wildfire response is formulated as a spatio-temporal decision process over a
816 evolving physical system governed by environmental forcing and operational
817 interventions. Unlike static planning problems, it requires sequential decision-
818 making under time-varying conditions, limited resources, and partial observability.
819 The wildfire environment is defined over a spatial domain $\Omega \subset \mathbb{R}^2$, discretized into
820 grid cells indexed by (x, y) . Within the computational framework introduced in
821 Chapter 4, this domain is represented as a structured grid $\mathcal{G}(x, y)$, enabling spatially
822 distributed state evolution.

At each discrete time step k , the system is characterized by internal variables such as burning status, available fuel mass, fire intensity, and time since ignition. These variables define the system state $S_k(x, y)$, representing the spatial distribution of wildfire conditions. Environmental and operational factors are explicitly excluded from the system state and are instead modeled as external inputs, ensuring a strict separation between physical dynamics and exogenous influences.

All external influences on wildfire dynamics are represented through the structured input set $\mathcal{F}_{in,k}$, which includes meteorological inputs $U_{Meteo,k}$, terrain-related inputs U_{Geo} , fuel-related inputs $U_{Fuel,k}$, ignition inputs $U_{Ign,k}$, and DSS-generated intervention inputs $U_{DSS,k}$. These inputs act as boundary conditions governing the evolution of the wildfire system. In addition to $\mathcal{F}_{in,k}$, the DSS process relies on a contextual input set $\mathcal{F}_{DSS,k}$, which consists of value-related inputs $U_{Val,k}$ and resource-related inputs $U_{Res,k}$. Unlike $\mathcal{F}_{in,k}$, which directly affects physical state evolution, $\mathcal{F}_{DSS,k}$ does not act on wildfire dynamics itself; instead, it constrains and informs the decision-generation process.

The evolution of the wildfire system is governed by the state transition: $S_{k+1}(x, y) = \Phi(S_k(x, y), \mathcal{F}_{in,k})$ where $\Phi(\cdot)$ denotes the state transition operator encoding fire propagation dynamics as a function of environmental conditions and intervention effects. Within this formulation, the DSS does not directly control the system state. Instead, it generates structured intervention inputs that influence system evolution indirectly through the external input layer. Wildfire response is therefore modeled as a sequential and partially observable intervention-generation process in which decisions are produced over time under evolving environmental conditions, incomplete information, and operational constraints.

The structural relationships among the physical system, observation, decision reasoning, and intervention introduced above are summarized graphically in Figure 2.1. The formal problem statement in Sections 2.2–2.5 builds on this structure.

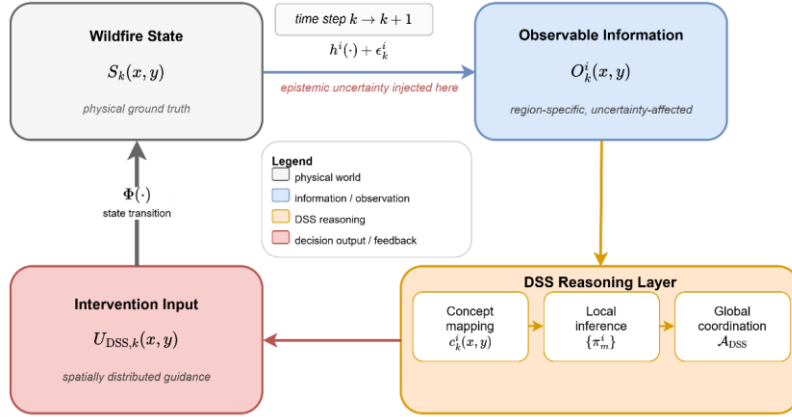


Figure 2.1 Closed-loop decision problem structure

Figure 2.1 shows the four structural components of the wildfire decision process and the causal relationships that link them. The physical system evolves according to the state transition operator $\Phi(\cdot)$, producing the wildfire state $S_k(x, y)$ at each discrete time step k . The observation mapping $h^i(\cdot)$ projects this state onto region-specific observable information $O_k^i(x, y)$ subject to epistemic uncertainty $\epsilon_k^i(x, y)$ that is injected exclusively at this stage. The DSS Reasoning Layer transforms the observations into the global intervention field $U_{DSS,k}(x, y)$ through three conceptual stages: concept mapping producing the concept vector $c_k^i(x, y)$, local inference producing the per-region decision $U_{DSS,k}^i(x, y)$, and global coordination producing the spatially complete intervention $U_{DSS,k}(x, y)$. The intervention re-enters the physical system through the external input set $F_{in,k}$ and advances the state via $\Phi(\cdot)$, closing the loop.

Two structural properties of this formulation are central to the remainder of the thesis: the DSS never accesses $S_k(x, y)$ directly, reasoning exclusively from $O_k^i(x, y)$, and it never modifies $S_k(x, y)$ directly, influencing evolution only through the external input layer. The detailed realisation of the DSS Reasoning Layer,

including its full pipeline, is deferred to Chapters 5 and 6; this chapter uses the closed-loop abstraction to derive the formal problem statement in Section 2.5.

2.2 System State and Observable Quantities

While the system state $S_k(x, y)$ represents the underlying physical condition of the wildfire environment, it is not directly observable to the DSS. Instead, situational awareness is obtained through observable information derived from sensors, remote observations, and operational reports.

The observable information at time step k is denoted by $O_k(x, y)$, representing a set of measurable or inferable quantities associated with each grid cell (x, y) . This observable information is modeled as a function of the underlying system state: $O_k(x, y) = h(S_k(x, y), \epsilon_k(x, y))$. The observation mapping $h: \mathcal{S} \rightarrow \mathcal{O}$ is a nonlinear transformation that extracts observable features from the underlying system state, potentially including spatial aggregation, filtering, and sensor-induced transformations. In addition, $\epsilon_k(x, y)$ represents epistemic uncertainty arising from incomplete, delayed, and noisy observations.

The uncertainty term $\epsilon_k(x, y)$ captures epistemic disturbance caused by sensing limitations, spatial coverage gaps, and reporting delays. This uncertainty is not assumed to follow a known probabilistic distribution. Instead, it represents structural incompleteness of available information and cannot be reduced through repeated measurement within a single time step.

Within this structure, the wildfire decision process can be expressed as a sequential information–decision mapping: $S_k(x, y) \rightarrow (O_k(x, y), \mathcal{F}_{DSS,k}) \rightarrow U_{DSS,k}(x, y)$. The DSS process operates on observable information $O_k(x, y)$ together with contextual inputs $\mathcal{F}_{DSS,k}$. At this stage, the DSS mapping π_{DSS} is introduced only in abstract form as a local decision-generation mapping from observable information and contextual inputs to region-specific intervention outputs.

894 2.3 Decision Variables, Uncertainty, and Feasible Intervention Space

895 The decision variable at time step k is denoted by $U_{DSS,k}(x, y)$, representing the
896 intervention-related parameters generated by DSS. The decision variable is defined
897 at the regional level as $U_{DSS,k}^i(x, y)$. The global intervention field $U_{DSS,k}(x, y)$ is
898 obtained through aggregation of all regional decisions. At the global level,
899 $U_{DSS,k}(x, y)$ denotes the coordinated intervention field obtained after local regional
900 decisions have been generated and aggregated.

901 The contextual input set is defined as $\mathcal{F}_{DSS,k} = \{U_{Val,k}, U_{Res,k}\}$. It provides value and
902 resource-related data that constrains feasible intervention space. The admissible
903 intervention configurations are: $U_{DSS,k}(x, y) \in \mathcal{U}_{DSS}(O_k(x, y), \mathcal{F}_{DSS,k})$. The feasible
904 intervention space $\mathcal{U}_{DSS}$ is a bounded, context-dependent set defined by operational
905 constraints including resource availability, accessibility, and temporal feasibility.

906 Formally, $\mathcal{U}_{DSS} \subseteq [0,1]^M$ where each dimension corresponds to an intervention type.
907 Decision-making is performed under epistemic uncertainty, since the feasible
908 intervention space depends on incomplete and evolving observable information:
909 $S_k(x, y) \rightarrow O_k(x, y) \rightarrow U_{DSS,k}(x, y)$, $U_{DSS,k}(x, y) \subset \mathcal{F}_{in,k}$. The generated
910 intervention field is designed to be compatible with the external input structure
911 defined in the simulation framework. The feasible intervention space defines
912 admissible actions but does not address their spatial organization.

913 2.4 Multi-Level Decision Requirements in Wildfire Management

914 Wildfire response decisions are spatially distributed and must be generated across
915 multiple regions of the operational domain. The spatial domain Ω is partitioned into
916 N disjoint regions Ω_i , such that: $\Omega = \bigcup_{i=1}^N \Omega_i$, $\Omega_i \cap \Omega_j = \emptyset$ ($i \neq j$). Each region Ω_i
917 represents a localized operational area characterized by region-specific wildfire
918 conditions, observable information, and contextual constraints.

919 Since the global observation process $h(\cdot)$ defined in Section 2.2 operates over the
 920 entire spatial domain Ω , the regional decomposition of the decision problem requires
 921 a corresponding decomposition of the observation process. The region-specific
 922 observation function $h^i(\cdot)$ is defined as the restriction of $h(\cdot)$ to the spatial
 923 subdomain Ω_i , producing region-specific observable information as $O_k^i(x, y) =$
 924 $h^i(S_k(x, y), \epsilon_k^i(x, y))$, where $\epsilon_k^i(x, y)$ denotes the epistemic uncertainty associated
 925 with region Ω_i at time step k .

926 Global observation process is recovered as the union of all regional observations:
 927 $O_k(x, y) = h(S_k(x, y), \epsilon_k) = \sum_{i=1}^N \mathbb{1}_{(x, y) \in \Omega_i} \cdot h^i(S_k(x, y), \epsilon_k^i)$. This decomposition
 928 ensures that each local DSS agent reasons exclusively from its own regionally
 929 restricted observation, preserving spatial locality of information access.

930 At time step k , the observable information and contextual inputs associated with
 931 region Ω_i are denoted by $O_k^i(x, y)$ and $\mathcal{F}_{DSS, k}^i$, respectively. Based on this
 932 information, region-specific intervention parameters are generated locally as:
 933 $U_{DSS, k}^i(x, y) = \pi_{DSS}^i(O_k^i(x, y), \mathcal{F}_{DSS, k}^i)$. This formulation reflects the requirement
 934 that decisions must be generated locally in response to spatially heterogeneous
 935 conditions. Fire behavior, resource availability, and asset exposure vary across the
 936 domain, making globally uniform decisions both infeasible and operationally
 937 ineffective. The mapping π_{DSS}^i is shared across all regions and represents a common
 938 decision policy applied to region-specific inputs. Spatial variability in decisions
 939 arises from differences in local observations and contextual inputs, rather than from
 940 region-specific policy definitions.

941 However, local decision generation alone is insufficient. Wildfire response involves
 942 shared and limited resources, and decisions made independently across regions may
 943 lead to conflicts, inconsistencies, or inefficient resource allocation. For example,
 944 multiple regions may simultaneously require the same limited resources, or local
 945 decisions may conflict with broader containment or protection priorities.

946 To address this, a global coordination mechanism is introduced to integrate region-
 947 specific decisions into a coherent intervention strategy. The global decision structure
 948 is defined as: $U_{DSS,k}(x, y) = \mathcal{A}_{DSS} \left(U_{DSS,k}^1(x, y), \dots, U_{DSS,k}^N(x, y) \right)$ where $\mathcal{A}_{DSS}(\cdot)$
 949 denotes the spatial coordination and aggregation operator that integrates regional
 950 decision outputs into a globally consistent intervention configuration.

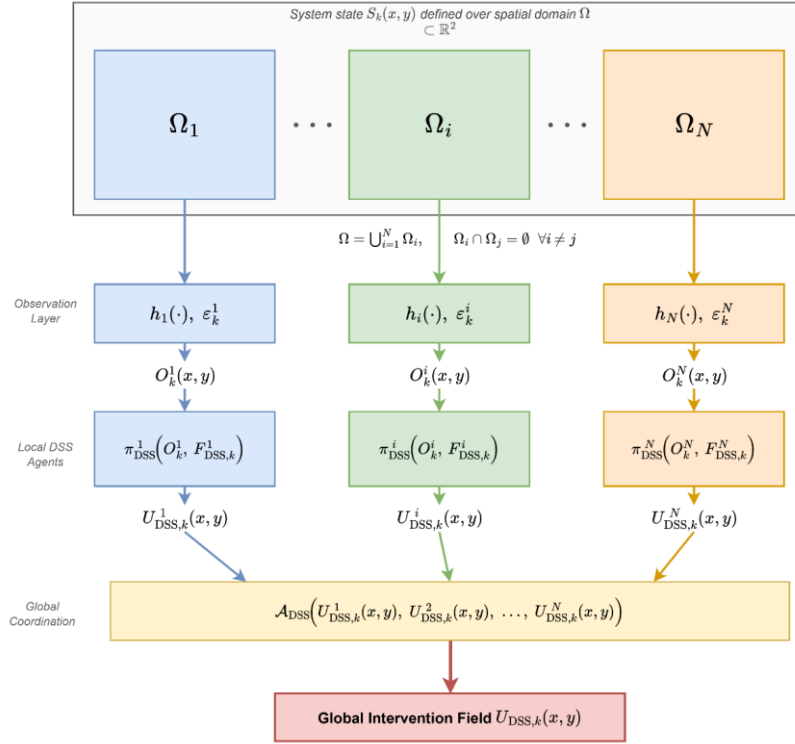
951 The mapping π_{DSS}^i operates strictly at regional level and generates region-specific
 952 intervention outputs on the basis of local observations and contextual inputs. It does
 953 not define the global intervention field directly. Global decision consistency is
 954 established only after the regional outputs are integrated through aggregation $\mathcal{A}_{DSS}$.

955 In addition, decisions within each region are not restricted to a single action. Instead,
 956 multiple intervention types may be activated simultaneously with varying degrees of
 957 priority. The regional decision output is represented as a multi-action structure:
 958 $U_{DSS,k}^i(x, y) = \{u_{k,1}^i(x, y), u_{k,2}^i(x, y), \dots, u_{k,M}^i(x, y)\}$, $u_{k,m}^i \in [0,1]$ where $u_{k,m}^i$
 959 denotes the priority or activation level of intervention type m in region Ω_i , and M
 960 represents the number of intervention types considered within the framework. This
 961 representation enables the activation of multiple intervention strategies, reflecting
 962 the non-exclusive and context-dependent nature of wildfire response actions.

963 The overall decision structure hence consists of three coupled components: local
 964 decision generation based on region-specific information, multi-action
 965 representation of intervention strategies, and global coordination ensuring
 966 consistency across regions. This multi-level formulation ensures that the DSS
 967 maintains both local responsiveness and global coherence under dynamic, resource-
 968 constrained, and partially observable conditions.

969 Figure 2.2 depicts the hierarchically distributed decomposition in three layered
 970 structural components. The spatial partition $\Omega = \cup_{i=1}^N \Omega_i$ is shown with three
 971 representative regions Ω_1, Ω_i , and Ω_N . In the observation layer, each $h_i(\cdot)$ operates
 972 on the restriction of $S_{k(x,y)}$ to Ω_i and injects the region-specific epistemic uncertainty
 973 ε_k^i , producing the regional observable $O_k^i(x, y)$. The local DSS agents apply the

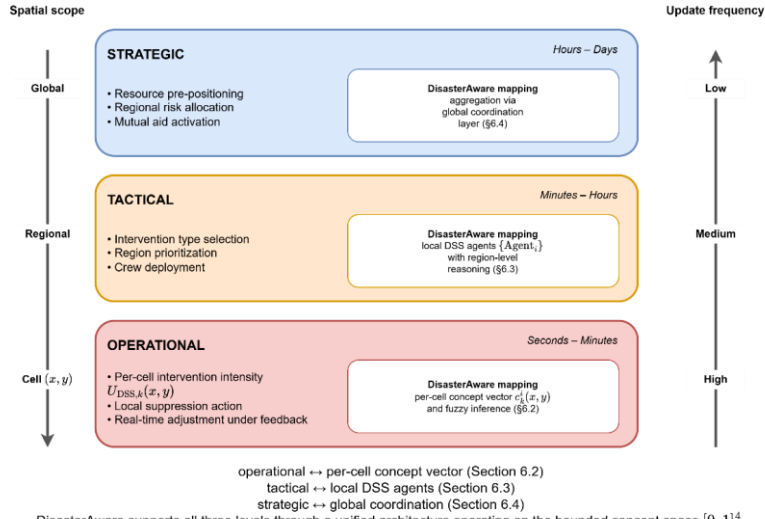
974 shared policy π_{DSS}^i to its region-specific observation and contextual input to yield
 975 the local intervention; $U_{DSS,k}^i(x,y)$ the global coordination operator $\mathcal{A}_{DSS}$ then
 976 aggregates all N regional outputs into the spatially coherent intervention field
 977 $U_{DSS,k}(x,y)$.



978

979 Figure 2.2 Spatial observations, local decision mappings, and global aggregation

980 The three decision levels that the distributed structure introduced above must
 981 accommodate, together with their temporal and spatial scales and the DisasterAware
 982 architectural elements that realize them, are summarized in Figure 2.3.



983

984 Figure 2.3 Multi-level decision hierarchy for wildfire management

985 Wildfire response decisions operate across three distinct temporal and spatial scales.
986 At the *strategic* level, decisions span hours to days and cover the full spatial domain
987 G : resource pre-positioning, regional risk allocation across the N regions $\{\Omega_i\}_{i=1}^N$,
988 and mutual aid activation. At the *tactical* level, decisions span minutes to hours and
989 operate at the region scale Ω_i : intervention type selection among the M available
990 types, region prioritisation, and crew deployment. At the *operational* level, decisions
991 span seconds to minutes and operate at the individual grid cell (x, y) : per-cell
992 intervention intensity $U_{DSS,k}(x, y)$, local suppression action, and real-time
993 adjustment under feedback propagated through the closed loop of Figure 2.1.

994 Figure 2.1 shows how DisasterAware addresses all three levels through a single
995 unified architecture operating on the bounded concept space $[0, 1]^4$. The operational
996 level is realized by the per-cell concept vector $c_k^i(x, y)$ and fuzzy inference of
997 Section 6.2; the tactical level by the local DSS agents of Section 6.3; and the strategic
998 level by the global coordination operator $\mathcal{A}_{DSS}$ of Section 6.4. The spatial scope
999 arrow on the left of the figure (Global → Regional → Cell) and the update frequency

arrow on the right (Low $\rightarrow$ Medium $\rightarrow$ High) move in opposite directions: coarser spatial scopes admit slower update cycles, while finer spatial scopes require more frequent updates. This inverse relationship is not an incidental design choice but a structural property of the wildfire decision problem that any admissible DSS architecture must respect, motivating the layered realization of DisasterAware developed in Chapter 6.

2.5 Formal Problem Statement

This section formulates the wildfire DSS problem as a discrete-time, partially observable, and spatially distributed decision process over an evolving wildfire environment. The formulation is consistent with the Simulation Framework presented in Chapter 4 and preserves the structural separation between system dynamics, observation, and decision generation.

The wildfire system evolves over the spatial domain according to: $S_{k+1}(x, y) = \Phi(S_k(x, y), \mathcal{F}_{in,k})$ where $S_k(x, y)$ denotes the system state and $\mathcal{F}_{in,k}$ represents the structured external input set. Since the system state is not directly observable, decision-making therefore relies on observable information, modeled as $O_k(x, y) = h(S_k(x, y), \epsilon_k(x, y))$ where $h(\cdot)$ is the observation process and $\epsilon_k(x, y)$ represents epistemic uncertainty associated with incomplete and imperfect observation.

Based on observable information and contextual inputs $\mathcal{F}_{DSS,k}$, decision generation is performed in a hierarchically distributed manner over spatial regions Ω_i . Region-specific intervention outputs are generated as: $U_{DSS,k}^i(x, y) = \pi_{DSS}^i(O_k^i(x, y), \mathcal{F}_{DSS,k}^i)$. Each regional decision is represented as a multi-action structure: $U_{DSS,k}^i(x, y) = \{u_{k,1}^i(x, y), u_{k,2}^i(x, y), \dots, u_{k,M}^i(x, y)\}$, $u_{k,m}^i \in [0, 1]$ where $u_{k,m}^i(x, y)$ denotes the activation level of intervention type m in region Ω_i , and M is the number of intervention types.

1025 Formally, the regional DSS mapping is defined as $\pi_{DSS}^i: (O_k^i, \mathcal{F}_{DSS,k}^i) \rightarrow U_{DSS,k}^i$,
 1026 where O_k^i denotes region-specific observable information and $\mathcal{F}_{DSS,k}^i$ denotes the
 1027 corresponding contextual decision inputs. The explicit five-stage realization of π_{DSS}^i
 1028 is established in Chapter 5.

1029 Global coordination integrates regional decisions into a consistent intervention field:
 1030 $U_{DSS,k}(x, y) = \mathcal{A}_{DSS}(U_{DSS,k}^1(x, y), \dots, U_{DSS,k}^N(x, y))$ where $\mathcal{A}_{DSS}(\cdot)$ is the
 1031 aggregation operator enforcing consistency across regions under shared constraints.

1032 The DSS mapping π_{DSS}^i operates strictly at the regional level and generates region-
 1033 specific intervention outputs. It does not define the global intervention field directly.
 1034 The global intervention field is constructed exclusively through the aggregation
 1035 operator $\mathcal{A}_{DSS}(\cdot)$. The generated intervention field is subject to feasibility and
 1036 structural constraints: $U_{DSS,k}(x, y) \in \mathcal{U}_{DSS}(O_k(x, y), \mathcal{F}_{DSS,k})$, $U_{DSS,k}(x, y) \subset \mathcal{F}_{in,k}$.
 1037 These constraints ensure that decisions remain operationally admissible and
 1038 influence system evolution only through the external input layer.

1039 Decision quality is evaluated through a multi-criteria function: $Q_{DSS,k} =$
 1040 $f_{eval}(O_k(x, y), \mathcal{F}_{DSS,k}, U_{DSS,k}(x, y))$ with the satisfaction condition: $Q_{DSS,k} \geq \eta_k$
 1041 where $\eta_k \in [0,1]$ is a time-varying decision satisfaction threshold.

1042 The decision quality function is not defined as a classical optimization objective, but
 1043 as a multi-criteria evaluation measure used to assess the admissibility and
 1044 effectiveness of generated intervention configurations. Furthermore, the function
 1045 f_{eval} is not part of the intervention-generation mapping π_{DSS} . It operates as a post-
 1046 decision evaluation mechanism computed exclusively from observable information
 1047 O_k , contextual inputs $\mathcal{F}_{DSS,k}$, and the generated intervention field $U_{DSS,k}$, thereby
 1048 preserving the structural principle that no DSS component accesses the true state S_k
 1049 directly.

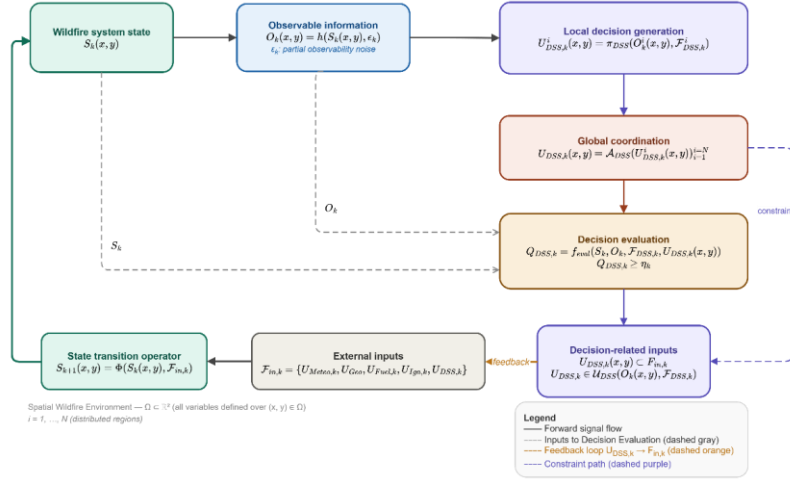


Figure 2.4 Conceptual architecture of the DisasterAware DSS

The overall structure of the DSS, including the interaction between system dynamics, observation, and decision generation, is illustrated in Figure 2.4. The system operates as a closed-loop, hierarchically distributed decision pipeline over a spatial wildfire environment $\Omega \subset \mathbb{R}^2$. At each time step k , the wildfire system state $S_k(x, y)$ is fed into an observation function $h(\cdot)$, which produces the observable information $O_k(x, y)$ under the influence of partial observability noise $\epsilon_k(x, y)$, acknowledging that the true system state is never fully accessible.

This observable data is then passed to the local decision generation stage, where N spatially distributed DSS agents each independently produce a region-specific intervention input $U_{DSS,k}^i$ through their local policy π_{DSS}^i , conditioned on their respective observations and contextual constraints. These local decisions are subsequently aggregated by the global coordination operator $\mathcal{A}_{DSS}$ into a single, globally consistent intervention field $U_{DSS,k}$.

The resulting global decision is then assessed by the decision evaluation function f_{eval} , which takes the observable information, contextual factors, and the generated intervention as inputs, and checks whether the multi-criteria decision quality score

1068 $Q_{DSS,k}$ meets the minimum satisfaction threshold η_k . The generated decisions feed
 1069 into the decision block, which enforces two structural constraints: the intervention
 1070 must remain a subset of the external input space $\mathcal{F}_{in,k}$, and must lie within the
 1071 feasibility set $\mathcal{U}_{DSS}$. $U_{DSS,k}$ itself is also a component of $\mathcal{F}_{in,k}$, creating a feedback
 1072 loop through which decisions re-enter the system as external inputs alongside
 1073 meteorological, geographical, and fuel-related data.

1074 These combined external inputs drive the state transition operator $\Phi(\cdot)$, which
 1075 advances the system to its next state $S_{k+1}(x, y)$, closing the decision loop. Taken
 1076 together, the architecture formalizes wildfire decision support not as a classical
 1077 optimization problem, but as a distributed, constraint-driven, and uncertainty-aware
 1078 intervention generation process under partial observability. The mathematical
 1079 formulation of the wildfire DSS problem is summarized in Equation (1).

$$\begin{aligned} S_{k+1}(x, y) &= \Phi(S_k(x, y), \mathcal{F}_{in,k}) \\ O_k(x, y) &= h(S_k(x, y), \epsilon_k^i(x, y)) \end{aligned}$$

Local decision generation:

$$U_{DSS,k}^i(x, y) = \pi_{DSS}^i(O_k^i(x, y), \mathcal{F}_{DSS,k}^i)$$

Global coordination:

$$U_{DSS,k}(x, y) = \mathcal{A}_{DSS} \left(U_{DSS,k}^1(x, y), \dots, U_{DSS,k}^N(x, y) \right) \quad (1)$$

subject to

$$\begin{aligned} U_{DSS,k}(x, y) &\subset \mathcal{F}_{in,k} \\ U_{DSS,k}(x, y) &\in \mathcal{U}_{DSS}(O_k(x, y), \mathcal{F}_{DSS,k}) \end{aligned}$$

Decision evaluation:

$$\begin{aligned} Q_{DSS,k} &= f_{eval} \left(O_k(x, y), \mathcal{F}_{DSS,k}, U_{DSS,k}(x, y) \right) \\ Q_{DSS,k} &\geq \eta_k \end{aligned}$$

Each equation in (1) corresponds to a distinct structural component of the DSS: state evolution, observation, local decision generation, and global coordination. In detail, the equation (1) states that wildfire decision support is posed as the generation of feasible intervention inputs from uncertain observations of an evolving physical system, under region-specific constraints and global coordination requirements. This interpretation is central to the thesis because it distinguishes the problem from classical optimization-based control and establishes the formal basis for the hierarchically distributed DSS architecture developed in DisasterAware.

The formulated DSS problem does not conform to classical optimal control or Markov decision process frameworks. The decision space is not explicitly enumerable, and the system operates under partial observability, where the true state is not fully accessible. Furthermore, the decision process involves evolving constraints and multi-criteria evaluation that cannot be consistently reduced to a single global objective function. Accordingly, decision-making is not defined as the optimization of a predefined objective over a fixed action space, but as the generation of feasible and context-dependent intervention configurations under uncertainty. This perspective defines the structural nature of the problem and distinguishes it from classical decision-theoretic formulations.

Remark 2.1 (Computational Intractability of the Wildfire DSS Problem). The problem defined in Equation (1) is computationally intractable under any exact optimization formulation, even when simplified. Consider a discretized variant in which each intervention type $m \in \{1, \dots, M\}$ admits L discrete activation levels per grid cell. The joint decision space at a single time step has cardinality $L^{|G| \times M}$, which grows exponentially in both grid resolution and intervention dimensionality. For a modest configuration of $|G| = 2500$ cells (50×50), $M = 3$ intervention types, and $L = 5$ levels, the single-step action space exceeds 5^{7500} , a number that renders exhaustive search, branch-and-bound, or dynamic programming infeasible by many orders of magnitude.

Partial observability compounds this intractability. Since the true system state $S_k(x,y)$ is not directly accessible, optimal decision-making would require maintaining and updating a belief distribution over the full state space at every time step, a requirement that is well known to be PSPACE-hard even for finite-state partially observable Markov decision processes (Papadimitriou & Tsitsiklis, 1987).

The wildfire DSS problem further departs from standard POMDP assumptions: the state space is continuous, the observation noise $\epsilon_k^i(x,y)$ lacks a known distributional form, the feasibility set $\mathcal{U}_{DSS}$ evolves dynamically with resource depletion and fire progression, and the evaluation function f_{eval} is multi-criteria rather than scalar-additive. These properties collectively establish that the wildfire DSS problem, as formulated in Equation (1), does not admit tractable exact solutions under any known computational framework. This intractability provides the formal motivation for the satisfaction-based decision paradigm adopted in this thesis: rather than seeking a globally optimal intervention policy, the framework generates feasible and context-sensitive intervention configurations that satisfy the quality condition $Q_{DSS,k} \geq \eta_k$ at each time step, replacing an intractable global optimization with a tractable local satisficing process.

2.6 Scope and Assumptions

The formulation presented in this chapter defines a structured abstraction of wildfire decision support under partial observability. To ensure mathematical consistency, computational tractability, and alignment with the Simulation Framework, a set of modeling assumptions is introduced. These assumptions explicitly define the scope and limitations of the proposed framework. The system is modeled as a discrete-time, grid-based spatio-temporal process in which physical dynamics, observation, and decision generation are treated as structurally distinct components. The assumptions summarized in Table 2.1 establish the boundaries of the model and ensure consistency between the problem formulation, the simulation framework, and the DSS architecture developed in subsequent chapters.

1136 Table 2.1 Scope and modeling assumptions of the wildfire DSS formulation

<i>Category</i>	<i>Assumption / Scope Definition</i>
<i>Contextual Input Structure</i>	The contextual input set $\mathcal{F}_{DSS,k}$ includes value-related and resource-related information used for decision evaluation and feasibility constraints.
<i>Decision Paradigm</i>	The problem is not formulated as an optimal control problem but as a constraint-driven, satisfaction-based decision generation.
<i>Decision Structure</i>	Decision-making is spatially distributed across Ω_i , with local decisions coordinated through a global aggregation mechanism.
<i>Decision Variables</i>	Operational interventions are represented by $U_{DSS,k}(x,y)$, as a spatial decision field generated through distributed local decisions and coordinated into a global intervention field.
<i>Execution Synchrony</i>	N regional agents execute their local DSS pipelines synchronously within each discrete time step k . Inter-agent communication occurs through the global coordination layer $\mathcal{A}_{DSS}$; no direct agent-to-agent message passing is modeled.
<i>External Input Structure</i>	All exogenous influences are represented as $\mathcal{F}_{in,k}$, including meteorology, terrain, fuel, ignition, and DSS-generated inputs.
<i>Observation Model</i>	Observable information $O_k(x,y)$ represents the system state partially and uncertainly through the observation process $h(\cdot)$.
<i>Observation Noise Boundedness</i>	The epistemic uncertainty term $\epsilon_k^i(x,y)$ in the observation model is assumed to be bounded such that $O_k^i(x,y) \in [0,1]^4$ for all $(x,y) \in \Omega_i$ and all k . No distributional assumptions (e.g., Gaussianity) are imposed on $\epsilon_k^i(x,y)$; the fuzzy reasoning mechanism is designed to operate without such assumptions.

<i>Category</i>	<i>Assumption / Scope Definition</i>
<i>Operational Constraints</i>	Feasible decisions are constrained by resource availability, suppression capacity, terrain accessibility, and travel-time.
<i>Regional Partition Stationarity</i>	The spatial partition $\Omega = \cup_{i=1}^N \Omega_i$ is defined at initialization and remains fixed throughout the simulation horizon. Dynamic repartitioning in response to evolving fire conditions is not supported in the current formulation.
<i>Spatial Representation</i>	The wildfire environment is defined over a discretized spatial domain $\Omega \subset \mathbb{R}^2$, partitioned into grid cells indexed by (x, y) . Each cell is assumed to be internally homogeneous.
<i>System Immutability</i>	DSS does not directly modify the system state. All intervention effects are applied through the external input set $\mathcal{F}_{in,k}$.
<i>System State Definition</i>	The state $S_k(x, y)$ includes only internal wildfire variables such as burning status, fuel load, intensity, and ignition time.
<i>Temporal Representation</i>	System evolution and decision-making are modeled in discrete time steps indexed by k .
<i>Uncertainty Representation</i>	The uncertainty term $\epsilon_k(x, y)$ represents epistemic uncertainty and is not assumed to follow a known probabilistic distribution.

1137

1138 The three assumptions of execution synchrony, regional partition stationarity, and
1139 observation noise boundedness are structural simplifications that enable the formal
1140 guarantees established in Chapters 5 and 6 (specifically, Proposition 5.3 and
1141 Proposition 5.4, and the boundedness of $\mathcal{A}_{DSS}$ in Remark 6.4). Relaxation of these
1142 assumptions, particularly dynamic repartitioning and asynchronous agent execution,
1143 constitutes a direction for future work identified in Section 8.2.

Remark 2.2 (Notational Conventions). The notation used throughout this thesis follows four systematic conventions that reduce the cognitive effort required to parse multi-index expressions.

Convention 1 (Temporal-spatial-regional indexing). Subscript k denotes the time step and (x, y) denotes the grid cell. Superscript i identifies the operational region Ω_i . For example, $c_{k,h}^i(x, y)$ denotes the h -th concept value at cell (x, y) , in region i , at time step k . When context is unambiguous, the spatial argument (x, y) is suppressed for readability.

Convention 2 (Observation hat). A hat accent ($\hat{\cdot}$) denotes an observed estimate of a state variable, as distinct from its true value. For instance, $\hat{B}_k^i$ is the observed burning status in region i , derived from the true state B_k through the observation process $h(\cdot)$.

Convention 3 (Confidence bar). An overbar ($\bar{\cdot}$) denotes a confidence-gated quantity, obtained by blending the current estimate with a persistence-based prior according to the observation confidence coefficient. For example, $\bar{c}_{k,h}^i$ is the confidence-weighted concept value, while $c_{k,h}^i$ is the raw (unweighted) concept value.

Convention 4 (Subscript hierarchy). Function-level symbols use descriptive subscripts separated by commas: $f_{feat}, f_{fuzzy}, f_{concept,h}, f_{agg}, f_{eval}$. Intervention-type-specific quantities are indexed by m (e.g., $u_{k,m}^i$ for the candidate degree of type m). Criterion-specific quantities are indexed by j (e.g., $q_{j,k}$ for the j -th quality criterion at step k).

These conventions ensure that every symbol in the thesis can be decoded from its visual structure alone: the base letter identifies the physical quantity, the subscript identifies the time and context, the superscript identifies the region, and the accent identifies the epistemic status (observed, confidence-gated, or true).

1168

CHAPTER 3

1169

BACKGROUND INFORMATION

1170 This chapter provides the theoretical background for the DisasterAware framework.
1171 It establishes the fundamental principles of DSS, introduces fuzzy logic as a
1172 reasoning paradigm under epistemic uncertainty, and analyzes the limitations of
1173 classical decision-making approaches in dynamic and partially observable
1174 environments. Furthermore, it examines the structural properties of decision
1175 processes in wildfire management, including multi-level decision hierarchies,
1176 distributed architectures, and spatio-temporal decision dependencies. These
1177 foundations provide the theoretical grounding for the DSS design and architecture
1178 presented in Chapters 5 and 6. This chapter is organized as follows.

1179 Section 3.1 introduces the fundamental components of DSS and establishes the
1180 conceptual foundations required for analyzing decision processes in complex and
1181 uncertain operational environments. It defines the core dimensions of DSS design,
1182 including decision structuring, knowledge representation, uncertainty handling,
1183 timeliness, and human-machine interaction.

1184 Section 3.2 presents fuzzy logic as a reasoning paradigm suitable for decision
1185 support under epistemic uncertainty. It outlines the structure of fuzzy inference
1186 systems (FIS), including fuzzification, rule-based knowledge representation,
1187 inference mechanisms, and defuzzification, and explains how fuzzy logic enables
1188 approximate and interpretable reasoning over imprecise inputs.

1189 Section 3.3 analyzes the structural limitations of multi-criteria decision-making
1190 (MCDM) and multi-attribute group decision-making (MAGDM) approaches. It
1191 formalizes their underlying assumptions and demonstrates why these frameworks
1192 are not well suited for dynamic, partially observable, and time-critical environments.

1193 Section 3.4 examines decision levels and hierarchical structures in operational
1194 decision making. It introduces a formal taxonomy based on temporal horizon, spatial
1195 scope, epistemic uncertainty, and automation feasibility, and discusses their
1196 implications for DSS design.

1197 Section 3.5 compares centralized and distributed DSS architectures from both
1198 structural and computational perspectives, focusing on communication complexity,
1199 decision latency, scalability, and robustness under partial observability.

1200 Finally, Section 3.6 presents the spatio-temporal nature of decision making in
1201 wildfire management. It analyzes how spatial heterogeneity, temporal dynamics, and
1202 partial observability jointly constrain decision processes and derives the structural
1203 requirements for DSS operating over dynamic environments.

1204 3.1 Core Elements of Decision Support Systems

1205 DSS operating in complex and high-stakes environments must satisfy a set of
1206 fundamental structural requirements that extend beyond classical analytical
1207 capabilities. Rather than being defined by a single algorithmic capability, DSS
1208 effectiveness emerges from the coordinated interaction of multiple foundational
1209 components. These components jointly determine how decisions are framed,
1210 interpreted, supported, and ultimately enacted in practice. The following sections
1211 describe the core dimensions that shape operational DSS design, each of which
1212 addresses a distinct but interrelated aspect of decision support.

1213 Operational DSS implementations are *adaptive decision structuring mechanisms*
1214 whose scope and effectiveness depend on how well they align decision objectives,
1215 information processing, and organizational practice (Power & Heavin, 2017; Arnott
1216 & Pervan, 2005). Traditional DSS literature highlights the necessity of articulating
1217 the decision context and the intended decision scope, while recognizing that decision
1218 objectives should evolve adaptively through iterative interaction between the
1219 decision maker and the system (Sprague & Carlson, 1982; Power, Decision support

systems: Concepts and resources for managers, 2002). Recent research shows that adaptivity is a core design principle, particularly in environments characterized by rapidly changing conditions and incomplete information (Arnott & Pervan, 2005). Hence, DSS should adopt adaptive structures that go beyond optimizing predefined alternatives, continuously restructuring the decision space to accommodate shifting priorities, emerging constraints, and evolving interpretations of decision relevance in response to operational feedback and contextual dynamics.

A fundamental dimension of DSS is the *representation of data and knowledge*, including how raw observations are transformed into operationally meaningful information for decision-making. Decision makers do not typically act directly on raw data or model outputs; instead, they rely on interpreted information that reflects context, relevance, and confidence. (Simon, 1960; Turban, 2011). In fact, recent studies claim that the practical value of advanced analytics and DSS tools depends not only on data availability, but also on how heterogeneous data is contextualized, integrated, and presented to human decision makers (Sharda, Delen, & Turban, 2018; McNeese, Schelble, Canonico, & Demir, 2021). Thus, data representation serves as an epistemic bridge between raw sensor inputs and higher-level decision reasoning.

Beyond data availability, *uncertainty* plays a central role in shaping DSS in complex and dynamic domains such as natural disasters (i.e., wildfire, flood, etc.). It is known that real-world decision environments are rarely fully observable, and decision makers must operate with incomplete, delayed, or ambiguous information (Simon, 1996; French, Maule, & Papamichail, 2019; Walker, et al., 2003). In such contexts, uncertainty is not only a secondary concern but also a distinguishing property of the decision environment. The sources of uncertainty directly influence (i) the choice of reasoning methods, (ii) the degree of automation that can be safely applied, and (iii) the degree to which human involvement must be retained in the loop.

Another fundamental dimension of DSS is *timeliness* as the success value of a decision often depends on when it is made rather than on optimality alone (Klein, 2017). Recent research in human-centered and naturalistic decision making

1249 emphasizes that, in time-critical situations, DSS must balance analytical depth with
1250 responsiveness, acknowledging the cognitive and operational limits of real-world
1251 decision environments (Endsley, Situation awareness misconceptions and
1252 misunderstandings, 2015; French, Maule, & Papamichail, 2019). Studies on human-
1253 machine teaming further show that timely and contextually appropriate actions are
1254 often preferable to delayed but analytically superior solutions under uncertainty and
1255 time pressure. As a result, time constraints significantly shape DSS by limiting
1256 exhaustive analysis and favoring adaptive reasoning strategies.

1257 In high-risk and high-consequence environments *explainability* plays a crucial role
1258 because the decision makers must be able to understand and justify system-supported
1259 recommendations to trust and effectively use them. As emphasized by Ribeiro et al.
1260 (2016) explanations play a critical role in building trust by allowing users to assess
1261 the plausibility and relevance of individual recommendations in specific situations.
1262 Similarly, Guidotti et al. (2018) highlight that explainability is essential not only for
1263 transparency, but also for accountability and human-machine collaboration.

1264 This perspective is further supported by human-centered XAI research, which shows
1265 that explanations tailored to user needs improve calibrated trust, situational
1266 awareness, and effective collaboration between humans and intelligent systems
1267 (Sanneman & Shah, 2020; Abdul, Vermeulen, Wang, Lim, & Kankanhalli, 2018).
1268 Without explainability, even technically sound decision support outputs may be
1269 ignored, misinterpreted, or misapplied; thus, explainability must be treated not as an
1270 auxiliary feature but as a core requirement that directly supports effective human-
1271 machine collaboration in operational decision-making.

1272 Effective DSS are designed to assist human reasoning by structuring information,
1273 highlighting trade-offs, and supporting the sensemaking process, rather than
1274 replacing human decision makers (Endsley, Toward a Theory of Situation
1275 Awareness, 1995; Ross, Shafer, & Klein, 2006). Therefore, human-machine
1276 interaction is a core element of DSS in complex and time-critical environments. In
1277 this context, DSS contributes to maintaining situation awareness and facilitating

context-sensitive judgment under uncertainty. Recent studies show that effective human-machine interaction enhances human cognitive capabilities while preserving human authority over final decisions by providing shared situational awareness, commonly referred to as a Common Operational Picture (COP), and adaptive decision making in dynamic and uncertain environments (McNeese, Schelble, Canonico, & Demir, 2021; Endsley, Situation awareness misconceptions and misunderstandings, 2015).

Finally, decisions are shaped not only by cognitive considerations but also by practical constraints related to available resources and costs in real-world operations. In other words, the decision alternatives must be evaluated in terms of trade-offs between effectiveness, risk, and operational feasibility. This requirement reflects the principle of operational feasibility under resource constraints, in turn, DSS must prioritize actions that are executable within real operational limits rather than those that are analytically optimal but impractical as discussed by Keeney and Raiffa (1976) and Power (2007). Recent studies further confirm that accounting for resource availability, cost, and implementation constraints is essential for DSS adoption and effective use in dynamic, high-risk environments (French, Maule, & Papamichail, 2019; Arnott & Pervan, 2005).

Collectively, these core elements shape how decision problems are framed, how information is interpreted, and how decisions are enacted under real-world operational constraints such as uncertainty, time pressure, and limited resources. In contrast to conventional DSS that emphasize static models or centralized optimization, as introduced in Chapter 1 and formalized in Chapters 4–6, DisasterAware utilizes these dimensions through a layered and adaptive architecture.

Specifically, decision framing and reasoning are supported via concept-based fuzzy reasoning that transforms heterogeneous low-level inputs into semantically meaningful decision constructs. Operational adaptability is achieved through parallel regional agents (a shared reasoning policy evaluated independently on each region's

1306 local observations and contextual inputs) enabling timely and location-specific
1307 responses without a centralized inferential bottleneck..

1308 At a higher level, an AI-supported governance layer provides interpretive oversight,
1309 consistency checking, and explanatory support, ensuring that system outputs remain
1310 transparent, accountable, and aligned with human decision-making processes.

1311 Table 3.1 summarizes these core dimensions, clarifying their functional roles in DSS
1312 design and highlighting how each is explicitly realized within the proposed
1313 framework. These dimensions collectively define the structural requirements that
1314 any DSS must satisfy in dynamic and partially observable environments such as
1315 wildfire response. As formalized in Chapter 2, these requirements directly motivate
1316 the design of the DisasterAware architecture.

1317 Table 3.1 DSS core dimensions with DisasterAware architectural realization

Core DSS Dimension	Why It Matters	DisasterAware Realization
Adaptive Decision Structuring	Operational conditions evolve; fixed alternatives become invalid	Open decision space; decisions emerge from concept configurations
Epistemic Knowledge Representation	Raw data is not decision-ready; inputs must become semantically meaningful	Concept-based fuzzy layers compressing observations into four decision concepts
Uncertainty-Aware Reasoning	Information is incomplete, delayed, and conflicting	Fuzzy inference over graded belief and confidence values
Timely Decision Guidance	Decision value decays under time pressure	Distributed local agents enabling rapid, context-specific inference

Core DSS Dimension	Why It Matters	DisasterAware Realization
Explainable Decision Support	Operators must understand and justify outputs to trust them	Concept-level traceability from input data to final recommendation
Human-in-the-Loop Sensemaking	Human judgment cannot be removed from high-risk decisions	Operator-centered guidance interface, not action prescription
Operational Feasibility	Decisions must respect resource, cost, and safety limits	Cost-aware fuzzy evaluation of risk–effectiveness–resource trade-offs
AI Governance	System-level oversight and consistency checking are required	AI layer for coherence analysis, conflict detection, and audit support

3.2 Fuzzy Logic as a Reasoning Paradigm for Decision Support

Given the DSS problem defined in Chapter 2, which operates under partial observability, epistemic uncertainty, and continuously evolving system conditions, the choice of reasoning paradigm becomes a structural design decision rather than a methodological preference. Fuzzy logic provides a practical and theoretically grounded reasoning paradigm for DSS operating under such conditions (Zadeh, Fuzzy sets, 1965; Thompson & Calkin, 2011). Unlike Boolean logic, which assumes that system states and observations can be represented using crisp binary values such as true or false, fuzzy logic allows graded representation through membership in linguistic categories such as Very Low, Low, Medium, High, and Very High. This enables a continuous and operationally meaningful representation of real-world conditions rather than a forced binary abstraction (Pacheco, 2015).

This representational flexibility is particularly suitable for operational DSS, where decision makers must act under time pressure, limited information, and evolving

1332 situational constraints. Data streams originating from sensors, models, and human
1333 reports are often incomplete, asynchronous, or mutually inconsistent. Fuzzy logic
1334 enables the integration of heterogeneous inputs within a unified reasoning
1335 framework without requiring premature hard thresholds or precise probabilistic
1336 models, which are often unavailable or unjustified in practice (Vásquez, 2021). This
1337 property directly addresses the limitations identified in Chapter 1, particularly the
1338 need for robust reasoning over incomplete observations while maintaining
1339 transparency in decision generation.

1340 Specifically in wildfire management, these characteristics become critical.
1341 Measurements derived from remote sensing, in situ sensors, and predictive fire
1342 behavior models are frequently affected by noise, latency, spatial limitations, and
1343 model simplifications. At the same time, expert reasoning in wildfire operations is
1344 predominantly expressed in qualitative terms, such as high spread potential,
1345 moderate suppression feasibility, or low confidence detection (Thompson & Calkin,
1346 2011; Soualah, 2024). Fuzzy logic aligns naturally with this form of linguistic
1347 reasoning by enabling the direct formalization of expert knowledge through rule-
1348 based structures, without requiring rigid numerical thresholds.

1349 Beyond representation, fuzzy logic supports approximate reasoning in decision
1350 contexts where complete knowledge and statistically robust uncertainty models are
1351 not available. By operating on graded evidence and partial truth values, fuzzy
1352 inference mechanisms can produce meaningful decision guidance even when inputs
1353 are sparse, partially conflicting, or only weakly reliable. This capability allows DSS
1354 to remain operationally useful in conditions where probabilistic or optimization-
1355 based approaches may fail due to data scarcity, model instability, or excessive
1356 computational requirements (Pacheco, 2015; Vásquez, 2021).

1357 From a DSS design perspective, fuzzy logic functions not merely as a mathematical
1358 tool, but as a cognitive bridge between raw data, expert judgment, and actionable
1359 decisions. Its emphasis on rule transparency, structured reasoning, and human-
1360 readable knowledge representation directly supports trust, explainability, and

1361 effective human-machine collaboration in high-risk operational domains such as
1362 wildfire management. This cognitive alignment is a key design principle in
1363 DisasterAware, where decision reasoning is intentionally structured to reflect
1364 human-like interpretation of context-dependent information.

1365 3.2.1 Core building blocks of fuzzy inference

1366 In the context of DisasterAware, the fuzzy inference system is not treated as a generic
1367 mapping mechanism, but as the computational realization of the DSS mapping π_{DSS}
1368 introduced in Chapter 2. A fuzzy inference system (FIS) is a rule-based reasoning
1369 framework that maps uncertain, imprecise, or linguistically expressed inputs into
1370 interpretable decision outputs.

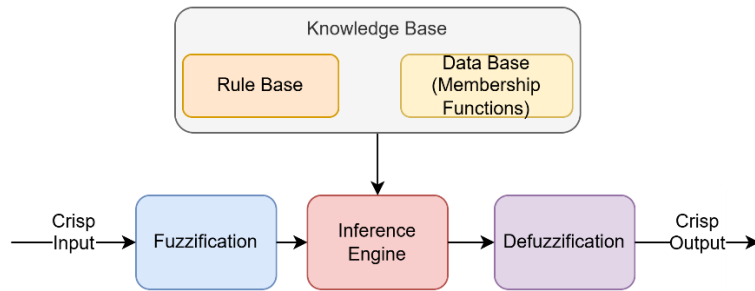
1371 Unlike classical logical or probabilistic inference, FIS enables reasoning under
1372 partial knowledge and epistemic uncertainty. Formally, a FIS defines a nonlinear
1373 mapping from an input space to an output space through linguistic variables,
1374 membership functions, and inference rules (Zadeh, Fuzzy sets, 1965; Pacheco, 2015)

1375 Let the crisp input vector be defined as (2) where each x_i represents a measurable
1376 or estimated system variable, such as wind speed, fuel moisture, slope, or detection
1377 confidence. The output variable is denoted as (3) which typically corresponds to a
1378 decision concept (e.g., Spread Risk, Operational Priority, Alert Level), rather than a
1379 raw numerical control signal.

$$x = (x_1, x_2, \dots, x_n) \in \mathbb{R}^n \quad (2)$$

$$y \in \mathbb{R} \quad (3)$$

1380 In the context of the DSS introduced in Chapter 2, the input vector x represents
1381 observable quantities derived from $O_k(x, y)$, while the output variable y corresponds
1382 to an intermediate or final DSS quantity rather than a direct control action.



1383
1384 Figure 3.1 Fuzzy inference system block diagram

1385 As illustrated in Figure 3.1, a standard fuzzy inference system consists of four
1386 fundamental building blocks (i) Fuzzification, (ii) Rule Base, (iii) Inference Engine
1387 and (iv) Defuzzification. Each block contributes a distinct functional role in
1388 transforming raw data into DSS outputs.

1389 3.2.2 Fuzzification and Data Base (Membership Functions)

1390 In the wildfire DSS problem defined in Chapter 2, decision-making must operate on
1391 observations that are incomplete, delayed, and inherently imprecise. Under such
1392 conditions, the transformation of raw observations into a structured representation
1393 that preserves uncertainty becomes a fundamental requirement. Fuzzification is
1394 therefore introduced not merely as a standard fuzzy-logic operation, but as a
1395 mechanism for encoding epistemic uncertainty into a form that can be directly
1396 utilized by the DSS process.

1397 Fuzzification is the process of transforming crisp input values into fuzzy
1398 representations. Each input variable x_i is associated with a linguistic variable X_i ,
1399 defined by a finite set of linguistic terms (4) where each A_{ij} denotes a fuzzy set
1400 such as *Low*, *Medium*, or *High*. Each fuzzy set A_{ij} is characterized by a membership
1401 function which assigns a degree of membership to the crisp input (5) where X_i

1402 denotes the linguistic variable associated with input x_i , A_{ij} represents its j -th
 1403 linguistic term, and $\mu_{A_{ij}}(x_i)$ defines the degree of membership of x_i in that term.

$$X_i = \{A_{i1}, A_{i2}, \dots, A_{im_i}\} \quad (4)$$

$$\mu_{A_{ij}}: \mathbb{R} \rightarrow [0,1] \quad (5)$$

$$\mu_{A_{ij}}(x_i) = \text{degree to which } x_i \text{ satisfies linguistic concept } A_{ij}$$

1404 The fuzzification stage replaces binary classification with **graded truth values**,
 1405 allowing the system to represent ambiguity, vagueness, and measurement
 1406 uncertainty. This property is critically important for representing ambiguity,
 1407 incomplete knowledge, and epistemic uncertainty in real-world decision
 1408 environments, where exact system states are not directly observable.

1409 In a FIS, the fuzzification process operates on the definitions provided by the Data
 1410 Base, which stores the universe of discourse for each input variable, the set of
 1411 associated linguistic terms, and the parameters of their corresponding membership
 1412 functions. For each input variable x_i , the Data Base contains the tuple (6). Thus,
 1413 fuzzification performs the numerical transformation from crisp observations to
 1414 graded memberships, while the Data Base defines the semantic structure within
 1415 which these memberships become decision-relevant linguistic representations.

$$\mathcal{D}_i = \left\langle X_i, \{A_{i1}, A_{i2}, \dots, A_{im_i}\}, \{\mu_{A_{i1}}, \mu_{A_{i2}}, \dots, \mu_{A_{im_i}}\} \right\rangle \quad (6)$$

1416 In Figure 3.2, a wind speed of 22 km/h falls precisely in the transition region between
 1417 the Low and Medium categories, yielding equal membership degrees in both terms.
 1418 A single crisp input $x = 22$ km/h is mapped across the three fuzzy sets (Low,
 1419 Medium, High), with membership degrees $\mu_{Low}(22) = 0.50$, $\mu_{Medium}(22) = 0.50$,
 1420 and $\mu_{High}(22) = 0.00$, summing to 1.00. This example highlights a key advantage
 1421 of fuzzification in DSS: a single observation can simultaneously support multiple
 1422 linguistic interpretations without enforcing rigid threshold-based classifications.
 1423 This example illustrates how fuzzification preserves uncertainty information by
 1424 allowing multiple simultaneous linguistic interpretations of a single observation.

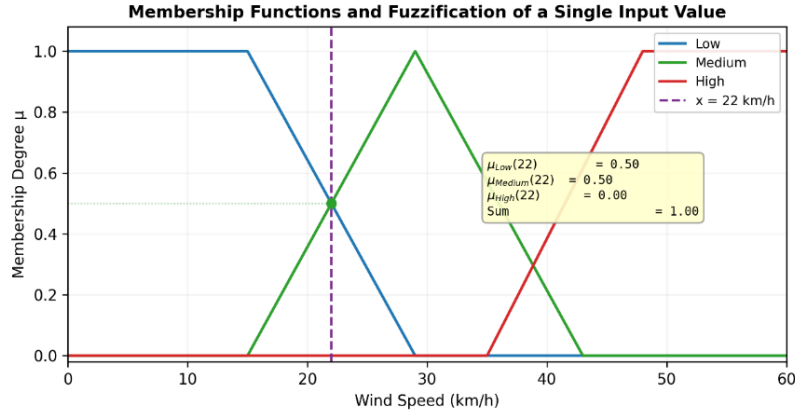


Figure 3.2 Shoulder–Triangular–Shoulder membership functions, fuzzification

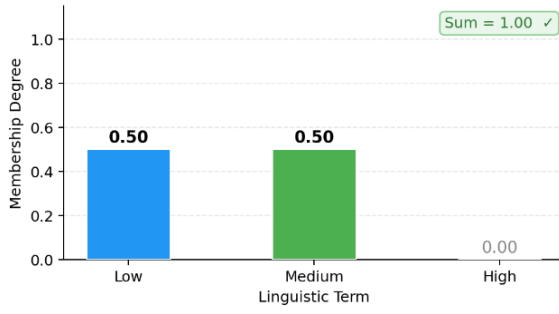


Figure 3.3 Membership degrees produced by fuzzification

More generally, fuzzification allows a single crisp input x_i to activate multiple fuzzy sets simultaneously. For a given x_i , the fuzzified representation is the vector (7) where $\sum_{j=1}^{m_i} \mu_{A_{ij}}(x_i)$ is not constrained to equal one in the general formulation; however, the DisasterAware framework enforces a partition-of-unity design (Proposition 5.2) guaranteeing that the sum equals one at every point.

$$\boldsymbol{\mu}(x_i) = [\mu_{A_{i1}}(x_i), \mu_{A_{i2}}(x_i), \dots, \mu_{A_{im_i}}(x_i)] \quad (7)$$

Overlapping membership representations allow a single observation to support multiple linguistic interpretations, which is essential for modeling ambiguity in real-world decision environments. In the DisasterAware framework, this principle is implemented through a partition-of-unity structure, as formalized in Chapter 5. The choice of membership shape directly influences the behavior, robustness, and interpretability of the fuzzy inference system. Commonly used forms include triangular, trapezoidal, Gaussian, and sigmoidal functions, each offering distinct advantages for different application contexts.

The specific choice of membership function shapes affects both the interpretability and robustness of the fuzzy inference system. In general, piecewise-linear functions are preferred in DSS applications due to their computational simplicity and ease of interpretation. While various forms such as trapezoidal and Gaussian functions can be used, the key requirement in DSS contexts is not the specific functional form, but the ability to represent gradual transitions between linguistic concepts in a stable and interpretable manner.

Figure 3.2 and Figure 3.3 emphasize the key contribution of fuzzification within a DSS: numerical inputs are not treated as exact or definitive, but as evidence supporting multiple linguistic interpretations to varying degrees. This representation forms the foundation for subsequent fuzzy rule evaluation and inference, ultimately enabling decision guidance under epistemic uncertainty.

In DSS applications, fuzzification plays a critical role in aligning numerical sensor data with human reasoning processes. By operating on linguistic concepts rather than hard thresholds, fuzzification improves tolerance to sensor noise, reduces sensitivity to minor measurement fluctuations, and enables smoother transitions between decision regions. This makes the decision logic more robust and interpretable, particularly in operational domains such as wildfire management, where inputs are uncertain and expert knowledge is commonly expressed in qualitative terms rather than precise numerical rules (Thompson & Calkin, 2011). In this thesis, fuzzification serves as the primary mechanism for encoding epistemic uncertainty, ensuring that

1463 incomplete and imprecise observations can be systematically incorporated into the
1464 DSS process without requiring probabilistic assumptions.

1465 In summary, the fuzzification and Data Base design adopted in this work deliberately
1466 balances interpretability and robustness. Trapezoidal membership functions are well
1467 suited for encoding stable operating ranges, while Gaussian functions model gradual
1468 transitions around prototypical values. In the DisasterAware framework, a
1469 piecewise-linear shoulder–triangular–shoulder design is adopted, which combines
1470 the flat-core property of trapezoidal functions with computational simplicity and
1471 partition-of-unity guarantees, as formalized in Chapter 5.

1472 While the generic formulation in this section uses x_i as the input variable and
1473 $\mu_{A_{ij}}(x_i)$ as the membership function notation following standard fuzzy-logic
1474 convention, the DisasterAware framework adopts a structured feature representation
1475 denoted as $z_{k,l}^i(x, y)$, where l indexes feature dimensions, i denotes the spatial
1476 region, and k represents the time step. This distinction separates feature variables
1477 from spatial coordinates and ensures consistency with the spatio-temporal DSS
1478 formulation introduced in Chapter 2.

1479 3.2.3 Rule Base and Knowledge Representation

1480 Within a fuzzy-logic framework, this mapping π_{DSS} is realized through a structured
1481 rule base that encodes domain knowledge in an explicit and interpretable form. The
1482 rule base constitutes the core mechanism through which expert knowledge and
1483 domain heuristics are encoded within a fuzzy inference system. It represents
1484 knowledge in the form of linguistic IF–THEN rules, allowing qualitative human
1485 reasoning to be formalized and executed computationally.

1486 Each fuzzy rule R defines a local mapping between conditions observed in the input
1487 space and an inferred decision outcome. A generic fuzzy rule R can be expressed as
1488 (8) where A_i^R denotes the antecedent fuzzy set associated with input variable x_i in
1489 rule k , B^R denotes the consequent fuzzy set associated with the output variable y .

$$R: \text{IF } x_1 \text{ is } A_1^R \wedge x_2 \text{ is } A_2^R \wedge \dots \wedge x_n \text{ is } A_n^R \text{ THEN } y \text{ is } B^R \quad (8)$$

In DSS contexts, these rules do not merely represent logical relationships but act as localized reasoning units that approximate incident-command reasoning documented in operational literature under uncertainty. In Equation (8), A_i^R denotes the antecedent fuzzy set associated with input variable x_i in rule R , while B^R represents the consequent fuzzy set defining the output implication of the rule. This structure enables explicit encoding of expert knowledge and supports traceable reasoning, which is essential for high-risk DSS environments.

Each rule represents **local reasoning**, valid over a specific region of the input space where its antecedent conditions hold to a sufficient degree. Importantly, no single rule is intended to describe the system behavior globally. Instead, the overall system behavior emerges from the collective activation and aggregation of multiple partially fired rules, each contributing proportionally to the final inference result. In the context of wildfire DSS, fuzzy rules naturally reflect the qualitative reasoning used by experienced operators and analysts. An example is:

IF *Wind Speed* is High AND *Fuel Moisture* is Low AND *Slope* is Steep THEN *Spread Risk* is Very High.

Such rules directly reflect the linguistic reasoning patterns used by human decision-makers in wildfire management, where precise numerical thresholds are rarely available and decisions must be made under partial observability.

From a knowledge representation perspective, the rule base operates in conjunction with the data base (membership functions) to form the knowledge base of the FIS. While the Data Base defines the semantic meaning and numerical realization of linguistic concepts, the rule base defines the inferential structure that governs how these concepts are combined to produce decisions. This separation enables modularity, interpretability, and maintainability: membership functions can be refined independently of rules, and rules can be added or modified without altering the underlying data representation.

1515 The explicit, linguistic nature of the rule base also provides traceability and
1516 explainability, which are essential properties for DSS operating in high-risk and
1517 safety-critical domains. Each inferred decision can be traced back to a set of activated
1518 rules and their corresponding antecedent conditions, facilitating validation, operator
1519 trust, and post-event analysis. For these reasons, the rule base serves as a formalized
1520 representation of expert reasoning, bridging human knowledge and algorithmic

1521 In the DisasterAware framework, this structure enables decision generation to
1522 remain both interpretable and adaptable, ensuring that intervention guidance can be
1523 traced back to explicit reasoning steps while operating under partial observability
1524 (Pacheco, 2015). This property directly addresses the explainability requirement
1525 identified in Chapter 1 (G4) and forms the basis of the interpretable decision-
1526 generation mechanism adopted in the DisasterAware framework.

1527 3.2.4 Inference Engine and Rule Evaluation

1528 Within the DSS formulation introduced in Chapter 2, the inference engine provides
1529 the computational mechanism through which the mapping π_{DSS} is realized under
1530 uncertainty. The inference engine is the computational core of a FIS. Its primary role
1531 is to evaluate the rule base using fuzzified inputs and to synthesize a coherent fuzzy
1532 output that reflects the combined influence of multiple, partially satisfied rules.

1533 Unlike crisp logical systems, the inference engine operates on graded truth values,
1534 enabling approximate reasoning under epistemic uncertainty. Mathematically, the
1535 inference process can be decomposed into three sequential stages: **rule activation**,
1536 **implication**, and **aggregation**. This type of decomposition is canonical in Mamdani-
1537 type FIS and provides a transparent and interpretable reasoning mechanism for DSS
1538 applications (Mamdani, 1974).

3.2.4.1 Rule Activation (Antecedent Evaluation)

In the rule activation stage, a fuzzy rule R is defined as $R: IF\ x_1\ is\ A_1^R \wedge x_2\ is\ A_2^R \wedge \dots \wedge x_n\ is\ A_n^R\ THEN\ y\ is\ B^R$, while the crisp input vector is $\mathbf{x} = (x_1, x_2, \dots, x_n)$. After fuzzification, each input x_i is represented by its membership degree in the antecedent fuzzy set A_i^R , denoted as $\mu_{A_i^R}(x_i) \in [0,1]$. The inference engine computes the **rule firing strength** α_R , which quantifies the degree to which the antecedent conditions of rule R are satisfied. This is achieved using a fuzzy conjunction operator (t-norm) in (9) and (10) where α_R denotes the firing strength of rule R , computed from the membership degrees of its antecedent conditions. The firing strength $\alpha_R \in [0,1]$ allows rules to be **partially activated**, reflecting the fact that real-world conditions rarely satisfy all antecedent clauses perfectly.

$$\text{Minimum t-norm: } \alpha_R = \min_{i=1, \dots, n} \left(\mu_{A_i^R}(x_i) \right) \quad (9)$$

$$\text{Product t-norm: } \alpha_R = \prod_{i=1}^n \mu_{A_i^R}(x_i) \quad (10)$$

3.2.4.2 Implication (Consequent Modulation)

Once the firing strength α_R is obtained, the inference engine applies an implication operator to the consequent fuzzy set B^R . In Mamdani-type inference, the most widely used implication mechanism scales the consequent membership function according to the firing strength (11) where $\mu'_{B^R}(y)$ represents the scaled consequent membership function after applying the implication operator.

$$\mu'_{B^R}(y) = \alpha_R \cdot \mu_{B^R}(y) \quad (11)$$

This operation preserves the semantic shape of the consequent fuzzy set while proportionally adjusting its influence. Rules that are weakly satisfied contribute marginally to the output, whereas strongly satisfied rules exert greater influence. This implication can be interpreted as a weighting operation in function space, where each rule contributes to a scaled fuzzy set rather than a crisp outcome. This ensures

smooth transitions in the output space and avoids discontinuities that would otherwise arise from binary rule evaluation. In DSS, this mechanism ensures that stronger evidence contributes more significantly to the final decision, while weaker or uncertain evidence is still considered without being discarded.

3.2.4.3 Aggregation

The inference engine aggregates all rule-specific fuzzy outputs into a single fuzzy set since multiple rules may be simultaneously activated (12) where $\vee$ denotes the maximum operator and the union is taken over all rules R in the rule base.

$$\mu_B(y) = \bigvee_R \mu'_{B^R}(y) \quad (12)$$

This aggregation combines all rule contributions into a single fuzzy output set representing the overall inference result. The aggregated fuzzy set B represents the system's overall reasoning outcome prior to defuzzification. Importantly, aggregation does not resolve conflicts explicitly; instead, it allows overlapping and competing rules to coexist through graded membership values. This mechanism naturally reflects how decision-makers integrate multiple, potentially conflicting sources of information, making it particularly suitable for a DSS operating under uncertainty and partial observability. This inference process provides a structured and interpretable mechanism for transforming fuzzified observations into DSS outputs, forming the DSS mapping defined in Chapter 5.

3.2.5 Defuzzification and Decision Output

Defuzzification is the final stage of a FIS when a crisp output is required, responsible for transforming the aggregated fuzzy output set B into a crisp value y^* that can be directly used in decision-making or control actions. While FIS operates on graded membership values and linguistic abstractions, defuzzification provides a single representative output that summarizes the overall reasoning outcome. $\mu_B(y)$ denotes

1585 the membership function obtained after rule aggregation. The defuzzification
1586 process defines a mapping (13) where y^* represents the crisp decision output.

$$\mathcal{D} : \mu_B(y) \rightarrow y^* \in \mathbb{R} \quad (13)$$

1587 However, in many DSS applications, especially those involving human-in-the-loop
1588 operation, producing a single crisp value is not always necessary or desirable. The
1589 most widely used defuzzification technique is the **centroid**, or **center of gravity**,
1590 method. It computes the weighted average of the output domain, using the
1591 aggregated membership function as a weighting factor (14). This formulation
1592 computes the centroid of the output distribution, treating the membership function
1593 as a weighting function over the output domain.

$$y^* = \frac{\int y \cdot \mu_B(y) dy}{\int \mu_B(y) dy} \quad (14)$$

1594 While such methods provide smooth and continuous outputs, they inherently
1595 compress the underlying uncertainty into a single representative value. Centroid
1596 defuzzification can be interpreted as computing the expected value of the output
1597 variable under a possibility distribution induced by fuzzy inference. This
1598 interpretation reinforces its suitability for approximate reasoning under uncertainty
1599 (Mendel, 2017). In addition to the centroid method, there are several alternative
1600 defuzzification techniques listed in Table 3.2.

1601 Table 3.2 Commonly used defuzzification techniques

Defuzzification Methods	Explanations
Mean of maxima	Computes the average of all output values where $\mu_B(y)$ reaches its maximum.
Largest of maxima	Selects the largest value of y corresponding to the maximum membership degree.

Defuzzification Methods	Explanations
Weighted average	Often used in Sugeno-type fuzzy systems, where rule consequents are crisp values and defuzzification reduces to a weighted sum (Takagi & Sugeno, 1985).

1602

1603 These methods are computationally simpler than centroid defuzzification but are
 1604 generally more sensitive to local maxima and less responsive to the overall shape of
 1605 the fuzzy output. As a consequence, they may introduce discontinuities or abrupt
 1606 decision changes, which are undesirable in many DSS applications.

1607 In many operational DSS contexts, particularly those involving high uncertainty and
 1608 HIL decision making, defuzzification is not always mandatory. Instead of producing
 1609 a single crisp value, the system may present the fuzzy output directly in linguistic
 1610 form (e.g., *Low Risk*, *Moderate Risk*, *High Risk*), allowing human operators to
 1611 interpret and contextualize the result. This approach enhances transparency, trust,
 1612 and cognitive alignment between the DSS and its users (Thompson & Calkin, 2011).
 1613 In the DisasterAware framework, defuzzification is treated as optional, allowing the
 1614 system to preserve fuzzy outputs for improved interpretability and HIL interaction.

1615 **3.2.6 Structural Alignment Between Fuzzy Logic and DisasterAware**

1616 The design principles of fuzzy logic exhibit a strong structural and conceptual
 1617 alignment with the DisasterAware decision support framework. At its core,
 1618 DisasterAware is conceived as an adaptive, concept-driven DSS that must operate
 1619 under uncertainty, heterogeneous data sources, and evolving operational conditions.
 1620 Fuzzy logic directly supports these requirements by enabling graded reasoning,
 1621 linguistic abstraction, and rule-based inference without assuming complete or
 1622 precise information. Rather than forcing premature numerical certainty, fuzzy

inference preserves epistemic uncertainty and supports decision making based on partial, local, and context-dependent evidence.

This property is fundamental to DisasterAware, where decisions are not selected from a closed set of predefined alternatives but instead emerge dynamically from configurations in the concept space. Furthermore, the modularity of FIS allows DisasterAware to run the same inference pipeline in parallel across regions, each instance conditioned on its own observations and contextual inputs, while higher-level fusion at the tactical layer integrates the resulting regional outputs.

By separating fuzzification, rule evaluation, inference, and defuzzification, fuzzy logic provides DisasterAware with a flexible reasoning backbone that supports explainability, scalability, and human-centered operation across multiple decision levels and spatio-temporal scales. This alignment confirms that fuzzy logic is not an arbitrary modeling choice, but a direct consequence of the structural requirements of the wildfire DSS problem defined in Chapter 2 as summarized in Table 3.3.

Table 3.3 Alignment between fuzzy logic components and DisasterAware

Fuzzy Logic Component	Role in Classical Fuzzy DSS	Alignment in DisasterAware
Linguistic variables	Represent vague concepts (Low, Medium, High)	Used as decision concepts (e.g., risk level, confidence, priority) rather than raw indices
Fuzzification	Maps crisp inputs to fuzzy sets	Enables semantic abstraction of heterogeneous sensor, model, and human inputs
Membership functions	Encode tolerance and uncertainty	Support robust reasoning under noisy, delayed, or incomplete data
Rule base	Encodes expert IF–THEN knowledge	Implements domain knowledge as explainable reasoning logic at local and global levels

Fuzzy Logic Component	Role in Classical Fuzzy DSS	Alignment in DisasterAware
Rule activation (graded firing)	Partial rule satisfaction	Enables approximate reasoning and avoids weak threshold-based decisions
Aggregation of rules	Combines multiple rule outputs	Supports emergent decisions from interacting factors
Defuzzification	Produces crisp outputs	Treated as optional and context-dependent; often deferred or omitted to preserve interpretability
Interpretability	Human-readable reasoning	Directly supports operator trust, explainability, and governance
Modularity	Independent fuzzy components	Enables distributed local decision makers and hierarchical fusion
Non-probabilistic uncertainty handling	No reliance on full probability models	Matches epistemic uncertainty dominant in disaster operations
Static rule evaluation	Fixed rule base	Extended in DisasterAware via adaptive weighting and feedback mechanisms
Closed decision space	Predefined outputs	Replaced by open decision space with emergent decisions

1638 3.3 Limitations of Classical MCDM and MAGDM Approaches

1639 Multi-Criteria Decision Making (MCDM) methods constitute a well-established
1640 class of decision analysis frameworks for evaluating and ranking a finite set of
1641 alternatives across multiple, potentially conflicting criteria (Hwang & Yoon, 1981;
1642 Triantaphyllou, 2000; Zavadskas & Turskis, 2011). Their Multi-Attribute Group

1643 Decision Making (MAGDM) extensions incorporate consensus mechanisms for
 1644 settings involving multiple decision makers (Xu, 2015). However, within the DSS
 1645 problem presented in Chapter 2, the structural assumptions underlying these methods
 1646 introduce fundamental limitations. This section analyzes these limitations and
 1647 demonstrates why classical MCDM and MAGDM frameworks are not compatible
 1648 with dynamic, partially observable, and time-critical decision environments.

1649 3.3.1 Formal MCDM structure

1650 In the canonical MCDM formulation, a decision problem is represented by a decision
 1651 matrix $\mathbf{X} \in \mathbb{R}^{m \times n}$, where m denotes the number of predefined alternatives
 1652 $\{A_1, A_2, \dots, A_m\}$ and n denotes the number of evaluation criteria $\{C_1, C_2, \dots, C_n\}$. Each
 1653 entry x_{ij} encodes the crisp performance value of alternative A_i with respect to
 1654 criterion C_j . A weight vector $\mathbf{w} = (w_1, w_2, \dots, w_n)^T$ subject to (15) encodes the
 1655 relative importance of each criterion. Here, w_j denotes the weight assigned to
 1656 criterion C_j , representing its relative importance in the decision process. The final
 1657 ranking is derived through a method-specific aggregation function $S: \mathbb{R}^{m \times n} \times \mathbb{R}^n \rightarrow$
 1658 $\mathbb{R}^m$, yielding a scalar score $S(A_i)$ for each alternative.

$$\sum_{j=1}^n w_j = 1, w_j \geq 0 \forall j \quad (15)$$

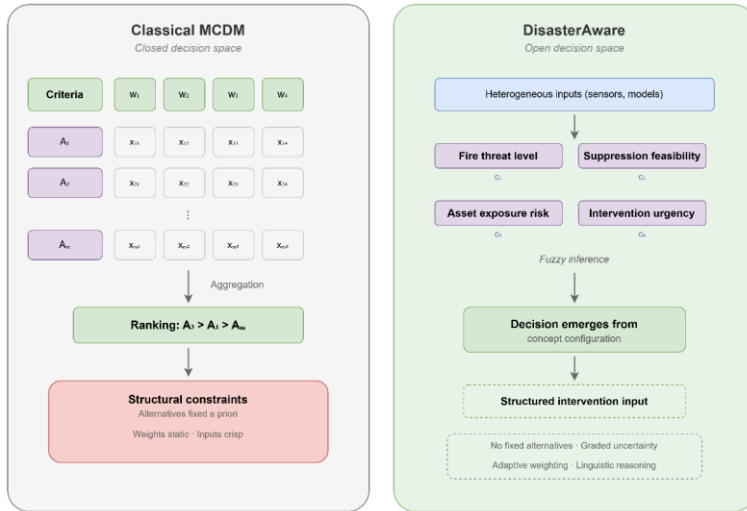
1659 In Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS), for
 1660 instance, the performance matrix is first normalized as (16) and weighted values
 1661 are computed as $v_{ij} = w_j \cdot r_{ij}$. In this expression, r_{ij} represents the normalized
 1662 performance value of alternative A_i with respect to criterion C_j .

$$r_{ij} = \frac{x_{ij}}{\sqrt{\sum_{i=1}^m x_{ij}^2}} \quad (16)$$

1663 The positive ideal solution $A^+ = \{v_j^+ = \max_i v_{ij}\}$ and negative ideal solution $A^- =$
1664 $\{v_j^- = \min_i v_{ij}\}$ are defined, and each alternative is scored by its relative closeness
1665 (17). The value C_i denotes the relative closeness of alternative A_i to the ideal
1666 solution, which is used to rank alternatives.

$$C_i = \frac{D_i^-}{D_i^+ + D_i^-}, \text{ where } D_i^\pm = \sqrt{\sum_{j=1}^n (v_{ij} - v_j^\pm)^2} \quad (17)$$

1667 In the Analytic Hierarchy Process (AHP), weights are derived from a pairwise
1668 comparison matrix $A = [a_{ij}]$ where $a_{ij} = \frac{w_i}{w_j}$, by computing the principal
1669 eigenvector w $Aw = \lambda_{\max} w$. Consistency is assessed by the Consistency Ratio
1670 $CR = \frac{CI}{RI}$, where $CI = \frac{\lambda_{\max} - n}{n-1}$ and RI is a random index dependent on matrix size
1671 (Saaty, 1980).



1672
1673 Figure 3.4 Closed vs open decision space: classical MCDM and DisasterAware
1674 As illustrated in Figure 3.4, classical MCDM operates within a structurally closed
1675 decision space: a fixed set of alternatives $\{A_1, \dots, A_m\}$ is evaluated against predefined

criteria using crisp performance values x_{ij} and a static weight vector, ultimately producing a deterministic ranking. This distinction is central to the proposed approach, as DisasterAware eliminates the assumption of predefined alternatives and instead generates decisions dynamically through concept-based reasoning.

The entire reasoning chain from input to output is constrained by the assumption that all relevant alternatives are known and quantifiable prior to evaluation. In contrast, DisasterAware operates within an open decision space in which no alternatives are enumerated in advance.

Heterogeneous sensor and model inputs are transformed through fuzzy inference into a set of decision concepts, such as Spread Risk, Operational Priority, and Alert Level, whose configuration collectively drives the emergence of a structured intervention input. This architecture eliminates the closed-alternative assumption and replaces static aggregation with adaptive, linguistically grounded reasoning. These structural differences form the basis of the limitations analyzed in the following subsections.

3.3.2 Limit 1 – Dependence on a Closed and Predefined Alternative Set

The most fundamental structural constraint is that all alternatives $\{A_1, \dots, A_m\}$ must be enumerated prior to evaluation (Comes, Hiete, Wijngaards, & Schultmann, 2011; Triantaphyllou, 2000). This assumption is incompatible with wildfire response, where feasible interventions are not fixed but evolve dynamically with fire behavior, resource availability, and operational constraints. Neither TOPSIS, AHP, nor their extensions support the emergence of new alternatives from operational dynamics; the decision space is closed by design.

3.3.3 Limit 2 – Crisp Performance Values and Weight Assignment

The decision matrix $\mathbf{X}$ requires precise, fully observed values x_{ij} . Classical MCDM methods are fundamentally designed for deterministic settings, where criteria values

are treated as fully observable and precisely quantifiable (Bellman & Zadeh, 1970). In practice, wildfire decision environments are characterized by epistemic uncertainty arising from measurement noise, sensor latency, model simplifications, and partially conflicting information streams (Walker, et al., 2003; French, Maule, & Papamichail, 2019). Interval-valued and fuzzy extensions to MCDM such as fuzzy TOPSIS using triangular or trapezoidal fuzzy numbers $\tilde{x}_{ij} = (l_{ij}, m_{ij}, u_{ij})$ have been proposed to partially address this limitation (Chen & Hwang, 1992).

However, these adaptations confine uncertainty to predefined numerical intervals and do not support the continuous, graded, and linguistically expressed uncertainty inherent in real-time operational sensing. Similarly, the weight vector $\mathbf{w}$ is typically assigned a priori or elicited through static pairwise comparisons, precluding dynamic reweighting in response to operational feedback. This limitation is particularly critical in wildfire environments, where observations are inherently incomplete and cannot be reliably represented as precise numerical values.

3.3.4 Limit 3 – Computational Scalability Under Real-Time Constraints

Classical MCDM methods carry computational overhead that grows with the number of alternatives and criteria. AHP's pairwise comparison procedure requires $\frac{n(n-1)}{2}$ comparisons per decision level, and the consistency constraint $CR < 0.10$ becomes increasingly difficult to satisfy as n grows. In large-scale operational scenarios involving spatially distributed resources, multiple simultaneous active fire zones, and dynamic prioritization, the computational cost of classical MCDM becomes prohibitive for real-time decision support (Triantaphyllou, 2000). Furthermore, the centralized aggregation mechanism inherent in most MCDM methods is structurally ill-suited to distributed operational architectures requiring local decisions without global communication. This computational burden is incompatible with real-time DSS requirements, where decisions must be generated within strict time constraints.

3.3.5 Limit 4 – MAGDM Consensus Latency and Cognitive Bias

In MAGDM frameworks, individual expert weight vectors $\mathbf{w}^{(k)}$ are aggregated into a group weight vector, typically as (18). This aggregation assumes synchronized decision cycles and a priori assignment of expert authority weights ω_k . In time-critical response, the latency of consensus formation is operationally unacceptable, and the normative rational choice model assumed by MAGDM diverges sharply from the pattern-recognition-based decision making observed in experienced operators under time pressure (Klein, 2017; Tversky & Kahneman, 1974). This limitation is particularly problematic in wildfire response, where decision latency directly reduces operational effectiveness due to rapidly evolving conditions.

$$\bar{w}_j = \sum_{k=1}^K \omega_k \cdot w_j^{(k)}, \text{ where } \sum_{k=1}^K \omega_k = 1 \quad (18)$$

3.3.6 Limit 5 – Opacity and Limited Explainability

While MCDM methods produce ranked alternatives or scalar scores, the reasoning pathway from input data to final recommendation is often opaque to end users. In AHP, a final priority score aggregates contributions from multiple levels of pairwise comparison in ways that are difficult to trace without dedicated analysis tools. In TOPSIS, the distances $D_i^\pm$ summarize high-dimensional performance differences to a single closeness coefficient, losing the criterion-level interpretability that operators require for validation and override (Guidotti, et al., 2018). This opacity undermines trust and limits operational adoption in high-risk domains.

Table 3.4 demonstrates that these limitations are consistent across different MCDM variants, indicating that they are inherent to the underlying decision paradigm rather than specific to individual methods.

1749 Table 3.4 MCDM/MAGDM method limitations in wildfire DSS context

Attribute	Description
AHP	
<i>Mechanism</i>	Pairwise comparison → priority eigenvector
<i>Assumption</i>	Alternatives, criteria, and expert judgments are stable and enumerable.
<i>Manifestation in DSS</i>	Pairwise consistency breaks under time pressure; the alternative set cannot accommodate emergent response options.
TOPSIS	
<i>Mechanism</i>	Distance from positive/negative ideal solution
<i>Assumption</i>	Crisp performance values are available; the ideal solution is known a priori.
<i>Manifestation in DSS</i>	Distances are undefined under epistemic uncertainty; the ideal solution shifts dynamically with fire evolution.
VIKOR	
<i>Mechanism</i>	Compromise ranking via Q-index
<i>Assumption</i>	A stable aspiration level exists for each criterion.
<i>Manifestation in DSS</i>	Aspiration levels for suppression effectiveness and resource use cannot be defined a priori in evolving fires.
ELECTRE	
<i>Mechanism</i>	Outranking via concordance / discordance thresholds
<i>Assumption</i>	Crisp threshold parameters are available for each criterion pair.
<i>Manifestation in DSS</i>	Binary outranking relations are inappropriate for graded, probabilistic observations; threshold calibration is infeasible.
PROMETHEE	
<i>Mechanism</i>	Preference functions per criterion
<i>Assumption</i>	Preference function shape and parameters are calibrated.

Manifestation in DSS	Missing or delayed sensor data renders preference functions uncalibrated; the method collapses under sparse input.
Fuzzy MCDM Extensions	
Mechanism	Interval or triangular fuzzy numbers for uncertainty representation
Assumption	Uncertainty is representable by predefined fuzzy intervals.
Manifestation in DSS	Interval bounds require domain calibration that is rarely available; the approach does not support distributed, real-time inference.
MAGDM (Consensus)	
Mechanism	Aggregated group preference weights
Assumption	Synchronized expert availability and stable expert authority weights.
Manifestation in DSS	Consensus cycles are incompatible with real-time operational timelines; weight assignment is arbitrary under dynamic role allocation.

1750

1751 Collectively, these limitations are structural rather than incidental, arising directly
1752 from the closed, static, and fully observable assumptions embedded in classical
1753 MCDM formulations. Hence, these methods are fundamentally incompatible with
1754 the open, dynamic, and partially observable decision space defined in Chapter 2.
1755 These limitations motivate the architectural departure adopted in DisasterAware,
1756 where decision generation is formulated as a distributed, uncertainty-aware
1757 reasoning process over an open decision space rather than a ranking problem over
1758 predefined alternatives.

1759 3.4 Decision Levels and Decision Scope in Operational Decision Support

1760 Operational decision making in complex and dynamic environments such as wildfire
1761 response is inherently hierarchical, involving multiple decision levels with distinct

temporal, spatial, and informational characteristics. These differences impose structural requirements on DSS design, as a single reasoning mechanism cannot simultaneously satisfy the responsiveness required at the operational level and the analytical depth required at higher levels.

3.4.1 Formal Taxonomy of Decision Levels

The initial taxonomy of decision-making was established by Anthony (1965), who identified three hierarchically nested levels (i.e., strategic planning, management control, and operational control) distinguished primarily by their temporal horizon and degree of environmental uncertainty. Simon (1960) complemented this framework by classifying decisions along a structural dimension: structured decisions are fully specifiable by algorithms; unstructured decisions require human judgment and cannot be reduced to routine procedures; semi-structured decisions exhibit elements of both and represent the primary domain of DSS intervention. Gorry and Scott Morton (1971) synthesized these frameworks, producing a widely adopted classification matrix in which decision types are mapped against organizational levels, establishing the theoretical basis for level-differentiated DSS.

To formalize these distinctions, each decision level DL_k can be characterized as a tuple of four scope parameters (19) where $THR_k \in \mathbb{R}_{>0}$ denotes the temporal horizon of decisions at level k , SCP_k denotes the spatial scope of decision influence, $EPI_k \in [0,1]$ denotes the normalized epistemic uncertainty level, and $AUT_k \in [0,1]$ denotes the automation feasibility index, the degree to which DSS can be safely delegated to algorithmic reasoning without requiring direct human override.

$$DL_k = (THR_k, SCP_k, EPI_k, AUT_k) \quad (19)$$

Across the three canonical levels, these parameters follow consistent monotonic orderings (20) reflecting the general principle that as decision horizon shortens and spatial scope narrows, uncertainty decreases and automation becomes feasible (Anthony, 1965; French, Maule, & Papamichail, 2019). These relations indicate that

1788 as decision levels move from strategic to operational, temporal horizons decrease,
 1789 uncertainty is reduced, and the feasibility of automated decision-making increases.

$$\begin{aligned} \text{THR}_{\text{str}} &> \text{THR}_{\text{tac}} > \text{THR}_{\text{opr}} \\ \text{EPI}_{\text{str}} &> \text{EPI}_{\text{tac}} > \text{EPI}_{\text{opr}} \\ \text{AUT}_{\text{opr}} &> \text{AUT}_{\text{tac}} > \text{AUT}_{\text{str}} \end{aligned} \quad (20)$$

1790 3.4.2 Temporal Urgency and Decision Value Decay

1791 A critical and often underappreciated dimension of multi-level DSS is the concept
 1792 of decision value decay: the utility of a decision degrades as a function of elapsed
 1793 time relative to the operational context in which it is needed (Klein, 2017). This
 1794 relationship can be formalized as a level-specific exponential decay function (21)
 1795 where $V_k(d, \Delta t)$ is the residual value of decision d at level k after a delay of Δt , V_0
 1796 is the initial decision value at zero delay, and $\lambda_{dec,k} > 0$ is the level-specific decay
 1797 rate. This property is particularly critical in wildfire response, where delayed
 1798 decisions may lose their operational relevance due to rapidly evolving fire dynamics.

$$V_k(d, \Delta t) = V_0 \cdot e^{-\lambda_{dec,k} \cdot \Delta t} \quad (21)$$

1799 At the operational level, $\lambda_{dec,opr} \gg \lambda_{dec,str}$: a resource redeployment decision that is
 1800 delayed by minutes in an active wildfire front may lose all operational relevance,
 1801 whereas a strategic resource allocation decision may retain its value over hours or
 1802 days. This asymmetry directly motivates the architectural separation of decision
 1803 levels in operational DSS: a system that applies strategic-level analytical depth to
 1804 operational-level decisions will consistently generate outputs that are correct but
 1805 temporally obsolete (Endsley, Toward a Theory of Situation Awareness, 1995;
 1806 French, Maule, & Papamichail, 2019).

3.4.3 Decision Scope and Information Flow

As illustrated in Figure 3.5, decision levels are not independent: they form a coupled hierarchy in which aggregated observational data flows upward from the operational level, while decision guidance and resource directives flow downward from the strategic level. A layered information flow in which local observations are aggregated upward while coordination and policy-level guidance propagate downward (Turban, 2011; Power, Decision support systems: Concepts and resources for managers, 2002).

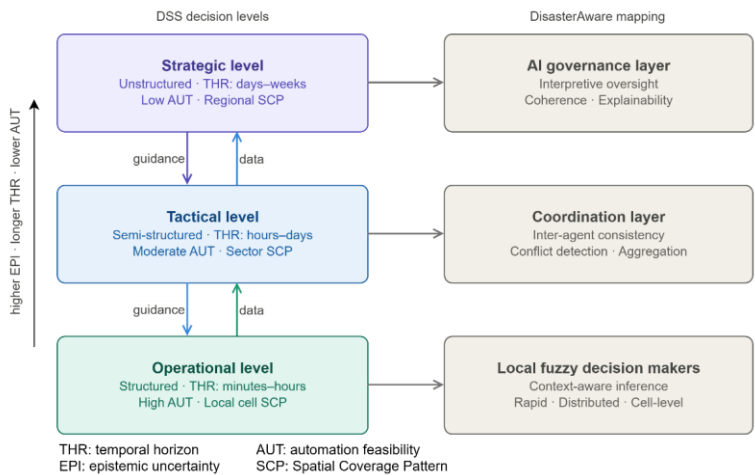


Figure 3.5 Decision levels hierarchy and DisasterAware mapping

This hierarchical structure directly motivates the need for a layered DSS architecture capable of supporting both local responsiveness and global coordination. Each decision level operates with a distinct role, ranging from real-time situational awareness at the operational level to coordination at the tactical level and interpretive oversight at the strategic level.

The properties of the three decision levels and the corresponding DSS design requirements are summarized in Table 3.5. This mapping is not merely descriptive

but prescriptive, as it directly determines the architectural decomposition of the DisasterAware framework into operational, tactical, and strategic layers.

Table 3.5 Decision level characteristics and DSS requirements

Dimension	Strategic	Tactical	Operational
Temporal horizon THR_k	Days–weeks	Hours–days	Minutes–hours
Spatial scope SCP_k	Regional	Sector	Local cell
Epistemic uncertainty EPI_k	High	Moderate	Lower
Automation feasibility AUT_k	Low	Moderate	High
Decay rate $\lambda_{dec,k}$	Slow	Moderate	Rapid
Primary DSS role	Advisory and interpretive	Coordinative and recommendatory	Inferential and directive
DisasterAware realization	AI governance layer	Inter-agent coordination	Local distributed fuzzy DMs

3.4.4 Implications for DSS Architecture

The level taxonomy has direct and non-trivial architectural consequences. A DSS that conflates decision levels by applying a single reasoning engine uniformly across all temporal and spatial scales will systematically fail at the extremes: it will be slow at the operational level and too coarse at the strategic level. This observation motivates level-differentiated DSS architectures in which distinct reasoning mechanisms are deployed at each level, coupled through well-defined information interfaces (Gorry & Scott Morton, 1971; Arnott & Pervan, 2005).

The degree of human involvement must be calibrated to each level. At the strategic level, where uncertainty is high, decisions are unstructured, and consequences are

1837 long-lasting, human judgment is irreplaceable and DSS output should remain
1838 advisory. At the operational level, where decay rate is high and decisions are
1839 relatively structured, a higher degree of algorithmic autonomy is both feasible and
1840 necessary to maintain decision responsiveness (Klein, 2017; Endsley, Situation
1841 awareness misconceptions and misunderstandings, 2015).

1842 As depicted in Figure 3.5, three functionally distinct layers correspond to the three
1843 decision levels: distributed local fuzzy decision makers operating at the operational
1844 level provide rapid, cell-level inference under real-time constraints; a coordination
1845 layer at the tactical level ensures inter-agent consistency and conflict resolution; and
1846 an AI-supported governance layer at the strategic level provides interpretive
1847 oversight, explanatory output, and policy-level coherence checking.

1848 This architecture ensures that the decay rate constraint $\lambda_{dec,k}$ is respected at each
1849 level without sacrificing the analytical depth required at higher levels of the
1850 hierarchy. DisasterAware adopts a level-differentiated architecture in which distinct
1851 reasoning mechanisms are deployed at each decision level while maintaining
1852 structured information flow across layers.

1853 Overall, the multi-level structure of decision-making imposes strict architectural
1854 requirements on DSS design, including the need for distributed local reasoning,
1855 hierarchical coordination, and level-specific decision mechanisms. These
1856 requirements are directly addressed in the DisasterAware framework, whose
1857 architecture is explicitly designed to operate across multiple decision levels while
1858 preserving both responsiveness and global coherence.

1859 3.5 Centralized and Distributed Decision Support Architectures

1860 The architecture of a DSS, whether it is centralized or distributed, has fundamental
1861 implications for system responsiveness, scalability, and operational feasibility in
1862 dynamic environments such as wildfire response. This distinction is not merely an
1863 implementation detail but a structural design choice that directly determines whether

the system can operate effectively under partial observability, communication constraints, and rapidly evolving conditions (Wooldridge, 2009; Sycara, 1998). Within the distributed paradigm, two structural variants are distinguished: a strict peer-to-peer form in which agents exchange observations with immediate neighbors to reach local decisions without any central coordinator, and a hierarchical form in which agents reason independently and a coordination operator assembles their outputs without itself performing inference. The next sections formalize these three paradigms and examine their communication, latency, and scalability properties.

3.5.1 Formal Definitions

In centralized DSS, a single decision authority Δ_c receives state observations from all n field agents $\{a_1, a_2, \dots, a_n\}$, aggregates them into a global state estimate, and issues decisions back to each agent (22) where o_i denotes the local observation of agent a_i and d_i the resulting decision issued to that agent. The central authority maintains a global world model and is the sole source of decision output.

$$d_i = \Delta_c \left(\bigcup_{i=1}^n o_i \right), \forall i \in \{1, \dots, n\} \quad (22)$$

In a strict peer-to-peer distributed DSS architecture, each agent a_i is equipped with a local decision maker δ_i that operates exclusively on its own observation o_i and the observations of its neighborhood $\mathcal{N}(i)$ (23)

$$d_i = \delta_i(o_i, \{o_j : j \in \mathcal{N}(i)\}) \quad (23)$$

No global state is required. Each δ_i is a self-contained reasoning unit that produces a locally valid decision without waiting for or depending on a central authority. The neighborhood $\mathcal{N}(i)$ is typically defined by spatial adjacency or communication range, establishing a sparse interaction topology rather than a fully connected graph.

In a hierarchically distributed DSS architecture, each agent a_i is equipped with a local decision maker δ_i that operates exclusively on its own observation o_i , without

1887 exchanging information with other agents. A separate coordination operator $\mathcal{A}$,
 1888 which is explicitly non-inferential, assembles the collection of local decisions into a
 1889 globally consistent output U (24).

$$d_i = \delta_i(o_i), \forall i \in \{1, \dots, N\}; U = \mathcal{A}(d_1, d_2, \dots, d_N) \quad (24)$$

1890 3.5.2 Communication Complexity and Decision Latency

1891 The communication cost per decision cycle differs fundamentally between the
 1892 architectures. In the centralized case, every agent must transmit its observation to the
 1893 center and receive a decision in return, yielding a per-cycle message count of (25).

$$M_c = 2n \Rightarrow M_c \in O(n) \quad (25)$$

1894 In the distributed case, each agent communicates only with its $|\mathcal{N}(i)|$ neighbors.
 1895 Assuming a bounded neighborhood size $|\mathcal{N}(i)| \leq K$ for all i (26) while per-agent
 1896 communication cost remains $O(K)$, independent of n . This distinction becomes
 1897 critical at scale: as n grows, the centralized authority becomes a communication
 1898 bottleneck, whereas each distributed agent's local load remains constant.

$$M_d = \sum_{i=1}^n |\mathcal{N}(i)| \leq nK \Rightarrow M_d \in O(n) \quad (26)$$

1899 In the hierarchically distributed case, each agent transmits its completed local
 1900 decision to the coordinator in a single one-way message, and the coordinator's output
 1901 is consumed by the simulation layer rather than returned to agents. The per-cycle
 1902 message count is therefore (27). Although the total scales as $O(n)$, the per-agent
 1903 load is $O(1)$, and the transmitted messages carry completed decisions rather than raw
 1904 observations, so no agent blocks on observations from any other agent.

$$M_h = n \rightarrow M_h \in O(n) \quad (27)$$

1905 Equations (25), (26) and (27) indicate that total communication grows as $O(n)$ in
 1906 all three architectures. The per-agent load, however, differs: unbounded at the center
 1907 in the centralized case, bounded by neighborhood size K in the peer-to-peer case,

and constant in the hierarchical case. The more fundamental distinction concerns what the messages carry: in the centralized case they carry observations and the central authority performs all inference, whereas in both distributed variants agents complete their local reasoning before any transmission, so the $O(n)$ message count does not translate into an inferential bottleneck.

Decision latency follows a parallel pattern. The end-to-end latency of a centralized decision cycle is (28) where τ_{collect} is the time to receive all n observations, τ_{agg} is global aggregation time, and $\tau_{\text{distribute}}$ is the time to issue decisions to all agents.

$$\tau_c = \tau_{\text{collect}} + \tau_{\text{agg}} + \tau_{\text{distribute}} \quad (28)$$

In contrast, the latency of a distributed local decision is (29) where τ_{local} is local inference time and $\tau_{\text{nb_exchange}}$ is the time to exchange observations with neighbors. Since $\tau_d \ll \tau_c$ in practice, particularly under communication constraints and large-scale deployments, distributed architectures are structurally better aligned with the temporal urgency constraints formalized in Section 3.4.

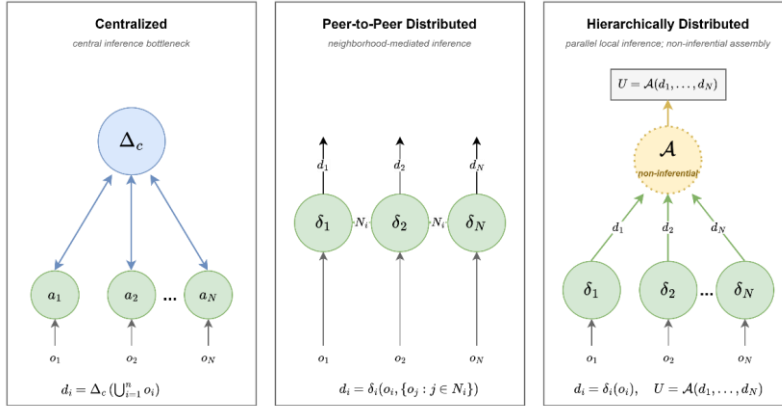
$$\tau_d = \tau_{\text{local}} + \tau_{\text{nb_exchange}} \quad (29)$$

In the hierarchically distributed case, the decision cycle comprises parallel local inference across all N agents, one-way transmission of completed decisions to the coordinator, and non-inferential spatial assembly (30). Because τ_{local} executes in parallel across agents rather than serially at a central inference node, and because τ_{assembly} covers only deterministic spatial assignment and resource normalization rather than fuzzy inference, the effective latency τ_h is dominated by a single local inference pass. This contrasts sharply with the centralized τ_c in (28), which serializes collection, inferential aggregation, and decision distribution.

$$\tau_h = \tau_{\text{local}} + \tau_{\text{transmit}} + \tau_{\text{assembly}} \quad (30)$$

As illustrated in Figure 3.6, the centralized architecture routes all state reports and decisions through a single node as formalized in Equation (22). This means that no local decision can be issued until the central authority has received observations from all agents, completed global aggregation, and distributed decisions back, a sequential

1933 dependency chain that accumulates latency at every stage as captured by τ_c in
 1934 Equation (28).



1935
 1936 Figure 3.6 Centralized, P2P distributed, and hierarchically distributed DSS

1937 In contrast, the peer-to-peer distributed architecture allows each local DM δ_i to infer
 1938 independently within its neighborhood as formalized in Equation (23), with no
 1939 mandatory path through a central bottleneck. Since δ_i requires only o_i and the
 1940 observations of its bounded neighborhood $\mathcal{N}(i)$, the decision cycle reduces to the
 1941 local inference time plus a single neighborhood exchange step, yielding the
 1942 significantly lower latency τ_d defined in Equation (29).

1943 In the hierarchically distributed architecture, each δ_i infers independently using only
 1944 o_i , and the completed local decisions are assembled by the coordinator $\mathcal{A}$ as
 1945 formalized in Equation (24). Since $\mathcal{A}$ performs no inference of its own, there is no
 1946 inferential dependency on a central authority. The decision cycle reduces to parallel
 1947 local inference plus a non-inferential assembly step, yielding the latency τ_h defined
 1948 in Equation (30).

1949 This architectural difference is not a performance characteristic but a structural
 1950 consequence of the dependency graph. In the centralized case, every agent's decision
 1951 is blocked by the slowest reporting agent and the central inferential aggregation step.

1952 In the peer-to-peer distributed case, each agent's decision is blocked only by its own
 1953 inference and its immediate neighbors. In the hierarchically distributed case, each
 1954 agent's decision is blocked only by its own inference, since the coordinator's non-
 1955 inferential assembly introduces no inferential dependency across agents.

1956 Under the rapid decay rates $\lambda_{dec,opr}$ characteristic of the operational decision level,
 1957 this difference directly determines whether decision outputs retain operational value
 1958 or arrive too late to influence evolving fire conditions. Table 3.6 summarizes the key
 1959 architectural differences across dimensions relevant to operational DSS design in
 1960 disaster management.

1961 Table 3.6 demonstrates that distributed architectures provide structural advantages
 1962 in terms of latency, scalability, and robustness relative to centralized systems,
 1963 making them more suitable for dynamic and partially observable environments.

1964 Table 3.6 Comparison of centralized and distributed DSS architectures

Dimension	Centralized	Peer-to-Peer Distributed	Hierarchically Distributed
Comm. per cycle	$O(n)$, all agents report to center	$O(K)$ per agent, neighborhood exchange	$O(n)$ total, $O(1)$ per agent, one-way to coordinator
Decision latency	τ_c $= \tau_{collect} + \tau_{agg}$ $+ \tau_{distribute}$	$\tau_d = \tau_{local} + \tau_{nb}$	τ_h $= \tau_{local}$ $+ \tau_{transmit}$ $+ \tau_{assembly}$
Scalability	Degrades with n ; central bottleneck	Constant per-agent load	Constant per-agent inference; non-inferential assembly

Dimension	Centralized	Peer-to-Peer Distributed	Hierarchically Distributed
Single point of failure	Yes; center failure halts all decisions	No; graceful degradation	Partial; local decisions survive coordinator failure
Global optimality	Potentially achievable	Locally optimal; global not guaranteed	Locally optimal; global consistency via assembly
Observability requirement	Full global state required	Local plus neighborhood state	Local observation only
Uncertainty handling	Global model absorbs all uncertainty	Local inference under partial observability	Per-region local inference under partial observability
Fault tolerance	Low	High	Moderate; high at agent level, bounded at coordinator
Human oversight coupling	Single operator interface	Requires coordination layer	Coordination layer plus AI governance overlay
Realization in DisasterAware	AI governance layer (oversight only)	Not used	Local fuzzy DMs plus $\mathcal{A}_{DSS}$ coordination

3.5.3 Limitations of centralized architecture in wildfire DSS

In the specific context of wildfire disaster management, the centralized architecture exhibits structural failure modes that are not merely performance concerns but fundamental incompatibilities. Wildfire environments are characterized by rapidly

evolving local conditions (i.e., fire front dynamics, fuel state changes, wind shifts) that require decision updates on timescales of minutes or less. A centralized system that must collect global state before issuing any local decision cannot satisfy the temporal decay constraint $\lambda_{dec,opr}$ derived in Section 3.4 (Comes, Hiete, Wijngaards, & Schultmann, 2011).

Furthermore, communication infrastructure in active wildfire zones is frequently degraded by terrain, smoke, and equipment damage, rendering the $O(n)$ communication requirement of centralized architectures operationally fragile. Finally, the requirement for a fully observed global state is inconsistent with the partial observability that characterizes real wildfire environments (Walker, et al., 2003; French, Maule, & Papamichail, 2019). These limitations indicate that centralized architecture is not only inefficient but structurally incompatible with the requirements of real-time wildfire decision support.

3.5.4 Alignment with DisasterAware

DisasterAware realizes the hierarchically distributed architecture of Equation (24). Each local fuzzy agent operates on its own regional observation alone, with no inter-agent communication ($N_i = \emptyset$ for all i). The global coordination operator $\mathcal{A}_{DSS}$ composes the N regional intervention fields through deterministic spatial assignment and resource normalization; $\mathcal{A}_{DSS}$ is non-inferential and carries no reasoning workload. The per-cycle message count and end-to-end latency follow Equations (27) and (30). The latency τ_h is dominated by a single parallel local inference pass and satisfies the temporal decay constraint $\lambda_{dec,opr}$ of the operational decision level.

The architecture realizes the three decision levels established earlier in this chapter. The N local fuzzy agents implement the operational level through parallel inference at regional and cell scale. $\mathcal{A}_{DSS}$ implements the tactical level through global spatial assembly and resource conflict resolution. The AI Governance Layer implements the strategic level through rule-level traceability, audit logging, and operator override.

Completed local decisions flow upward through the assembly stage, and the assembled intervention field influences subsequent observations, closing the decision loop across levels. The architectural realization of these components, including the formal specification of $\mathcal{A}_{DSS}$, the concept mapping pipeline, and the AI Governance Layer, is developed in the subsequent chapters.

3.6 Spatio-Temporal Nature of Decision Making in Wildfire Management

Wildfire behavior constitutes a highly nonlinear spatio-temporal dynamical process in which environmental conditions, terrain properties, fuel characteristics, and suppression actions interact across space and time. This structure imposes strict requirements on DSS implementations, as decisions must be generated continuously over a spatial domain and updated at each time step under partial observability (Rothermel, 1972; Finney, 2006; Sullivan, 2009a; Sullivan, 2009b; Sullivan, 2009c).

Unlike decision problems in bounded, stationary environments, wildfire management requires decisions that are inherently indexed to both spatial location and discrete time step, a property that imposes specific and non-negotiable requirements on the design of any DSS intended to support wildfire response.

3.6.1 Spatial Heterogeneity and Decision Implications

Wildfire behavior is spatially heterogeneous: fire spread rate, intensity, and suppression feasibility vary across the geographic domain as functions of local terrain slope, fuel type and load, wind field, and accessibility. Consequently, decision variables such as resource deployment, suppression priority, and evacuation routing cannot be defined globally but must be indexed to individual spatial locations.

Formally, any decision-relevant quantity must be treated as a spatial field defined over the discretized grid domain $\mathcal{G}$, partitioned into $N_x \times N_y$ cells. For instance, the system state $S_k(x, y)$, the intervention field $U_{DSS,k}(x, y)$, and the observable

information $O_k(x, y)$ are all spatially indexed, emphasizing that decision variables must be treated as spatial fields rather than scalar values. This requirement can be expressed in general form as (31), where $\xi_k(x, y)$ denotes an arbitrary decision-relevant quantity.

$$\xi_k: \mathcal{G} \rightarrow \mathbb{R}, \xi_k = \xi_k(x, y), (x, y) \in \mathcal{G} \quad (31)$$

This field representation stands in fundamental contrast to aggregate MCDM formulations in which a single performance value x_{ij} is assigned to an alternative without spatial indexing. A DSS that ignores spatial heterogeneity will systematically misallocate resources by treating spatially distinct fire conditions as equivalent, a critical failure mode in operational wildfire management (Finney, 2006; Thompson & Calkin, 2011).

3.6.2 Temporal Non-Stationarity and the State Transition Imperative

Wildfire evolution is temporally non-stationary: the fire state at time $k + 1$ is not predictable from time-independent models but depends on the current state S_k and current environmental forcing $\mathcal{F}_{in,k}$ through a structured transition mechanism. This implies that the DSS must operate in discrete-time lockstep with the physical simulation, re-evaluating decision concepts at each step k rather than computing a static decision at initialization. The general form of the spatio-temporal state transition is (32) where Φ denotes the state transition operator encoding fire spread, fuel consumption, and combustion dynamics, and $\mathcal{F}_{in,k}$ encompasses all external environmental and intervention inputs at step k .

$$S_{k+1} = \Phi(S_k, \mathcal{F}_{in,k}) \quad (32)$$

This formulation implies that a DSS must operate synchronously with system evolution, continuously updating decisions based on the evolving state trajectory. This is elaborated further in Chapter 4, which establishes that wildfire state is never static, it is a trajectory through a high-dimensional state space whose evolution is jointly governed by physical dynamics and operational interventions.

2046 3.6.3 Grid-Based State Representation and Decision Indexing

2047 The wildfire state S_k is represented as a collection of local cell states $s_k(x, y)$ over
2048 the grid domain. Each cell state captures the local fire condition through a structured
2049 set of variables, burning status, available fuel load, local intensity, and time elapsed
2050 since ignition, whose values directly determine the decision relevance of each cell at
2051 each time step. This cell-level representation enables the DSS to index decisions
2052 spatially: a cell with high burning status and high intensity requires immediate
2053 suppression attention, while an adjacent unburned cell with high fuel load and wind-
2054 aligned position represents an imminent risk requiring preemptive action. The
2055 decision concepts formalized in Chapter 5, such as Spread Risk and Operational
2056 Priority, are computed per cell and per time step, inheriting the spatio-temporal
2057 indexing structure of the underlying state representation. This structure directly
2058 supports cell-level decision reasoning, enabling localized intervention strategies that
2059 adapt to spatially varying wildfire conditions.

2060 3.6.4 Coupling Between Fire Dynamics and Decision Scope

2061 The spatial scope SCP_k introduced in Section 3.4 is not an independent design
2062 parameter but is structurally determined by the spatial resolution of the state
2063 representation and the temporal horizon THR_k at each decision level. At the
2064 operational level, decisions are indexed to individual grid cells or small
2065 neighborhoods, matching the cell-level granularity of the state vector. At the tactical
2066 level, decisions span contiguous sectors, spatial aggregations of cells sharing similar
2067 fire behavior, enabling inter-agent coordination without requiring cell-level detail.
2068 At the strategic level, decisions apply to the full regional domain and are informed
2069 by spatially aggregated summary statistics rather than cell-level state.

2070 As illustrated in Figure 3.7, the left panel shows the grid domain G with spatially
2071 heterogeneous cell states at a single time step (burning, burned-out, at-risk,
2072 unaffected), while the right panel shows the discrete-time state evolution from $k =$

0 through intervention and controlled phases, illustrating how the DSS observes $O_k(x, y)$ and issues $U_{DSS,k}$ at each step. The lower panel shows how decision scope SCP_k scales with decision level. This structure illustrates that decision-making cannot be separated from the underlying spatio-temporal dynamics, reinforcing the need for integrated simulation–decision frameworks.

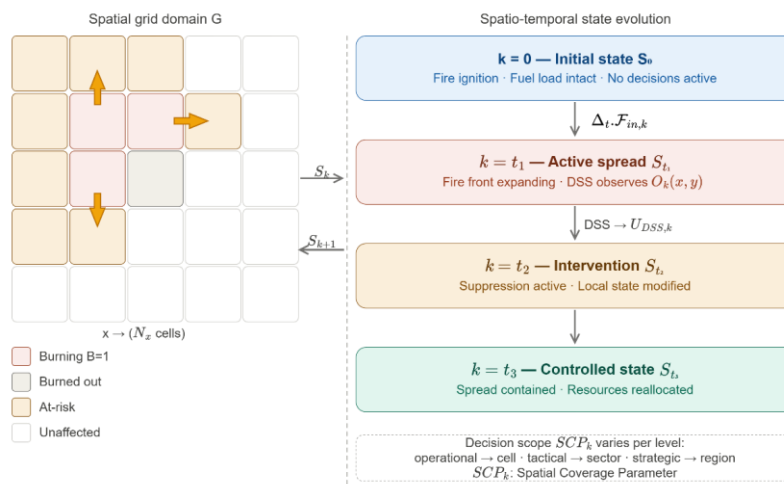


Figure 3.7 Spatio-temporal structure of wildfire DSS decision making

This multi-scale coupling means that spatial structure is not merely a visualization property but a load-bearing element of the decision architecture: the scope of each decision is determined by the spatial resolution at which the state is observable and actionable. This coupling between spatial resolution and decision scope directly how DSS mechanisms must be structured across different levels of DSS architecture.

3.6.5 Partial Observability and Spatial Uncertainty

In operational wildfire environments, the state S_k is never fully observable across the entire domain. Sensor coverage is spatially sparse, remote sensing data is subject to cloud occlusion and update latency, and human observers provide information only

for accessible terrain. These data are not generated internally by the simulator but are acquired from external sources such as meteorological services, geographic information systems (GIS), remote sensing platforms, and operational databases (Kyzirakos, et al., 2012). The result is a partially observed state field in which cell-level confidence varies across the domain (33) where $O_k \subseteq \mathcal{G}$ denotes the set of observable cells at step k , $\hat{S}_k(x, y)$ is the estimated state, and $\tilde{S}_k(x, y)$ is an inferred estimate for unobserved cells.

$$\hat{S}_k(x, y) = S_k(x, y) \cdot \mathbb{1}_{(x, y) \in O_k} + \tilde{S}_k(x, y) \cdot \mathbb{1}_{(x, y) \notin O_k} \quad (33)$$

This partial observability is a fundamental property of the wildfire decision environment that any DSS must accommodate structurally, not merely as an edge case. It directly motivates the use of fuzzy inference in DisasterAware: rather than requiring precise cell-level state values, the fuzzy decision layer operates on graded membership degrees that explicitly represent the uncertainty associated with partially observed or model-estimated state variables (Thompson & Calkin, 2011; Walker, et al., 2003).

Table 3.7 summarizes these spatio-temporal properties that impose strict structural requirements on DSS design, particularly the need for spatially distributed reasoning, time-synchronous decision updates, and uncertainty-aware inference mechanisms.

Table 3.7 Spatio-temporal properties of wildfire dynamics

Property	Description	DSS Design Implication	DisasterAware Realization
Spatial heterogeneity	Fire behavior varies continuously across $\mathcal{G}$ as a function of terrain, fuel, and wind	Decision variables must be spatially indexed fields, not scalar alternatives	Cell-level fuzzy decision makers indexed to grid coordinates (x, y)

Property	Description	DSS Design Implication	DisasterAware Realization
Temporal non-stationarity	State S_k evolves through transition at every time step k	DSS must re-evaluate decisions at each discrete time step	Event-triggered and step-synchronous inference
Multi-scale coupling	Decision scope depends on aggregation level	Different reasoning granularity per decision level	Operational: cell · Tactical: sector · Strategic: region
Partial observability	$O_k(x, y) = h(S_k, \epsilon_k)$ only a subset of cells is observable	Reasoning must tolerate missing and uncertain state information	Fuzzy membership degrees encode observation confidence
Nonlinear spread dynamics	Spread rate R_{spread} is nonlinear in wind, slope, fuel moisture	Predictive horizon is short; decisions must be rapidly revised	Short-horizon inference with adaptive reweighting
State–decision coupling	DSS outputs $U_{DSS,k}$ feed back into $\mathcal{F}_{in,k}$	Decision effects must be modeled as state-influencing inputs, not direct overrides	$U_{DSS,k}$ enters simulation as structured external input via $\mathcal{F}_{in,k}$

2107

2108 The spatio-temporal properties described in this section establish the structural
2109 requirements that the DisasterAware Simulation Framework must satisfy. The grid
2110 domain $\mathcal{G}$, the state vector $s_k(x, y)$, the state transition operator Φ , and the external
2111 input structure $\mathcal{F}_{in,k}$, all formalized in Chapter 4, are direct architectural responses
2112 to the requirements of spatial indexing, temporal non-stationarity, and partial
2113 observability identified here.

2114 Similarly, the strict separation between physical state evolution and decision logic
2115 enforced in Chapter 4 is motivated by the state-decision coupling property: decisions
2116 must enter the simulation as structured external inputs through $U_{DSS,k}$, never as direct
2117 overwrites of S_k , in order to preserve physical consistency and simulation integrity.

2118 Specifically, a DSS must operate over spatially indexed state representations, update
2119 decisions synchronously with system evolution, and explicitly account for partial
2120 observability. These requirements directly motivate the simulation framework
2121 presented in Chapter 4 and the distributed DSS architecture developed in Chapters 5
2122 and 6.

2123

CHAPTER 4

2124

DISASTER-AWARE SIMULATION FRAMEWORK

2125 This chapter defines the structural, computational, and physical foundations of the
2126 DisasterAware Simulation Framework. Its primary objective is to formalize how
2127 wildfire dynamics are represented and computed before introducing the cognitive
2128 reasoning mechanisms of the Decision Support System in the subsequent chapter.
2129 The simulation layer constitutes the physical backbone of the entire framework;
2130 therefore, its structure, assumptions, and constraints must be explicitly defined and
2131 rigorously bounded.

2132 Wildfire evolution is modeled as a spatio-temporal process over a discretized spatial
2133 domain. The geographic region under analysis is partitioned into grid cells, each
2134 representing a localized fire condition described by structured state variables. These
2135 state variables capture whether combustion is active, how much combustible fuel
2136 remains, how intense the local fire behavior is, and how long combustion has
2137 persisted. By organizing wildfire behavior into a state-based representation, the
2138 framework moves beyond purely geometric spread visualization and instead treats
2139 fire propagation as a constrained dynamical system.

2140 The evolution of the system is influenced by multiple categories of external forcing.
2141 Meteorological drivers such as wind speed and humidity affect spread acceleration
2142 and combustion sustainability. Terrain properties such as slope and aspect introduce
2143 directional bias and amplification effects. Fuel characteristics regulate ignition
2144 thresholds and burn duration. In addition, operational intervention parameters alter
2145 local fuel availability through suppression actions. These influences are explicitly
2146 separated from the internal wildfire state in order to preserve architectural clarity and
2147 computational stability.

2148 A key design principle of the Simulation Framework is the strict separation between
2149 physical state evolution and decision logic. This layer is responsible for modeling
2150 wildfire behavior under environmental and operational inputs. It does not perform
2151 reasoning, optimization, or policy selection. This separation ensures that high-level
2152 decision mechanisms run on stable and physically consistent environment. Human
2153 interaction and scenario manipulation are permitted only through controlled input-
2154 level modifications, never through direct state overwriting.

2155 In addition to defining wildfire propagation mechanics, this chapter establishes how
2156 interaction, visualization, and structural coupling with the DSS are organized within
2157 the architecture. The Decision Support System is introduced here only to clarify its
2158 structural interface with the simulation environment. Its internal fuzzy reasoning
2159 structure, layered decision synthesis, and optimization mechanisms are intentionally
2160 deferred to the next chapter. The remainder of this chapter is organized as follows:

2161 Section 4.1 presents the layered architectural structure and inter-layer data flow of
2162 the DisasterAware framework.

2163 Section 4.2 details and formalizes the external input sources, discrete-time state
2164 representation, state variables, hybrid fire spread transition mechanism, and
2165 interaction and scenario control.

2166 Section 4.3 specifies the external data layers, (i.e., meteorological, terrain, fuel,
2167 value-at-risk, resource) that drive the simulation.

2168 Section 4.4 describes the UI Dashboard as a supervisory interface for visualization
2169 and controlled scenario manipulation.

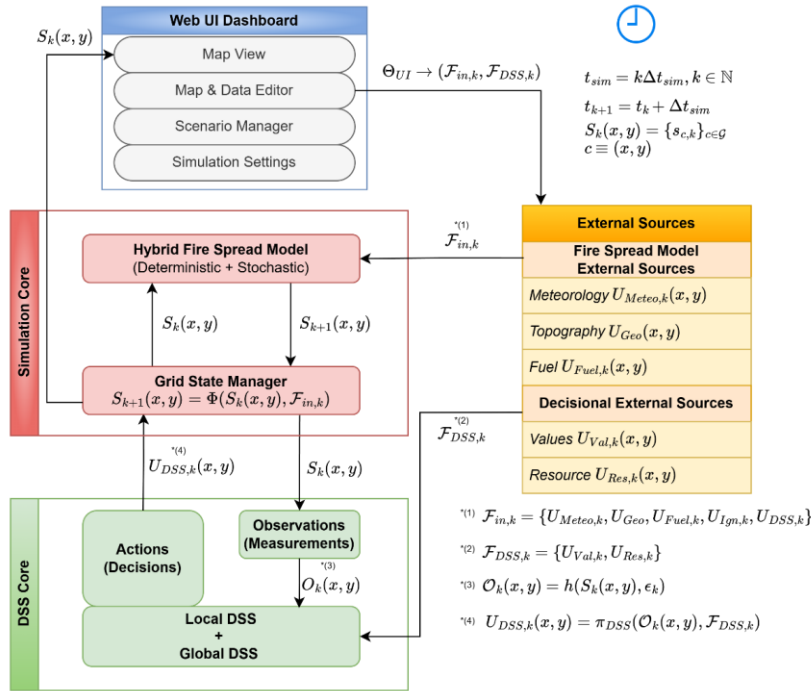
2170 Section 4.5 introduces the structural positioning of the DSS Core within the
2171 framework, deferring the design specifications to Chapter 5 and the architectural
2172 realization to Chapter 6.

2173 Section 4.6 synthesizes the chapter's architectural and mathematical contributions
2174 and transitions to the DSS design specifications in Chapter 5.

2175 Through this structured progression, Chapter 4 establishes the physically grounded
 2176 computational environment required for reliable, interpretable, and scalable decision
 2177 support in wildfire management.

2178 4.1 Overall System Block Diagram

2179 Figure 4.1 presents the architectural structure of the DisasterAware Simulation
 2180 Framework. The system is designed as a discrete-time, state-driven, layered DSS
 2181 architecture with a clear separation between input acquisition, physical state
 2182 evolution, decision evaluation, and human interaction. The architectural objective is
 2183 to preserve physical consistency within wildfire simulation while enabling adaptive
 2184 and interpretable decision-making.



2185
 2186 Figure 4.1 DisasterAware system block diagram

2187 The architecture is organized into four principal layers: External Data Sources,
 2188 Simulation Core, Decision Support System (DSS) Core, and Web UI Dashboard.
 2189 Each layer performs a strictly defined function and communicates through structured
 2190 interfaces rather than direct state manipulation. This separation ensures physical
 2191 consistency, operational robustness, and architectural transparency.

2192 The **External Data Sources** layer provides structured exogenous inputs including
 2193 meteorological, terrain, fuel, and decisional parameters. Detailed formulation of
 2194 these inputs is provided in Section 4.2.1. These inputs act as boundary conditions for
 2195 the simulation. Environmental inputs are independent of the system state, while
 2196 decision-driven inputs generated by the DSS depend on observable quantities
 2197 derived from the system state.

2198 At the center of the framework lies the **Simulation Core**, which serves as the
 2199 authoritative computational engine. Wildfire propagation is modeled over a spatial
 2200 grid where each cell contains a structured representation of local fire conditions. The
 2201 complete wildfire state includes burning status, available combustible mass, local
 2202 fire intensity proxy, and time since ignition. This state is maintained and updated at
 2203 every simulation time step. The Grid State Manager enforces physical constraints
 2204 such as non-negativity of fuel load and bounded combustion behavior. No external
 2205 module, including the DSS or the UI, is permitted to directly overwrite internal state
 2206 variables. All modifications must occur through controlled parameter influence,
 2207 preserving simulation integrity.

2208 The **Decision Support System Core** operates as a cognitive layer built on top of the
 2209 Simulation Core. The DSS operates on observable quantities $O_k(x, y)$ derived from
 2210 the system state through the observation process. This observation layer extracts
 2211 structured indicators from the raw state variables, such as fire front behavior,
 2212 acceleration zones, exposure metrics, and operational accessibility conditions.

2213 The DSS can evaluate conditions at every simulation time step, but decision
 2214 activation may follow event-triggered conditions. It may operate in an event-driven
 2215 manner, producing intervention strategies only when predefined thresholds or risk

2216 conditions are triggered. When a decision is generated, it is translated into
2217 operational parameters that influence subsequent state transitions indirectly through
2218 the simulation engine.

2219 The **Web UI Dashboard** forms the operational boundary between human operators
2220 and the computational layers. Unlike the Simulation Core, which is time-
2221 synchronous, the UI layer operates asynchronously. The operator can pause the
2222 simulation; resume execution; roll back to a stored checkpoint; restart from initial
2223 conditions; inject new ignition scenarios; modify operational resource availability;
2224 and cancel an active simulation.

2225 Importantly, the UI never directly modifies internal state variables. Instead, it adjusts
2226 permissible configuration parameters that affect future simulation steps. This ensures
2227 that human intervention remains controlled, traceable, and reversible.

2228 The data flow between layers follows a strict directional structure:

- 2229 • External inputs feed the Simulation Core at each time step.
- 2230 • The Simulation Core updates the authoritative wildfire state.
- 2231 • The DSS observes the current state through observable information $O_k(x, y)$,
2232 uses contextual decision inputs $\mathcal{F}_{DSS,k}$, evaluates risk, and generates
2233 structured intervention recommendations when required.
- 2234 • The UI visualizes both physical state evolution and DSS outputs while
2235 allowing controlled scenario manipulation.

2236 As shown in Figure 4.1, this layered organization enforces a clear separation between
2237 wildfire physics, decision cognition, and human supervision. The architecture
2238 guarantees: (i) Physical consistency of fire propagation; (ii) Modular extensibility;
2239 (iii) Horizontal scalability; (iv) Traceable intervention logic; (v) Robust HIL control.

2240 By maintaining strict boundaries between simulation dynamics and decision
2241 reasoning, the framework establishes a stable computational foundation upon which

distributed fuzzy reasoning and adaptive coordination mechanisms can be constructed in subsequent chapters.

4.2 Mathematical Model of the Simulation Core

This section formalizes the discrete-time wildfire simulation dynamics implemented in the DisasterAware Simulation Core. The objective is to represent wildfire propagation over the spatial grid $\mathcal{G}(x, y)$ using a structured state-space formulation that clearly separates spatial domain definition, physical propagation dynamics, external forcing, and controlled user interaction.

The Simulation Core operates as a discrete-time state evolution system defined over a spatially discretized domain. At each simulation step, the system receives environmental forcing inputs, updates the wildfire state over the grid, and produces the next state through a structured transition mechanism. The Simulation Core is organized around four functional components: the external input structure, the spatial grid and state representation, the wildfire state-transition mechanism, and the interaction layer for controlled scenario manipulation. Together, these components define how wildfire dynamics are represented, updated, and influenced within the DisasterAware simulation environment.

The external input structure defines meteorological, terrain, fuel, ignition, and decision-related inputs in Section 4.2.1. The spatial representation defines the discretized grid and the wildfire state variables associated with each cell in Section 4.2.2. The transition mechanism governs burning persistence, spatial propagation, fuel consumption, and intensity evolution in Section 4.2.3. The interaction layer enables admissible user intervention without direct modification of the internal wildfire state in Section 4.2.4.

2266 4.2.1 External Input Sources

2267 The external input $\mathcal{F}_{in,k}$ represents all parameters that influence wildfire evolution
 2268 at discrete time step k . These inputs include both exogenous environmental factors
 2269 and decision-driven intervention parameters generated by the DSS. These inputs act
 2270 as boundary conditions and forcing mechanisms for the state transition process.
 2271 Environmental inputs are updated independently of the system state, whereas
 2272 decision-driven inputs depend on observable quantities derived from the state. At
 2273 each discrete simulation step k , the external input set is updated and provided to the
 2274 Simulation Core.

2275 External inputs can be categorized into three distinct groups: (i) Environmental
 2276 input: including meteorological, terrain, and fuel-related factors that govern physical
 2277 wildfire dynamics; (ii) Exogenous event inputs: such as ignition events, which
 2278 trigger fire initiation independently of system evolution; (iii) Decision-driven inputs:
 2279 intervention parameters generated by the DSS, representing operational actions
 2280 influencing wildfire evolution.

2281 $\mathcal{F}_{in,k}$ (34) represents all external parameters that influence wildfire dynamics,
 2282 namely meteorology $U_{Meteo,k}$ (35), topography U_{Geo} (36), fuel $U_{Fuel,k}$ (37),
 2283 ignition $U_{Ign,k}$ (38) and inputs generated by DSS $U_{DSS,k}$ (39). This structured
 2284 separation ensures that environmental forcing, ignition triggering, and operational
 2285 intervention remain conceptually and functionally distinct.

$$\mathcal{F}_{in,k} = \{U_{Meteo,k}, U_{Geo}, U_{Fuel,k}, U_{Ign,k}, U_{DSS,k}\} \quad (34)$$

2286 In addition to the simulation input set $\mathcal{F}_{in,k}$, the DSS process relies on a distinct
 2287 contextual input set defined as $\mathcal{F}_{DSS,k} = \{U_{Val,k}, U_{Res,k}\}$. This set represents value-
 2288 related and resource-related information used by the DSS for decision evaluation.
 2289 Unlike $\mathcal{F}_{in,k}$, which directly influences the state transition operator, $\mathcal{F}_{DSS,k}$ does not
 2290 affect wildfire dynamics directly. Instead, it provides contextual information that
 2291 guides decision-generation within the DSS layer.

2292 Meteorological inputs $U_{Meteo,k}$ (35) are spatio-temporal environmental drivers that
 2293 directly influence ignition sustainability, rate of spread, fuel drying, and combustion
 2294 intensity. These fields are updated at every discrete simulation step to ensure
 2295 dynamic adaptation of wildfire behavior to evolving environmental conditions.

$$U_{Meteo,k}(x,y) = \begin{bmatrix} W_{temp,k}(x,y) \\ W_{rh,k}(x,y) \\ W_{ws,k}(x,y) \\ W_{wd,k}(x,y) \\ W_{gust,k}(x,y) \\ W_{prec,k} \end{bmatrix}$$

$W_{temp,k}(x,y)$: Air temperature field (35)

$W_{rh,k}(x,y)$: Relative humidity field

$W_{ws,k}(x,y)$: Wind speed field

$W_{wd,k}(x,y)$: Wind direction field

$W_{gust,k}(x,y)$: Wind gust field

$W_{prec,k}$: Precipitation (temporal form)

2296 Terrain parameters U_{Geo} (36) define static spatial constraints over which wildfire
 2297 propagation occurs. Unlike meteorological variables, these parameters remain
 2298 constant during short-horizon simulation but modulate directional spread,
 2299 acceleration, and accessibility.

$$U_{Geo}(x,y) = \begin{bmatrix} G_{elev}(x,y) \\ G_{slope}(x,y) \\ G_{aspect}(x,y) \\ G_{access}(x,y) \end{bmatrix}$$

$G_{elev}(x,y)$: Elevation (36)

$G_{slope}(x,y)$: Terrain slope

$G_{aspect}(x,y)$: Terrain orientation

$G_{access}(x,y)$: Accessibility level

2300 Fuel-related inputs $U_{Fuel,k}$ (37) describe the combustible material available at each
 2301 grid cell. $F_{type}(x,y)$ is the vegetation class and $F_{load,0}(x,y)$ is the available

2302 combustible mass at initial time. Fuel moisture $F_{moist,k}(x, y)$ is treated as an
 2303 exogenous spatio-temporal input field updated from meteorological drivers or
 2304 scenario configurations.

$$U_{Fuel,k}(x, y) = \begin{bmatrix} F_{type}(x, y) \\ F_{load,0}(x, y) \\ F_{moist,k}(x, y) \end{bmatrix} \quad (37)$$

$F_{type}(x, y)$: Vegetation/fuel class

$F_{load,0}(x, y)$: Available combustible mass at initial time at $k = 0$

$F_{moist,k}(x, y)$: Surface fuel moisture content

2305 Ignition is modeled separately from fire intensity. It represents the activation of
 2306 combustion at a specific grid cell and time step. Ignition may originate from
 2307 predefined initial conditions, external events, or spotting effects. It functions as a
 2308 triggering mechanism within the transition dynamics and does not constitute a
 2309 persistent state variable.

$$U_{Ign,k}(x, y) = [I_{Ign,k}(x, y)] \quad (38)$$

$I_{Ign,k}(x, y)$: Binary ignition injection indicator

2310 The DSS layer generates decision-driven intervention parameters $U_{DSS,k}$ (39) based
 2311 on observed system conditions. These parameters do not represent physical
 2312 measurements but structured representations of operational actions, which are
 2313 incorporated into the external input set to influence wildfire dynamics.

$$U_{DSS,k}(x, y) = \begin{bmatrix} R_{cap,k}(x, y) \\ R_{avail,k}(x, y) \\ R_{eff,k}(x, y) \\ R_{time,k}(x, y) \end{bmatrix} \quad (39)$$

$R_{cap,k}(x, y)$: Available suppression capacity

$R_{avail,k}(x, y)$: Operational availability status of resources

$R_{eff,k}(x, y)$: Suppression effectiveness

$R_{time,k}(x, y)$: Estimated travel time to reach grid cell (x, y)

2314 By structuring the external input subsystem into meteorological forcing, terrain
 2315 constraints, fuel properties, ignition triggers, and operational intervention
 2316 parameters, the architecture preserves a clear separation between environmental
 2317 dynamics, fire initiation, and decision-driven mitigation. This modular organization
 2318 guarantees that physical wildfire behavior remains consistent while enabling
 2319 adaptive and controlled intervention strategies.

2320 4.2.2 Spatio-Temporal State Transition

2321 The discretized grid map $\mathcal{G}$ defines the computational approximation of the
 2322 continuous geographic wildfire environment. Each grid cell corresponds to a spatial
 2323 coordinate pair (x, y) and represents a localized fire condition (40).

$$\mathcal{G} = \{ (x, y) \mid x = 1, \dots, N_x, y = 1, \dots, N_y \}$$

N_x is the number of grid cells along the horizontal spatial axis, (40)

N_y is the number of grid cells along the vertical spatial axis,

$(x, y) \in \mathcal{G}$ denotes a discrete grid cell coordinate.

2324 At each simulation step, the wildfire state over the grid is represented by S_k . The
 2325 system state is defined as the collection of local cell states over the domain (41).

$$S_k = \{s_k(x, y) \mid (x, y) \in \mathcal{G}\}$$

$$s_k = \begin{bmatrix} B_k(x, y) \\ F_{load,k}(x, y) \\ I_k(x, y) \\ \tau_k(x, y) \end{bmatrix} \quad (41)$$

$B_k(x, y)$: Burning status of grid (x, y)

$F_{load,k}(x, y)$: Available combustible mass

$I_k(x, y)$: Local fire intensity proxy

$\tau_k(x, y)$: Time since ignition

2326 The state vector includes only internal wildfire variables. Environmental factors and
 2327 operational interventions are not part of the state and are represented as external
 2328 inputs influencing the state transition.

By representing wildfire evolution as a grid-based state-space system, the framework enables localized propagation analysis; cell-level intervention control; consistent discrete-time state evolution; and direct integration with decision-layer observation mechanisms. This structured state representation forms the foundation for the hybrid transition operator introduced in the subsequent section.

Figure 4.2 illustrates the three-step relationship between the continuous geographic domain Ω , its discretization into the grid $\mathcal{G}$ of Equation (40), and the four-component cell state vector $s_k(x, y)$ of Equation (41) that serves as the authoritative state representation throughout the Simulation Core.

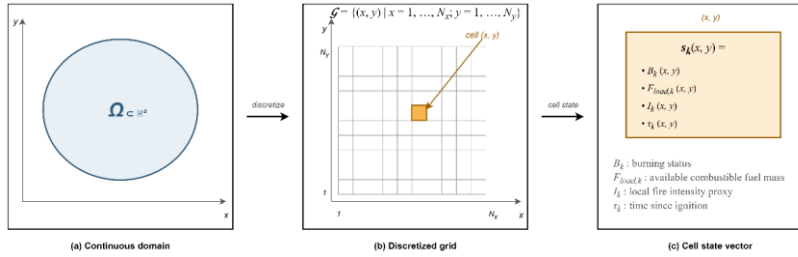


Figure 4.2 Grid domain representation and per-cell state structure

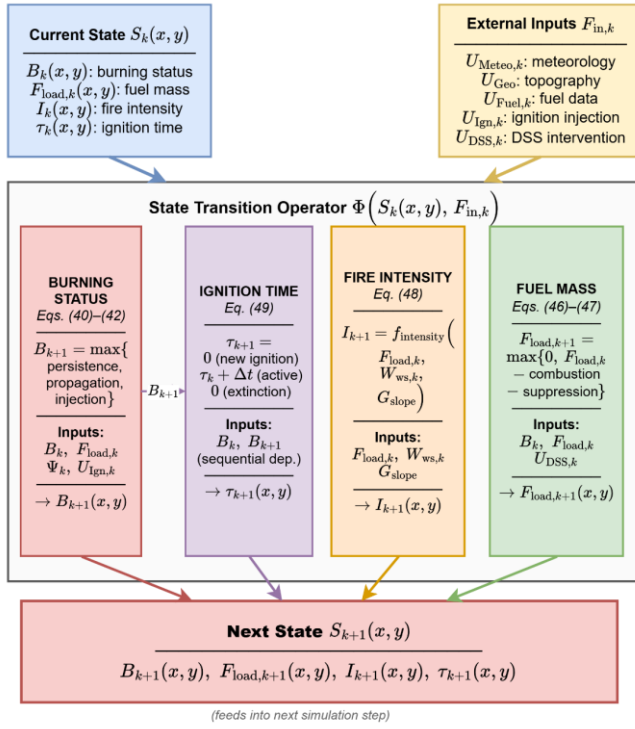
4.2.3 Hybrid Fire Spread Model

The hybrid model integrates deterministic combustion dynamics with stochastic variability to represent realistic wildfire behavior. The global state evolution (42) is governed by the transition operator where $\mathcal{F}_{in,k}$ (34) includes meteorological forcing, topography, fuel conditions, ignition injection, and DSS-generated intervention parameters. Φ denotes the structured state transition operator composed of burning status evolution, fuel mass evolution, and fire intensity update as defined in the subsequent subsections. The transition operator itself is deterministic. Stochastic variability may arise only through exogenous input fields (e.g., wind gust fluctuations), which are treated as externally provided spatio-temporal inputs.

$$S_{k+1}(x, y) = \Phi(S_k(x, y), F_{in,k}) \quad (42)$$

$$F_{in,k} = \{U_{Meteo,k}, U_{Geo}, U_{Fuel,k}, U_{Ign,k}, U_{DSS,k}\}$$

2350 The transition for each cell satisfies the coupled equations for each grid cell $(x, y) \in$
 2351 $\mathcal{G}$, and the update consists of three coupled components: *burning status evolution*,
 2352 *fuel mass evolution*, and *fire intensity evolution*. Decision-driven inputs $U_{DSS,k}$
 2353 influence the transition ensuring that intervention effects are propagated indirectly
 2354 through the physical model.



2355

2356 Figure 4.3 Hybrid fire spread transition mechanism

2357 Figure 4.3 decomposes the hybrid transition operator Φ into four parallel update
 2358 mechanisms, each corresponding to one component of the cell-level state vector
 2359 $s_k(x, y) = (B_k, F_{load,k}, I_k, \tau_k)$. The Burning Status mechanism (43)–(45)

2360 combines persistence, propagation, and external ignition injection to produce B_{k+1} .
 2361 The Fuel Mass mechanism (49)–(50) applies combustion-driven depletion,
 2362 conditioned on $B_k = 1$, and suppression-driven reduction, driven by $U_{DSS,k}$, under
 2363 a non-negativity constraint, yielding $F_{load,k+1}$. The Fire Intensity mechanism (51)
 2364 evaluates the intensity function $f_{intensity}$ over current fuel, wind speed, and terrain
 2365 slope to produce I_{k+1} , intentionally excluding any feedback from I_k itself in order
 2366 to preserve the role of intensity as a purely descriptive proxy. The Ignition Time
 2367 mechanism (52) updates τ_{k+1} according to the ignition-initialization, active-
 2368 burning, or extinction rule, and exhibits a distinctive sequential dependency on both
 2369 B_k and the just-computed B_{k+1} , indicated in the figure by the dashed horizontal
 2370 arrow. Exogenous forcing enters the operator exclusively through the external input
 2371 set $F_{in,k}$, preserving the architectural separation between physical dynamics and
 2372 decision-driven inputs established in Section 4.1.

2373 **Burning Status Evolution ($B_k(x, y) \rightarrow B_{k+1}(x, y)$)**

2374 The burning status $B_k(x, y) \in \{0,1\}$ indicates whether the cell is actively burning at
 2375 time step k . The burning status at the next step B_{k+1} (43) is determined by three
 2376 mechanisms: persistence of ongoing burning B_{k+1}^{pers} (44), propagation from
 2377 neighboring burning cells, and external ignition injection $I_{Ign,k}(x, y)$. Combining
 2378 these three mechanisms, the burning status update rule is defined as (43).

$$B_{k+1}(x, y) = \max \left(B_{k+1}^{pers}(x, y), B_{k+1}^{prop}(x, y), I_{Ign,k}(x, y) \right) \quad (43)$$

2379 This formulation ensures that a burning cell persists while fuel remains available;
 2380 neighboring burning cells may activate a new cell if sufficient spread influence
 2381 accumulates; external ignition acts as an independent triggering mechanism.

2382 The persistence term B_{k+1}^{pers} ensures that a cell that is already burning remains burning
 2383 as long as fuel exists(44) where $\mathbf{1}[\cdot]$ denotes the indicator function. To avoid
 2384 numerical persistence under vanishing residual fuel mass, a small extinction
 2385 threshold $\epsilon_{fuel_{load}}$ is used.

$$B_{k+1}^{pers}(x, y) = B_k(x, y) \cdot \mathbf{1}[F_{load,k}(x, y) > \epsilon_{fuel_{load}}] \quad (44)$$

$$\epsilon_{fuel_{load}} = 10^{-4}$$

2386 Propagation to cell (x, y) occurs if the accumulated influence exceeds a critical
 2387 ignition threshold Θ_{ign} , where Θ_{ign} is a model parameter representing the minimum
 2388 spread intensity required to activate combustion in a new cell (45).

$$B_{k+1}^{prop}(x, y) = \mathbf{1}[\Psi_k(x, y) > \Theta_{ign}] \quad (45)$$

$\Theta_{ign} \in (0,1)$: calibrated threshold relative to base spread rate

2389 Wildfire spread to neighboring cells is modeled through local propagation influence.
 2390 $\mathcal{N}^8(x, y)$ denotes the 8-connected neighborhood of cell (x, y) . The total propagation
 2391 influence received by cell (x, y) at time step k $\Psi_k(x, y)$ is defined (46) where rate
 2392 of spread $R_{spread,k}(x, y)$ is (47). The rate-of-spread mapping $f_{ros}(\cdot)$ is instantiated
 2393 using a semi-empirical Rothermel-type formulation (Rothermel, 1972). Detailed
 2394 parameterization is provided in Appendix A.

$$\Psi_k(x, y) = \frac{1}{8} \sum_{(i,j) \in \mathcal{N}^8(x,y)} B_k(i, j) \cdot R_{spread,k}(i, j) \cdot g_{dir,k}^{(i,j) \rightarrow (x,y)} \quad (46)$$

$$R_{spread,k}(x, y) = f_{ros} \left(\begin{matrix} W_{ws,k}(x, y), W_{wd,k}(x, y), \\ G_{slope}(x, y), G_{aspect}(x, y), \\ F_{moist,k}(x, y), F_{type}(x, y) \end{matrix} \right)$$

f_{ros} : function rate-of-spread (Appendix A)

$W_{ws,k}(x, y)$: wind speed field

$W_{wd,k}(x, y)$: wind direction field (47)

$G_{slope}(x, y)$: terrain slope

$G_{aspect}(x, y)$: terrain orientation

$F_{moist,k}(x, y)$: fuel surface moisture

$F_{type}(x, y)$: fuel type

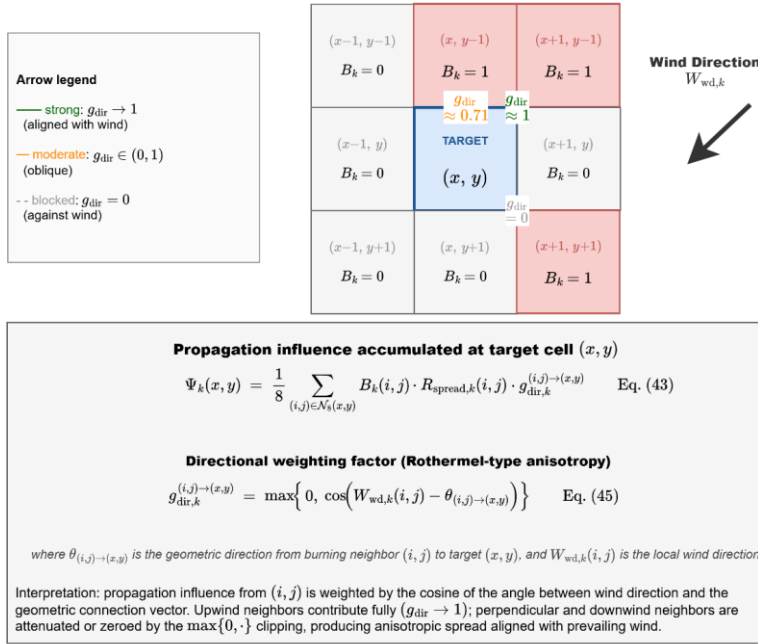
2395 The scalar magnitude $R_{spread,k}(x, y)$ does not determine spread direction.
 2396 Directional influence (48) is introduced when propagating fire from a burning

neighbor (i, j) to target cell (x, y). $\theta_{(i,j) \rightarrow (x,y)}$ is the geometric direction between cells. This ensures anisotropic spread aligned with wind direction.

$$g_{dir,k}^{(i,j) \rightarrow (x,y)} = \max(0, \cos(W_{wd,k}(i,j) - \theta_{(i,j) \rightarrow (x,y)})) \quad (48)$$

8-connected neighborhood $N_8(x, y)$ and wind-dependent directional propagation into target cell (x, y)

Filled red cells indicate burning neighbors ($B_k = 1$); arrow thickness reflects directional weight $g_{dir} \in [0, 1]$



2399

2400 Figure 4.4 Wind-dependent directional propagation into target cell

2401 Figure 4.4 depicts the spatial structure of the propagation influence accumulation.
 2402 The 3×3 stencil centered on the target cell (x, y) represents the 8-connected
 2403 neighborhood $N_8(x, y)$ defined in (46); the three cells shaded in red denote burning
 2404 neighbors with $B_k(i, j) = 1$, while the remaining five are non-burning. The wind
 2405 direction $W_{wd,k}$, represented by the large arrow in the upper-right region of the
 2406 figure, establishes the axis of anisotropy.

2407 For each burning neighbor, a propagation arrow is drawn toward the target cell with
 2408 line weight proportional to its directional weight $g_{dir,k}^{(i,j) \rightarrow (x,y)}$: the arrow from the
 2409 northeast neighbor is aligned with the wind vector and therefore receives full weight
 2410 $g_{dir} \rightarrow 1$; the arrow from the northern neighbor is oblique and receives an
 2411 intermediate weight $g_{dir} \approx 0.71$; and the arrow from the southeastern neighbor is
 2412 oriented opposite to the wind and is entirely eliminated, $g_{dir} = 0$, by the $\max\{0, \cdot\}$
 2413 clipping. The aggregate propagation influence $\Psi_k(x, y)$ computed through (46) is
 2414 therefore not a symmetric average over the 8-neighborhood but a wind-aligned
 2415 weighted sum, and the ignition condition $\Psi_k(x, y) > \theta_{ign}$ (45) is consequently
 2416 satisfied preferentially along the direction of prevailing wind. This mechanism
 2417 realizes the Rothermel-type directional bias that is central to operationally realistic
 2418 wildfire propagation.

2419 External ignition injection $I_{Ign,k}(x, y) \in \{0, 1\}$ directly activates combustion
 2420 independent of neighbor influence. Ignition injection may represent lightning strikes,
 2421 human-caused ignition, or scenario-based ignition input from the UI layer.

2422 It is important to emphasize that ignition injection $I_{Ign,k}(x, y)$ is not part of the
 2423 persistent state vector. It is an external triggering input defined under $U_{Ign,k}(x, y)$.
 2424 Ignition injection is a scenario-level forcing that sets $B_{k+1}(x, y) = 1$ regardless of
 2425 local conditions; if $F_{load,k}(x, y)$ is insufficient, extinction follows naturally via the
 2426 persistence condition in the next step. In contrast, the local fire intensity proxy
 2427 $I_k(x, y)$ represents combustion strength after ignition has occurred and does not
 2428 participate in the activation condition. This separation preserves conceptual clarity
 2429 between fire initiation and fire magnitude.

2430 Through this formulation, burning status evolution is explicitly linked to spatial
 2431 coupling, fuel availability, and ignition forcing, thereby establishing a physically
 2432 interpretable and DSS compatible propagation mechanism.

2433 Propagation dynamics are governed exclusively by the rate-of-spread mapping
 2434 $R_{spread,k}(x, y)$. The local fire intensity proxy does not directly influence activation

conditions; instead, it modulates suppression effectiveness and decision-layer interpretation. This separation preserves structural clarity between spread mechanics and combustion strength representation. Thus, intensity affects wildfire dynamics indirectly through suppression resistance and decision-layer interpretation, rather than directly controlling fire propagation.

The rate-of-spread mapping $R_{spread,k}(x, y)$ is independent of instantaneous fuel mass $F_{load,k}(x, y)$. Fuel availability influences spread only indirectly through the burning persistence condition, preventing propagation from fuel-depleted cells. This structural separation avoids double-counting mitigation effects.

2444 **Fuel Mass Evolution** ($F_{load,k}(x, y) \rightarrow F_{load,k+1}(x, y)$)

Fuel mass evolution governs the depletion of combustible material within each cell as a consequence of two distinct mechanisms: combustion-driven consumption and intervention-driven suppression. Unlike the burning status variable, which represents the activation of combustion, the fuel mass variable represents the remaining energy potential available for future fire propagation. Therefore, accurate modeling of fuel evolution is essential to ensure physical consistency, numerical stability, and interpretable interaction between wildfire dynamics and decision-driven mitigation. Combustion-driven consumption and suppression-driven reduction are applied within the same discrete step k as a simultaneous update, represented by the combined terms in (49):

$$F_{load,k+1}(x, y) = \max \left(0, \left(\begin{array}{c} F_{load,k}(x, y) \\ -F_{load,k}(x, y) B_k(x, y) F_{burn,k}(x, y) \\ -F_{red,k}(x, y) \end{array} \right) \right)$$

$B_k(x, y) \in \{0,1\}$: the burning status indicator,

$F_{burn,k}(x, y) \in [0,1]$: the fractional burn rate (Appendix B), (49)

$F_{load,k}(x, y)$: the currently available fuel mass.

$F_{red,k}(x, y) \in [0,1]$: intervention-based fuel reduction

Constraints: $0 \leq F_{load,k+1}(x, y) \leq 1$

2455 Fuel cannot become negative and cannot increase unless explicitly modeled. This
 2456 formulation explicitly separates combustion-induced depletion from suppression-
 2457 induced reduction while enforcing non-negativity of fuel mass.

2458 Fuel consumption due to combustion occurs only when the cell is actively burning.
 2459 The term $B_k(x, y) F_{load,k}(x, y) F_{burn,k}(x, y)$ represents the fraction of available fuel
 2460 consumed during the current simulation step. The multiplication by $B_k(x, y)$ ensures
 2461 that fuel depletion due to combustion occurs only in actively burning cells. If the cell
 2462 is not burning, no combustion-driven reduction takes place. This preserves physical
 2463 realism and prevents unintended fuel decay in inactive regions. The burn fraction
 2464 $F_{burn,k}(x, y)$ captures the local combustion intensity per time step and can be
 2465 interpreted as a discretized approximation of heat-release-driven fuel consumption.
 2466 The suppression-induced fuel reduction term $F_{red,k}(x, y)$ is computed from
 2467 decision-driven inputs $U_{DSS,k}(x, y)$ and is not an independent external input.

2468 On the other hand, operational intervention modifies the fuel state through the
 2469 suppression term (50). This mapping converts operational capability and
 2470 reachability into a normalized fuel reduction effect. The suppression term may
 2471 represent water or retardant application; mechanical fuel removal; controlled burn
 2472 operations; and resource-constrained mitigation strategies.

$$F_{red,k}(x, y) = f_{supp} \left(\begin{matrix} R_{cap,k}(x, y), R_{avail,k}(x, y), R_{eff,k}(x, y), \\ R_{time,k}(x, y), G_{access}(x, y) \end{matrix} \right)$$

f_{supp} : Suppression to fuel reduction map (Appendix B)

$R_{cap,k}$: available suppression capacity

$R_{avail,k}$: operational availability status of resources (50)

$R_{eff,k}(x, y)$: suppression effectiveness at cell (x, y)

$R_{time,k}(x, y)$: estimated travel time to reach (x, y)

$G_{access}(x, y)$: accessibility level at grid cell (x, y)

2473 Importantly, suppression does not directly overwrite the burning state. Instead, it
 2474 modifies the available fuel mass, which subsequently influences combustion
 2475 persistence and propagation through the transition operator. This separation

2476 guarantees architectural clarity between physical wildfire dynamics and decision-
2477 driven intervention mechanisms.

2478 The function $f_{supp}(\cdot)$ is a suppression-to-mitigation mapping that converts operational
2479 capability and reachability into a normalized fuel-reduction (or mitigation) effect.

2480 **Fire Intensity Evolution** $I_k(x, y) \rightarrow I_{k+1}(x, y)$

2481 The local fire intensity proxy evolves according to (51). The function $f_{intensity}(\cdot)$
2482 approximates local fire strength as a function of available fuel, wind forcing, and
2483 terrain gradient. $I_{k+1}(x, y)$ denotes the updated intensity at the next time step; I_k
2484 does not appear as an argument of $f_{intensity}$.

$$I_{k+1}(x, y) = f_{intensity} \left(F_{load,k}(x, y), W_{ws,k}(x, y), G_{slope}(x, y) \right)$$

$f_{intensity}$: local fire strength function (Appendix C)

$F_{load,k}(x, y)$: the available combustible mass (51)

$W_{ws,k}(x, y)$: wind speed

$G_{slope}(x, y)$: terrain slope

2485 The fire intensity proxy serves as an intermediate physical-descriptive variable with
2486 the following roles: it does not control ignition activation; it does not directly modify
2487 the spread operator; it influences suppression effectiveness through the intervention
2488 mapping; and it provides structured features to the Decision Support System. Thus,
2489 intensity is a bounded combustion-strength indicator that links environmental
2490 forcing to operational response without introducing thermodynamic complexity.

2491 **Ignition Time Evolution** $\tau_k(x, y) \rightarrow \tau_{k+1}(x, y)$

2492 The ignition time variable $\tau_k(x, y)$ represents the time elapsed since the most recent
2493 ignition event at grid cell (x, y) . Unlike fuel mass or intensity, τ_k does not influence
2494 combustion physics directly. Instead, it serves as a temporal memory state that
2495 preserves the burning history of the cell. τ_k is a useful state for decision features and
2496 explainability within the DSS layer. A deterministic update is (52)

$$\tau_{k+1}(x, y) = \begin{cases} 0, & B_{k+1} = 1 \wedge B_k = 0 \\ \tau_k + \Delta t, & B_{k+1} = 1 \wedge B_k = 1 \\ 0, & B_{k+1} = 0 \end{cases} \quad (52)$$

2497 This update has three distinct operational meanings:

- 2498 1. **Ignition Initialization:** When a cell transitions from non-burning to burning
2499 $B_k = 0, B_{k+1} = 1$, the ignition clock is initialized: $\tau_{k+1} = 0$. This marks the
2500 start of a new combustion phase.
 - 2501 2. **Active Burning Progression:** When a cell remains in the burning state $B_k =$
2502 $1, B_{k+1} = 1$, the ignition time increases linearly: $\tau_{k+1} = \tau_k + \Delta t$. Thus, τ_k
2503 measures the residence time of active combustion at the cell.
 - 2504 3. **Extinction Reset:** When combustion ceases $B_{k+1} = 0$, the ignition clock
2505 resets to 0. It ensures extinguished cells carry no residual temporal memory.
- 2506 τ_k does not feed back into the burning activation condition in the present model,
2507 preserving the separation between physical fire dynamics and temporal descriptors.

2508 4.2.4 Mathematical Formulation of Interaction and Scenario Control

2509 The Interaction and Scenario Control mechanism enables controlled human
2510 intervention within the DisasterAware framework while preserving the physical
2511 consistency of wildfire dynamics. In contrast to conventional simulation systems
2512 where user actions may directly alter system states, the proposed framework enforces
2513 strict state immutability and represents interaction as a structured transformation
2514 applied to the input and decision-context layers.

2515 The external simulation input set is $\mathcal{F}_{in,k} = \{U_{Meteo,k}, U_{Geo}, U_{Fuel,k}, U_{Ign,k}, U_{DSS,k}\}$
2516 and decision-context set is $\mathcal{F}_{DSS,k} = \{U_{Val,k}, U_{Res,k}\}$. User interaction is modeled
2517 through an operator as (53) where $\Theta_{UI}(\cdot)$ represents controlled modifications
2518 applied through the UI. This operator enables the user to intervene in both the
2519 simulation input layer and the decision-context layer without introducing a separate
2520 input category.

$$(\mathcal{F}_{in,k}^*, \mathcal{F}_{DSS,k}^*) = \Theta_{UI}(\mathcal{F}_{in,k}, \mathcal{F}_{DSS,k}) \quad (53)$$

2521 **Definition 4.1 (User Interaction Operator).** The user interaction operator Θ_{UI} is a
 2522 structured input transformation mapping that produces admissible modifications of
 2523 the external input set and decision-context set (54).

$$\begin{aligned} \Theta_{UI}: \mathcal{F}_{in} \times \mathcal{F}_{DSS} &\rightarrow \mathcal{F}_{in}^* \times \mathcal{F}_{DSS}^* \\ (\mathcal{F}_{in,k}^*, \mathcal{F}_{DSS,k}^*) &= \Theta_{UI}(\mathcal{F}_{in,k}, \mathcal{F}_{DSS,k}) \end{aligned} \quad (54)$$

2524 The operator acts independently on five admissible inputs. For each component, the
 2525 admissible transformation is defined by a bounded perturbation set in Table 4.1. All
 2526 unmodified components pass through unchanged: $U_{Geo}^* = U_{Geo}$, $U_{Fuel,k}^* = U_{Fuel,k}$.

2527 Table 4.1 Admissible transformations of interaction operator

Component	Admissible transformation	Constraint
Ignition injection $U_{Ign,k}$	$U_{Ign,k}^*(x, y) = U_{Ign,k}(x, y) + \delta_{Ign}(x, y)$	$U_{Ign,k}^* \in \{0,1\}$
Meteorological input $U_{Meteo,k}$	$U_{Meteo,k}^* = U_{Meteo,k} + \delta_{Meteo}$	$U_{Meteo,k}^* \in \mathcal{U}_{Meteo}$
Resource parameters $U_{DSS,k}$	$U_{DSS,k}^*(x, y) = U_{DSS,k}(x, y) + \delta_{DSS}(x, y)$	$U_{DSS,k}^* \in [0,1]$
Value context $U_{Val,k}$	$U_{Val,k}^*(x, y) = \sigma_{Val} \cdot U_{Val,k}(x, y)$	$\sigma_{Val} \in [0,1],$ $U_{Val,k}^* \in [0,1]$
Resource availability $U_{Res,k}$	$U_{Res,k}^*(x, y) = U_{Res,k}(x, y) + \delta_{Res}(x, y)$	$U_{Res,k}^* \in [0,1]$

2528

2529 **Remark 4.1 (State Immutability):** The operator Θ_{UI} acts exclusively on the
 2530 external input set $\mathcal{F}_{in,k}$ and decision-context set $\mathcal{F}_{DSS,k}$. The system state $S_k(x, y)$ is
 2531 not in the domain of Θ_{UI} , $\frac{\partial S_k}{\partial \Theta_{UI}} = 0$. All user influence therefore propagates through
 2532 physically consistent state transitions $S_{k+1} = \Phi(S_k, \mathcal{F}_{in,k}^*)$, preserving the causal
 2533 structure and state immutability established in Section 4.1.

2534 **Remark 4.2 (Temporal Scope).** Θ_{UI} modifies only future state transitions. It does
 2535 not alter the current state S_k or any previously realized state $S_j, j < k$. This ensures
 2536 that scenario exploration and operational overrides are prospective, not retrospective.

2537 The modified decision-context set influences decision generation $U_{DSS,k} =$
 2538 $\pi_{DSS}(O_k, \mathcal{F}_{DSS,k}^*)$ and the modified simulation input set governs system evolution
 2539 $S_{k+1} = \Phi(S_k, \mathcal{F}_{in,k}^*)$. A fundamental constraint of the framework is that user
 2540 interaction does not directly modify the system state $\frac{\partial S_k}{\partial \Theta_{UI}} = 0$. Thus, all user
 2541 influence propagates through admissible transformations of input and context
 2542 variables, preserving the causal structure of the system.

2543 Within this formulation, user interaction supports multiple operational functions:

- 2544 • **Scenario intervention:** modification of ignition inputs $U_{Ign,k}$ to inject or
 2545 alter fire events
- 2546 • **Environmental what-if analysis:** controlled perturbation of meteorological
 2547 inputs $U_{Meteo,k}$
- 2548 • **Operational override:** adjustment of resource-related parameters $U_{DSS,k}$
- 2549 • **Priority reconfiguration:** modification of value-based context $U_{Val,k}$
- 2550 • **Resource reallocation:** modification of resource availability $U_{Res,k}$

2551 These interaction capabilities allow the system to operate as a **HIL DSS**, where the
 2552 user can dynamically influence both the operational context and the evolution of the
 2553 simulated environment.

2554 Importantly, interaction does not bypass the Decision Support System. Any
 2555 modification applied to $\mathcal{F}_{DSS,k}$ is incorporated into the decision-generation process
 2556 through $\pi_{DSS}(\cdot)$, ensuring that all intervention signals remain consistent with the
 2557 decision architecture. This design preserves three key architectural principles: (i)

2558 • **State immutability:** wildfire dynamics remain governed exclusively by the
 2559 transition operator

2560 • **Layer separation:** simulation, decision reasoning, and interaction remain
 2561 structurally distinct

2562 • **Traceability:** all user interventions are explicitly represented as input-level
 2563 transformations

2564 Through this formulation, the Interaction and Scenario Control mechanism enables
 2565 flexible scenario exploration and operational override while maintaining the
 2566 integrity, interpretability, and modularity of the DisasterAware framework.

2567 **4.3 External Data Sources**

2568 This section describes the data sources that instantiate the external input structures
 2569 defined in Section 4.2.1.

2570 **4.3.1 Fire Spread Model External Data Layer**

2571 The Fire Spread Model External Data Sources comprise all heterogeneous inputs
 2572 required to initialize, drive, and update the wildfire simulation engine. These data
 2573 are not generated internally by the simulator but are acquired from external sources
 2574 such as meteorological services, geographic information systems (GIS), remote
 2575 sensing platforms, and operational databases.

2576 This layer provides environmental and contextual information that governs fire
 2577 behavior and enables DSS functionality. Conceptually, the data layer is structured
 2578 into two primary categories:

- 2579 1. **Temporal Environmental Drivers:** These include meteorological variables
 2580 that vary over time and influence ignition probability, rate of spread, fire
 2581 intensity, and fuel moisture evolution.

2. **Spatial Physical Constraints:** These include terrain and fuel-related parameters that define static or slowly varying spatial properties affecting combustion dynamics and propagation patterns.

The separation of these categories ensures modularity and clarity within the system architecture. Environmental forcing, spatial constraints, and operational priorities remain logically distinct, allowing the fire spread model to operate independently from decision logic while still enabling integrated risk-based analysis.

At each discrete time step Δt , the Simulation Core retrieves relevant temporal inputs (e.g., meteorology), combines them with spatial layers (terrain and fuel), and updates the grid state accordingly. Decision-related inputs are treated as external parameters influencing the simulation, while their generation mechanism is defined in the DSS architecture in Chapter 6. This layered structure ensures that environmental forcing, spatial constraints, and operational inputs remain explicitly separated.

4.3.1.1 Meteorology Data Layer

This layer provides dynamic atmospheric conditions that directly influence fire ignition probability, rate of spread, fire intensity, spotting behavior, plume dynamics, and fuel moisture evolution. These variables listed in Table 4.2 are primarily **time-dependent** and may optionally be **spatially dependent** when high-resolution gridded weather fields are available.

From a Decision Support System (DSS) perspective, the meteorological data layer is particularly suitable because it enables short-horizon predictive modeling under discrete time steps (Δt_{sim}), supports what-if scenario analysis, and allows rapid recalibration of fire behavior in response to updated field observations. By explicitly modeling weather variables as spatio-temporal inputs, the simulator can dynamically adapt to changing environmental conditions, thereby providing operationally relevant and actionable forecasts for intervention planning and resource allocation.

2608 Table 4.2 Meteorology data layer parameters

Parameter	Type	Unit	Purpose in Simulation
Air Temperature $W_{\text{temp},k}(x, y)$	Spatio-temporal	°C	Represents ambient air temperature, influencing fuel drying rate, ignition thresholds, and fire intensity potential.
Relative Humidity $W_{\text{rh},k}(x, y)$	Spatio-temporal	%	Represents atmospheric moisture content and directly regulates live and dead fuel moisture levels.
Wind Speed $W_{\text{ws},k}(x, y)$	Spatio-temporal	m/s	Serves as the primary driver of fire spread rate and significantly increases the rate of spread under high-speed conditions.
Wind Direction $W_{\text{wd},k}(x, y)$	Spatio-temporal	rad	Determines the directional propagation of the fire front and controls anisotropic spread behavior.
Wind Gust Speed $W_{\text{gust},k}(x, y)$	Spatio-temporal	m/s	Represents short-term wind fluctuations and stochastic acceleration in fire propagation.
Precipitation $W_{\text{prec},k}(x, y)$	Temporal	mm/h	Increases fuel moisture and can suppress, delay, or weaken active combustion processes.

2609 4.3.1.2 Topography Data Layer

2610 The Topography Data layer represents the static spatial characteristics of the terrain
2611 over which wildfire propagation occurs. Unlike meteorological variables, which
2612 evolve over time, topographic variables remain constant throughout the simulation
2613 but act as deterministic spatial modifiers of fire behavior. In a grid-based simulation
2614 framework, terrain properties influence rate of spread, directional bias, wind
2615 modulation, responder accessibility, and local acceleration effects.

From a Decision Support System (DSS) perspective, accurate terrain modeling enables identification of natural fire corridors, acceleration zones, defensible ridgelines, and operational access constraints. The geometry (topography) parameters are listed in Table 4.3.

Table 4.3 Geometry (topography) data layer parameters

Parameter	Type	Unit	Purpose in Simulation
Elevation $G_{\text{elev}}(x, y)$	Spatial	m	Represents the altitude of each grid cell and influences local wind behavior, vegetation type distribution, environmental gradients.
Slope $G_{\text{slope}}(x, y)$	Spatial	rad	Represents terrain and increases fire spread rate in the uphill direction due to flame heat transfer effects.
Aspect $G_{\text{aspect}}(x, y)$	Spatial	rad	Represents slope orientation and influences solar radiation exposure, fuel drying patterns, and directional fire propagation bias.
Accessibility Index $G_{\text{access}}(x, y)$	Spatial	-{0,1}	Represents terrain traversability and influences responder mobility, intervention timing, and operational feasibility.

4.3.1.3 Fuel Data Layer

The Fuel Data layer represents the combustible material available within each grid cell. These parameters determine ignition probability, fire intensity, burn duration, and the potential transition from surface fire to crown fire. Unlike topographic variables, fuel properties may evolve over time due to combustion and intervention actions. Fuel load decreases as burning progresses, and fuel moisture varies according to meteorological conditions.

From a Decision Support System (DSS) perspective, this layer is essential because operational actions such as fuel reduction and fire-suppression directly modify fuel-related parameters, enabling action-aware simulation and what-if scenario analysis. The fuel parameters are listed in Table 4.4.

Table 4.4 Fuel data layer parameters

Parameter	Type	Unit	Purpose in Simulation
Fuel Type $F_{\text{type}}(x, y)$	Spatial	–	Defines the vegetation within each grid cell and determines combustion characteristics such as burn rate and heat release behavior.
Fuel Load $F_{\text{load},k}(x, y)$	Spatio-temporal	$\{0,1\} \frac{kg}{m^2}$ (norm.)	Represents the available combustible mass in each grid cell and directly controls fire intensity, burn duration, and energy release.
Fuel Moisture $F_{\text{moist},k}(x, y)$	Spatio-temporal	– (mass fraction)	Represents moisture of surface fuels and determines ignition probability and combustion sustainability.

4.3.2 Decisional External Sources

The Decisional External Sources layer represents non-physical inputs that directly influence the decision-making process within the distributed Decision Support System (DSS). Unlike meteorological, topographic, and fuel layers that govern fire dynamics, this layer provides contextual information required for evaluating operational priorities and constraints within the DSS process. It therefore determines how simulation outcomes are interpreted and acted upon.

This layer consists of two main components: the Value Data layer and the Resource Data layer. They are used to transform physical fire propagation outputs into actionable decision variables. While the Simulation Core estimates fire behavior

over space and time, the DSS evaluates consequences and operational feasibility. In particular, the Value Data layer defines spatial exposure and asset importance, enabling risk-based prioritization. The Resource Data layer defines available suppression capabilities and logistical constraints, enabling feasible intervention planning. Through this separation, the system maintains a clear distinction between fire behavior modeling and decision optimization.

All variables in this layer are either spatially dependent or temporally dependent, and they may dynamically influence intervention strategies without altering the underlying physical fire spread equations. From a DSS perspective, Decisional External Sources enable risk quantification based on exposed assets; protection priority ranking; resource optimization; feasibility assessment of intervention strategies; and what-if scenario comparison under operational constraints. These decisional inputs collectively define the contextual input set $\mathcal{F}_{DSS,k}$ used by the DSS. This architectural separation ensures that physical simulation outputs remain independent of strategic policy choices, while still allowing decision logic to operate on updated system states at each discrete time step Δt .

4.3.2.1 Value Data Layer

The Values-at-Risk Data layer represents exposed assets, infrastructure, and socioeconomic elements that may be affected by wildfire propagation. Unlike meteorological, topographic, or fuel layers that primarily describe fire behavior, this layer defines the potential consequences of fire spread.

In DSS context, the objective is not only to simulate fire propagation but also to evaluate its impact on critical assets and prioritize protection strategies. Thus, this layer enables risk quantification, intervention prioritization, and resource allocation. The value data layer parameters are listed in Table 4.5. All variables in this layer are spatially dependent and static during short-term operational simulations, although their priority weights may be dynamically adjusted based on operational objectives.

2670 Table 4.5 Value data layer parameters

Parameter	Type	Unit	Purpose in Simulation
Building Footprint $V_{bld}(x, y)$	Spatial	$\{0,1\}$	Represents the presence of residential or commercial structures and enables identification of wildland–urban zones.
Critical Facility Index $V_{crit}(x, y)$	Spatial	$\{0,1\}$	Identifies high-priority assets such as hospitals, power plants, fuel depots, and water infrastructure.
Population Density $V_{pop}(x, y)$	Spatial	person/km ²	Represents exposed population per grid cell and supports evacuation and safety prioritization.
Accessibility to Evacuation Routes $V_{evac}(x, y)$	Spatial	m	Indicates proximity to evacuation infrastructure and supports emergency planning decisions.
Protection Priority Score $V_{prio}(x, y)$	Spatial	–	Aggregated weighting factor combining multiple exposure indicators to rank cells by importance.

2671

2672 The protection priority score $V_{prio}(x, y)$ aggregates the four value indicators into a
2673 single scalar index that ranks each grid cell by its overall protection importance. The
2674 aggregation is defined as a normalized weighted sum (55) where $V_{pop}^{norm} =$
2675 $\frac{V_{pop}}{\max V_{pop}}$ is the min–max normalized population density, and $V_{evac}^{norm} = \frac{V_{evac}}{\max V_{evac}}$ is
2676 the inverted and normalized evacuation distance (higher value indicates closer
2677 proximity to evacuation routes, hence higher priority).

$$V_{prio}(x, y) = w_{bld} \cdot V_{bld}(x, y) + w_{crit} \cdot V_{crit}(x, y) + w_{pop} \cdot V_{pop}^{norm}(x, y) + w_{evac} \cdot V_{evac}^{norm}(x, y) \quad (55)$$

2678 The weights satisfy $w_{bld} + w_{crit} + w_{pop} + w_{evac} = 1$, with default values
2679 $w_{bld} = 0.20$, $w_{crit} = 0.40$, $w_{pop} = 0.25$, $w_{evac} = 0.15$.

2680 The weight assignment reflects the operational priority hierarchy: critical facilities
2681 ($w_{crit} = 0.40$) receive the highest weight because their loss has cascading systemic
2682 consequences (e.g., hospitals, power infrastructure, water supply). Population
2683 density ($w_{pop} = 0.25$) ranks second because human life exposure scales with
2684 density. Building footprint ($w_{bld} = 0.20$) captures structural asset density.
2685 Evacuation accessibility ($w_{evac} = 0.15$) receives the lowest weight because it
2686 represents a logistical constraint rather than a direct measure of asset value; however,
2687 it remains non-zero because locations with poor evacuation access require protective
2688 intervention earlier in the decision timeline.

2689 Since all input variables and weights are bounded in $[0, 1]$ and the weights sum to
2690 unity, $V_{prio}(x, y) \in [0, 1]$ by construction, which preserves compatibility with the
2691 feature extraction mapping f_{feat} defined in Chapter 5. The sensitivity of the DSS to
2692 variations in these weights is evaluated in Section 7.6.3.

2693 4.3.2.2 Resource Data Layer

2694 The Resource Data layer represents the operational capabilities available for wildfire
2695 intervention. While the fire behavior model estimates how fire evolves under
2696 environmental conditions, this layer defines the constraints and effectiveness of
2697 suppression actions.

2698 In a Decision Support System context, resources are not passive elements; they
2699 actively modify the state of the simulation through intervention. Therefore, this layer
2700 enables feasibility analysis, dispatch prioritization, and action-aware scenario
2701 evaluation. Unlike topographic variables, resource parameters are generally spatio-
2702 temporal and may change during simulation due to movement, deployment, or
2703 depletion. The resource data layer parameters are listed in Table 4.6.

2704 Table 4.6 Resource data layer parameters

Parameter	Type	Unit	Purpose in Simulation
Suppression Capacity $R_{cap,k}(x, y)$	Spatio-temporal	m ³ water equivalent / step	Represents available suppression capability such as water volume or manpower at grid x, y and at time instance t .
Travel Time $R_{time,k}(x, y)$	Spatio-temporal	min or h	Represents estimated time required to reach a grid cell from current position.
Suppression Efficiency $R_{eff,k}(x, y)$	Spatio-temporal	$\{0,1\}$	Quantifies how effectively a resource reduces fuel load or fire intensity.
Resource Availability $R_{avail,k}(x, y)$	Spatio-temporal	$\{0,1\}$	Indicates whether the resource is currently deployable.

2705 4.4 UI Dashboard

2706 The UI Dashboard serves as the supervisory interface between the human operator,
 2707 the Simulation Core, and the DSS Core. It provides three functions: real-time
 2708 visualization of the wildfire state, controlled scenario manipulation, and structured
 2709 human-in-the-loop interaction. The dashboard presents the spatial distribution of key
 2710 state variables, including burning status $B_k(x, y)$, fuel load $F_{load,k}(x, y)$, fire
 2711 intensity $I_k(x, y)$, and ignition time $\tau_k(x, y)$, overlaid with value-at-risk layers and
 2712 DSS intervention outputs, enabling operators to interpret both physical fire dynamics
 2713 and decision recommendations within a unified spatial representation.

2714 The architectural constraint governing all UI interaction is that the dashboard does
 2715 not directly modify the internal state vector S_k . Instead, operator actions, including
 2716 ignition injection, resource reconfiguration, and meteorological perturbations for
 2717 what-if analysis, modify admissible components of the external input set $\mathcal{F}_{in,k}$,

2718 producing a controlled configuration $\mathcal{F}_{in,k}^*$. The subsequent state transition is
 2719 therefore computed as: $S_{k+1}(x, y) = \Phi(S_k(x, y), \mathcal{F}_{in,k}^*)$. This input-level-only
 2720 interaction principle ensures that all operator modifications influence wildfire
 2721 evolution exclusively through the state transition operator Φ , preserving physical
 2722 consistency, traceability, and the architectural separation between system dynamics
 2723 and human supervisory control established in Section 4.1.

2724 4.5 Decision Support System Core

2725 This section defines only the structural role and interface of the DSS within the
 2726 simulation framework. Its internal specification and architectural realization are
 2727 presented in Chapters 5 and 6, respectively. The Decision Support System (DSS)
 2728 Core constitutes the cognitive layer of the DisasterAware framework, responsible for
 2729 transforming observable information into structured intervention recommendations.

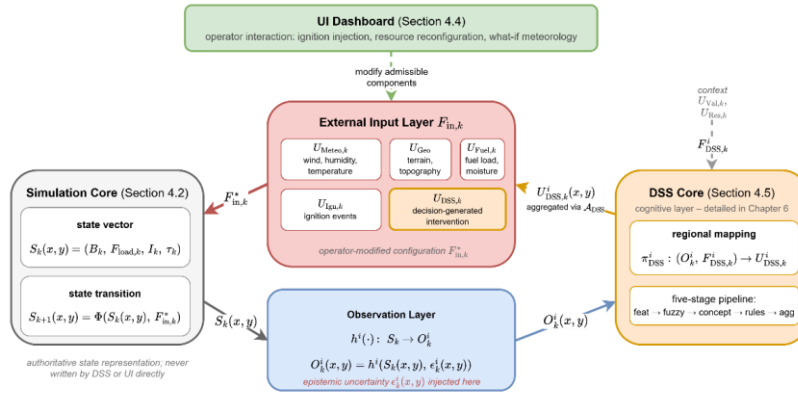
2730 While the Simulation Core governs the physical evolution of wildfire dynamics
 2731 through the state transition operator $S_{k+1}(x, y) = \Phi(S_k(x, y), \mathcal{F}_{in,k})$, the DSS
 2732 operates on observable quantities $O_k(x, y) = h(S_k(x, y), \epsilon_k)$ derived from the
 2733 simulation state through the observation process, rather than directly accessing the
 2734 full system state. Observation uncertainty $\epsilon_k(x, y)$ is incorporated within the DSS
 2735 through uncertainty-aware reasoning mechanisms operating on $O_k(x, y)$.

2736 The DSS evaluates decision-relevant indicators derived from $O_k(x, y)$ and
 2737 contextual inputs $\mathcal{F}_{DSS,k}$ to generate structured intervention inputs $U_{DSS,k}$ that
 2738 influence subsequent system evolution. DSS outputs are not control actions but
 2739 structured external inputs affecting system evolution.

2740 The DSS interaction follows the (56). This preserves the architectural separation
 2741 between state evolution and decision generation. All decision impact must propagate
 2742 through admissible modifications of the external input set. The information and
 2743 control flow captured by Equation (56) is summarized graphically in Figure 4.5,

which makes the data pathways and the architectural separation between simulation dynamics and decision reasoning visually explicit.

$$\begin{aligned}
S_k(x, y) &\rightarrow O_k(x, y), \mathcal{F}_{DSS,k} \rightarrow U_{DSS,k}(x, y) \rightarrow \Phi(\cdot) \rightarrow S_{k+1}(x, y) \\
O_k(x, y) &= h(S_k(x, y), \epsilon_k) \\
U_{DSS,k}(x, y) &= \pi_{DSS}^i(O_k(x, y), \mathcal{F}_{DSS,k}) \subset \mathcal{F}_{in,k}
\end{aligned} \tag{56}$$



Data flow at each time step k : $S_k \rightarrow (O_k^i, F_{DSS,k}^i) \rightarrow U_{DSS,k}^i \subset F_{in,k} \rightarrow \Phi(\cdot) \rightarrow S_{k+1}$ [Equation (53)].

The DSS Core never writes to $S_k(x, y)$ directly; all intervention effects propagate through the external input set $F_{in,k}$, preserving the architectural separation established in Section 4.1.

Figure 4.5 DSS–Simulation Core Interface

The Figure 4.5 decomposes the framework into five architectural elements and the data flow that connects them at every simulation time step k .

- The **Simulation Core** maintains the authoritative state vector $S_k(x, y) = (B_k, F_{load,k}, I_k, \tau_k)$ and advances it through the state transition operator Φ .
- The **Observation Layer** applies the region-specific observation function $h^i(\cdot)$ to project the state onto observable information $O_k^i(x, y) = h^i(S_k(x, y), \epsilon_k^i(x, y))$, injecting the epistemic uncertainty $\epsilon_k^i(x, y)$ at this stage and at this stage only.

- 2757 • The **DSS Core** executes the regional mapping $\pi_{\text{DSS}}^i: (O_k^i, F_{\text{DSS},k}^i) \rightarrow U_{\text{DSS},k}^i$
 2758 on the region-specific observation and the contextual inputs $F_{\text{DSS},k}^i =$
 2759 $(U_{\text{Val},k}, U_{\text{Res},k})$, producing the regional decision field $U_{\text{DSS},k}^i(x, y)$; the
 2760 detailed five-stage pipeline that realizes this mapping is the subject of
 2761 Chapter 6.
- 2762 • Regional decisions from the N agents are aggregated through the
 2763 coordination operator $\mathcal{A}_{\text{DSS}}$ into the global intervention field, which is
 2764 injected into the **External Input Layer** $F_{\text{in},k}$ as the $U_{\text{DSS},k}$ component
 2765 alongside the four exogenous inputs $U_{\text{Meteo},k}$, U_{Geo} , $U_{\text{Fuel},k}$, and $U_{\text{Ign},k}$.
- 2766 • The **UI Dashboard** can modify admissible components of $F_{\text{in},k}$ through the
 2767 operator interaction operator θ_{UI} , yielding the operator-modified
 2768 configuration $F_{\text{in},k}^*$ that drives the next state transition $S_{k+1}(x, y) =$
 2769 $\Phi(S_k(x, y), F_{\text{in},k}^*)$.

2770 The Figure 4.5 makes explicit the architectural principle established in Section 4.1
 2771 and reinforced throughout this chapter: the DSS Core never writes to $S_k(x, y)$
 2772 directly, reasoning exclusively from $O_k^i(x, y)$ and influencing evolution exclusively
 2773 through the external input layer. This preserves the strict separation between physical
 2774 dynamics and decision reasoning that underlies the subsequent design specifications
 2775 of Chapter 5 and the runtime architecture of Chapter 6.

2776 In summary, within the Simulation Framework, the DSS Core is a cognitively
 2777 distinct module that interprets observable system representations and produces input-
 2778 level intervention signals, maintaining modularity and interpretability.

2779 4.6 Summary of the Simulation Framework

2780 This chapter establishes the structural and mathematical foundations of the
 2781 DisasterAware Simulation Framework, defining wildfire evolution as a discrete-
 2782 time, grid-based state-space system coupled with an external DSS.

2783 At the core of the framework, wildfire dynamics are represented by the state vector
 2784 $S_k(x, y)$ defined over the spatial domain $\mathcal{G}$, where each grid cell encodes combustion
 2785 status, available fuel, intensity, and temporal progression. The system evolves
 2786 according to the discrete-time transition model $S_{k+1}(x, y) = \Phi(S_k(x, y), \mathcal{F}_{in,k})$
 2787 where $\Phi(\cdot)$ represents the physically grounded wildfire propagation dynamics, and
 2788 $\mathcal{F}_{in,k}$ denotes the structured external input set.

2789 The external input structure integrates multiple categories of influence, including
 2790 meteorological forcing, terrain constraints, fuel properties, ignition triggers, and
 2791 decision-driven intervention parameters. These inputs are explicitly separated from
 2792 the internal state variables, ensuring that wildfire evolution remains governed by
 2793 physical processes rather than direct external manipulation.

2794 The transition mechanism $\Phi(\cdot)$ combines burning persistence, spatial propagation,
 2795 fuel consumption, and intensity evolution within a unified and deterministic update
 2796 structure. Stochastic variability is introduced only through exogenous input fields,
 2797 preserving numerical stability and interpretability of the system.

2798 A key architectural principle established is the strict separation between simulation
 2799 dynamics and decision-making. The Simulation Core maintains the authoritative
 2800 system state, while all external influences, including human interaction and DSS
 2801 outputs, are introduced exclusively through the input set $\mathcal{F}_{in,k}$. This enforces the
 2802 causal structure $S_k(x, y) \rightarrow O_k(x, y), \mathcal{F}_{DSS,k} \rightarrow U_{DSS,k}(x, y) \subset \mathcal{F}_{in,k} \rightarrow \Phi(\cdot) \rightarrow$
 2803 $S_{k+1}(x, y)$ which ensures that DSS mechanisms do not directly alter the physical
 2804 state but influence system evolution through structured input-level modifications.

2805 The Decision Support System (DSS) is therefore positioned as a cognitively distinct
 2806 layer that operates on observable information $O_k(x, y)$ together with contextual
 2807 inputs $\mathcal{F}_{DSS,k}$ and generates intervention-related inputs $U_{DSS,k}(x, y)$, while the
 2808 Simulation Core remains responsible for physically consistent state evolution.
 2809 Similarly, the UI layer provides controlled human interaction by modifying
 2810 admissible input parameters without violating state immutability.

2811 Through this formulation, the DisasterAware Simulation Framework achieves a
2812 clear separation between physical modeling, decision reasoning, and human
2813 interaction, while maintaining a unified and mathematically consistent system
2814 representation.

2815

CHAPTER 5

2816

DISASTER-AWARE SYSTEM SPECIFICATIONS

2817 This chapter establishes the formal design specification of the Disaster-Aware DSS,
2818 which is defined through a structured sequence of transformations: observable
2819 wildfire information is mapped into scalar features, fuzzified into linguistic
2820 membership values, abstracted into situational concepts, evaluated through decision
2821 rules, and aggregated into intervention outputs. Each of these transformations
2822 involves explicit design choices, the selection of which features to extract, how to
2823 define the linguistic terms, which concepts to form, and how to construct the decision
2824 rules. These choices are not arbitrary; they follow from the operational requirements
2825 of wildfire management, the formal properties of the simulation environment defined
2826 in Chapter 4, and the decision-theoretic principles established in Chapter 2.

2827 The purpose of this chapter is to establish the formal design specification of the
2828 Disaster-Aware DSS. While Chapter 6 addresses the question of *how* the system
2829 operates, this chapter addresses the question of *why* each design element was chosen.
2830 Every parameter, function, and structural decision presented here is grounded in one
2831 or more of three foundations: the operational context of wildfire response, the
2832 mathematical structure of the Chapter 4 simulation framework, or established
2833 principles from the fuzzy reasoning and decision support literature reviewed in
2834 Chapter 3. This chapter is organized as follows.

2835 Section 5.1 establishes the operational decision context by characterizing temporal,
2836 spatial, and epistemic uncertainty dimensions of DSS decisions, defining four
2837 fundamental operational questions, and justifying fuzzy reasoning mechanism.

2838 Section 5.2 derives the input space, specifies the fuzzy variable and membership
2839 functions, formalizes the four situational concepts and their construction mappings,
2840 and proves the rule complexity reduction achieved through concept abstraction.

2841 Section 5.3 specifies the complete rule base, including concept construction rules for
2842 all concepts, intervention decision rules for each action types, the action aggregation
2843 mechanism f_{agg} , and the decision quality criteria with their evaluation weights.

2844 Section 5.4 synthesizes the principal results of the chapter and positions the
2845 DisasterAware DSS specification as the formal foundation for the architectural
2846 realization presented in Chapter 6.

2847 **5.1 Operational Decision Context**

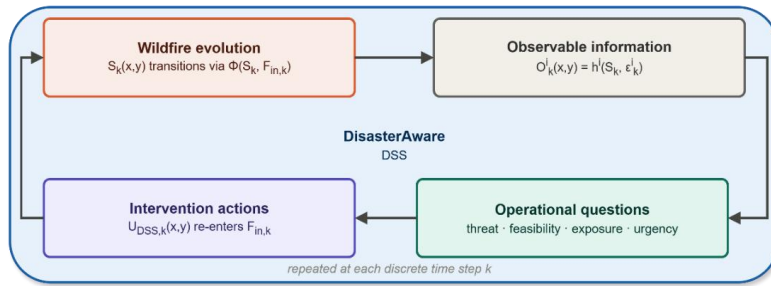
2848 Wildfire response operations unfold within a decision space that is simultaneously
2849 dynamic, spatially distributed, resource-constrained, and informationally partial.
2850 Incident commanders and field coordination units must evaluate rapidly evolving
2851 fire conditions, assess the exposure of high-value assets and human settlements,
2852 determine the operational feasibility of suppression interventions, and coordinate the
2853 deployment of limited resources across geographically dispersed areas.

2854 These activities must be performed continuously, under strict time pressure, and on
2855 the basis of information that is inherently partial, temporally delayed, and subject to
2856 epistemic uncertainty. Understanding this operational context is essential because
2857 every design decision in the DisasterAware DSS, from the choice of features to the
2858 structure of the rule base, is grounded in the requirements that this context imposes.

2859 **5.1.1 The Temporal Structure of Wildfire Decisions**

2860 The formal problem statement introduced in Chapter 2 characterizes wildfire
2861 response as a discrete-time sequential decision process indexed by k . At each time
2862 step k , three events occur in succession. The wildfire system evolves from state
2863 $S_k(x, y)$ to state $S_{k+1}(x, y)$ through the state transition operator $\Phi(S_k(x, y), \mathcal{F}_{in,k})$,
2864 reflecting the combined effect of environmental forcing and intervention inputs. The
2865 Observation Layer then derives, for each operational region Ω_i , a partial and

2866 uncertainty-affected representation of the current state as $O_k^i(x, y) =$
 2867 $h^i(S_k(x, y), \epsilon_k^i)$. Finally, the DSS generates intervention inputs $U_{DSS,k}(x, y)$ based
 2868 on this observable information and the contextual decision inputs $\mathcal{F}_{DSS,k}$, and these
 2869 inputs re-enter the external input set $\mathcal{F}_{in,k}$ to influence the subsequent state transition.



2870
 2871 Figure 5.1 Sequential cycle of the DisasterAware DSS

2872 This sequential structure, depicted in Figure 5.1, implies that the quality of each
 2873 decision step directly conditions both the state of the wildfire system and the
 2874 informational basis available for subsequent decisions.

2875 A delayed or incorrect intervention at time step k may allow fire spread that renders
 2876 feasible containment positions inaccessible at time step $k + 1$, or may consume
 2877 suppression resources that would have been critical at a later stage. This temporal
 2878 coupling means that the DSS must not optimize for a single time step in isolation but
 2879 must generate adaptive guidance that remains consistent with the evolving
 2880 operational context across successive time steps.

2881 5.1.2 The Spatial Structure of Wildfire Decisions

2882 In addition to its temporal dimension, wildfire response decision-making is
 2883 inherently spatial. Fire propagation, asset distribution, terrain configuration, resource
 2884 availability, and suppression effectiveness all vary continuously across the spatial
 2885 domain $\mathcal{G}$. A suppression strategy that is appropriate at one location may be

2886 infeasible or counterproductive at an adjacent location due to differences in terrain
2887 accessibility, fuel load, or proximity to high-value assets. This spatial heterogeneity
2888 has a direct consequence for system design: a single centralized decision mechanism
2889 that produces a uniform intervention policy for the entire domain cannot adequately
2890 capture the local variability that characterizes real wildfire incidents.

2891 This observation is formalized in Chapter 2, Section 2.4, through the regional
2892 decomposition of the decision problem. The spatial domain is partitioned into N
2893 operational regions $\Omega = \bigcup_{i=1}^N \Omega_i$, each associated with locally observable
2894 information $O_k^i(x, y)$ and region-specific decision context $\mathcal{F}_{DSS, k}^i$. Local decisions
2895 are generated independently within each region and subsequently coordinated
2896 through the global coordination mechanism $\mathcal{A}_{DSS}$. The distributed multi-agent
2897 DisasterAware DSS is the direct architectural realization of this requirement.

2898 5.1.3 The Epistemic Uncertainty Challenge

2899 A distinguishing characteristic of wildfire decision environments, compared to other
2900 planning or control domains, is the epistemic character of the uncertainty that
2901 pervades operational information. Walker et al. (2003) distinguish between aleatory
2902 uncertainty, which arises from inherent stochasticity in physical processes, and
2903 epistemic uncertainty, which arises from incomplete knowledge of the system state.
2904 In wildfire response, while both types of uncertainty are present, epistemic
2905 uncertainty is the more operationally consequential of the two.

2906 Fire boundary estimates derived from satellite imagery or aerial reconnaissance are
2907 subject to temporal delays of minutes to hours, and to spatial resolution limitations
2908 that may obscure individual cell-level conditions. Fuel moisture content can only be
2909 estimated from meteorological models or remote sensing indices, not directly
2910 measured at the spatial resolution required for cell-level decision support. The
2911 operational status, location, and remaining capacity of suppression resources are
2912 often known only approximately at central coordination nodes, as field units may

2913 operate in areas with degraded communication coverage. Terrain accessibility is
2914 nominally defined through digital elevation data, but actual accessibility at a given
2915 moment may be affected by smoke conditions, fire proximity, or temporary route
2916 closures that have not been reported to the coordination center.

2917 As established in Chapter 4, Section 4.2.1, the observation process $O_k^i(x, y) =$
2918 $h^i(S_k(x, y), \epsilon_k^i(x, y))$ makes this epistemic incompleteness explicit through the
2919 uncertainty term $\epsilon_k^i(x, y)$. Importantly, $\epsilon_k^i(x, y)$ is not modeled as a stochastic noise
2920 term with a known probability distribution. Such a characterization would itself
2921 require knowledge that is typically unavailable in operational settings, specifically
2922 the statistical properties of sensing uncertainties under field conditions. Rather,
2923 $\epsilon_k^i(x, y)$ represents a structural form of incomplete knowledge that cannot be reduced
2924 to a probability distribution without additional assumptions.

2925 This observation directly motivates the adoption of fuzzy inference as the primary
2926 reasoning mechanism: rather than requiring precise probabilistic models of
2927 observation uncertainty, fuzzy logic enables the system to represent and reason with
2928 graded degrees of belief derived directly from imprecise observations. This
2929 motivation is consistent with the theoretical background presented in Chapter 3 and
2930 with prior work on epistemic uncertainty management in decision support (Mendel,
2931 2017; Walker, et al., 2003).

2932 **5.1.4 The Four Fundamental Operational Questions**

2933 Despite the apparent complexity of wildfire DSS, the decision-making activity of
2934 experienced field commanders is structured around a small number of fundamental
2935 operational questions. This observation is grounded in naturalistic decision-making
2936 research (Klein, 2017), which shows that expert practitioners in high-tempo
2937 environments do not engage in exhaustive multi-criteria optimization but rather
2938 rapidly classify situations according to recognizable patterns and generate responses
2939 appropriate to each pattern.

2940 In the wildfire DSS literature, Thompson and Calkin (2011), Pacheco (2015), and
2941 Martell (2007) identify the core operational concerns of incident commanders as
2942 threat assessment, resource feasibility, asset prioritization, and response timing. The
2943 concerns can be categorized as four questions.

2944 **5.1.4.1 Q1: How severe is the fire threat at this location?**

2945 The first question concerns the current and projected destructive potential of the fire
2946 at a given grid cell within operational region Ω_i at step k . A severe fire threat exists
2947 when local combustion intensity is high, when available fuel is sufficient to sustain
2948 prolonged burning, and when the propagation influence from neighboring burning
2949 cells is strong.

2950 Each condition is observable, at least partially, through the state variables maintained
2951 by the Simulation Core: the local fire intensity proxy $I_k(x, y)$ reflects combustion
2952 strength as defined in Equation (51); and the propagation influence $\Psi_k(x, y)$ defined
2953 in Equation (46) captures the directional spread potential from the 8-connected
2954 neighborhood. A high fire threat level indicates that suppression or containment
2955 actions may be urgently required, while a low threat level permits resource
2956 conservation or reallocation to higher-priority locations.

2957 In operational practice, this assessment is never expressed in precisely quantified
2958 terms. Field commanders and aerial observation crews report conditions such as
2959 "extreme fire behavior", "active crown fire involvement", or "moderate surface
2960 spread" rather than precise values of combustion rate or intensity.

2961 This linguistic character of threat assessment is a fundamental feature of the
2962 operational context, not a deficiency to be corrected. It reflects the irreducible
2963 epistemic uncertainty of field observation and the limited time available for precise
2964 measurement. The fuzzy inference mechanism of DSS is designed to operate in this
2965 same linguistic register.

2966 **5.1.4.2 Q2: Can we effectively intervene at this location?**

2967 The second question concerns the operational feasibility of deploying available
2968 suppression resources to the location in question. Even when the fire threat at a grid
2969 cell (x, y) is severe, intervention may be infeasible for several reasons. Resources
2970 may not be able to reach the location within an operationally meaningful time
2971 window, either because of physical distance or because of terrain obstacles that
2972 preclude safe access. The capacity of available resources may be insufficient relative
2973 to the suppression requirement implied by local fuel load and fire intensity.
2974 Resources may already be committed to other high-priority locations and therefore
2975 unavailable for redeployment.

2976 The relevant information for this question is contained in the resource-related subset
2977 of the external input set $U_{DSS,k}$: the estimated travel time $R_{time,k}(x, y)$, the
2978 operational availability status $R_{avail,k}(x, y)$, and the suppression effectiveness
2979 parameter $R_{eff,k}(x, y)$ as defined in Section 4.2.1.

2980 Additionally, the terrain accessibility index $G_{access}(x, y)$ from the topography data
2981 layer conditions the physical reachability of any given grid cell. The available fuel
2982 load $F_{load,k}(x, y)$ provides an indication of the suppression effort required. In
2983 combination, these quantities determine whether an intervention at location (x, y) is
2984 feasible in the operational sense, that is, whether it can be executed with available
2985 resources within a time frame that is consistent with the fire dynamics.

2986 **5.1.4.3 Q3: What is the exposure of assets at this location?**

2987 The third question concerns the risk to high-value assets that may be threatened by
2988 fire spread originating at or propagating through the current location. Assets include
2989 residential and commercial structures, critical infrastructure facilities such as
2990 hospitals and power plants, water supply installations, and population centers. Their
2991 spatial distribution and protection priority are encoded in the value data layer $U_{val,k}$,

specifically through the protection priority score $V_{prio}(x, y)$ and the critical facility index $V_{crit}(x, y)$ defined in Chapter 4, Section 4.3.2.

The asset exposure at a given location depends not only on the presence of high-value assets in the immediate vicinity but also on the direction and momentum of fire spread. A fire front propagating rapidly in the direction of a populated area represents a higher exposure risk than a comparable fire front propagating away from it, even if the distance to the assets is the same. This directional dependence means that asset exposure cannot be assessed from asset distribution alone but requires information about fire propagation dynamics, specifically the propagation influence field $\Psi_k(x, y)$ and the burning status of neighboring cells.

5.1.4.4 Q4: How urgent must action be taken?

The fourth question synthesizes the responses to the preceding three questions in the context of temporal dynamics. Urgency depends not only on the current severity of the threat, the feasibility of intervention, and the exposure of assets, but also on how rapid conditions are changing and whether delays will substantially reduce the effectiveness of the available intervention options.

A fire front that has been burning continuously for many time steps may have progressively depleted local fuel reserves, causing the rate of spread to naturally decrease. A newly ignited fire front in an area with high fuel load and favorable wind conditions may be accelerating rapidly, with each time step of delay substantially reducing the accessible containment perimeter.

This temporal dimension of urgency is captured in the simulation framework by the ignition time variable $\tau_k(x, y)$, defined in Chapter 4, Equation (52), which records the time elapsed since combustion was initiated at each grid cell. A high value of $\tau_k(x, y)$ indicates a fire that has been burning for a sustained period, while a value near zero indicates a recently initiated fire front. The relationship between $\tau_k(x, y)$ and intervention urgency is non-trivial and context-dependent, motivating the

3019 construction of a dedicated urgency concept rather than the use of τ_k directly as a
 3020 decision input.

3021 5.1.5 Connecting Operational Questions to Design Decisions

3022 The four operational questions identified above provide the complete motivational
 3023 foundation for the design of the DisasterAware DSS. Table 5.1 summarizes the
 3024 connection between each question, the information it requires, and the Chapter 4
 3025 state variables that provide this information.

3026 Table 5.1 Operational questions, information requirements, and sources

	Required Information	Source
Q1: Fire Threat Severity	Local combustion intensity, propagation influence	$I_k(x, y), \Psi_k(x, y)$
Q2: Suppression Feasibility	Travel time, availability, accessibility, fuel resistance	$R_{time,k}, R_{avail,k}, G_{access}, F_{load,k}$
Q3: Asset Exposure Risk	Asset priority, fire spread direction and proximity	$V_{prio}(x, y), \Psi_k(x, y), B_k$
Q4: Intervention Urgency	Temporal fire progression, composite threat context	$\tau_k(x, y), c_1, c_2, c_3$

3027

3028 This table has two important structural properties. First, each question maps to a
 3029 distinct and identifiable subset of the Chapter 4 state variables and contextual inputs.
 3030 This one-to-one correspondence confirms that the information required to answer the
 3031 four operational questions is fully contained within the simulation framework
 3032 developed in Chapter 4, with no additional external data sources required. Second,
 3033 the questions are not mutually independent of their information requirements: the
 3034 spread potential $\Psi_k(x, y)$ contributes to both Q1 and Q3, and Q4 depends on the
 3035 answers to Q1, Q2, and Q3 as well as on $\tau_k(x, y)$. This interdependence motivates

3036 the hierarchical concept formation mechanism introduced in Section 5.2.3, in which
3037 Q4 is constructed as a second-order concept over the outputs of Q1, Q2, and Q3.

3038 The four operational questions correspond directly to the four situational concepts of
3039 the DisasterAware DSS: Fire Threat Level (c_1), Suppression Feasibility (c_2), Asset
3040 Exposure Risk (c_3), and Intervention Urgency (c_4). This correspondence is not
3041 incidental. The concept space of the DSS is derived from a structured analysis of the
3042 operational questions that wildfire incident commanders must answer in order to
3043 make effective intervention decisions. By grounding the concept space in operational
3044 practice, the DSS ensures that its internal representations are semantically
3045 meaningful to the human operators who rely on its outputs.

3046 5.1.6 Why Fuzzy Reasoning Is the Appropriate Mechanism

3047 The four questions share a common characteristic: the answers are inherently
3048 imprecise and require graded assessments rather than binary classifications. A fire
3049 threat is not simply "high" or "low" but may exhibit intermediate degrees of severity
3050 that depend on the relative contributions of intensity, fuel, and spread potential.
3051 Suppression feasibility is not binary but varies continuously with resource travel
3052 time, terrain accessibility, and the ratio of suppression capacity to suppression
3053 requirement. Asset exposure risk depends on a gradient of proximity and
3054 directionality rather than on a sharp threshold. Intervention urgency combines
3055 multiple continuous factors through a context-dependent weighing mechanism.

3056 These characteristics make threshold-based reasoning fundamentally inappropriate
3057 for this domain. A system that classifies fire threat as "high" if and only if $I_k(x, y) >$
3058 θ for some fixed threshold θ would fail to capture the graded nature of the underlying
3059 concept and would produce discontinuous decision outputs that are sensitive to small
3060 perturbations in the observed value near the threshold. This brittleness is particularly
3061 problematic when observations are subject to epistemic uncertainty, as small
3062 observation ambiguity could cause large discontinuities in decision outputs.

3063 Fuzzy logic provides a reasoning paradigm that directly addresses these limitations.
3064 By representing operational concepts as graded membership values rather than
3065 binary classifications, FIS can reason over the continuous spectrum of wildfire
3066 conditions without requiring precise boundary definitions (Zadeh, Fuzzy sets, 1965;
3067 Mendel, 2017).

3068 The membership values produced by fuzzification are inherently robust to small
3069 perturbations in observed values, as the membership functions provide a smooth
3070 mapping from observations to concept activations. Moreover, fuzzy reasoning is
3071 transparent and interpretable: the membership values and rule activations can be
3072 inspected by human operators at any point in the reasoning chain, providing a natural
3073 form of explainability that is difficult to achieve with optimization-based or machine
3074 learning approaches.

3075 The selection of fuzzy logic as the primary reasoning mechanism is not arbitrary but
3076 follows from a structured evaluation of available uncertainty reasoning paradigms
3077 against the operational requirements established in Sections 5.1.1 through 5.1.4.
3078 Three principal alternatives are considered: Bayesian probabilistic reasoning,
3079 Dempster–Shafer evidence theory, and neural network–based approaches.

3080 **Bayesian probabilistic reasoning** provides a mathematically rigorous framework
3081 for updating beliefs under uncertainty through conditional probability distributions
3082 (Pearl, 1988). However, its application to the wildfire DSS context presents two
3083 fundamental obstacles. First, Bayesian inference requires the specification of prior
3084 probability distributions over the state space, which presupposes statistical
3085 knowledge about the frequency and distribution of wildfire conditions that is rarely
3086 available at the spatial and temporal resolution required for cell-level decision
3087 support (Thompson & Calkin, 2011; Morgan & Henrion, 1990).

3088 In the DisasterAware formulation, the epistemic uncertainty term ϵ_k^i in the
3089 observation model $O_k^i(x, y)$ explicitly represents incomplete knowledge whose
3090 statistical properties are unknown, precisely the condition under which Bayesian
3091 inference loses its theoretical guarantees. Second, maintaining and updating joint

3092 probability distributions over the six-dimensional feature space at every grid cell and
3093 time step would impose computational costs that are incompatible with the real-time
3094 operational constraints of wildfire response, particularly in the distributed multi-
3095 agent architecture where each regional agent must independently generate
3096 intervention outputs within a single simulation time step.

3097 **Dempster–Shafer (D-S) evidence theory** extends probabilistic reasoning by
3098 allowing the representation of ignorance through belief and plausibility intervals,
3099 rather than requiring point probability assignments (Shafer, 1976). While this
3100 capacity to represent partial ignorance aligns with the epistemic uncertainty
3101 challenge of wildfire decision-making, D-S theory introduces two practical
3102 limitations in the DSS context.

3103 Dempster’s rule of combination can produce counterintuitive results when evidence
3104 sources are highly conflicting, a condition that arises naturally when regional agents
3105 observe the same fire front from different vantage points with different degrees of
3106 observational degradation (Zadeh, 1986). Furthermore, D-S theory does not provide
3107 a natural mechanism for linguistic representation of operational concepts such as
3108 “high fire threat” or “limited suppression feasibility,” which are central to the
3109 interpretability requirement (Contribution C4) and to the alignment of DSS outputs
3110 with the reasoning patterns of field commanders identified in Section 5.1.4.

3111 **Neural network–based approaches**, including deep reinforcement learning and
3112 transformer architectures, have demonstrated strong predictive performance in
3113 wildfire spread modeling (Jain, et al., 2020). However, their application as decision
3114 reasoning mechanisms, rather than predictive components, conflicts directly with the
3115 explainability requirement that is central to this thesis (Gap G4, Contribution C4).

3116 The internal representations of neural networks are opaque by design, and post-hoc
3117 explainability methods such as SHAP or LIME provide statistical attributions rather
3118 than the causal rule-level traceability that the DisasterAware governance layer
3119 requires (Ribeiro, Singh, & Guestrin, 2016). Moreover, training neural network–
3120 based decision policies requires large datasets of expert wildfire response decisions

that are not systematically available, and the resulting policies cannot be inspected, validated, or overridden at the rule level, capabilities that are essential for the operator trust and regulatory compliance requirements of emergency management systems (McNeese, Schelble, Canonico, & Demir, 2021).

Fuzzy logic addresses all three limitations simultaneously. It operates without requiring prior probability distributions, representing epistemic uncertainty through graded membership values that are derived directly from observable information (Zadeh, Fuzzy sets, 1965; Mendel, 2017). Its rule-based structure provides intrinsic traceability: every intervention recommendation can be traced back through specific rule activations, concept values, and feature observations, as formalized in the Explainability Layer of Section 6.5.

Its computational cost scales linearly with the number of rules and grid cells, satisfying the real-time constraint of the distributed architecture. Finally, its linguistic variables align naturally with the operational vocabulary of wildfire incident commanders, supporting the human-machine collaboration objective that distinguishes DisasterAware from purely automated decision systems. Table 5.2 summarizes this comparative assessment.

Table 5.2 Comparative assessment of uncertainty reasoning paradigms for DSS

Criterion	Bayesian	Dempster–Shafer	Neural Networks	Fuzzy Logic
Operates without prior distributions	✗	✓	✓	✓
Handles epistemic uncertainty explicitly	Partial	✓	✗	✓
Rule-level traceability	✗	✗	✗	✓
Linguistic interpretability	✗	✗	✗	✓

Criterion	Bayesian	Dempster– Shafer	Neural Networks	Fuzzy Logic
Real-time computational feasibility	✗	✓	Partial	✓
No training data requirement	✓	✓	✗	✓
Operator override at rule level	✗	✗	✗	✓

3139

3140 This comparison establishes that fuzzy logic is not merely a convenient modeling
3141 choice but the reasoning paradigm that most directly satisfies the architectural
3142 requirements of a distributed, explainable, and uncertainty-aware DSS operating
3143 under partial observability.

3144 The applicability of fuzzy inference to wildfire decision support has been established
3145 in prior literature (Vásquez, 2021; Soualah, 2024). However, existing applications
3146 have typically applied fuzzy inference directly to raw environmental measurements,
3147 without the concept abstraction layer that is central to the DisasterAware design.

3148 When fuzzy inference is applied directly over $\mathcal{N}_{feat}$ features each with $\mathcal{N}_\mu$ linguistic
3149 terms, the number of rules required to characterize the decision space grows as
3150 $\mathcal{N}_\mu^{\mathcal{N}_{feat}}$. For the feature space and linguistic variable design established in Sections
3151 5.2.1 and 0, this would yield an unmanageably large rule base that is difficult to
3152 construct, validate, and maintain.

3153 The concept abstraction mechanism introduced in Section 5.2.3 addresses this
3154 limitation directly by reducing the effective dimensionality of the rule base from
3155 $\mathcal{N}_{feat}$ to $\mathcal{N}_{concept}$, a reduction whose magnitude and significance are formally
3156 characterized through a complexity proposition.

3157 5.2 Feature Space Design and Formal Derivation

3158 The DSS processing pipeline involves three sequential design decisions: the
3159 selection and derivation of scalar features (feature extraction), the definition of
3160 linguistic variables and membership functions (fuzzification), and the construction
3161 of higher-level situational concepts from fuzzified inputs (concept construction).
3162 Each of these transformations involves formal design decisions that determine the
3163 informational content, linguistic expressiveness, and computational tractability of
3164 the DSS. The present section formalizes these decisions, provides analytical
3165 justification for each design choice, and establishes a formal result on the rule
3166 complexity reduction achieved through concept abstraction.

3167 5.2.1 Feature Extraction Mapping and Scalar Indicators

3168 The feature extraction mapping f_{feat} transforms the observable information $O_k^i(x, y)$
3169 into a vector of $\mathcal{N}_{feat}$ scalar indicators. Each feature must satisfy three design
3170 criteria. First, it must be fully computable from the state variables and contextual
3171 inputs defined in Chapter 4, without requiring additional external data sources not
3172 already present in the simulation framework. Second, it must be bounded in the
3173 interval $[0, 1]$, which enables a uniform fuzzy variable design in which a single
3174 parameterization framework applies to all features. Third, it must carry operationally
3175 meaningful information that directly contributes to answering one or more of the
3176 four operational questions identified in Section 5.1.4. These criteria collectively
3177 ensure that the feature space is complete with respect to the operational requirements,
3178 internally consistent with the Chapter 4 framework and appropriately scaled for
3179 fuzzy variable design.

3180 The analysis of operational information requirements presented in Table 5.1 reveals
3181 that answering all four operational questions requires information from the full state
3182 vector $\{B_k, F_{load,k}, I_k, \tau_k\}$, from the propagation dynamics captured by Ψ_k , and from
3183 the contextual inputs $U_{val,k}$ and $U_{DSS,k}$. No single state variable is sufficient to

3184 answer any of the four questions in isolation; each question requires a combination
3185 of physical state information and operational context. The following six features are
3186 derived from this analysis.

3187 **5.2.1.1 Feature 1: Fire Intensity**

3188 The fire intensity feature is defined directly from the local fire intensity proxy
3189 maintained by the Simulation Core: $z_{k,1}^i(x, y) = I_k(x, y)$ where $I_k(x, y)$ is defined
3190 by Chapter 4, Equation (51) as $f_{intensity}(F_{load,k}, W_{ws,k}, G_{slope})$. The intensity
3191 proxy is bounded in $[0, 1]$ by construction through the normalization embedded in
3192 $f_{intensity}$, satisfying the boundedness criterion without additional transformation.

3193 This feature directly addresses operational question Q1 (fire threat severity) by
3194 quantifying the local combustion strength at grid cell (x, y) . A high value of $z_{k,1}^i$
3195 indicates intense fire behavior associated with high heat release rates and sustained
3196 combustion, which is a primary indicator of fire threat in operational classification
3197 schemes (Rothermel, 1972; Thompson & Calkin, 2011).

3198 **5.2.1.2 Feature 2: Fuel Load**

3199 The fuel load feature is defined as the currently available combustible mass:
3200 $z_{k,2}^i(x, y) = F_{load,k}(x, y)$ where $F_{load,k}(x, y)$ is maintained by the Simulation Core
3201 and updated according to Chapter 4, Equation (49). The variable is bounded in $[0, 1]$
3202 by construction.

3203 Within the context of the four operational questions, fuel load carries dual
3204 informational content. In relation to Q1 (fire threat), a high fuel load indicates
3205 sustained combustion potential. In relation to Q2 (suppression feasibility), fuel load
3206 represents suppression effort: a cell with high remaining fuel requires greater
3207 suppression capacity to achieve meaningful fuel reduction, while a cell with depleted
3208 fuel may be approaching natural extinction. This dual role is resolved at the concept

formation stage, where fuel load contributes with different weights and at different linguistic levels depending on the concept being formed.

5.2.1.3 Feature 3: Spread Potential

The spread potential feature is derived from the propagation influence field defined in Chapter 4, Equation (46): $z_{k,3}^i(x, y) = \min\left(1, \frac{\Psi_k(x, y)}{\Psi_{ref}}\right)$ where $\Psi_k(x, y) = \frac{1}{8} \sum_{(i,j) \in \mathcal{N}^8(x, y)} B_k(i, j) \cdot R_{spread, k}(i, j) \cdot g_{dir, k}^{(i,j) \rightarrow (x, y)}$ is the weighted propagation influence from the 8-connected neighborhood, and Ψ_{ref} is a reference normalization constant set to the characteristic propagation influence under high-wind, high-fuel conditions. The clipping operation $\min(1, \cdot)$ ensures $z_{k,3}^i \in [0, 1]$ whether or not Ψ_k exceeds Ψ_{ref} under extreme co-alignment of fuel, wind, slope, and aspect effects. The spread potential feature addresses both Q1 (fire threat severity, through the momentum of the advancing fire front) and Q3 (asset exposure risk, through the spatial direction of propagation relative to high-value assets). The directional weighting factor $g_{dir, k}^{(i,j) \rightarrow (x, y)}$ defined in Chapter 4, Equation (48) ensures that $z_{k,3}^i$ captures not only the magnitude of spread potential but also its spatial orientation, which is essential for asset exposure assessment.

5.2.1.4 Feature 4: Asset Exposure

The asset exposure feature combines the spatial distribution of high-value assets with the proximity of active fire fronts: $z_{k,4}^i(x, y) = V_{prio}(x, y) \cdot \bar{B}_k^\delta(x, y)$ where $V_{prio}(x, y) \in [0, 1]$ is the protection priority score from the value data layer $U_{val, k}$, defined in Section 4.3.2, and $\bar{B}_k^\delta(x, y)$ is the mean burning status over the δ -extended neighborhood of cell (x, y) : $\bar{B}_k^\delta(x, y) = \frac{1}{|\mathcal{N}^\delta(x, y)|} \sum_{(i,j) \in \mathcal{N}^\delta(x, y)} B_k(i, j)$ with $\mathcal{N}^\delta(x, y)$ denoting the set of cells within radius δ of (x, y) . The product structure of $z_{k,4}^i$ reflects the multiplicative nature of asset exposure risk: a high-priority asset

located far from any burning cells carries low immediate exposure ($\bar{B}_k^\delta \approx 0$), while even a moderate-priority asset in the immediate vicinity of an active fire front carries significant exposure ($\bar{B}_k^\delta$ close to 1). The result is bounded in $[0, 1]$ since both factors are bounded in $[0, 1]$. This feature directly addresses Q3 (asset exposure risk).

5.2.1.5 Feature 5: Resource Accessibility

The resource accessibility feature aggregates three operational constraints on suppression deployment: $z_{k,5}^i(x, y) = R_{avail,k}(x, y) \cdot \exp(-\beta_t \cdot R_{time,k}(x, y)) \cdot G_{access}(x, y)$ where $R_{avail,k}(x, y) \in \{0, 1\}$ is the operational availability status of suppression resources define in Section 4.2.1, $R_{time,k}(x, y)$ is the estimated travel time to grid cell (x, y) , $\beta_t > 0$ is a time-decay coefficient that controls how rapidly accessibility decreases with travel time, and $G_{access}(x, y) \in [0, 1]$ is the terrain accessibility index from the topography data layer defined in Section 4.3.1.

The multiplicative structure of $z_{k,5}^i$ reflects the conjunction of three necessary conditions for effective suppression deployment: resources must be available ($R_{avail,k} = 1$), must be reachable within an operationally meaningful time window (captured by the exponential decay), and the terrain must permit safe access ($G_{access} > 0$). If any of these conditions are violated, the product approaches zero, indicating negligible suppression accessibility regardless of the other factors. The exponential decay model for travel time is consistent with the operational reality that suppression effectiveness decreases rapidly as travel time increases relative to the characteristic fire spread timescale. This feature directly addresses Q2 (suppression feasibility). The parameter β_t is a calibration constant in units of inverse time steps.

3255 5.2.1.6 Feature 6: Temporal Urgency

3256 The temporal urgency feature transforms the ignition time variable $\tau_k(x, y)$ into a
3257 bounded urgency indicator: $z_{k,6}^i(x, y) = 1 - \exp\left(-\lambda \cdot \frac{\tau_k(x, y)}{\tau_{ref}}\right)$ where $\tau_k(x, y)$ is
3258 defined by Chapter 4, Equation (52), τ_{ref} is a reference time scale representing the
3259 characteristic duration of a fire spread event, and $\lambda > 0$ is a shape parameter
3260 controlling the rate at which urgency saturates. The result is bounded in $[0, 1]$: $z_{k,6}^i =$
3261 0 when $\tau_k = 0$ (newly initiated fire) and $z_{k,6}^i \rightarrow 1$ as $\tau_k \rightarrow \infty$.

3262 The exponential transformation is motivated by the observation that urgency does
3263 not grow linearly with time. A fire that has been burning for a very short time ($\tau_k \approx$
3264 0) presents maximal urgency because it may still be containable with limited
3265 resources, whereas a fire that has been burning for an extended period ($\tau_k \gg \tau_{ref}$)
3266 may have already expanded beyond containable limits, making urgency less
3267 sensitive to further delays.

3268 The transformation $1 - e^{-\frac{\lambda \tau}{\tau_{ref}}}$ captures this saturating behavior while remaining
3269 monotonically increasing. This feature directly addresses Q4 (intervention urgency)
3270 by providing a temporally sensitive measure of how long the current fire front has
3271 been active at each grid cell.

3272 5.2.1.7 Summary of the Feature Space

3273 The six features are summarized in Table 5.3, together with their operational
3274 question mapping, source, and normalization method. The feature vector is therefore
3275 $\mathcal{N}_{feat} = 6$. This value is not chosen to minimize or maximize the number of features
3276 but is the minimum number of features required to provide sufficient information to
3277 answer all four operational questions identified in Section 5.1.4.

3278 Formally, no proper subset of $\{z_{k,1}^i, \dots, z_{k,6}^i\}$ can address all four operational
 3279 questions simultaneously, since each question requires information from at least one
 3280 feature that is not shared with the others through Q2-specific features $z_{k,5}^i$ and Q4-
 3281 specific feature $z_{k,6}^i$.

3282 Table 5.3 Feature definitions, sources, and operational question assignments

Feature	Symbol	Operational Question	Source	Normalization
Fire intensity	$z_{k,1}^i$	Q1 (threat severity)	I_k	Inherent [0, 1]
Fuel load	$z_{k,2}^i$	Q1, Q2 (threat, feasibility)	$F_{load,k}$	Inherent [0, 1]
Spread potential	$z_{k,3}^i$	Q1, Q3 (threat, exposure)	Ψ_k	Clipped normalization
Asset exposure	$z_{k,4}^i$	Q3 (exposure risk)	V_{prio}, B_k	Product structure
Resource accessibility	$z_{k,5}^i$	Q2 (suppression feasibility)	$R_{avail}, R_{time}, G_{access}$	Exponential product
Temporal urgency	$z_{k,6}^i$	Q4 (intervention urgency)	τ_k	Exponential transformation

3283
 3284 **Remark 5.1 (Exclusion of $R_{cap,k}$ and $R_{eff,k}$ from the Feature Space).** Two
 3285 parameters from the Resource Data Layer (Table 4.6), namely suppression capacity
 3286 $R_{cap,k}(x, y)$ and suppression efficiency $R_{eff,k}(x, y)$, are deliberately excluded from
 3287 the six-dimensional feature space. This exclusion reflects a structural design decision
 3288 rather than an oversight and is motivated by three considerations.

3289 *Architectural separation of reasoning and execution.* The feature space is designed
 3290 to support *decision reasoning*: determining what type and intensity of intervention
 3291 is appropriate given the current situational assessment. The parameters $R_{cap,k}$ and
 3292 $R_{eff,k}$ govern *decision execution*: they determine the physical magnitude of fuel
 3293 reduction achieved per unit of intervention intensity. Including execution parameters
 3294 in the reasoning layer would conflate two functionally distinct stages and violate the
 3295 separation between the DSS mapping π_{DSS}^i and the suppression function f_{supp}
 3296 established in Chapter 4, Equation (50).

3297 *Operational role.* Resource capacity and efficiency determine *how much* physical
 3298 suppression is achieved for a given intervention signal, not *whether* intervention
 3299 should be attempted. The question "Should we intervene at this location?" (answered
 3300 by the rule base) is distinct from "How much fuel reduction does this intervention
 3301 produce?" (f_{supp}).

3302 The feature z_5 (resource accessibility) already captures the feasibility of intervention
 3303 through $R_{avail,k}$, $R_{time,k}$, and G_{access} . Crucially, the binary deployability indicator
 3304 $R_{avail,k} \in \{0, 1\}$ (Table 4.6) reflects capacity constraints indirectly: when
 3305 suppression capacity at or near a grid cell is exhausted or unavailable, the
 3306 corresponding resource is not deployable and $R_{avail,k} = 0$, which drives z_5 to zero
 3307 regardless of travel time or terrain conditions. The capacity signal therefore enters
 3308 the feature space through the availability channel without requiring a dedicated
 3309 feature dimension.

3310 Adding $R_{cap,k}$ as a separate feature would introduce redundancy with this existing
 3311 pathway while shifting the reasoning layer toward execution-level concerns. $R_{eff,k}$,
 3312 by contrast, governs the translation of intervention intensity into physical fuel
 3313 reduction (i.e., a mapping that varies by fuel type, moisture, and terrain) and is thus
 3314 structurally aligned with the suppression function f_{supp} (50) rather than with the
 3315 situational assessment encoded in the feature space.

3316 *Computational role.* $R_{cap,k}$ is utilized in the Global Coordination Layer (Section 6.4)
 3317 during the resource normalization pass (Equations (102)–(107)), where the
 3318 aggregate intervention demand is scaled against available capacity C_k . $R_{eff,k}$ is
 3319 utilized in the suppression-to-fuel-reduction mapping f_{supp} (Equation (50) and
 3320 Appendix B), where intervention intensity is translated into physical fuel load
 3321 changes. Both parameters therefore enter the system at precisely the stage where
 3322 their physical meaning is operationally relevant.

3323 5.2.1.8 Calibration Parameters of the Feature Space

3324 The feature extraction mappings defined in Sections 5.2.1.3 through 5.2.1.6
 3325 introduce four calibration parameters that govern the normalization and sensitivity
 3326 of individual features. These parameters are not free variables; each has a physical
 3327 interpretation grounded in the simulation framework of Chapter 4 and a default value
 3328 derived from operational considerations.

3329 The membership function parameters $(p_{1,l}, p_{2,l}, p_{3,l})$ are feature-specific and reflect
 3330 the operational significance of different value ranges. The six features fall into three
 3331 parameterization groups based on their skewness profile.

3332 **Symmetric parameterization.** Fire intensity (z_1) and fuel load (z_2) use
 3333 approximately symmetric configurations: $(p_1, p_2, p_3) = (0.20, 0.50, 0.80)$ and
 3334 $(p_1, p_2, p_3) = (0.25, 0.50, 0.75)$, respectively. Both variables are already
 3335 normalized within the simulation framework through physically meaningful scaling
 3336 (49)–(51), making their operational interpretation approximately uniform across
 3337 $[0, 1]$. The midpoint $p_2 = 0.50$ centers the Medium term, reflecting the absence of
 3338 asymmetric operational significance in these scales.

3339 **Left-skewed parameterization.** Spread potential (z_3) and asset exposure (z_4) use
 3340 left-skewed configurations: $(p_1, p_2, p_3) = (0.20, 0.40, 0.70)$ and $(p_1, p_2, p_3) =$
 3341 $(0.15, 0.45, 0.75)$. The leftward shift of p_2 reflects the operational reality that even
 3342 moderate values of these features warrant elevated attention: a spread potential of

0.4 already represents significant fire front momentum, and moderate asset proximity to an active fire front demands precautionary intervention rather than delayed response.

Right-skewed parameterization. Resource accessibility (z_5) and temporal urgency (z_6) share the same right-skewed configuration $(p_1, p_2, p_3) = (0.30, 0.60, 0.85)$. For resource accessibility, the rightward shift reflects the multiplicative constraint structure of the feature: because three independent conditions (availability, travel time, terrain access) must be simultaneously satisfied, achieving High accessibility requires favorable alignment across all three factors, which occurs only in a narrow upper range. For temporal urgency, the same configuration captures the saturating nature of the urgency signal: early fire fronts require high urgency even at moderate τ values, while the distinction between Medium and High urgency becomes operationally meaningful only at advanced burn stages.

These parameters are design defaults derived from the operational analysis of Section 5.1. The critical design property is the relative ordering and skewness across features, which encodes operational judgment about which value ranges are significant. The absolute values are secondary to this structural property and may be refined through simulation-based sensitivity analysis as discussed in Chapter 7.

Remark 5.2 (Reproducibility). With the parameter values in Table 5.3, all six features $z_{k,1}^i, \dots, z_{k,6}^i$ are fully computable from Chapter 4 state variables and contextual inputs, without any unspecified free parameters. The feature extraction mapping f_{feat} is therefore completely defined.

5.2.2 Fuzzy Variable Design

The fuzzification stage transforms each scalar feature $z_{k,l}^i(x, y) \in [0,1]$ into a graded membership representation over a set of linguistic terms. The design of the fuzzy variables involves two decisions: the number of linguistic terms per feature and the

3369 shape and parameters of the membership functions. Each decision is grounded in
3370 both operational requirements and formal properties.

3371 5.2.2.1 Choice of Linguistic Term Count

3372 Each feature is assigned $\mathcal{N}_\mu = 3$ linguistic terms, denoted Low (L), Medium (M), and
3373 High (H). This choice is motivated by the following considerations. From the
3374 operational perspective, three terms correspond to the natural granularity of
3375 operational assessments in wildfire response. Field commanders and coordination
3376 centers consistently classify operational conditions into three actionable categories:
3377 conditions that do not require immediate intervention (Low), conditions that warrant
3378 monitoring and preparatory action (Medium), and conditions that require immediate
3379 and prioritized intervention (High) (Thompson & Calkin, 2011; Klein, 2017).
3380 Adding a fourth or fifth term would introduce finer distinctions that are not
3381 operationally meaningful given the epistemic uncertainty of field observations.

3382 From the formal perspective, increasing $\mathcal{N}_\mu$ has an exponential effect on rule base
3383 complexity, as established in the complexity analysis of Section 5.2.3. Specifically,
3384 the decision rule base complexity grows as $\mathcal{N}_\mu^{\mathcal{N}_{concept}}$. With $\mathcal{N}_{concept} = 4$ and $\mathcal{N}_\mu =$
3385 3, the maximum rule base size is 81 rules per intervention type. Increasing to $\mathcal{N}_\mu =$
3386 4 would yield 256 rules per intervention type, a factor of more than three increase
3387 in rule base size for a marginal gain in representational granularity.

3388 5.2.2.2 Membership Function Shapes

3389 For a feature l with parameter triplet $(p_{1,l}, p_{2,l}, p_{3,l})$ satisfying $0 \leq p_{1,l} < p_{2,l} <$
3390 $p_{3,l} \leq 1$, the three membership functions are defined as (57).

$$\begin{aligned}
\mu_{L,l}(z) &= \begin{cases} 1 & z \leq p_{1,l} \\ \frac{p_{2,l} - z}{p_{2,l} - p_{1,l}} & p_{1,l} < z \leq p_{2,l} \\ 0 & z > p_{2,l} \end{cases} \\
\mu_{M,l}(z) &= \begin{cases} 0 & z \leq p_{1,l} \text{ or } z \geq p_{3,l} \\ \frac{z - p_{1,l}}{p_{2,l} - p_{1,l}} & p_{1,l} < z \leq p_{2,l} \\ \frac{p_{3,l} - z}{p_{3,l} - p_{2,l}} & p_{2,l} < z < p_{3,l} \end{cases} \\
\mu_{H,l}(z) &= \begin{cases} 0 & z < p_{2,l} \\ \frac{z - p_{2,l}}{p_{3,l} - p_{2,l}} & p_{2,l} \leq z < p_{3,l} \\ 1 & z \geq p_{3,l} \end{cases}
\end{aligned} \tag{57}$$

3391 The Low function is a left-shoulder function that achieves full membership ($\mu = 1$)
3392 for all values below $p_{1,l}$ and decreases linearly to zero at $p_{2,l}$. The High function is
3393 the symmetric right-shoulder function. The Medium function is a symmetric
3394 triangular function peaking at $p_{2,l}$.

3395 The choice of shoulder functions for Low and High, rather than triangular functions
3396 for all three terms, is formally justified by the following consideration. At the
3397 boundary values $z = 0$ and $z = 1$, a triangular Low or High function would produce
3398 $\mu = 1$ only at the apex, which may not coincide with the boundary. The shoulder
3399 formulation guarantees that $\mu_{L,l}(z) = 1$ for all $z \leq p_{1,l}$ and $\mu_{H,l}(z) = 1$ for all $z \geq$
3400 $p_{3,l}$, ensuring that extreme observations are assigned full membership to the
3401 appropriate extreme linguistic term. This is consistent with the recommendations for
3402 bounded universes of discourse (Mendel, 2017).

3403 A fundamental formal property of this design is the partition of unity.

3404 **Proposition 5.1 (Partition of Unity):** *For any $z \in [0,1]$ and any feature l , the three*
3405 *membership functions satisfy*

$$\mu_{L,l}(z) + \mu_{M,l}(z) + \mu_{H,l}(z) = 1 \tag{58}$$

3406 *Proof:* The proof follows direct verification over the four disjoint intervals defined
3407 by the parameters (59).

For $z \leq p_{1,l}$: $\mu_{L,l} = 1$, $\mu_{M,l} = 0$, $\mu_{H,l} = 0$. Sum = 1.

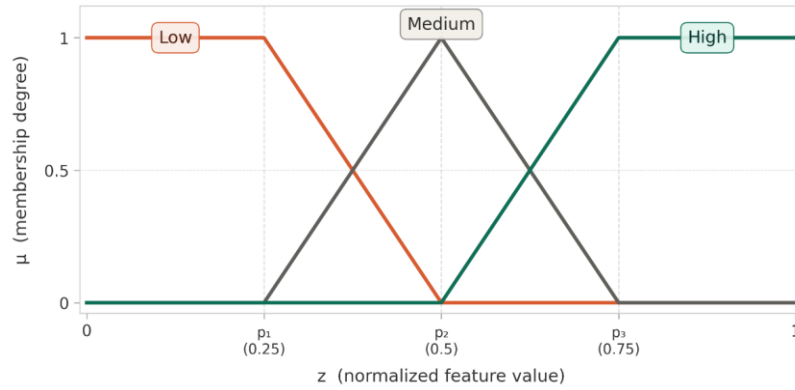
For $p_{1,l} < z \leq p_{2,l}$: $\mu_{L,l} = \frac{p_{2,l}-z}{p_{2,l}-p_{1,l}}$, $\mu_{M,l} = \frac{z-p_{1,l}}{p_{2,l}-p_{1,l}}$, $\mu_{H,l} = 0$. Sum = $\frac{p_{2,l}-z+z-p_{1,l}}{p_{2,l}-p_{1,l}} = 1$.

(59)

For $p_{2,l} < z < p_{3,l}$: $\mu_{L,l} = 0$, $\mu_{M,l} = \frac{p_{3,l}-z}{p_{3,l}-p_{2,l}}$, $\mu_{H,l} = \frac{z-p_{2,l}}{p_{3,l}-p_{2,l}}$. Sum = 1.

For $z \geq p_{3,l}$: $\mu_{L,l} = 0$, $\mu_{M,l} = 0$, $\mu_{H,l} = 1$. Sum = 1.

3408 The partition of unity ensures that the total evidential weight assigned across the
 3409 three linguistic terms is conserved for every observation. This prevents any feature
 3410 value from being ambiguously unassigned and ensures that the fuzzified
 3411 representation carries the same total information content as the original scalar
 3412 observation. Figure 5.2 illustrates the three membership functions for a
 3413 representative parameterization.



3414
 3415 Figure 5.2 Membership functions for a representative feature parameterization

3416 Left-shoulder (Low), triangular (Medium), and right-shoulder (High) membership
 3417 functions for a feature l with parameters $(p_1, p_2, p_3) = (0.25, 0.50, 0.75)$. The
 3418 partition of unity $\mu_L + \mu_M + \mu_H = 1$ holds for all $z \in [0, 1]$.

5.2.2.3 Parameter Specification

The parameters $(p_{1,l}, p_{2,l}, p_{3,l})$ are feature-specific and reflect the operational significance of different value ranges for each feature. Table 5.4 presents the default parameter values together with their justification.

Table 5.4 Membership function parameters for each feature

Feature	$p_{1,l}$	$p_{2,l}$	$p_{3,l}$	Parameterization Rationale
z_1(fire intensity)	0.20	0.50	0.80	Symmetric: intensity scale is approximately uniform in [0, 1]
z_2(fuel load)	0.25	0.50	0.75	Symmetric: fuel scale is approximately uniform; High above 0.75 implies ample combustion potential
z_3(spread potential)	0.20	0.40	0.70	Left-skewed: even moderate spread potential (0.4) represents significant fire front momentum operationally
z_4(asset exposure)	0.15	0.45	0.75	Left-skewed: conservative threshold; moderate asset proximity to fire warrants elevated attention
z_5(resource accessibility)	0.30	0.60	0.85	Right-skewed: multiplicative constraints make High accessibility rare; requires simultaneous availability, proximity, and terrain access.
z_6(temporal urgency)	0.30	0.60	0.85	Right-skewed: urgency signal saturates at advanced stages; early fire fronts require high urgency even at moderate τ values

These parameters are design defaults derived from the operational analysis of Section 5.1. They are intended as calibration starting points and may be refined through simulation-based sensitivity analysis or expert elicitation, as discussed in Chapter 7. The critical design property is the relative ordering and skewness of the

parameters across features, which encodes the operational judgment about which value ranges are operationally significant. The absolute parameter values are secondary to this structural property.

5.2.2.4 Parameter Calibration Methodology

The calibration parameters in Table 5.3 and the membership function parameters in Table 5.4 are established through a three-source calibration methodology that combines *domain knowledge*, *operational literature*, and *structural analysis* of the simulation framework. This methodology is consistent with established practice in FIS design, where parameter initialization is guided by expert knowledge and subsequently validated through sensitivity analysis (Mendel, 2017; Pedrycz, 1993).

Domain knowledge. The skewness patterns in Table 5.4 encode asymmetric operational significance: for spread potential (z_3), the left-skewed parameterization ($p_1 = 0.20, p_2 = 0.40$) reflects the operational reality that even moderate propagation influence constitutes significant tactical concern, because fire spread accelerates nonlinearly once a critical momentum threshold is exceeded (Rothermel, 1972). For resource accessibility (z_5), the right-skewed parameterization ($p_2 = 0.60, p_3 = 0.85$) reflects the multiplicative constraint structure of the feature itself: because three independent conditions (availability, travel time, terrain access) must be simultaneously satisfied, achieving High accessibility requires alignment across all three factors, which occurs only in a narrow range of the feature space.

Operational literature. The symmetric parameterizations for fire intensity (z_1) and fuel load (z_2) are grounded in the observation that these variables are already normalized within the simulation framework through physically meaningful scaling (49) and (51), making their operational interpretation approximately uniform across the $[0, 1]$ range. The three-level classification (Low, Medium, High) with symmetric boundaries is consistent with the operational fire behavior assessment scales used in wildfire DSS (Thompson & Calkin, 2011; Finney, 2006).

Structural analysis. The reference parameters $\Psi_{\text{ref}} = 1.0$ and $\tau_{\text{ref}} = 10\Delta t$ in Table 5.3 are calibration constants aligned with the operational linguistic scales of features z_3 and z_6 . Ψ_{ref} is the characteristic scale of Ψ_k under high-wind, high-fuel conditions and serves as the operational saturation point for the “High” linguistic term on z_3 . It is not an absolute upper bound on Ψ_k : under extreme co-alignment of fuel, wind, slope, and aspect effects, the unclipped rate of spread $R_{\text{spread},k}$ may exceed unity in cells per simulation step (Appendix A), and Ψ_k can correspondingly exceed Ψ_{ref} . The clipping operation $\min(1, \Psi_k/\Psi_{\text{ref}})$ in the definition of z_3 absorbs such cases structurally, ensuring $z_{k,3}^i \in [0,1]$. The temporal reference τ_{ref} is calibrated to the characteristic timescale over which a newly ignited cell reaches peak spread rate under nominal meteorological conditions in the Chapter 4 simulation model.

Importantly, the formal properties of the DSS pipeline, output boundedness (Proposition 5.3), partition of unity (Proposition 5.1), and Lipschitz continuity (Proposition 5.4), hold for *any* valid parameter assignment satisfying $0 \leq p_{1,l} < p_{2,l} < p_{3,l} \leq 1$, not only for the defaults presented here. The parameters affect the quantitative behavior of the system (which rules activate more strongly for a given observation) but do not alter its qualitative guarantees. The sensitivity of system performance to parameter variation is examined in Chapter 7, Section 7.6.

5.2.3 Concept Space Design and Rule Complexity Reduction

The fuzzification stage produces, for each grid cell (x, y) within region Ω_i , a membership matrix (60). This matrix has $\mathcal{N}_\mu \times \mathcal{N}_{\text{feat}} = 3 \times 6 = 18$ entries per grid cell per region. The concept construction step reduces this 18-dimensional membership representation to a 4-dimensional concept vector $\mathbf{c}_k^i(x, y) = \{c_{k,h}^i(x, y)\}_{h=1}^4$. This dimensionality reduction has both semantic and computational consequences, which are formalized in this section.

$$\boldsymbol{\mu}_k^i(x, y) = \{\mu_{k,n,l}^i(x, y)\}_{n \in \{L, M, H\}, l \in \{1, \dots, 6\}} \quad (60)$$

3481 5.2.3.1 Concept Construction Formulas

3482 Each situational concept $c_{k,h}^i(x, y)$ is defined as a normalized weighted combination
 3483 of selected membership values. Let $\mathcal{S}_h$ denote the index set of (n, l) pairs that
 3484 contribute to concept h , and let $w_{h,n,l} > 0$ denote the corresponding weights. The
 3485 concept value is (61).

$$c_{k,h}^i(x, y) = \frac{\sum_{(n,l) \in \mathcal{S}_h} w_{h,n,l} \cdot \mu_{k,n,l}^i(x, y)}{\sum_{(n,l) \in \mathcal{S}_h} w_{h,n,l}} \quad (61)$$

3486 Since $\mu_{k,n,l}^i \in [0,1]$ and $w_{h,n,l} > 0$, the normalized weighted average satisfies $c_{k,h}^i \in$
 3487 $[0,1]$ for all parameter configurations.

3488 5.2.3.1.1 Concept 1: Fire Threat Level

3489 Fire Threat Level (62) is high when EITHER fire intensity is linguistically High
 3490 OR spread potential is linguistically High, or both. Equal weighting reflects the
 3491 operational judgment that both combustion intensity and propagation momentum are
 3492 equally important indicators of fire threat. This formulation is consistent with the
 3493 threat classification scheme of Thompson and Calkin (2011), in which threat is
 3494 defined by the combined effect of fire behavior intensity and spread rate.

$$\mathcal{S}_1 = \{(H, 1), (H, 3)\}, w_{1,H,1} = 1.0, w_{1,H,3} = 1.0 \quad (62)$$

3495 5.2.3.1.2 Concept 2: Suppression Feasibility

3496 Suppression Feasibility is high when resource accessibility is High AND fuel load is
 3497 Low. The inclusion of $\mu_{1,2}$ (Low fuel membership) reflects the operational reality that
 3498 high fuel load increases the suppression effort required and therefore reduces
 3499 effective suppression feasibility even when resources are accessible. The weight
 3500 ratio of 1.0 : 0.7 reflects the judgment that resource accessibility is slightly more

3501 determinative of feasibility than fuel load, since resources that cannot reach the fire
3502 are ineffective regardless of fuel conditions (Martell, 2007).

$$\mathcal{S}_2 = \{(H, 5), (L, 2)\}, w_{2,H,5} = 1.0, w_{2,L,2} = 0.7 \quad (63)$$

3503 5.2.3.1.3 Concept 3: Asset Exposure Risk

3504 Asset Exposure Risk (64) is high when asset priority is High AND spread potential
3505 toward those assets is High. The weight 0.8 on spread potential reflects that asset
3506 exposure risk is primarily driven by asset presence (weight 1.0), with propagation
3507 dynamics as a modulating factor. A high-priority asset with moderate spread
3508 potential in its direction still warrants attention, but somewhat less urgently than one
3509 with high spread potential directly aligned toward it (Pacheco, 2015).

$$\mathcal{S}_3 = \{(H, 4), (H, 3)\}, w_{3,H,4} = 1.0, w_{3,H,3} = 0.8 \quad (64)$$

3510 5.2.3.1.4 Concept 4: Intervention Urgency (Second-Order)

3511 The Intervention Urgency concept (65) is a second-order concept that takes the
3512 first-order concept values as inputs together with the temporal urgency feature where
3513 $\mu_H(c_h)$ and $\mu_L(c_h)$ denote the High and Low memberships of the concept value $c_{k,h}^i$
3514 using the same membership defined in Section 5.2.2, applied to the concept values
3515 directly. This is valid because concepts are bounded in $[0, 1]$ by construction.

$$\begin{aligned} \mathcal{S}_4 &= \{(H, c_1), (L, c_2), (H, c_3), (H, z_6)\}, \\ w_{4,H,c_1} &= w_{4,L,c_2} = w_{4,H,c_3} = w_{4,H,z_6} = 1.0 \end{aligned} \quad (65)$$

3516 The second-order structure reflects the operational logic of urgency: intervention is
3517 most urgent when (i) the fire threat is High, (ii) suppression feasibility is Low,
3518 meaning the window for effective intervention is closing, (iii) asset exposure risk is
3519 High, and (iv) the temporal urgency signal is High indicating an advancing or
3520 recently ignited fire front. The equal weights reflect the judgment that all four
3521 contributors to urgency are operationally equivalent in the default configuration.

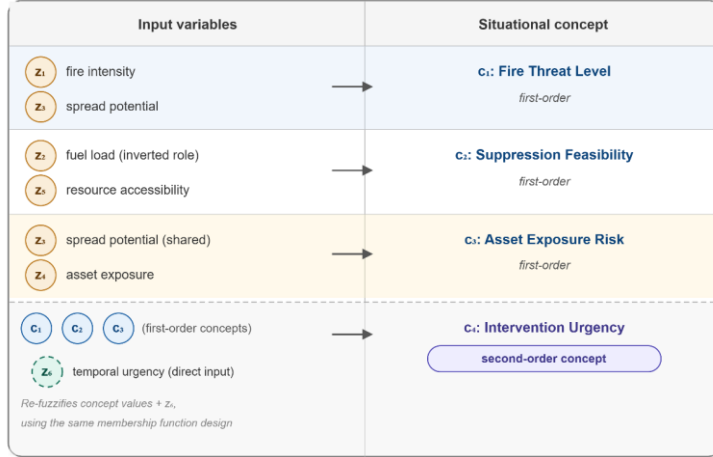


Figure 5.3 Concept formation structure

Figure 5.3 shows the concept formation structure, illustrating the two-tier hierarchy and the mapping from features to first-order concepts and from first-order concepts to the second-order urgency concept. It also shows the mapping from input features to first-order concepts (c_1, c_2, c_3) and from first-order concepts together with temporal urgency z_6 to the second-order concept c_4 . Feature z_3 (spread potential) contributes to both c_1 and c_3 with different semantic roles.

5.2.3.2 Rule Complexity Reduction via Concept Abstraction

The rule complexity of a fuzzy DSS is measured by the total number of rules required across all decision rule bases. This section formalizes the reduction in rule complexity achieved by the concept abstraction architecture relative to direct fuzzy inference.

Definitions: Let $\mathcal{N}_{feat} = 6$ be the number of input features, $\mathcal{N}_\mu = 3$ be the number of linguistic terms per variable, $\mathcal{N}_{concept} = 4$ be the number of situational concepts, and M be the number of intervention types. In a fuzzy rule base, each rule specifies a linguistic term for every input variable. The number of possible antecedent

3539 combinations, the maximum number of rules required for complete coverage, grows
 3540 exponentially with the number of input variables.

3541 5.2.3.3 Direct Fuzzy Inference

3542 Without concept abstraction, the decision mapping $\pi_{i,m}$ for each intervention type
 3543 m operates directly on the $\mathcal{N}_{feat} = 6$ features, each with $\mathcal{N}_\mu = 3$ linguistic terms.
 3544 The maximum rule base size for each intervention type is $\mathcal{N}_\mu^{\mathcal{N}_{feat}}$. Across M
 3545 intervention types, the total rule base size is (66):

$$|\mathcal{R}^{direct}| = M \cdot \mathcal{N}_\mu^{\mathcal{N}_{feat}} = M \cdot 3^6 = 729M \quad (66)$$

3546 5.2.3.4 Concept-Mediated Fuzzy Inference

3547 With concept abstraction, the decision rule bases operate over concepts rather than
 3548 raw features. Concept construction itself admits two equivalent representations: for
 3549 the first-order concepts c_1, c_2, c_3 the weighted aggregation formula of Equation (61
 3550) is used at runtime, and an inspectable rule form is provided by Table 5.6 - Table
 3551 5.8 (9 rules each) for human validation. The second-order concept c_4 , whose inputs
 3552 include three first-order concepts plus z_6 , is constructed directly through the
 3553 structured rule base of Table 5.9 (81 rules with a product fallback for non-
 3554 enumerated combinations). The total concept-construction rule count is therefore
 3555 $R_{CC} = 9 + 9 + 9 + 81 = 108$, *invariant of M*. The decision mapping $\pi_{i,m}$ for
 3556 each intervention type m operates on the $N_{concept} = 4$ concepts, each with $N_\mu = 3$
 3557 linguistic terms; the maximum decision-rule-base size per intervention type is
 3558 $N_\mu^{N_{concept}}$. Across M intervention types, the total decision rule base size is (67):

3559 With concept abstraction, concept construction is performed analytically through the
 3560 weighted aggregation formula, which requires no rules. The decision mapping $\pi_{i,m}$
 3561 for each intervention type m operates on the $\mathcal{N}_{concept} = 4$ concepts, each with $\mathcal{N}_\mu =$

3562 3 linguistic terms. The maximum rule base size per intervention type is $\mathcal{N}_\mu^{\mathcal{N}_{concept}}$.
 3563 Across M intervention types, the total decision rule base size is (67):

$$|\mathcal{R}^{concept}| = M \cdot \mathcal{N}_\mu^{\mathcal{N}_{concept}} = M \cdot 3^4 = 81M \quad (67)$$

3564 **Proposition 5.2: (Per-Intervention-Type Rule Complexity Reduction).** *The*
 3565 *concept-mediated architecture achieves a constant per-intervention-type (marginal)*
 3566 *maximum rule complexity reduction ratio that is independent of the number of*
 3567 *intervention types $M \geq 1$, as given in (68). The overall reduction ratio, which*
 3568 *accounts for the M -invariant concept-construction rule count R_{CC} , is given by*
 3569 *Equation (68a) and monotonically approaches κ_{marg} as $M \rightarrow \infty$. For the*
 3570 *DisasterAware configuration ($M = 3, R_{CC} = 108$), $\kappa_{marg} = 9$ and $\kappa_{total}(3) =$*
 3571 $\frac{2187}{351} \approx 6.2$.

$$\kappa_{marg} = N_\mu^{N_{feat} - N_{concept}} = 3^{6-4} = 9 \quad (68)$$

$$\kappa_{total}(M) = \frac{N_\mu^{N_{feat}} \cdot M}{R_{CC} + N_\mu^{N_{concept}} \cdot M} \quad (69)$$

3572 This result has a significant implication for system scalability. The marginal cost of
 3573 adding a new intervention type (i.e., the number of additional rules required to
 3574 support a new $\pi_{i,m}$ rule base) decreases from $\mathcal{N}_\mu^{N_{feat}} = 729$ rules in direct inference
 3575 to $\mathcal{N}_\mu^{N_{concept}} = 81$ rules in concept-mediated inference, a marginal reduction of
 3576 exactly 9 per intervention type (68). The system is therefore marginally scalable
 3577 with respect to the number of intervention types: each new intervention type can be
 3578 supported by a compact, independently designable rule base operating in concept
 3579 space, rather than a large rule base operating in raw feature space. The overall rule
 3580 count, including the M -invariant concept-construction base R_{CC} , grows as $R_{CC} +$
 3581 $N_\mu^{N_{concept}} \cdot M$ (69). Table 5.5 shows the total rule base sizes.

Table 5.5 Rule base complexity comparison of varying number of interventions

Intervention types M	$\ \mathcal{R}^{direct}\ $	$\ \mathcal{R}^{concept}\ $	Reduction ratio κ
1	729	81	9.0
3	2187	243	9.0
5	3645	405	9.0
10	7290	810	9.0
M	$729M$	$81M$	9.0

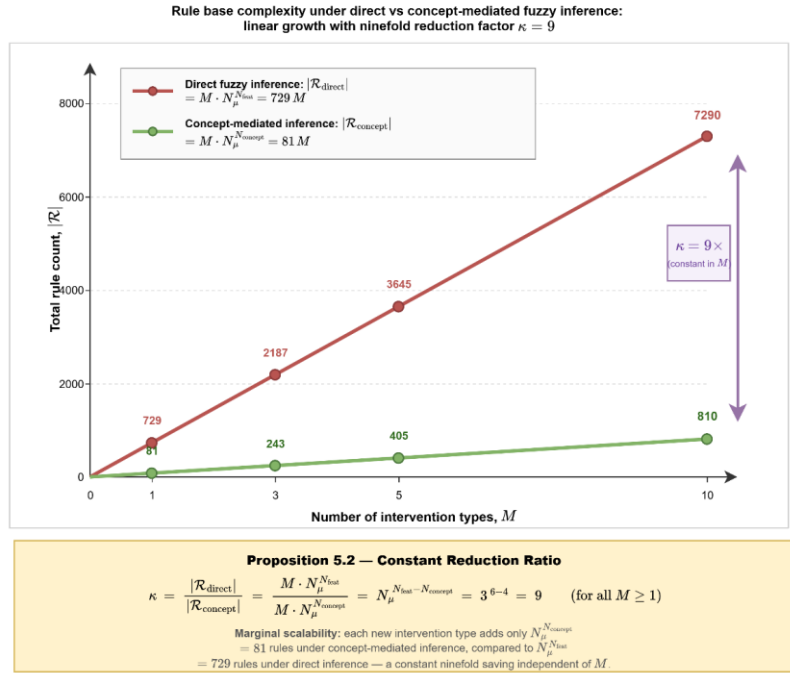


Figure 5.4 Rule base complexity under direct vs concept-mediated fuzzy inference

Figure 5.4 illustrates three structural properties of the concept-mediated architecture. First, both rule counts grow linearly in M , confirming that the architectural overhead of supporting additional intervention types scales gracefully in both formulations

rather than super-linearly. Second, the vertical gap between the two curves grows in absolute terms as M increases from $729 - 81 = 648$ rules at $M = 1$ to $7290 - 810 = 6480$ rules at $M = 10$ so the knowledge-engineering and validation burden avoided through concept abstraction grows proportionally to the operational scope of the DSS. Third, the reduction ratio $\kappa = \frac{|\mathcal{R}_{direct}|}{|\mathcal{R}_{concept}|}$ remains invariant at 9 for all values of $M \geq 1$, as proven in Proposition 5.2. This invariance is a direct consequence of the exponent difference $N_{feat} - N_{concept} = 2$ appearing identically in both numerator and denominator once M is factored out, leaving $\kappa = N_{\mu}^{N_{feat} - N_{concept}} = 3^2 = 9$ independent of M . Operationally, the ninefold *marginal* saving (per-intervention-type decision rule base) is a structural property of the concept abstraction layer that is invariant of M . The *overall* saving, which includes the M -invariant concept-construction rule count $R_{CC} = 108$, is $\kappa_{total}(3) \approx 6.2$ at $M = 3$ and approaches $\kappa_{marg} = 9$ asymptotically as the DSS is extended to richer intervention vocabularies in future work..

Remark 5.3 (Knowledge Engineering Modularity): An important secondary benefit of concept abstraction concerns the process of rule base construction and validation. Under complete antecedent coverage, direct fuzzy inference would require up to $729M$ rules defined over combinations of raw feature values such as "if fire intensity is High AND fuel load is Medium AND spread potential is Low AND asset exposure is High AND accessibility is Medium AND temporal urgency is High THEN...".

Such rules are difficult for human domain experts to evaluate and validate because the linguistic terms are defined over physically heterogeneous variables that are not directly comparable. In concept-mediated inference, the $81 M$ rules are defined over operationally meaningful concepts such as "if Fire Threat Level is High AND Suppression Feasibility is Low AND Asset Exposure Risk is High AND Intervention Urgency is High THEN deploy maximum suppression resources."

3616 These rules correspond directly to the operational reasoning vocabulary of wildfire
3617 incident commanders, enabling domain-expert review and iterative refinement
3618 within the operational planning process.

3619 **Remark 5.4 (Shared Feature Contribution):** Feature z_3 (spread potential)
3620 contributes to both concept c_1 and c_3 , with different semantic roles and different
3621 weights in each context. This shared contribution is not redundant but reflects the
3622 dual operational significance of spread potential identified in Section 5.1.2.

3623 The concept abstraction layer correctly encodes this dual role by constructing two
3624 distinct concept values from the same feature membership input, which would be
3625 impossible in a direct feature-to-decision mapping without explicitly encoding the
3626 duality in the rule base structure.

3627 5.3 Rule Base Specification

3628 The concept construction mechanism and the local decision mappings defined in
3629 Section 5.2 both rely on structured rule bases that encode operational knowledge in
3630 formal linguistic form. This section specifies these rule bases in their entirety.
3631 Throughout this chapter, the terms *rule* and *decision* carry distinct semantic and
3632 computational roles that must not be conflated. The 351 rules are not derived through
3633 a formal expert-elicitation protocol. They consolidate operational heuristics
3634 documented in the wildfire-management literature (Thompson & Calkin, 2011;
3635 Pacheco, 2015; Martell, 2007), and are treated here as a literature-synthesized
3636 operational doctrine that is subject to empirical validation in Chapter 7.

3637 A *rule* is a static IF–THEN linguistic mapping that encodes operational doctrine: the
3638 conditional reasoning that an experienced incident commander would apply when
3639 assessing a given combination of situational conditions. For example, the rule “IF
3640 Fire Threat Level is High AND Suppression Feasibility is Low AND Intervention
3641 Urgency is High, THEN Direct Suppression is Medium” encodes the operational
3642 doctrine that deploying suppression resources to a location that is operationally

3643 inaccessible would waste limited capacity, regardless of the severity of the threat.
3644 Rules are design-time artifacts: they are defined prior to simulation execution, do not
3645 change at runtime, and collectively represent the complete operational knowledge
3646 base of the DSS. The full rule base of 351 rules (Table 5.19) constitutes the entirety
3647 of the literature-synthesized operational doctrine embedded in the system.

3648 A *decision*, by contrast, is the runtime output of the DSS pipeline: a numerical
3649 intervention activation degree $u_{k,m}^i(x,y) \in [0,1]$ generated dynamically at each
3650 time step k and each grid cell (x,y) by evaluating the full rule base against the current
3651 concept vector. Semantically, a decision answers the operational question “how
3652 much of each intervention type should be activated at this specific location at this
3653 specific moment.” For instance, an output vector $U_{DSS} = [0.82, 0.10, 0.65]^T$
3654 directs that 82% of available suppression capacity should be allocated to this cell,
3655 preventive fuel reduction should remain minimal, and asset protection should be
3656 prioritized at 65% intensity. Decisions are ephemeral, context-dependent, and
3657 regenerated at every time step in response to the evolving wildfire state.

3658 In summary, rules encode *what to do under which conditions*; decisions encode *how*
3659 *much of each intervention type to activate here and now*. The mapping from rules to
3660 decisions is mediated by the five-stage composition of π_{DSS}^i (Definition 5.1): rules
3661 provide the reasoning structure, and the fuzzy inference mechanism produces graded,
3662 spatially distributed decisions by evaluating these rules against real-time
3663 observations.

3664 5.3.1 Concept Construction Rule Base

3665 Although the concept construction formulas presented in Section 5.2.3 define the
3666 weighted aggregation mechanism formally, the rule base provides an equivalent and
3667 complementary representation that makes the reasoning process transparent and
3668 inspectable by human operators. Each concept is defined by a set of IF-THEN rules
3669 in which the antecedent specifies linguistic conditions on fuzzified features, and the

consequent specifies the resulting concept activation level. The rule base and the weighted aggregation formula are consistent representations of the same mapping; the rule base form supports human validation and explainability, while the formula form supports efficient computation.

5.3.1.1 Concept Construction Rules for Fire Threat Level

The concept c_1 reflects the severity of the fire threat at a grid cell (x,y) , drawing from fire intensity (z_1) and spread potential (z_3). Nine rules cover all combinations of the two input terms. The rules are presented in Table 5.6. The output term "Very High" corresponds to $c_1 \approx 1$ and "Very Low" to $c_1 \approx 0$, with intermediate terms mapped uniformly across $[0, 1]$. The asymmetry between R1.3 and R1.7 reflects the operational judgment that high fire intensity with low spread is still a Medium threat (sustained combustion at fixed location requires attention), whereas low intensity with high spread is also a Medium threat (fast-moving but low-energy fire front). Both conditions warrant monitoring and preparatory action but not immediate maximum-priority response.

Table 5.6 Rule base for Fire Threat Level

Rule	IF z_1 is	AND z_3 is	THEN c_1 is
R1.1	High	High	Very High
R1.2	High	Medium	High
R1.3	High	Low	Medium
R1.4	Medium	High	High
R1.5	Medium	Medium	Medium
R1.6	Medium	Low	Low
R1.7	Low	High	Medium
R1.8	Low	Medium	Low
R1.9	Low	Low	Very Low

3686 5.3.1.2 Concept Construction Rules for Suppression Feasibility

3687 The concept c_2 reflects how effectively suppression resources can be deployed at the
 3688 target location, drawing from resource accessibility (z_5) and fuel load (z_2 , inverted
 3689 role, i.e., Low fuel means easier suppression). Nine rules cover all combinations. The
 3690 rules are presented in Table 5.7.

3691 The pattern encodes a two-factor feasibility model (Martell, 2007): High feasibility
 3692 requires both high accessibility and low fuel resistance. When accessibility is Low
 3693 but fuel load is also Low (R2.7), feasibility is Medium rather than Very Low because
 3694 the suppression effort required is small. For instance, even limited resources can be
 3695 effective against a fuel-depleted fire. In detail, this asymmetry is operationally
 3696 motivated and distinguishes this rule based on symmetric product formulation.

3697 Table 5.7 Rule base for Suppression Feasibility

Rule	IF z_5 is	AND z_2 is	THEN c_2 is
R2.1	High	Low	Very High
R2.2	High	Medium	High
R2.3	High	High	Medium
R2.4	Medium	Low	High
R2.5	Medium	Medium	Medium
R2.6	Medium	High	Low
R2.7	Low	Low	Medium
R2.8	Low	Medium	Low
R2.9	Low	High	Very Low

3698 5.3.1.3 Concept Construction Rules for Asset Exposure Risk

3699 The concept c_3 reflects the risk of protecting assets, drawing them from asset
 3700 exposure (z_4) and spread potential (z_3). Nine rules are presented in Table 5.8. The

stronger asymmetry between R3.3 (High asset, Low spread $\rightarrow$ Medium) and R3.7 (Low asset, High spread $\rightarrow$ Low) reflects the asset-first prioritization in risk assessment (Pacheco, 2015): High-priority assets are protected even when spread is currently low, because conditions can change rapidly within a single time step.

Table 5.8 Rule base for Asset Exposure Risk

Rule	IF z_4 is	AND z_3 is	THEN c_3 is
R3.1	High	High	Very High
R3.2	High	Medium	High
R3.3	High	Low	Medium
R3.4	Medium	High	High
R3.5	Medium	Medium	Medium
R3.6	Medium	Low	Low
R3.7	Low	High	Low
R3.8	Low	Medium	Very Low
R3.9	Low	Low	Very Low

5.3.1.4 Concept Construction Rules for Intervention Urgency (Second-Order)

The second-order concept c_4 takes the first-order concepts c_1, c_2, c_3 and the temporal urgency feature z_6 as inputs. With four inputs each at three linguistic levels, the maximum rule base size is $3^4 = 81$ rules. Rather than enumerating all 81, the rule base is structured around the operational logic identified in Section 5.1.4: urgency is highest when threat is high, feasibility is low (closing window), exposure is high, and time pressure is high.

Table 5.9 presents the dominant rules that determine c_4 under the most operationally consequential conditions. Rules not listed default to the product of the individual input concept values scaled to $[0, 1]$. Rules R4.1 and R4.2 encode the critical

3717 scenario of maximum urgency: severe fire threat, infeasible suppression, high asset
 3718 exposure, and active temporal pressure. Rule R4.5 captures the situation where threat
 3719 is high but both exposure and spread toward assets are low, yielding only Medium
 3720 urgency. Rule R4.8 captures the opposite extreme: low threat, high feasibility, and
 3721 low temporal pressure, the classic scenario where controlled intervention is possible
 3722 and not urgent.

3723 Table 5.9 Rule base for Intervention Urgency

Rule	IF c_1 is	AND c_2 is	AND c_3 is	AND z_6 is	THEN c_4 is
R4.1	High	Low	High	High	Very High
R4.2	High	Low	High	Medium	High
R4.3	High	Low	Medium	High	High
R4.4	High	Medium	High	High	High
R4.5	High	Low	Low	High	Medium
R4.6	Medium	Low	High	High	High
R4.7	Medium	Medium	Medium	High	Medium
R4.8	Low	High	Low	Low	Very Low
R4.9	Low	Low	Low	Low	Very Low

3724 5.3.2 Intervention Decision Rule Base

3725 The local decision mapping $\pi_{DSS,m}^i$ for each intervention type m takes the concept
 3726 vector $\mathbf{c}_k^i(x, y) = \{c_1, c_2, c_3, c_4\}$ and the contextual inputs $\mathcal{F}_{DSS,k}$ as inputs and
 3727 produces a candidate intervention degree $u_{k,m}^i(x, y) \in [0,1]$. The three intervention
 3728 types defined below constitute the complete decision output space of the
 3729 DisasterAware DSS. Each intervention type corresponds to an operationally distinct
 3730 category of wildfire response action and maps to specific physical parameters within
 3731 the Simulation Framework.

Table 5.10 specifies the operational meaning, the physical effect on the simulation state, and the Chapter 4 parameters through which each intervention modifies the system. The output of the DSS pipeline is not a binary activation of a single intervention type but a simultaneous, graded activation of all three types, represented as the multi-action vector $U_{DSS,k}^i(x, y) = [u_{k,1}^i, u_{k,2}^i, u_{k,3}^i]^T \in [0,1]^3$, where each component specifies the activation intensity of the corresponding intervention type at the given grid cell and time step.

Table 5.10 Intervention type definitions and operational semantics

m	Intervention Type	Operational Meaning	Physical Effect on Simulation	Parameter Affected
1	Direct Suppression	Deploy suppression resources (water, retardant, ground crews) to actively reduce combustion at the target location.	Fuel load is reduced through active suppression; fire intensity decreases at the intervention cell.	$F_{red,k}$ via f_{supp} ; $R_{cap,k}, R_{eff,k}$
2	Preventive Fuel Reduction	Pre-emptively reduce fuel load at locations not yet burning but at risk of ignition, through controlled burning or mechanical clearing.	Fuel load is reduced before fire arrival; effective containment perimeter is widened.	$F_{load,k}$ reduced directly
3	Asset Protection Priority	Reallocate resources toward high-value assets: evacuation support, access route clearance, defensive perimeter establishment.	Resource allocation structure within the intervention field is modified to prioritize asset-proximate cells.	Resource reallocation within $U_{DSS,k}$

3740

3741 The linguistic output levels used in the decision rule bases (Table 5.12 - Table 5.16)
 3742 correspond to singleton consequent values v_R that anchor the defuzzified
 3743 intervention degrees produced by the fuzzy inference engine (80). Table 5.11
 3744 specifies the numerical value and operational interpretation of each linguistic level.
 3745 These five levels define a uniform discretization of the [0,1] output space with equal
 3746 spacing of 0.25, providing the semantic bridge between the linguistic rule structure
 3747 and the numerical intervention output. The choice of five levels rather than three
 3748 (Low, Medium, High) at the output stage reflects the need for finer-grained
 3749 differentiation in intervention intensity: the distinction between "no action" (Very
 3750 Low) and "preparatory standby" (Low), and between "priority deployment" (High)
 3751 and "maximum commitment" (Very High), carries direct operational significance for
 3752 resource dispatch decisions.

3753 Table 5.11 Mapping of linguistic output levels to numerical singleton values

Linguistic Output	v_R	Operational Interpretation
Very Low	0.00	No intervention or minimal monitoring. Resources are not allocated to this cell for this intervention type.
Low	0.25	Limited resource allocation. Preparatory-level engagement; standby or low-priority deployment.
Medium	0.50	Moderate intervention. Balanced resource commitment; standard operational response.
High	0.75	Priority intervention. Majority of available capacity directed to this cell for this type.
Very High	1.00	Maximum intervention. Full available capacity allocated; highest operational urgency.

3754

3755 5.3.2.1 Intervention Type Specification

3756 The M intervention types represent the operationally distinct categories of action
3757 available to wildfire response teams. Based on the operational literature (Thompson
3758 & Calkin, 2011; Martell, 2007; Pacheco, 2015), three primary intervention types are
3759 defined for the DisasterAware framework:

3760 **$m = 1$: Direct Suppression:** Deployment of available suppression resources (water,
3761 retardant, ground crews) to actively reduce fuel load and combustion at the target
3762 location. This corresponds to the suppression capacity parameters $R_{cap,k}$ and $R_{eff,k}$
3763 in the Chapter 4 input structure.

3764 **$m = 2$: Preventive Fuel Reduction:** Pre-emptive reduction of fuel load at locations
3765 not yet burning but at risk of ignition, through controlled burning or mechanical
3766 treatment. This intervention reduces $F_{load,k}$ in advance of fire arrival and widens the
3767 effective containment perimeter.

3768 **$m = 3$: Asset Protection Priority:** Reallocation of resources toward the protection
3769 of high-value assets identified through $V_{prio}(x, y)$, including evacuation support,
3770 access route clearance, and defensive perimeter establishment. This intervention
3771 modifies the resource allocation structure within $U_{DSS,k}$.

3772 These three types correspond to $M = 3$, yielding a decision rule base of $3^4 = 81$
3773 rules per intervention type and a total rule base of 243 rules across all types,
3774 consistent with the complexity analysis of Proposition 5.2 ($81 M = 81 \times 3 = 243$).

3775 The fixed cardinality $M = 3$ reflects an operational design choice, not a structural
3776 constraint of the framework; the rule base is modular and additional intervention
3777 types can be introduced by appending rule subsets without architectural
3778 modification.

3779 **5.3.2.2 Decision Rules for m=1 Direct Suppression**

3780 Table 5.12 presents the principal rules for direct suppression mapping $\pi_{i,1}$. Full High
 3781 activation is indicated when both the fire threat and suppression feasibility are
 3782 favorable, the threat is real and the resources can reach it. Rule D1.4 captures a
 3783 critical operational constraint: even when fire threat is High and urgency is High, if
 3784 suppression feasibility is Low, the recommended direct suppression degree is only
 3785 Medium rather than High. Deploying resources to an inaccessible or highly fuel-
 3786 loaded location wastes capacity that may be needed elsewhere. This rule encodes the
 3787 operational principle of resource economy under infeasibility conditions.
 3788 Operational interpretation of principal rules is listed in Table 5.13.

3789 Table 5.12 Principal decision rules for Direct Suppression

Rule	IF c_1 is	AND c_2 is	AND c_4 is	THEN $u_{k,1}^t$ is
D1.1	High	High	High	Very High
D1.2	High	High	Medium	High
D1.3	High	Medium	High	High
D1.4	High	Low	High	Medium
D1.5	Medium	High	Medium	Medium
D1.6	Low	High	Low	Low
D1.7	High	Low	Low	Low
D1.8	Low	Low	Low	Very Low

3790

3791 Table 5.13 Operational interpretation for Direct Suppression

Rule	Operational Meaning
D1.1	Fire threat is severe, suppression resources can reach the location, and the situation is urgent. Deploy maximum suppression capacity immediately; this is the ideal intervention scenario.

Rule	Operational Meaning
D1.2	Fire threat is severe and resources are accessible, but urgency is moderate. Deploy at high level; the situation is serious but allows brief coordination before full commitment.
D1.3	Fire threat is severe and urgency is high, but suppression feasibility is only moderate. Deploy at high level with awareness that operational constraints may limit effectiveness.
D1.4	Fire threat is severe and urgency is high, but the location is operationally inaccessible (low feasibility). Reduce suppression to medium: deploying full capacity to an unreachable location wastes resources needed elsewhere.
D1.5	Moderate fire threat with good accessibility and moderate urgency. Standard-level engagement; balanced resource commitment appropriate.
D1.6	Low fire threat with good accessibility and low urgency. Minimal suppression: the fire poses little immediate danger and resources are better reserved.
D1.7	Severe fire threat but inaccessible location and low urgency. Minimal suppression: despite the threat, resources cannot be effectively deployed and time permits alternative strategies.
D1.8	Low threat, inaccessible location, low urgency. No meaningful suppression warranted; monitoring only.

3792 5.3.2.3 Decision Rules for m=2: Preventive Fuel Reduction

3793 Table 5.14 presents the principal rules for $\pi_{i,2}$. Preventive fuel reduction is most
3794 appropriate when threat is Medium-to-High, assets are exposed, but direct
3795 suppression feasibility is limited, the system recommends pre-emptive action ahead
3796 of the fire front.

3797 The contrast between D2.4 (Medium output despite High threat and High exposure)
3798 and D2.1 (Very High output when feasibility is Low) reflects the substitution logic

3799 of fuel reduction: when direct suppression is feasible (D2.4), fuel reduction is less
 3800 urgent because direct action can be taken. When direct suppression is infeasible
 3801 (D2.1), preventive fuel reduction becomes the primary available option and is
 3802 therefore given maximum priority.

3803 Table 5.14 Principal decision rules for Preventive Fuel Reduction

Rule	IF c_1 is	AND c_3 is	AND c_2 is	THEN $u_{k,2}^i$ is
D2.1	High	High	Low	Very High
D2.2	High	Medium	Low	High
D2.3	Medium	High	Low	High
D2.4	High	High	High	Medium
D2.5	Medium	Medium	Medium	Medium
D2.6	Low	High	Low	Medium
D2.7	Low	Low	High	Very Low
D2.8	Low	Low	Low	Very Low

3804

3805 Operational interpretation of principal rules is listed in Table 5.15.

3806 Table 5.15 Operational interpretation for Preventive Fuel Reduction

Rule	Operational Meaning
D2.1	Fire threat is high, assets are highly exposed, but direct suppression is infeasible. Maximum preventive fuel reduction: since suppression cannot reach the area, removing fuel ahead of the fire front is the only viable intervention.
D2.2	High threat, moderate asset exposure, low suppression feasibility. High fuel reduction as the primary available response; suppress fire momentum indirectly.
D2.3	Moderate threat but high asset exposure with low suppression feasibility. High fuel reduction to create defensive barriers around exposed assets before conditions worsen.

Rule	Operational Meaning
D2.4	High threat and high exposure, but suppression IS feasible. Only medium fuel reduction: since direct suppression can be deployed, fuel reduction serves as supplementary rather than primary strategy.
D2.5	All conditions moderate. Standard fuel reduction engagement as part of balanced response.
D2.6	Low threat but high exposure and low feasibility. Moderate fuel reduction as a precautionary measure: although threat is currently low, exposed assets justify preemptive preparation.
D2.7	Low threat, low exposure, high feasibility. Minimal fuel reduction: direct suppression is available if needed, and current conditions do not warrant preemptive action.
D2.8	Low threat, low exposure, low feasibility. No fuel reduction warranted; resources are better allocated elsewhere.

3807 5.3.2.4 Decision Rules for $m=3$: Asset Protection Priority

3808 Table 5.16 presents the principal rules for $\pi_{i,3}$. Asset protection priority is driven
3809 primarily by exposure risk and urgency, independently of current suppression
3810 feasibility. Rules D3.1 and D3.2 both yield Very High output regardless of whether
3811 fire threat is High or Medium, as long as asset exposure and urgency are both High.
3812 This reflects the unconditional protection priority for high-value assets under time
3813 pressure: the protection action must be initiated before the fire front arrives
3814 regardless of current combustion intensity, because once the fire reaches the asset
3815 area, the protection window will have closed.

3816 Table 5.16 Principal decision rules for Asset Protection Priority

Rule	IF c_3 is	AND c_4 is	AND c_1 is	THEN $u_{k,3}^t$ is
D3.1	High	High	High	Very High

Rule	IF c_3 is	AND c_4 is	AND c_1 is	THEN $u_{k,3}^t$ is
D3.2	High	High	Medium	Very High
D3.3	High	Medium	High	High
D3.4	High	Low	High	Medium
D3.5	Medium	High	High	High
D3.6	Medium	Medium	Medium	Medium
D3.7	Low	High	High	Low
D3.8	Low	Low	Low	Very Low

3817

3818 Operational interpretation of principal rules is listed in Table 5.17.

3819 Table 5.17 Operational interpretation for Asset Protection Priority

Rule	Operational Meaning
D3.1	Asset exposure is high, urgency is high, and fire threat is severe. Maximum asset protection: deploy evacuation support, establish defensive perimeters, and clear access routes immediately.
D3.2	Asset exposure and urgency are both high, but fire threat is only moderate. Still maximum protection: the protection window is closing regardless of current fire intensity, and assets must be secured before the front arrives.
D3.3	High asset exposure, moderate urgency, severe threat. High-level asset protection; urgency allows some coordination time but asset vulnerability demands priority action.
D3.4	High asset exposure but low urgency despite severe threat. Medium protection: the extended time window permits a measured response rather than emergency deployment.
D3.5	Moderate exposure, high urgency, severe threat. High protection: the time pressure combined with threat severity elevates protection priority even for moderately exposed assets.

Rule	Operational Meaning
D3.6	All conditions moderate. Standard asset protection as part of balanced multi-type response.
D3.7	Low asset exposure, high urgency, severe threat. Low protection: despite urgency and threat, the absence of high-value assets at this location means resources are better directed to higher-exposure cells.
D3.8	Low exposure, low urgency, low threat. No asset protection needed; monitoring status only.

5.3.3 Action Aggregation Function

Having generated a candidate intervention degree $u_{k,m}^i(x, y)$ for each of the $M = 3$ intervention types, the Local DSS Model must consolidate these into a single executable intervention parameter $U_{DSS,k}^i(x, y)$ through the aggregation function f_{agg} . The aggregation function addresses two sub-problems. The first is conflict resolution: different intervention types may recommend contradictory resource allocations for the same grid cell. The second is normalization: the output $U_{DSS,k}^i(x, y)$ must be expressed in units compatible with the external input structure $U_{DSS,k}$ defined in Chapter 4, specifically as parameters in the interval $[0, 1]$ that can be mapped to physical suppression quantities through the suppression function f_{supp} .

The aggregation function is defined as a priority-weighted combination that respects the operational hierarchy of intervention types (70):

$$U_{DSS,k}^i(x, y) = \frac{\sum_{m=1}^M \alpha_m \cdot u_{k,m}^i(x, y)}{\sum_{m=1}^M \alpha_m} \quad (70)$$

where $\alpha_m > 0$ is the priority weight assigned to intervention type m . Priority weights reflect the operational hierarchy: direct suppression is the primary response action, preventive fuel reduction is secondary, and asset protection priority is activated in parallel. The default configuration is (71):

$$\alpha_1 = 1.0, \alpha_2 = 0.7, \alpha_3 = 0.9 \quad (71)$$

The weight $\alpha_3 = 0.9$ for asset protection is set below 1.0 but above α_2 , reflecting the operational judgment that asset protection is nearly as urgent as direct suppression when triggered, but that direct suppression remains the primary operational objective when both are simultaneously active.

The weighted average formulation ensures that the output $U_{DSS,k}^i(x, y)$ remains bounded in $[0, 1]$, since all inputs are bounded and the weights are strictly positive. It also ensures that the aggregated output responds continuously and proportionally to changes in any of the individual intervention degrees, avoiding the discontinuities that would result from a maximum or minimum operator.

A key property of the weighted average aggregation is that it naturally resolves conflicts between intervention types through proportional blending rather than winner-takes-all selection. Consider the case where direct suppression is infeasible ($u_{k,1}^i \approx 0$) but preventive fuel reduction is highly recommended ($u_{k,2}^i \approx 1$). In a maximum-based aggregation, the output would be dominated by fuel reduction. In the weighted average formulation, the output is (72):

$$U_{DSS,k}^i = \frac{1.0 \times 0 + 0.7 \times 1.0 + 0.9 \times u_{k,3}^i}{1.0 + 0.7 + 0.9} \quad (72)$$

which distributes the intervention signal proportionally, allowing the simulation engine to allocate the available resource capacity across feasible actions rather than committing entirely to a single type.

Remark 5.5 (Interpretability of f_{agg}): The weighted average formulation of f_{agg} preserves the interpretability of the DSS output. Because each weight α_m and each candidate degree $u_{k,m}^i$ are individually inspectable, the provenance of any aggregated output can be traced back through the intervention degrees and the rule evaluations that produced them. This traceability is a direct realization of the explainability requirement of Contribution C4 from Chapter 1.

Remark 5.6 (Design Justification of f_{agg}). The priority-weighted average formulation of Equation (70) was selected over three alternative aggregation operators on the basis of four design criteria: output boundedness, conflict resolution behavior, continuity, and interpretability.

A *maximum operator* $f_{agg}^{max} = \max_m u_{k,m}^i$ would select the dominant intervention type at each cell, producing a winner-takes-all output that discards the contributions of lower-ranked but operationally relevant intervention types. This violates the multi-action design intent, because wildfire response at a single cell may require a blend of suppression, fuel management, and asset protection actions rather than a single dominant action.

A *minimum operator* $f_{agg}^{min} = \min_m u_{k,m}^i$ would be excessively conservative, suppressing the overall intervention intensity whenever any single intervention type has low activation, even if the remaining types strongly recommend action. A softmax-based operator is (73).

$$f_{agg}^{sm} = \sum_m u_{k,m}^i \cdot \frac{e^{\beta \alpha_m u_{k,m}^i}}{\sum_r e^{\beta \alpha_r u_{k,r}^i}} \quad (73)$$

would introduce a nonlinear temperature-controlled blending that approaches the maximum operator as $\beta \rightarrow \infty$ and the uniform average as $\beta \rightarrow 0$. While mathematically flexible, this formulation introduces an additional hyperparameter β whose calibration requires empirical tuning without a clear operational interpretation, and whose nonlinear weighting reduces the direct traceability of each intervention type's contribution to the final output.

The priority-weighted average of Equation (70) satisfies all four design criteria simultaneously: it inherits boundedness from the boundedness of its inputs and positive weights (Proposition 5.3); it resolves conflicts through proportional blending rather than selection or suppression; it is Lipschitz-continuous in all inputs (Proposition 5.4); and it preserves full traceability because each weight α_m and each candidate degree $u_{k,m}^i$ are individually inspectable. For these reasons, it is adopted

3886 as the default aggregation formulation. The sensitivity of the DSS output to
3887 variations in the priority weights $\alpha_1, \alpha_2, \alpha_3$ is evaluated in Chapter 7.

3888 **Remark 5.7 (Calibration Rationale for Priority Weights).** The default priority
3889 weights $\alpha_1 = 1.0, \alpha_2 = 0.7, \alpha_3 = 0.9$ are constrained by three operational
3890 ordering requirements rather than derived from a formal optimization procedure.

3891 *Constraint 1:* $\alpha_1 > \alpha_3 > \alpha_2$. Direct suppression (α_1) must dominate the
3892 aggregation because it is the primary operational action that directly reduces fire
3893 intensity at the intervention point. Asset protection (α_3) takes second priority
3894 because the consequences of failing to protect high-value assets are irreversible,
3895 whereas preventive fuel reduction (α_2) is a preparatory action whose benefit
3896 materializes over subsequent time steps rather than immediately. This ordering is
3897 consistent with the tactical priority structure documented in wildfire incident
3898 command protocols (Calkin, O'Connor, Thompson, & Stratton, 2021).

3899 *Constraint 2:* $\alpha_3 / \alpha_1 \geq 0.8$. The asset protection weight must remain close to the
3900 direct suppression weight, reflecting the operational principle that asset protection
3901 should not be significantly discounted relative to suppression when both are
3902 simultaneously active. A ratio below 0.8 would systematically under-prioritize asset
3903 protection in scenarios where high-value assets are threatened, which contradicts the
3904 protection mandate of wildfire incident management (Thompson & Calkin, 2011).

3905 *Constraint 3:* $\alpha_2 / \alpha_1 \in [0.5, 0.8]$. Preventive fuel reduction must carry sufficient
3906 weight to influence the aggregation when its activation is high but must not compete
3907 equally with direct suppression. A ratio below 0.5 would render fuel reduction
3908 operationally negligible; a ratio above 0.8 would blur the distinction between
3909 immediate suppression and preparatory action.

3910 The specific values (1.0, 0.7, 0.9) satisfy all three constraints and are positioned at
3911 the center of the admissible region defined by these inequalities. The qualitative
3912 behavior of the DSS, which intervention type dominates the aggregated output for a

3913 given concept vector, is invariant under perturbations that preserve the ordering
 3914 $\alpha_1 > \alpha_3 > \alpha_2$. The sensitivity to weight perturbation is evaluated in Section 7.6.3.

3915 5.3.4 Decision Quality Criteria and Evaluation Weights

3916 The decision quality evaluation function f_{eval} assesses the DSS output against four
 3917 operational criteria derived from the four operational questions of Section 5.1.4.
 3918 Each criterion measures the alignment between the intervention field $U_{DSS,k}(x, y)$
 3919 and the corresponding situational concept. Table 5.18 specifies the criteria, their
 3920 concept mappings, and the default weights.

3921 Table 5.18 Partial criterion scores of evaluation function

j	Criterion	Concept c_{hj}	Default weight w_j	Operational rationale
1	Fire spread mitigation	c_1 : Fire Threat Level	0.35	High-threat cells must receive proportional intervention
2	Asset risk reduction	c_3 : Asset Exposure Risk	0.30	High-value assets must be covered under threat
3	Resource efficiency	c_2 : Suppression Feasibility	0.20	Intervention deployed where operationally feasible
4	Operational timeliness	c_4 : Intervention Urgency	0.15	Urgent cells treated before conditions deteriorate

3922

3923 The weight ordering reflects the operational priority hierarchy: fire spread mitigation
 3924 ($w_1 = 0.35$) carries the highest weight as the primary safety objective, followed by
 3925 asset risk reduction ($w_2 = 0.30$). Resource efficiency ($w_3 = 0.20$) and operational
 3926 timeliness ($w_4 = 0.15$) are secondary but non-negligible. These weights are
 3927 calibration defaults and may be adjusted through simulation-based sensitivity

analysis as discussed in Chapter 7. The formal realization of f_{eval} as a runtime evaluation mechanism is presented in Section 6.4.4.

Remark 5.8 (Design Justification of f_{eval}). The decision quality function f_{eval} defined in Section 6.4.4, Equation (110)–(111), evaluates each partial criterion $q_{j,k}$ as the spatially averaged absolute deviation between the intervention field and the corresponding concept activation:

$$q_{j,k} = \frac{1}{|\mathcal{G}_k^+|} \sum_{(x,y) \in \mathcal{G}_k^+} \left(1 - |U_{DSS,k}(x,y) - c_{k,h_j}^i(x,y)|\right) \quad (74)$$

Three design choices in this formulation require justification.

Choice of the L1 (absolute deviation) metric. The local satisfaction term $1 - |U - c|$ was selected over the squared deviation $1 - (U - c)^2$ on the basis of interpretability and sensitivity uniformity. The L1 formulation penalizes all deviations linearly, meaning that a 0.1 deviation at any operating point produces the same quality penalty regardless of whether it occurs near zero, near one, or at the midpoint. The L2 alternative would under-penalize small deviations and over-penalize large ones, introducing a nonlinear weighting across the deviation range that lacks a clear operational justification in this context. Moreover, the L1 metric preserves the $[0,1]$ output range without requiring additional normalization, since $|U - c| \in [0,1]$ when both operands are bounded in $[0,1]$.

Choice of spatial averaging. The spatial mean over the active domain $\mathcal{G}_k^+$ produces a global summary score that reflects the average alignment between intervention and situational need across all operationally relevant cells. An alternative worst-case formulation would be dominated by the single cell with the largest deviation, making the global quality score fragile to localized outliers and insensitive to improvements across the rest of the domain. The spatial average ensures that quality improvements at any cell contribute proportionally to the global score, which is consistent with the distributed nature of the decision architecture.

$$q_{j,k}^{min} = \min_{(x,y) \in \mathcal{G}_k^+} \left(1 - \left| U_{DSS,k}(x,y) - c_{k,h_j}^i(x,y) \right| \right) \quad (75)$$

3953 *Choice of default weights.* The default weight configuration $w_1 = 0.35$, $w_2 = 0.30$,
 3954 $w_3 = 0.20$, $w_4 = 0.15$ reflects the operational priority hierarchy established in
 3955 Section 5.1.4, in which fire spread mitigation (Q1) is the primary safety objective
 3956 and intervention urgency (Q4) is a secondary timing concern. These weights are not
 3957 derived from formal optimization but from the structured expert judgment codified
 3958 in the operational question hierarchy. Their sensitivity to perturbation is evaluated
 3959 through simulation-based parameter sweeps in Section 7.6.4.

3960 **Remark 5.9 (Calibration Rationale for Decision Quality Weights).** The default
 3961 evaluation weights $w_1 = 0.35$, $w_2 = 0.30$, $w_3 = 0.20$, $w_4 = 0.15$ are derived
 3962 from two formal requirements and one operational principle.

3963 *Normalization requirement:* $\sum w_j = 1$. The weights form a convex combination,
 3964 ensuring that $Q_{DSS,k} \in [0,1]$ without additional normalization, which preserves
 3965 compatibility with the adaptive threshold η_k defined in Definition 6.4.

3966 *Priority ordering:* $w_1 > w_2 > w_3 > w_4$. This ordering reflects the operational
 3967 hierarchy codified in the four operational questions of Section 5.1.4. Fire spread
 3968 mitigation (w_1) carries the highest weight because uncontrolled fire spread poses the
 3969 most immediate and irreversible threat to life, property, and ecological systems.
 3970 Asset risk reduction (w_2) is the second priority because asset losses, once realized,
 3971 cannot be reversed. Resource efficiency (w_3) and operational timeliness (w_4) are
 3972 enabling criteria that support the primary objectives but do not independently
 3973 determine decision adequacy.

3974 *Non-dominance constraint:* $w_1 < 0.5$. No single criterion dominates the evaluation
 3975 function. This ensures that a decision cannot achieve an acceptable quality score
 3976 $Q_{DSS,k} \geq \eta_k$ by satisfying only the fire spread mitigation criterion while neglecting
 3977 asset protection, resource efficiency, and timeliness. In operational terms, a
 3978 suppression strategy that mitigates fire spread but ignores threatened assets or

depletes resources without regard to efficiency would not constitute an adequate operational decision.

Within the feasible region defined by these three constraints (normalization, ordering, non-dominance), the specific values (0.35, 0.30, 0.20, 0.15) are positioned to maintain a gradually decreasing priority gradient. The inter-weight spacing is deliberately narrow (0.05 between adjacent weights) to prevent any single criterion from overwhelming the quality assessment. The sensitivity of the decision acceptance condition $Q_{DSS,k} \geq \eta_k$ to weight perturbation is evaluated through systematic parameter sweeps in Chapter 7, Section 7.6.4.

5.3.5 Complete Rule Base Summary

Table 5.19 summarizes the complete rule base structure of the DisasterAware DSS, confirming consistency with the complexity analysis of Proposition 5.2. The total rule count of 351 compares to the 2187-rule complete-coverage bound of direct fuzzy inference ($729 \times M = 729 \times 3$), representing a reduction factor of $6.2 \times$ for $M = 3$ intervention types. As M increases, the reduction ratio approaches the theoretical limit of $9 \times$ established by Proposition 5.2.

Table 5.19 Summary and complexity verification

Rule base component	Input dimension	Rule count	Notes
c_1 : Fire Threat Level	$\mathcal{N}_\mu^2 = 9$	9	Full coverage
c_2 : Suppression Feasibility	$\mathcal{N}_\mu^2 = 9$	9	Full coverage
c_3 : Asset Exposure Risk	$\mathcal{N}_\mu^2 = 9$	9	Full coverage
c_4 : Intervention Urgency	$\mathcal{N}_\mu^4 = 81$	81	Principal rules shown; remainder by default

Rule base component	Input dimension	Rule count	Notes
$\pi_{i,1}$: Direct Suppression	$\mathcal{N}_\mu^4 = 81$	81	Principal rules shown
$\pi_{i,2}$: Preventive Fuel Reduction	$\mathcal{N}_\mu^4 = 81$	81	Principal rules shown
$\pi_{i,3}$: Asset Protection	$\mathcal{N}_\mu^4 = 81$	81	Principal rules shown
Total		351	729 $M = 2187$ direct inference

3996 5.3.6 Formal Composition of the Decision Mapping

3997 This section defines the mathematical composition of the local DSS mapping. It does
3998 not describe runtime execution order as an architectural process; runtime realization
3999 is presented in Chapter 6. Sections 5.3.1 through 5.3.4 define all component
4000 functions of the local DSS mapping: the concept construction mappings $f_{\text{concept},h}$
4001 (Section 5.2.3), the fuzzification mapping f_{fuzzy} (Section 5.2.2), the feature
4002 extraction mapping f_{feat} (Section 5.2.1), the intervention decision mappings $\pi_{i,m}$
4003 (Section 5.3.2), and the action aggregation function f_{agg} (Section 5.3.3). This section
4004 ties these components together as the explicit closed-form realization of the DSS
4005 mapping π_{DSS}^i introduced in Chapter 2.

4006 **Definition 5.1 (Formal Composition of π_{DSS}^i).** For operational region Ω_i at time
4007 step k , the regional DSS mapping is realized as the following five-stage sequential
4008 composition (76) where the concept vector $\mathbf{c}_k^i(x, y)$ is itself the three-stage

4009 composition $\mathbf{c}_k^i(x, y) = \left\{ f_{\text{concept},h} \left(f_{\text{fuzzy}} \left(f_{\text{feat}} \left(O_k^i(x, y) \right) \right) \right) \right\}_{h=1}^4$.

$$\begin{aligned}
U_{DSS,k}^i(x, y) &= \pi_{DSS}^i(O_k^i(x, y), \mathcal{F}_{DSS,k}) = f_{\text{agg}} \left(\{ \pi_{i,m}(\mathbf{c}_k^i(x, y), \mathcal{F}_{DSS,k}) \}_{m=1}^M \right) \\
&= f_{\text{agg}} \left(\left\{ \pi_{i,m} \left(\left\{ f_{\text{concept},h} \left(f_{\text{fuzzy}} \left(f_{\text{feat}} \left(O_k^i(x, y) \right) \right) \right) \right\}_{h=1}^4, \mathcal{F}_{DSS,k} \right) \right\}_{m=1}^M \right) \quad (76)
\end{aligned}$$

4010 The five stages, their input/output dimensions, and defining references are
 4011 summarized in Table 5.20.

4012 Table 5.20 Five-stage composition of decision mapping

Stage	Function	Input	Output	Codomain
1	f_{feat}	$O_k^i(x, y) \in \mathbb{R}^4$	$\mathbf{z}_k^i(x, y)$	$[0, 1]^6$ 6 scalar features
2	f_{fuzzy}	$\mathbf{z}_k^i(x, y) \in [0, 1]^6$	$\boldsymbol{\mu}_k^i(x, y)$	$[0, 1]^{18}$ 6 features $\times$ 3 terms = 18 membership values
3	$f_{\text{concept}, h}$	$\boldsymbol{\mu}_k^i(x, y) \in [0, 1]^{18}$	$\mathbf{c}_k^i(x, y)$	$[0, 1]^4$ 4 situational concept values
4	$\pi_{i, m}$	$\mathbf{c}_k^i(x, y) \in [0, 1]^4, \mathcal{F}_{DSS, k}$	$u_{k, m}^i(x, y)$	$[0, 1]$ Candidate degree per intervention type
5	f_{agg}	$\{u_{k, m}^i(x, y)\}_{m=1}^M \in [0, 1]^M$	$U_{DSS, k}^i(x, y)$	$[0, 1]$ Single aggregated intervention output

4013 5.3.6.1 Feature Extraction

4014 $f_{\text{feat}}: \mathbb{R}^4 \rightarrow [0, 1]^6$ maps the observable information vector $O_k^i(x, y) =$
 4015 $[\hat{B}_k^i, \hat{F}_{\text{load}, k}^i, \hat{I}_k^i, \hat{t}_k^i]^\top$ to a six-dimensional feature vector $\mathbf{z}_k^i(x, y) = [z_{k, 1}^i, \dots, z_{k, 6}^i]^\top$.
 4016 Each component is a scalar indicator derived from the state variables and contextual
 4017 inputs $\mathcal{F}_{DSS, k}$ (77).

$$\begin{aligned} z_{k, 1}^i &= \hat{I}_k^i(x, y) \\ z_{k, 2}^i &= \hat{F}_{\text{load}, k}^i(x, y) \end{aligned} \quad (77)$$

$$\begin{aligned}
z_{k,3}^i &= \min \left(1, \frac{\Psi_k(x,y)}{\Psi_{\text{ref}}} \right) \\
z_{k,4}^i &= V_{\text{prio}}(x,y) \cdot \bar{B}_k^\delta(x,y) \\
z_{k,5}^i &= R_{\text{avail},k}(x,y) \cdot \exp \left(-\beta_t \cdot R_{\text{time},k}(x,y) \right) \cdot G_{\text{access}}(x,y) \\
z_{k,6}^i &= 1 - \exp \left(-\lambda \cdot \frac{\hat{\tau}_k^i(x,y)}{\tau_{\text{ref}}} \right)
\end{aligned}$$

4018 All six features are bounded in $[0, 1]$ by construction or explicit normalization, as
4019 established in Section 5.2.1.

4020 5.3.6.2 Fuzzification

4021 $f_{\text{fuzzy}}: [0,1]^6 \rightarrow [0,1]^{18}$ maps each scalar feature $z_{k,l}^i(x,y)$ to a triplet of graded
4022 membership values over linguistic terms Low (L), Medium (M), and High (H) (78).

$$\mu_{k,n,l}^i(x,y) = f_{\text{fuzzy}} \left(z_{k,l}^i(x,y) \right), n \in \{L, M, H\}, l = 1, \dots, 6 \quad (78)$$

4023 The membership functions follow the piecewise-linear shoulder–triangular–shoulder
4024 design of Equation (57), parameterized by the feature-specific triplets
4025 $(p_{1,l}, p_{2,l}, p_{3,l})$ from Table 5.4. The output is the 3×6 membership matrix $\boldsymbol{\mu}_k^i(x,y)$
4026 defined in Equation (60). The partition-of-unity property $\mu_{L,l} + \mu_{M,l} + \mu_{H,l} = 1$
4027 holds for all $z \in [0,1]$ and all l , as established in Proposition 5.1.

4028 5.3.6.3 Concept Construction

4029 $f_{\text{concept},h}: [0,1]^{18} \rightarrow [0,1]$ (79) maps the membership matrix $\boldsymbol{\mu}_k^i(x,y)$ to the h -th
4030 situational concept value via the normalized weighted average of Equation (61):

$$c_{k,h}^i(x,y) = f_{\text{concept},h}(\boldsymbol{\mu}_k^i) = \frac{\sum_{(n,l) \in \mathcal{S}_h} w_{h,n,l} \cdot \mu_{k,n,l}^i(x,y)}{\sum_{(n,l) \in \mathcal{S}_h} w_{h,n,l}} \quad (79)$$

4031 The index sets $\mathcal{S}_h$ and weights $w_{h,n,l}$ for the four concepts, namely Fire Threat Level
4032 (c_1) , Suppression Feasibility (c_2) , Asset Exposure Risk (c_3) , and Intervention

Urgency (c_4) are specified in Equations (62)–(65). The concept vector $\mathbf{c}_k^i(x, y) = \{c_{k,h}^i(x, y)\}_{h=1}^4 \in [0,1]^4$ is defined in Equation (61).

5.3.6.4 Intervention Decision Mapping

$\pi_{i,m}: [0,1]^4 \times \mathcal{F}_{DSS} \rightarrow [0,1]$ maps the concept vector $\mathbf{c}_k^i(x, y)$ and contextual inputs $\mathcal{F}_{DSS,k}$ to a candidate intervention degree for type m . It is realized as a Mamdani-type fuzzy inference system over the 81-rule base $\pi_{i,m}$ defined in Section 5.3.2. Each rule R fires with activation strength $\alpha_R = \min_h \mu_{n_h}(c_{k,h}^i)$ and carries singleton consequent $v_R \in \{0.0, 0.25, 0.50, 0.75, 1.0\}$. The output is the activation-weighted average is (80) where the activation strength of each rule R with antecedent terms $n_h \in \{L, M, H\}$ is (81) and $v_R \in \{0.0, 0.25, 0.50, 0.75, 1.0\}$ is the singleton consequent of rule R .

$$u_{k,m}^i(x, y) = \pi_{i,m}(\mathbf{c}_k^i(x, y), \mathcal{F}_{DSS,k}) = \frac{\sum_{R \in \pi_{i,m}} \alpha_R \cdot v_R}{\sum_{R \in \pi_{i,m}} \alpha_R} \quad (80)$$

$$\alpha_R = \min_{h \in \{1,2,3,4\}} \mu_{n_h}(c_{k,h}^i(x, y)) \quad (81)$$

The membership function $\mu_{n_h}(\cdot)$ is the same piecewise-linear design of Equation (57), applied to concept values $c_{k,h}^i \in [0,1]$ directly. The full 81-rule bases for $m \in \{1,2,3\}$ are specified in Table 5.12, Table 5.14 and Table 5.16.

5.3.6.5 Action Aggregation

$f_{agg}: [0,1]^M \rightarrow [0,1]$ (82) maps the $M = 3$ candidate degrees to a single executable intervention parameter via the priority-weighted average of Equation (70) with default weights $\alpha_1 = 1.0$, $\alpha_2 = 0.7$, $\alpha_3 = 0.9$ as specified in Equation (71). The output is bounded in $[0, 1]$ since all inputs and weights are positive, and continuous in all $u_{k,m}^i$.

$$U_{DSS,k}^i(x,y) = f_{agg} \left(\{u_{k,m}^i(x,y)\}_{m=1}^M \right) = \frac{\sum_{m=1}^M \alpha_m \cdot u_{k,m}^i(x,y)}{\sum_{m=1}^M \alpha_m} \quad (82)$$

4053 **Proposition 5.3 (Boundedness of π_{DSS}^i).** For all $(x,y) \in \Omega_i$, all time steps k , and all
 4054 regions i , the output satisfies $U_{DSS,k}^i(x,y) \in [0,1]$.

4055 *Proof:* Stage 1 produces $\mathbf{z}_k^i \in [0,1]^6$ by construction, as established in Section 5.2.1
 4056 for all six features. Stage 2 maps $[0,1] \rightarrow [0,1]$ by the definition of the membership
 4057 functions in Equation (57). Stage 3 produces a normalized weighted average of
 4058 values in $[0,1]$ with strictly positive weights via Equation (61), hence $c_{k,h}^i \in [0,1]$
 4059 for all h . Stage 4 produces the activation-weighted average of singleton consequent
 4060 values $v_R \in \{0.0,0.25,0.50,0.75,1.0\} \subset [0,1]$ with non-negative activation
 4061 strengths, hence $u_{k,m}^i \in [0,1]$. Stage 5 applies Equation (70) with strictly positive
 4062 weights α_m , yielding a normalized weighted average of inputs in $[0,1]$, hence
 4063 $U_{DSS,k}^i \in [0,1]$.

4064 **Proposition 5.4 (Lipschitz Continuity of π_{DSS}^i).** The mapping $O_k^i(x,y) \mapsto$
 4065 $U_{DSS,k}^i(x,y)$ is Lipschitz-continuous with respect to the observable information, for
 4066 fixed $\mathcal{F}_{DSS,k}$.

4067 *Proof:* Stages 1–3 are piecewise-linear by construction. f_{feat} consists of bounded
 4068 affine and product operations (Section 5.2.1); f_{fuzzy} applies the piecewise-linear
 4069 shoulder–triangular–shoulder functions of Equation (57); $f_{\text{concept},h}$ applies the
 4070 normalized weighted average of Equation (61), whose denominator $\sum_{(n,l) \in S_h} w_{h,n,l}$
 4071 is a constant of design and whose numerator is linear in its inputs, yielding a linear
 4072 mapping in the membership matrix.

4073 For Stage 4, the candidate intervention degree $u_{k,m}^i$ of Equation (80) is a ratio whose
 4074 denominator depends on the concept vector. By the partition-of-unity property
 4075 (Proposition 5.1), $\sum_{n \in \{L,M,H\}} \mu_n(c_{k,h}^i) = 1$ for each concept $h \in \{1,2,3,4\}$. Because
 4076 the rule base $\pi_{i,m}$ enumerates every antecedent combination $(n_1, n_2, n_3, n_4) \in$
 4077 $\{L, M, H\}^4$ (Table 5.19), and because $\min(a, b) \geq a \cdot b$ for $a, b \in [0,1]$:

$$\sum_{R \in \mathcal{R}_{\pi^i, m}} \alpha_R = \sum_{(n_1, \dots, n_4)} \min_h \mu_{n_h}(c_{k,h}^i) \geq \sum_{(n_1, \dots, n_4)} \prod_{h=1}^4 \mu_{n_h}(c_{k,h}^i) = \prod_{h=1}^4 \left(\sum_{n_h} \mu_{n_h}(c_{k,h}^i) \right) \quad (83)$$

$$\prod_{h=1}^4 \left(\sum_{n_h} \mu_{n_h}(c_{k,h}^i) \right) = 1.$$

4078 The denominator of Equation (80) is therefore bounded below by 1, uniformly in
 4079 the concept vector. Hence $u_{k,m}^i$ is the ratio of a Lipschitz-continuous numerator to a
 4080 denominator strictly bounded away from zero, and is itself Lipschitz.

4081 Stage 5 applies the priority-weighted average of Equation (70) with constant
 4082 positive weights α_m ; it is linear in its inputs and therefore Lipschitz. The finite
 4083 composition of Lipschitz-continuous mappings is Lipschitz-continuous with
 4084 constant bounded by the product of the individual stage constants.

4085 **Remark 5.10 (Partial Observability Compliance).** The composed mapping π_{bss}^i
 4086 receives $O_k^i(x, y)$ and $\mathcal{F}_{DSS, k}$ as its only inputs. The true wildfire state $S_k(x, y)$ does
 4087 not enter at any stage. This enforces the partial observability constraint structurally
 4088 throughout the full pipeline, rather than only at the architectural boundary.

4089 5.4 Summary of DisasterAware Decision Support System Specifications

4090 This chapter establishes the formal design specification of the DisasterAware DSS,
 4091 grounding every design decision in the operational context of wildfire response, the
 4092 Chapter 4 simulation framework, and established fuzzy reasoning literature.

4093 Section 5.1 characterizes wildfire incident management through four structural
 4094 properties (temporal dynamics, spatial heterogeneity, epistemic uncertainty, and
 4095 resource constraints) and identifies four fundamental operational questions (i.e., fire
 4096 threat severity (Q1), suppression feasibility (Q2), asset exposure (Q3), and action
 4097 urgency (Q4)) that map directly onto the four situational concepts of the DSS.

4098 Section 5.2 formalizes the processing pipeline from raw observables to situational
 4099 concepts. Six scalar features, each bounded in $[0, 1]$, feed a three-term fuzzy variable

4100 design (Low, Medium, High) satisfying partition of unity. The resulting 18-
4101 dimensional membership representation is aggregated into four concept values, with
4102 Intervention Urgency operating as a second-order concept. Proposition 5.2
4103 establishes a ninefold *marginal* rule complexity reduction per intervention type (729
4104 $\rightarrow$ 81 rules), invariant of M ; at $M = 3$ the *overall* reduction is $\kappa_{total}(3) \approx 6.2$ (351
4105 = 108 concept-construction + 243 decision rules versus 2,187 under direct inference),
4106 approaching $\kappa_{marg} = 9$ asymptotically as M grows.

4107 Section 5.3 specifies the complete rule base (351 rules confirmed by Table 5.19) and
4108 defines the action aggregation function f_{agg} with priority weights $\alpha_1 = 1.0, \alpha_2 =$
4109 $0.7, \alpha_3 = 0.9$. Section 5.3.5 unifies all components as the five-stage composition
4110 of π_{DSS}^i (Definition 5.1), establishing output boundedness, Lipschitz continuity, and
4111 structural enforcement of partial observability. The decision quality function f_{eval}
4112 evaluates four operational criteria with weights $w_1 = 0.35, w_2 = 0.30, w_3 =$
4113 $0.20, w_4 = 0.15$; its runtime realization is presented in Section 6.4.4.

4114 The present chapter establishes the *complete mathematical specification* of the DSS:
4115 every function, parameter, and rule base is fully defined here. Chapter 6 does *not*
4116 introduce new mathematical content; it describes the *runtime organization* of these
4117 specifications within a layered, distributed agent architecture. A reader seeking
4118 mathematical definitions should consult this chapter; a reader seeking execution
4119 sequence and coordination mechanisms should consult Chapter 6.

4120

CHAPTER 6

4121

DISASTER-AWARE IMPLEMENTATION

4122 This chapter presents the Disaster-Aware Decision Support System as the cognitive
4123 layer of the overall framework. While Chapter 4 establishes the physically grounded
4124 Simulation Framework and Chapter 5 formalizes the design specifications of the
4125 DSS, the present chapter provides the architectural realization: it defines how
4126 observable wildfire information is transformed, step by step, into spatially
4127 distributed intervention guidance through a layered, distributed, concept-driven, and
4128 uncertainty-aware reasoning architecture. Chapter 5 answers *what* the DSS computes
4129 and *why* each design element is chosen; this chapter answers *how* those computations
4130 are sequenced across architectural layers, how data flows between them, and how
4131 distributed agents are coordinated at runtime. Chapter 5 remains the single
4132 authoritative source for every mathematical element; this chapter is the single
4133 authoritative source for layer decomposition, agent parallelism, data flow interfaces,
4134 global coordination, and closed-loop feedback.

4135 The DSS is defined not merely as an implementation component but as a structured
4136 reasoning architecture operating under partial observability. It does not directly
4137 access or modify the underlying wildfire state and does not produce direct control
4138 actions. Instead, it generates spatially distributed intervention representations that
4139 influence system evolution indirectly through the external input layer, preserving the
4140 separation between physical dynamics and decision reasoning. The approach
4141 decomposes decision generation into interconnected layers: observable information
4142 is transformed into decision-relevant features, abstracted into a reduced concept
4143 space, spatially decomposed into regional contexts, and evaluated through
4144 distributed local decision mechanisms. These local evaluations are then coordinated
4145 to produce a globally consistent intervention structure. The remainder of this chapter
4146 is organized as follows.

4147 Section 6.1 presents the overall layered architecture, introducing all principal
 4148 components and the mathematical interfaces in a structural overview. It describes the
 4149 Grid State Manager, Observation Layer, Concept Mapping and Abstraction Layer,
 4150 Local DSS Model, Global DSS, and the closed-loop structure that connects them.

 4151 Section 6.2 details the runtime execution of the three-stage transformation from raw
 4152 observable information to the concept vector $\mathbf{c}_k^i(x, y)$. It establishes the formal
 4153 properties of the transformation and provides an analytical treatment of how graded
 4154 membership encoding achieves robustness under partial observability.

 4155 Section 6.3 describes the fuzzy inference execution over the concept vector, the
 4156 evaluation of the intervention decision rule bases $\{\pi_{i,m}\}_{m=1}^M$, and the action
 4157 aggregation function f_{agg} that consolidates multi-type candidate degrees into a single
 4158 executable intervention signal per region.

 4159 Section 6.4 presents the global coordination mechanism $\mathcal{A}_{DSS}$ and the construction
 4160 of the spatial intervention $U_{DSS,k}(x, y)$ from the N region-level outputs, including
 4161 the resource normalization pass and the feedback injection into the Simulation Core.

 4162 Section 6.5 describes the auxiliary supervisory layer that provides traceability,
 4163 auditability, and operator override mechanisms without modifying the core pipeline.

 4164 Section 6.6 consolidates the architectural contributions of the chapter and connects
 4165 the established DSS structure and connects the DSS architecture to the simulation
 4166 and performance evaluation in Chapter 7.

 4167 Through this progression, the chapter establishes the DisasterAware DSS as a
 4168 layered and distributed reasoning architecture capable of generating adaptive,
 4169 interpretable, and operationally consistent intervention guidance under uncertainty.

4170 6.1 Architectural Overview of the DSS

4171 The DisasterAware DSS is the cognitive layer of the framework, responsible for
 4172 transforming observable wildfire information to spatially distributed, operationally

feasible, and interpretable intervention guidance. The architecture of the DSS in Figure 6.1 illustrates the structural organization of the system, the formulation, and the closed-loop coupling with the Simulation Core established in Chapter 4.

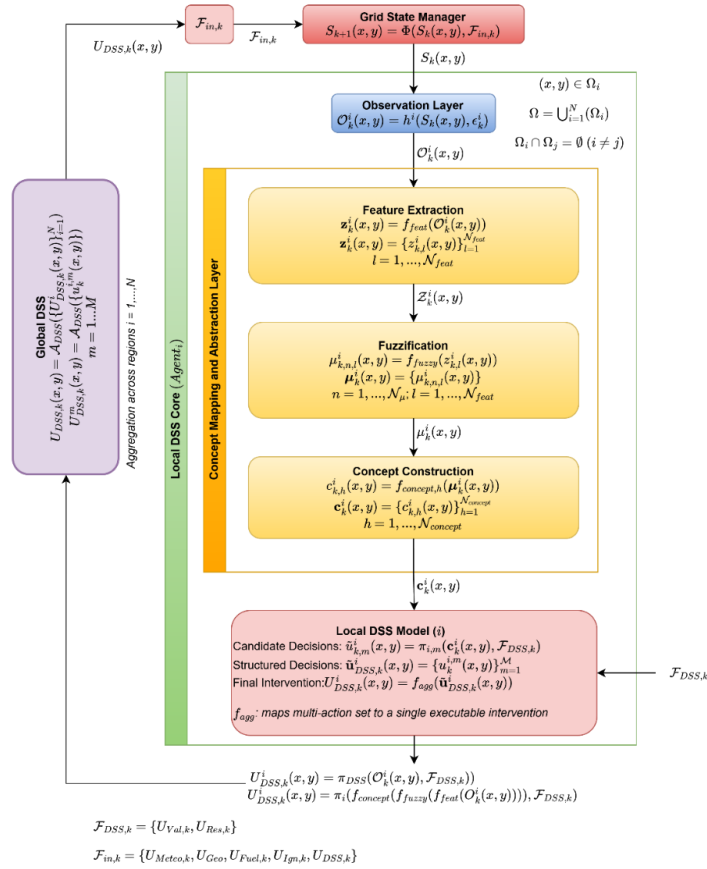


Figure 6.1 DSS system overview

The architectural design of the DSS is governed by three foundational principles that follow directly from the problem formulation of Chapter 2. First, the DSS operates under **partial observability**. In other words, it never accesses the true wildfire state $S_k(x,y)$ directly, but reasons exclusively from observable information subject to epistemic uncertainty. Second, the DSS generates **intervention inputs rather than**

4183 **control actions**; in other words, its outputs influence system evolution indirectly
 4184 through the external input layer $\mathcal{F}_{in,k}$, never by overwriting internal state variables.
 4185 Third, the DSS supports **multi-level decision generation**; in other words, local
 4186 context-aware reasoning is coordinated at a global level to produce a spatially
 4187 consistent intervention structure. These principles collectively define the boundary
 4188 conditions within which the DSS architecture is constructed.

4189 6.1.1 Grid State Manager and the Simulation Interface

4190 As shown in Figure 6.1, the architectural flow originates at the **Grid State Manager**,
 4191 which maintains the authoritative wildfire state $S_k(x, y)$ over the spatial domain $\mathcal{G}$
 4192 and governs its discrete-time evolution through the state transition $S_{k+1}(x, y) =$
 4193 $\Phi(S_k(x, y), \mathcal{F}_{in,k})$ where $\mathcal{F}_{in,k} = \{U_{Meteo,k}, U_{Geo}, U_{Fuel,k}, U_{Ign,k}, U_{DSS,k}\}$ is the
 4194 structured external input set defined in Chapter 4. The DSS interacts with this
 4195 operator exclusively through the decision-driven subset $U_{DSS,k}(x, y) \subset \mathcal{F}_{in,k}$, which
 4196 represents the intervention parameters generated by the decision layer. This interface
 4197 enforces the strict causal structure: $S_k(x, y) \rightarrow O_k^i(x, y), \mathcal{F}_{DSS,k} \rightarrow U_{DSS,k}(x, y) \subset$
 4198 $\mathcal{F}_{in,k} \rightarrow \Phi(\cdot) \rightarrow S_{k+1}(x, y)$, which ensures that all decision effects propagate
 4199 through physically consistent state transitions.

4200 6.1.2 Observation Layer

4201 Since the true wildfire state $S_k(x, y)$ is not directly accessible in operational
 4202 conditions, the first functional boundary of the DSS is the **Observation Layer**,
 4203 which produces a region-specific, uncertainty-affected representation of the current
 4204 system state. For each operational region Ω_i , the observation process is formally
 4205 defined as (84) where $h^i(\cdot)$ is the region-specific observation function that maps
 4206 internal state variables to measurable or estimable quantities, and ϵ_k^i represents the
 4207 epistemic uncertainty associated with region Ω_i at time step k .

$$O_k^i(x, y) = h^i(S_k(x, y), \epsilon_k^i) \quad (84)$$

4208 The uncertainty term ϵ_k^i captures the combined effect of sensing limitations, spatial
 4209 coverage gaps, temporal delays in field reporting, and measurement noise. It is not
 4210 modeled as a stochastic noise term with a known distribution rather, it represents the
 4211 structural incompleteness of knowledge that motivates the use of fuzzy reasoning in
 4212 subsequent layers. The observable information $O_k^i(x, y)$ provides the exclusive
 4213 informational basis for all downstream reasoning within region Ω_i . It may include
 4214 estimated burning extent, fire front propagation indicators, fuel depletion estimates,
 4215 wind-adjusted spread potential, and proximity-to-value metrics derived from the
 4216 state variables $\{B_k, F_{load,k}, I_k, \tau_k\}$ together with the contextual inputs $\mathcal{F}_{DSS,k}$.

4217 6.1.2.1 Observation Function

4218 The region-specific observation function $h^i(\cdot)$ introduced in Equation (84) is
 4219 formally defined as a structured state projection operator subject to spatially varying
 4220 coverage weights and bounded epistemic disturbance. The function maps the four-
 4221 dimensional wildfire state vector to an observable information vector of equal
 4222 dimension, restricted to the spatial domain of region Ω_i (85).

$$h^i: \mathbb{R}^4 \times \mathcal{E}^i \rightarrow \mathbb{R}^4, (x, y) \in \Omega_i \quad (85)$$

4223 For each grid cell $(x, y) \in \Omega_i$, the observable information vector is defined
 4224 component-wise as (86):

$$O_k^i(x, y) = \begin{bmatrix} \hat{B}_k^i(x, y) \\ \hat{F}_{load,k}^i(x, y) \\ \hat{I}_k^i(x, y) \\ \hat{\tau}_k^i(x, y) \end{bmatrix} = \begin{bmatrix} \theta_{B,k}^i(x, y) \cdot B_k(x, y) + \epsilon_{B,k}^i(x, y) \\ \theta_{F,k}^i(x, y) \cdot F_{load,k}(x, y) + \epsilon_{F,k}^i(x, y) \\ \theta_{I,k}^i(x, y) \cdot I_k(x, y) + \epsilon_{I,k}^i(x, y) \\ \theta_{\tau,k}^i(x, y) \cdot \tau_k(x, y) + \epsilon_{\tau,k}^i(x, y) \end{bmatrix} \quad (86)$$

4225 For cells outside the region, $O_k^i(x, y) = \mathbf{0}_4$ for all $(x, y) \notin \Omega_i$, enforcing spatial
 4226 locality of the observation process.

6.1.2.2 Observability Weights

Each scalar weight $\theta_{j,k}^i(x, y) \in [0, 1]$, for $j \in \{B, F, I, \tau\}$, encodes the degree to which state component j at cell (x, y) is accessible to region Ω_i at time step k . $\theta_{j,k}^i(x, y) = 1$ denotes full observability under nominal conditions; $\theta_{j,k}^i(x, y) = 0$ indicates complete inaccessibility due to coverage failure or data unavailability. Table 6.1 lists the physical interpretation and typical sensing source for each weight.

Table 6.1 Observability weights: interpretation and sensing source

Weight	State component	$\theta = 1$ (full observability)	$\theta = 0$ (no observation)	Typical source
$\theta_{B,k}^i$	Burning status B_k	Active burning reliably detected	Complete coverage gap	Satellite or aerial imagery
$\theta_{F,k}^i$	Fuel load $F_{\text{load},k}$	Fuel state precisely known	No fuel data available	Pre-fire fuel maps, field survey
$\theta_{I,k}^i$	Fire intensity I_k	Intensity directly measured	Sensor occlusion	Thermal infrared imaging
$\theta_{\tau,k}^i$	Ignition time τ_k	Time-since-ignition recorded	Unknown ignition time	Event log, field report

In the nominal simulation setting used throughout this thesis, all weights are set to unity, $\theta_{j,k}^i(x, y) = 1$ for all $(x, y) \in \Omega_i$ and all j , representing ideal sensing conditions. Degraded coverage scenarios are encoded by reducing individual weights for affected cells.

6.1.2.3 Epistemic Disturbance Model

The disturbance vector $\epsilon_k^i(x, y) = [\epsilon_{B,k}^i, \epsilon_{F,k}^i, \epsilon_{I,k}^i, \epsilon_{\tau,k}^i]^\top$ represents the bounded epistemic uncertainty remaining after coverage scaling. It is not modeled as a

4242 stochastic noise process. Instead, each component is bounded by a maximum
 4243 disturbance magnitude $\bar{\epsilon}_j^i > 0$ (87).

$$\epsilon_{j,k}^i(x, y) \in [-\bar{\epsilon}_j^i, \bar{\epsilon}_j^i], j \in \{B, F, I, \tau\} \quad (87)$$

4244 Additionally, whenever $\theta_{j,k}^i(x, y) = 0$, the disturbance is defined to be zero,
 4245 $\epsilon_{j,k}^i(x, y) = 0$, so that the j -th entry of $O_k^i(x, y)$ vanishes and unambiguously
 4246 encodes loss of observability for state component j at cell (x, y) rather than
 4247 generating a spurious noise reading.

4248 The disturbance magnitude is governed by per-component observation confidence
 4249 coefficients that aggregate four independent degradation factors: the spatially
 4250 varying observability weight $\theta_{j,k}^i(x, y) \in [0, 1]$ of Section 6.1.2.2, the normalized
 4251 sensor coverage density $\rho_k^i(x, y) \in [0, 1]$ reflecting the fraction of active sensing
 4252 assets providing data at cell (x, y) , an exponential decay $\exp(-\lambda_{\text{conf}} \cdot \Delta t_{\text{rep},k}^i(x, y))$
 4253 in the elapsed time since the most recent field observation or sensor report at (x, y)
 4254 with $\lambda_{\text{conf}} > 0$ governing the decay rate, and the regional communication reliability
 4255 $\gamma_k^i \in [0, 1]$ encoding data-link integrity between field sensors and the DSS:

$$\text{conf}_{j,k}^i(x, y) = \min(\theta_{j,k}^i(x, y), \rho_k^i(x, y), \exp(-\lambda_{\text{conf}} \cdot \Delta t_{\text{rep},k}^i(x, y)), \gamma_k^i) \quad (88)$$

$j \in \{B, F, I, \tau\}$

4256 The cell-level scalar confidence used by downstream layers is the minimum of the
 4257 component-specific confidences, reflecting the conservative principle that the
 4258 reliability of the observation vector is limited by its least reliable channel:

$$\text{conf}_k^i(x, y) = \min_{j \in \{B, F, I, \tau\}} \text{conf}_{j,k}^i(x, y) \quad (89)$$

4259 The effective per-component disturbance magnitude decreases as component
 4260 confidence increases,

$$|\epsilon_{j,k}^i(x, y)| \leq (1 - \text{conf}_{j,k}^i(x, y)) \cdot \bar{\epsilon}_j^i. \quad (90)$$

When $\theta_{j,k}^i(x, y) = 1$ and all remaining factors equal unity, $\text{conf}_{j,k}^i = 1$ and the disturbance vanishes. When $\theta_{j,k}^i(x, y) = 0$, $\text{conf}_{j,k}^i = 0$ jointly with the structural constraint $\epsilon_{j,k}^i = 0$ introduced following Equation (87), so that the unreliability of component j is signaled simultaneously through a zero j -th entry of the observation vector $O_k^i(x, y)$ and a zero component confidence $\text{conf}_{j,k}^i(x, y)$. In the nominal case $\theta_{j,k}^i(x, y) = 1$ for all j , (89) reduces to $\text{conf}_k^i(x, y) = \min \left(\rho_k^i(x, y), \exp \left(-\lambda_{\text{conf}} \cdot \Delta t_{\text{rep},k}^i(x, y) \right), \gamma_k^i \right)$, recovering the strict-sensing baseline.

The global observation function $h(\cdot)$ introduced in Chapter 2 is recovered from region-specific functions through spatial aggregation. Since the operational regions $\{\Omega_i\}_{i=1}^N$ form a disjoint partition of $\mathcal{G}$, each grid cell belongs to exactly one region, and the global observable is (91):

$$O_k(x, y) = h(S_k(x, y), \epsilon_k) = \sum_{i=1}^N \mathbf{1}_{(x,y) \in \Omega_i} \cdot h^i(S_k(x, y), \epsilon_k^i) \quad (91)$$

This relationship confirms that Equation (84) constitutes the architectural instantiation of the abstract observation model stated in Equation (1).

The four components of $O_k^i(x, y)$ serve as direct inputs to the feature extraction stage of Section 6.2.1. The mapping $\mathbf{z}_k^i(x, y) = f_{\text{feat}} \left(O_k^i(x, y) \right)$ defined in (85) derives the six operational features used in all subsequent reasoning stages exclusively from $\{\hat{B}_k^i, \hat{F}_{\text{load},k}^i, \hat{I}_k^i, \hat{\tau}_k^i\}$ together with the contextual inputs $\mathcal{F}_{\text{DSS},k}$.

6.1.3 Local DSS Core

Each operational region Ω_i is served by a dedicated reasoning unit, denoted **Agent_i**, which constitutes the Local DSS Core. As shown in Figure 6.1, the Local DSS Core is internally organized into two functionally distinct sublayers: the **Concept Mapping and Abstraction Layer** and the **Local DSS Model**. The separation

4283 between these sublayers reflects the fundamental distinction between situation
4284 understanding (what is happening?) and decision generation (what should be done?).

4285 **Remark 6.1 (Shared Policy Across Regions).** The regional DSS mapping π_{DSS}^i is
4286 functionally and parametrically identical across all regions $i = 1, \dots, N$: the feature
4287 extraction mapping f_{feat} , the fuzzification operator f_{fuzzy} , the concept construction
4288 mappings $\{f_{concept,h}\}_{h=1}^4$ (including all 108 concept-construction rules and the
4289 weights specified in Equations (62)–(65), the intervention decision mappings
4290 $\{\pi_{i,m}\}_{m=1}^M$ with their 243 rules, and the action aggregation function f_{agg} are shared.
4291 The default membership parameters (Table 5.4), priority weights (**Remark 5.7**), and
4292 decision quality weights (**Remark 5.9**) are likewise common to all regional agents.
4293 Spatial variability in the generated global intervention field $U_{DSS,k}(x,y)$ arises
4294 exclusively from region-specific inputs: the regional observation $O_k^i(x,y)$, the
4295 contextual inputs $F_{DSS,k}^i$, the observation confidence $conf_k^i(x,y)$, and the region-
4296 local concept memory $c_{prior,h}^i(x,y)$ used in the confidence-gated modulation of
4297 Section 6.2.4. This shared-policy design is a deliberate architectural choice rather
4298 than a limitation: it yields a uniformly validatable decision system, supports third-
4299 party certification against a single rule base, and avoids the combinatorial tuning
4300 burden of region-specific rule tailoring. The architecture remains extensible to
4301 region-specific specialization (e.g., distinct rule sets for wildland–urban interface
4302 versus wilderness zones) through per-region instantiation of π_{DSS}^i , without
4303 modification to the global coordination mechanism $\mathcal{A}_{DSS}^{sp}$ or the Simulation Core.

4304 6.1.3.1 Concept Mapping and Abstraction Layer

4305 The purpose of the Concept Mapping and Abstraction Layer is to transform the raw
4306 observable information $O_k^i(x,y)$ into a compact, interpretable, and reasoning-
4307 amenable situational description. This transformation proceeds through three
4308 sequential stages.

4309 6.1.3.1.1 Feature Extraction

4310 The first stage extracts a structured set of $\mathcal{N}_{feat}$ scalar features from the observable
4311 information through the feature extraction mapping f_{feat} , defined in (77). Each
4312 feature $z_{k,l}^i(x, y)$ is a scalar-valued indicator that captures a specific operationally
4313 relevant aspect of the current wildfire situation within region Ω_i at grid cell (x, y) .
4314 The feature vector $\mathbf{z}_k^i$ is defined over the same spatial domain as the observable
4315 information and is evaluated at every active grid cell within Ω_i . The design of the
4316 feature set and the six feature definitions are presented in Section 5.2.1.

4317 6.1.3.1.2 Fuzzification

4318 The second stage transforms each scalar feature $z_{k,l}^i(x, y)$ into a graded membership
4319 representation over $\mathcal{N}_\mu$ linguistic terms through the fuzzification mapping f_{fuzzy} ,
4320 defined in Equation (78). The membership value $\mu_{k,n,l}^i(x, y) \in [0,1]$ expresses the
4321 degree to which the feature $z_{k,l}^i(x, y)$ belongs to linguistic term n (e.g., the degree to
4322 which local fire intensity is **high**, or the degree to which suppression travel time is
4323 **critical**). This graded representation is the primary mechanism through which
4324 epistemic uncertainty ϵ_k^i is encoded in the reasoning process: rather than requiring
4325 precise crisp thresholds that would be unreliable under uncertain observations, the
4326 fuzzy membership structure allows the system to reason over partial and ambiguous
4327 evidence in a principled manner. The membership function designs and linguistic
4328 variable definitions are presented in Section 5.2.2, Equation (57).

4329 6.1.3.1.3 Concept Construction

4330 The third stage aggregates the fuzzified feature memberships into a reduced set of
4331 $\mathcal{N}_{concept}$ higher-level situational concepts through the concept construction
4332 mappings $f_{concept,h}$, defined in Equation (79), with index sets and weights specified

in Equations (62)–(65). Each concept $c_{k,h}^i(x, y) \in [0,1]$ represents a composite situational assessment that captures a semantically meaningful operational condition such as **imminent threat to high-value assets**, **suppression feasibility under current resource availability**, or **active fire front with high spread potential**.

The concept vector $\mathbf{c}_k^i(x, y)$ operates in a space of dimension $\mathcal{N}_{concept}$, which is substantially smaller than the $\mathcal{N}_{feat} \times \mathcal{N}_\mu$ dimensional space of the full membership representation. This dimensionality reduction constitutes the architectural mechanism through which rule complexity of the subsequent decision layer is controlled. The analytical treatment of this reduction is provided in Section 5.2.3.

6.1.3.2 Local DSS Model

The concept vector $\mathbf{c}_k^i(x, y)$ is passed to the **Local DSS Model**, which is responsible for generating region-level intervention guidance. The Local DSS Model operates through three sequential stages as shown in Figure 6.1.

6.1.3.2.1 Candidate Decision Generation

For each intervention type $m \in \{1, \dots, M\}$, a decision mapping $\pi_{i,m}$ evaluates the situational concept vector jointly with the contextual decision inputs $\mathcal{F}_{DSS,k}$ to produce a candidate intervention degree $u_{k,m}^i(x, y)$, as defined in Equation (80). The contextual input set $\mathcal{F}_{DSS,k} = \{U_{Val,k}, U_{Res,k}\}$ encodes the value-at-risk distribution and the operational resource constraints as defined in Chapter 2. Its inclusion in the decision mapping ensures that intervention candidates are evaluated not only with respect to the current fire situation but also with respect to what is operationally feasible and strategically prioritized.

Each $\pi_{i,m}$ is realized as a fuzzy inference system whose rule base encodes domain knowledge about appropriate intervention responses to identified situational concepts. The candidate degree $u_{k,m}^i(x, y) \in [0,1]$ expresses the degree to which

4358 intervention type m is warranted at grid cell (x, y) within region Ω_i at time step k .
 4359 It does not constitute a binary decision but a graded recommendation, consistent with
 4360 the satisfaction-based decision paradigm introduced in Chapter 2. The specification
 4361 of $\pi_{i,m}$ and its associated rule base are presented in Section 5.3.2, and its runtime
 4362 execution is detailed in Section 6.3.

4363 6.1.3.2.2 Structured Decision Assembly

4364 The candidate degrees across all M intervention types are organized into a structured
 4365 multi-action representation (92). This multi-action structure reflects the non-
 4366 exclusive nature of wildfire response operations, where multiple intervention
 4367 strategies (i.e., such as direct suppression, fuel reduction, and evacuation support)
 4368 may need to be activated simultaneously with varying degrees of priority.

$$\mathbf{u}_{DSS,k}^i(x, y) = \{u_{k,m}^i(x, y)\}_{m=1}^M \quad (92)$$

4369 6.1.3.2.3 Final Intervention Generation

4370 The multi-action vector is mapped to a single executable intervention parameter
 4371 through the aggregation function f_{agg} , defined in (70), to resolve any conflicts or
 4372 redundancies among simultaneously activated intervention types and produce an
 4373 output $U_{DSS,k}^i(x, y)$ expressed in terms compatible with the external input $U_{DSS,k}$.

4374 6.1.4 Global DSS Coordination

4375 The region-level intervention parameters generated by all N local agents are
 4376 consolidated within the **Global DSS** coordination layer. As illustrated in Figure 6.1,
 4377 the global coordination mechanism produces both a spatially distributed global
 4378 intervention field and action-type-specific global outputs (93) where $\mathcal{A}_{DSS}(\cdot)$ is the

4379 global coordination and fusion mechanism that integrates locally generated decisions
4380 into a globally consistent intervention structure.

$$\begin{aligned} U_{DSS,k}(x, y) &= \mathcal{A}_{DSS}(\{U_{DSS,k}^i(x, y)\}_{i=1}^N) \\ U_{DSS,k}^m(x, y) &= \mathcal{A}_{DSS}(\{u_{k,m}^i(x, y)\}_{i=1}^N), m = 1, \dots, M \end{aligned} \quad (93)$$

4381 Global aggregation serves two purposes. First, it resolves spatial inconsistencies that
4382 arise when locally optimal decisions in neighboring regions conflict with one
4383 another. For example, when two adjacent regions independently request the same
4384 suppression resource that cannot be simultaneously deployed in both locations.
4385 Second, it ensures that the global intervention field $U_{DSS,k}(x, y)$ is defined over the
4386 entire spatial domain $\mathcal{G}$, providing a complete spatially distributed intervention
4387 representation rather than a collection of disjoint regional outputs.

4388 The formalization of $\mathcal{A}_{DSS}$ and the spatial field generation mechanisms is presented
4389 in Section 6.4. The resulting global intervention field $U_{DSS,k}(x, y)$ is injected back
4390 into the external input set $\mathcal{F}_{in,k}$, closing the feedback loop between the DSS and the
4391 Simulation Core.

4392 6.1.5 Closed-Loop Structure

4393 The complete decision cycle illustrated in Figure 6.1 constitutes a closed-loop
4394 interaction between the cognitive DSS layer and the physical simulation layer. At
4395 each discrete time step k , the cycle proceeds as follows. The Grid State Manager
4396 updates the wildfire state $S_k(x, y)$ through the transition operator $\Phi(\cdot)$. The
4397 Observation Layer derives partial, uncertainty-affected observations $O_k^i(x, y)$ for
4398 each region. Each $Agent_i$ processes its regional observation through the Concept
4399 Mapping and Abstraction Layer and the Local DSS Model, producing a region-level
4400 intervention output $U_{DSS,k}^i(x, y)$. The Global DSS aggregates these outputs into the
4401 intervention field $U_{DSS,k}(x, y)$, which is injected into $\mathcal{F}_{in,k}$ to influence the next state
4402 transition $S_{k+1}(x, y)$.

4403 This closed-loop structure enables the DSS to update intervention guidance at each
4404 time step in response to evolving wildfire conditions. Rather than converging to a
4405 fixed policy, the DSS generates context-sensitive guidance from updated
4406 observations under partial observability.

4407 **6.2 Concept Mapping and Abstraction Layer**

4408 This section describes the runtime execution of the Concept Mapping and
4409 Abstraction Layer within the DSS architecture. At each simulation time step k and
4410 for each spatial region Ω_i , the layer transforms observable information $O_k^i(x, y)$ into
4411 a structured concept vector $\mathbf{c}_k^i(x, y)$ through a sequence of deterministic
4412 transformations. The transformation follows a three-stage pipeline consisting of
4413 feature extraction, fuzzification, and concept construction. All mappings, parameter
4414 definitions, and semantic interpretations of features and concepts are formally
4415 defined in Chapter 5. This section focuses exclusively on how these mappings are
4416 executed during runtime. The design rationale and parameter choices for each stage
4417 are established in Sections 5.2.1–5.2.3; this section specifies their runtime execution
4418 sequence and inter-layer data flow.

4419 **6.2.1 From Observation to Concept Vector: Runtime Execution**

4420 At each discrete time step k , every local DSS agent Agent_i receives the observable
4421 information $O_k^i(x, y)$ from the Observation Layer for all grid cells $(x, y) \in \Omega_i$ within
4422 its assigned region. The Concept Mapping and Abstraction Layer processes this input
4423 through three sequential computational stages that are applied independently at every
4424 active grid cell. Figure 6.2 illustrates this pipeline and its outputs at each stage.

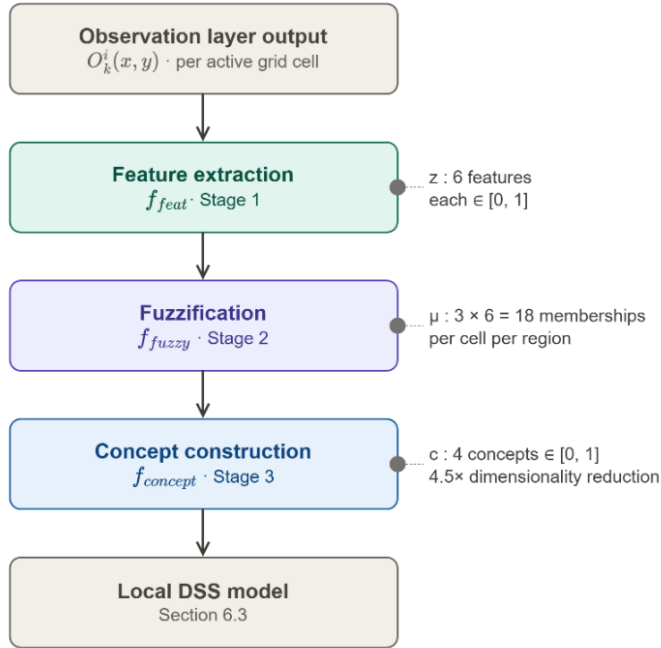


Figure 6.2 The 3-stage pipeline of the concept mapping and abstraction layer

6.2.1.1 Stage 1 Feature Extraction

The feature extraction mapping f_{feat} projects the observable information onto the six-dimensional feature vector $\mathbf{z}_k^i(x, y) \in [0, 1]^6$ whose components and derivations are defined in Section 5.2.1 Equation (77). All six features are bounded in $[0, 1]$ either by construction (for z_1, z_2) or by explicit normalization (for z_3 through z_6), which is a necessary precondition for the fuzzification stage that follows. This stage executes with no inter-agent communication and no memory of previous time steps: f_{feat} is a stateless, spatially local mapping evaluated independently at each (x, y) . The computational cost scales linearly with the number of active cells in Ω_i .

4436 6.2.1.2 Stage 2 Fuzzification

4437 The fuzzification mapping f_{fuzzy} transforms each scalar feature into a triplet of graded
4438 membership values over the linguistic terms Low (L), Medium (M), and High (H),
4439 producing the 3×6 membership matrix $\mu_k^i(x, y)$ with 18 entries per grid cell. The
4440 membership functions follow the piecewise-linear shoulder–triangular–shoulder
4441 design established in Section 5.2.2 Equation (57), parameterized by the feature-
4442 specific triplets $(p_{1,l}, p_{2,l}, p_{3,l})$ from Table 5.4. Like feature extraction, fuzzification
4443 is a stateless per-cell operation. The partition-of-unity property $\mu_{L,l} + \mu_{M,l} + \mu_{H,l} =$
4444 1 established in Proposition 5.1 holds at every grid cell and every time step, ensuring
4445 that no observation is left ambiguously unassigned.

4446 6.2.1.3 Stage 3 Concept Construction

4447 The concept construction mappings $\{f_{\text{concept},h}\}_{h=1}^4$ aggregate selected entries of the
4448 membership matrix into four situational concepts via the normalized weighted
4449 average defined in Section 5.2.3 Equation (61). The index sets $\mathcal{S}_h$ and weights
4450 $\{w_{h,n,l}\}$ for Fire Threat Level (c_1), Suppression Feasibility (c_2), Asset Exposure Risk
4451 (c_3), and Intervention Urgency (c_4) are specified in Equations (62)–(65). The
4452 concept vector $\mathbf{c}_k^i(x, y) = \{c_{k,h}^i(x, y)\}_{h=1}^4$ is the output of the Concept Mapping and
4453 Abstraction Layer and serves as the input to the DSS Model described in Section 6.3.

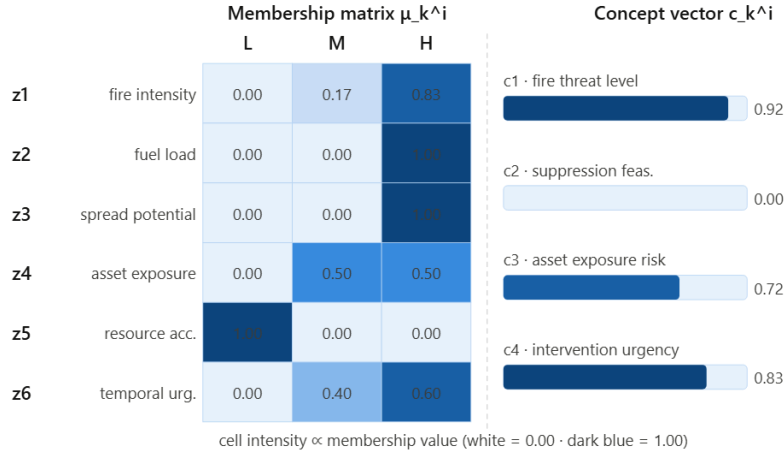


Figure 6.3 Membership matrix and concept vector

Figure 6.3 presents a worked numerical example of the complete pipeline, using a high-urgency wildfire scenario with parameter values $z_1 = 0.75$ (high fire intensity), $z_2 = 0.80$ (high fuel), $z_3 = 0.70$ (maximum spread potential), $z_4 = 0.60$ (moderate-high asset exposure), $z_5 = 0.20$ (low resource accessibility), and $z_6 = 0.75$ (high temporal urgency).

6.2.2 Formal Properties of the Transformation

The end-to-end transformation $O_k^i(x, y) \mapsto \mathbf{c}_k^i(x, y)$ satisfies three formal properties that are essential to the correctness and tractability of the DSS at runtime.

Property 6.1 (Output Boundedness). *For all grid cells $(x, y) \in \Omega_i$, all time steps k , and all agents i , the concept vector satisfies $c_{k,h}^i(x, y) \in [0, 1]$ for $h = 1, \dots, 4$.*

This follows from the stage-by-stage boundedness argument established in the proof of Proposition 5.3. Its architectural consequence is that the Local DSS Model can apply the same membership function design to concept inputs as to feature inputs, without requiring additional normalization.

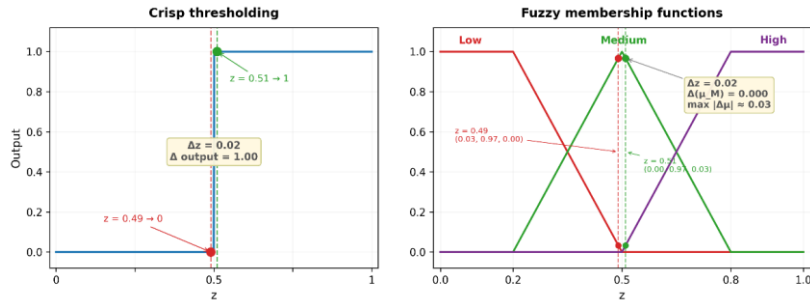
4470 **Property 6.2 (Partition of Unity).** *For each feature l and any $z \in [0,1]$, the*
 4471 *fuzzification stage satisfies $\mu_{L,l}(z) + \mu_{M,l}(z) + \mu_{H,l}(z) = 1$. This is Proposition 5.1,*
 4472 *proven in Section 5.2.2. Within the runtime loop, this property guarantees that no*
 4473 *feature value is ambiguously unassigned and no double-counting occurs, ensuring*
 4474 *that the concept construction stage operates on a well-defined probability-like*
 4475 *structure at every grid cell and every time step.*

4476 **Property 6.3 (Dimensionality Reduction).** *The transformation reduces the per-cell*
 4477 *representation from $\mathcal{N}_\mu \times \mathcal{N}_{feat} = 18$ dimensions to $\mathcal{N}_{concept} = 4$ dimensions. The*
 4478 *rule complexity reduction achieved by this architecture, from $3^6 = 729$ to $3^4 = 81$*
 4479 *rules per intervention type, is formally established in Proposition 5.2. Its architectural*
 4480 *significance is that each local agent executes a compact 81-rule inference pass per*
 4481 *intervention type rather than a 729-rule pass, directly controlling the per-step*
 4482 *computational cost of the distributed DSS.*

4483 6.2.3 Epistemic Encoding Under Partial Observability

4484 The Observation Layer introduces epistemic uncertainty ϵ_k^i into $O_k^i(x, y)$ through
 4485 sensor noise, coverage gaps, and estimation error, as formalized in Section 6.1.2. A
 4486 central design requirement of the Concept Mapping and Abstraction Layer is that
 4487 this uncertainty must be propagated into the concept vector in a principled manner,
 4488 rather than suppressed through crisp thresholding.

4489 A crisp alternative to fuzzification would assign each feature $z_{k,l}^i$ to exactly one
 4490 linguistic term based on hard boundaries: for instance, $z_{k,1}^i > 0.5 \Rightarrow$ High fire
 4491 intensity, otherwise Low or Medium. Under this approach, a feature value of 0.49
 4492 and a value of 0.51 would produce categorically different concept inputs, despite
 4493 being indistinguishable within the margin of observational uncertainty ϵ_k^i . This
 4494 sensitivity to boundary placement is precisely the failure mode that the graded
 4495 membership representation is designed to avoid. Figure 6.4 contrasts this behavior
 4496 with the fuzzy membership approach adopted in this system.



4497

4498 Figure 6.4 Crisp threshold and fuzzy membership

4499 Crisp thresholding versus fuzzy membership functions for observations $z = 0.49$
 4500 and $z = 0.51$ are illustrated in Figure 6.4. Left: a $\Delta z = 0.02$ difference at the crisp
 4501 boundary at $z = 0.50$ produces a categorical output jump of $\Delta = 1.00$. Right: the
 4502 same input difference maps to a change of $\Delta(\mu_M) \approx 0.03$ under the fuzzy
 4503 representation, preserving continuity in the concept space.

4504 The fuzzy representation resolves this by assigning both values 0.49 and 0.51, to
 4505 partially overlapping linguistic terms. At $z = 0.49$ with parameters $(p_1, p_2, p_3) =$
 4506 $(0.20, 0.50, 0.80)$, the observation falls in the Low–Medium transition zone
 4507 $(p^1 < z < p^2)$, and the membership triplet evaluates to $(\mu_L, \mu_M, \mu_H) =$
 4508 $(0.033, 0.967, 0.000)$; at $z = 0.51$, the observation crosses into the Medium–High
 4509 transition zone $(p_2 < z < p_3)$, yielding to $(0.000, 0.967, 0.033)$. The concept
 4510 values computed from these two inputs differ by at most 0.02 in any weighted
 4511 average which is a smooth, bounded response to a small perturbation in the input.
 4512 This smoothness is formalized by the Lipschitz continuity of the full pipeline
 4513 established in Proposition 5.4, which guarantees that small observational
 4514 uncertainties produce proportionally small perturbations in the concept vector.
 4515 Crucially, the dominant membership μ_M remains unchanged at 0.967, and the
 4516 maximum change in any single component is $\max|\Delta\mu| \approx 0.03$, a smooth, bounded
 4517 response to a small perturbation in the input."

4518 This Lipschitz property has a direct operational consequence for the DSS: the
 4519 intervention guidance produced by the Local DSS Model is robust to sensor-level
 4520 perturbations in $O_k^i(x, y)$. The system does not exhibit decision discontinuities at
 4521 linguistic boundaries, and the degree of uncertainty in each feature is implicitly
 4522 encoded in the distribution of membership values across the three terms. A feature
 4523 observed with high confidence near its core region (e.g., $z_{k,1}^i = 0.90$, giving $\mu_H \approx$
 4524 1.0) produces a sharply concentrated concept contribution, while a feature with
 4525 ambiguous value near a boundary (e.g., $z_{k,1}^i = 0.50$, giving $\mu_M = 1.0, \mu_L = \mu_H = 0$)
 4526 produces a contribution that activates only the transitional term, naturally reducing
 4527 its influence on strongly-polarized decision rules.

4528 The concept abstraction layer therefore functions simultaneously as a dimensionality
 4529 reducer, a linguistic encoder, and an epistemic buffer translating the uncertain, high-
 4530 dimensional observational state into a compact, semantically interpretable, and
 4531 uncertainty-aware representation that the Local DSS Model can act upon without
 4532 requiring assumptions of precise measurement.

4533 6.2.4 Confidence-Gated Concept Modulation

4534 The concept vector $c_k^i(x, y)$ produced by the construction mappings of (62)–(65)
 4535 is derived from the fuzzified membership matrix, which in turn depends on the
 4536 observable information $O_k^i(x, y)$. However, the observation model of Section 6.1.2
 4537 introduces the observation confidence coefficient $conf_k^i(x, y) \in [0,1]$ in Equation (89),
 4538 which quantifies the reliability of the information available at each grid cell.
 4539 The concept construction mappings of Equations (62)–(65) do not yet incorporate
 4540 this reliability measure: they treat all cells identically regardless of whether their
 4541 underlying observations are highly reliable or severely degraded.

4542 To close this gap, the concept vector is modulated by a confidence-gating mechanism
 4543 that blends the computed concept values with a persistence-based prior in proportion
 4544 to the local observation confidence. This mechanism ensures that concept values

4545 derived from low-confidence observations are attenuated toward their most recent
 4546 reliable estimate, while concept values derived from high-confidence observations
 4547 are passed through without modification.

4548 **Definition 6.1 (Confidence-Gated Concept Vector).** For each concept index $h \in$
 4549 $\{1, 2, 3, 4\}$, region Ω_i , grid cell (x, y) , and time step k , the confidence-gated concept
 4550 value is defined as:

$$\bar{c}_{k,h}^i(x, y) = \text{conf}_k^i(x, y) \cdot c_{k,h}^i(x, y) + (1 - \text{conf}_k^i(x, y)) \cdot c_{prior,h}^i(x, y) \quad (94)$$

4551 where $\text{conf}_k^i(x, y)$ is the observation confidence coefficient of Equation (79),
 4552 $c_{k,h}^i(x, y)$ is the concept value computed by Equations (62)–(65), and $c_{prior,h}^i(x, y)$
 4553 is the concept prior, defined as:

$$c_{prior,h}^i(x, y) = \{\bar{c}_{k-1,h}^i(x, y), \text{ if } k \geq 1\} \quad (95)$$

$$c_{prior,h}^i(x, y) = \{0.5, \text{ if } k = 0\} \quad (96)$$

4554 The prior is the previous-step confidence-gated concept value when available, and
 4555 the neutral midpoint value 0.5 at the initial time step. This choice ensures that no
 4556 prior information bias is introduced at initialization, and that the persistence
 4557 mechanism carries forward the most recent reliable assessment when current
 4558 observations are degraded.

4559 **Remark 6.2 (Boundedness Preservation).** Since $\text{conf}_k^i(x, y) \in [0, 1]$,
 4560 $c_{k,h}^i(x, y) \in [0, 1]$ by Proposition 5.3, and $c_{prior,h}^i(x, y) \in [0, 1]$ by induction, the
 4561 confidence-gated concept value satisfies $\bar{c}_{k,h}^i(x, y) \in [0, 1]$ for all k, h, i , and (x, y) .
 4562 The boundedness guarantee of Property 6.1 is therefore preserved.

4563 **Remark 6.3 (Operational Interpretation).** When $\text{conf}_k^i(x, y) = 1$, the
 4564 observation is fully reliable, and the concept value passes through unmodified:
 4565 $\bar{c}_{k,h}^i(x, y) = c_{k,h}^i(x, y)$. When $\text{conf}_k^i(x, y) = 0$, the observation carries no
 4566 information and the system falls back entirely to the previous assessment:
 4567 $\bar{c}_{k,h}^i(x, y) = c_{prior,h}^i(x, y)$. In intermediate cases, the mechanism produces a
 4568 smooth blend that degrades gracefully with decreasing confidence, consistent with

the Lipschitz continuity established in Proposition 5.4. The confidence-gated concept vector is therefore:

$$\bar{\mathbf{c}}_k^i(x, y) = [\bar{c}_{k,1}^i, \bar{c}_{k,2}^i, \bar{c}_{k,3}^i, \bar{c}_{k,4}^i]^T \in [0,1]^4 \quad (97)$$

All subsequent stages of the DSS pipeline, specifically the fuzzy inference of Section 6.3 and the decision quality evaluation of Section 6.4.4, operate on the confidence-gated concept vector $\bar{\mathbf{c}}_k^i(x, y)$ rather than the raw concept vector $\mathbf{c}_k^i(x, y)$. This substitution is transparent to the downstream stages because the boundedness, partition-of-unity compatibility, and dimensionality reduction properties established in Section 6.2.3 are all preserved under the gating operation.

6.3 Local Decision Generation and Action Aggregation

The Local DSS Model constitutes the inferential core of each local agent Agent_i . Receiving the concept vector $\mathbf{c}_k^i(x, y)$ produced by the Concept Mapping and Abstraction Layer at every active grid cell, it executes two successive operations: a fuzzy inference pass that evaluates the intervention decision rule bases $\{\pi_{i,m}\}_{m=1}^M$ defined in Section 5.3.2, and an action aggregation step that combines the resulting candidates through the function f_{agg} (Section 5.3.3) into a single executable intervention parameter $U_{DSS,k}^i(x, y)$. The output of this section is the per-cell regional decision field that is passed to the Global Coordination Layer described in Section 6.4. The rule base structure and aggregation weights are defined in Sections 5.3.1–5.3.3; this section describes the inference execution, activation propagation, and output assembly within the local agent architecture.

Figure 6.5 illustrates the two-stage structure of the Local DSS Model. The concept vector $\mathbf{c}_k^i(x, y)$ from Section 6.2 fans into three parallel fuzzy rule bases, each producing a candidate intervention degree $u_{k,m}^i$. The action aggregation function f_{agg} combines these into the local output $U_{DSS,k}^i(x, y)$ passed to Global DSS described in Section 6.4.

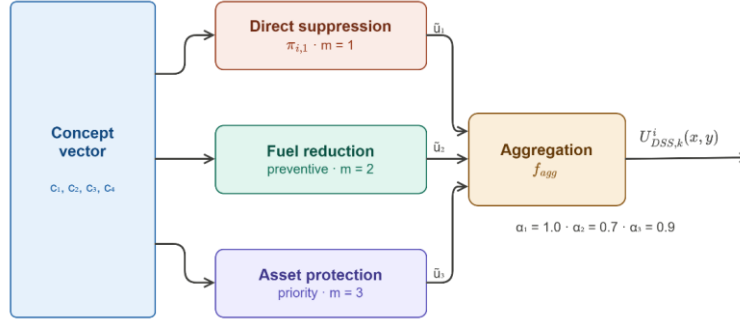


Figure 6.5 Runtime inference pipeline of the Local DSS Model

6.3.1 Fuzzy Inference over the Concept Vector

At each time step k , the inference engine processes every active cell $(x, y) \in \Omega_i$ independently. For each intervention type m , the engine traverses the $3^4 = 81$ -rule base $\pi_{i,m}$ and evaluates the activation strength of each rule R given the current concept vector $\mathbf{c}_k^i(x, y) = \{c_1, c_2, c_3, c_4\}$.

6.3.1.1 Rule Activation

Each rule $R \in \pi_{i,m}$ has an antecedent that specifies a linguistic term $n_h \in \{L, M, H\}$ for each of the four concepts. The activation strength α_R is computed via the min-operator conjunction defined in (81), applying the piecewise-linear membership functions used during fuzzification (Section 5.2.2) directly to concept values. This reuse is valid because concept values are bounded in $[0, 1]$ by Property 6.1.

6.3.1.2 Aggregation to Candidate Degree

Each rule carries a singleton consequent value v_R mapped from the five output terms defined in Section 5.3.2. The candidate intervention degree $u_{k,m}^i(x, y)$ is computed

4610 as the activation-weighted average over all rules in $\pi_{i,m}$ (80). The denominator is
4611 strictly positive because the partition-of-unity property (Property 6.2) guarantees that
4612 every concept value activates at least one linguistic term with nonzero degree, and
4613 the full 81-rule base ensures complete antecedent coverage.

4614 This inference pass executes independently for each of the three intervention types,
4615 yielding three candidate degrees $u_{k,1}^i, u_{k,2}^i, u_{k,3}^i$ at every active cell (x, y) . The
4616 principal rules driving each intervention type under operationally critical scenarios
4617 are specified in Table 5.12, Table 5.14, and Table 5.16, respectively.

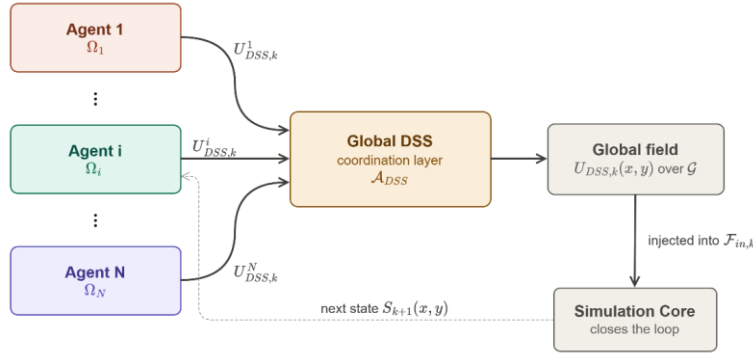
4618 6.3.2 Action Aggregation and Local Output Formation

4619 The three candidate degrees may recommend contradictory resource allocations for
4620 the same grid cell, and their combined physical realization must remain within the
4621 feasible intervention space $\mathcal{U}_{DSS}$. The action aggregation function f_{agg} resolves
4622 these conflicts through the priority-weighted combination defined in Equation (70)
4623 with the default weights specified in Equation (71). The weighted average
4624 formulation preserves $U_{DSS,k}^i(x, y) \in [0,1]$ and ensures that the aggregated output
4625 responds continuously and proportionally to changes in any individual candidate
4626 degree, avoiding the hard discontinuities that would result from a maximum- or
4627 threshold-based selector.

4628 A key architectural property of this formulation is its interpretability. As noted in
4629 Remark 5.5, the provenance of any aggregated output $U_{DSS,k}^i(x, y)$ can be traced
4630 back through the three candidate degrees and the rule evaluations that produced
4631 them, satisfying the explainability requirement of Contribution C4 from Chapter 1.
4632 Each layer of the computation, from concept values to rule activations to candidate
4633 degrees to aggregated output, is expressed in the same $[0, 1]$ space, enabling a
4634 consistent and human-readable audit trail.

4635 The result of Section 6.3 is the per-cell decision field $U_{DSS,k}^i(x,y)$ over the region
 4636 Ω_i . This field represents the local agent's intervention recommendation at every
 4637 active grid cell at time step k , expressed as a normalized intensity in the feasible
 4638 intervention space. It is passed to the Global Coordination Layer in Section 6.4,
 4639 which aggregates the per-region fields $\{U_{DSS,k}^i\}_{i=1}^N$ across all agents into the global
 4640 intervention field $U_{DSS,k}(x,y)$ that is injected into the simulation.

4641 6.4 Global Coordination and Spatial Field Generation



4642

4643 Figure 6.6 Global coordination pipeline

4644 At each time step k , the N local agents deliver their per-region intervention fields
 4645 $\{U_{DSS,k}^i(x,y)\}_{i=1}^N$ to the Global DSS coordination layer. The purpose of this layer is
 4646 twofold: to assemble these disjoint regional fields into a single spatially complete
 4647 intervention field $U_{DSS,k}(x,y)$ defined over the full domain $\mathcal{G}$, and to resolve any
 4648 resource conflicts that arise when the aggregate intervention demand exceeds the
 4649 available physical capacity. Both operations are deterministic; $\mathcal{A}_{DSS}$ itself is non-
 4650 inferential, with all fuzzy reasoning performed upstream by the local agents. The
 4651 output of this layer is the global intervention field that is injected into the external
 4652 input set $\mathcal{F}_{in,k}$, closing the feedback loop with the Simulation Core as illustrated in
 4653 Figure 6.6, summarizing the global coordination. The formal definition of f_{agg} and

the decision quality function f_{eval} are given in Sections 5.3.3 and 5.3.4; this section specifies the spatial assembly procedure, resource normalization pass, and feedback injection mechanism.

Definition 6.2 (Global Coordination and Spatial Assembly Mechanism). The global coordination mechanism $\mathcal{A}_{DSS}$ maps the collection of N regional intervention fields to a single spatially complete, resource-feasible global intervention field. It is defined as the sequential composition of three operations (98) where $\mathcal{A}_{DSS}^{sp}(\cdot)$ is the **spatial assembly operator** and $\mathcal{N}_{Res,k}(\cdot)$ is the **resource normalization operator**, defined as follows.

$$\begin{aligned} U_{DSS,k}(x, y) &= \mathcal{A}_{DSS}(\{U_{DSS,k}^i(x, y)\}_{i=1}^N) \\ &= \mathcal{N}_{Res,k}(\mathcal{A}_{DSS}^{sp}(\{U_{DSS,k}^i(x, y)\}_{i=1}^N)) \end{aligned} \quad (98)$$

Spatial Assembly: For each grid cell $(x, y) \in \mathcal{G}$, the assembled field is constructed via the bijective partition assignment (99).

$$\mathcal{A}_{DSS}^{sp}(\{U_{DSS,k}^i\}_{i=1}^N)(x, y) = \begin{cases} U_{DSS,k}^i(x, y) & (x, y) \in \Omega_i, i = 1, \dots, N \\ 0 & (x, y) \notin \bigcup_{i=1}^N \Omega_i \end{cases} \quad (99)$$

Resource Normalization $\mathcal{N}_{Res,k}$: Let $D_k = \sum_{(x,y) \in \mathcal{G}} U_{DSS,k}(x, y)$ denote the integrated intervention demand and C_k the available capacity. The normalization operator is (100).

$$\mathcal{N}_{Res,k}(U_{DSS,k}(x, y)) = \begin{cases} U_{DSS,k}(x, y) & D_k \leq C_k \\ U_{DSS,k}(x, y) \cdot \frac{C_k}{D_k} & D_k > C_k \end{cases} \quad (100)$$

$\mathcal{A}_{DSS}$ produces per-intervention-type global fields through the type decomposition (101) applying the same spatial assembly and normalization operations independently for each intervention type m , as specified in Equation (93).

$$U_{DSS,k}^m(x, y) = \mathcal{A}_{DSS}(\{u_{k,m}^i(x, y)\}_{i=1}^N), m = 1, \dots, M \quad (101)$$

Remark 6.4 (Boundedness of $\mathcal{A}_{DSS}$). The output satisfies $U_{DSS,k}(x, y) \in [0, 1]$ everywhere. The assembly step inherits boundedness from the regional fields

4673 ($U_{DSS,k}^i \in [0,1]$ by Proposition 5.3). The normalization step multiplies by $\frac{c_k}{d_k} \leq 1$
 4674 when triggered, which cannot increase any cell value above 1, and preserves non-
 4675 negativity.

4676 **Remark 6.5 (Spatial Completeness).** The assembled field is defined over the entire
 4677 domain $\mathcal{G}$. Every cell receives exactly one regional value by the partition property
 4678 $\mathcal{G} = \bigcup_{i=1}^N \Omega_i$, $\Omega_i \cap \Omega_j = \emptyset$ for $i \neq j$. No cell is left undefined.

4679 The following subsections formalize the three operational steps of the $\mathcal{A}_{DSS}$
 4680 mechanism. The N local agents deliver per-region intervention fields $U_{DSS,k}^i(x, y)$ to
 4681 the Global DSS coordination layer, which assembles and normalizes them into the
 4682 global intervention field $U_{DSS,k}(x, y)$ over the full domain $\mathcal{G}$. This field is injected
 4683 into the external input set $\mathcal{F}_{in,k}$, closing the feedback loop with the Simulation Core.

4684 6.4.1 Spatial Assembly of Regional Fields

4685 The spatial partition of the operational domain satisfies $\mathcal{G} = \bigcup_{i=1}^N \Omega_i$, $\Omega_i \cap \Omega_j = \emptyset$
 4686 for $i \neq j$, as established in Section 2.4. Because every active grid cell (x, y) belongs
 4687 to exactly one region Ω_i , the base assembly step of $\mathcal{A}_{DSS}$ is a direct bijective
 4688 assignment (102):

$$U_{DSS,k}(x, y) = U_{DSS,k}^i(x, y) \text{ for all } (x, y) \in \Omega_i, i = 1, \dots, N \quad (102)$$

4689 This direct assembly is well-defined and produces a globally complete field without
 4690 ambiguity or overlap, by virtue of the partition property. The assembled field inherits
 4691 the boundedness property from its constituent regional fields: since $U_{DSS,k}^i(x, y) \in$
 4692 $[0,1]$ for all i and all (x, y) by the action aggregation guarantee of Section 6.3, the
 4693 global field satisfies $U_{DSS,k}(x, y) \in [0,1]$ everywhere.

4694 In addition to the scalar global field, the assembly step produces action-type-specific
 4695 global fields $U_{DSS,k}^m(x, y)$ for $m = 1, \dots, M$ by decomposing each agent's candidate
 4696 degrees prior to aggregation, as specified in Equation (93). These per-type fields

4697 support intervention-specific resource allocation in the simulation layer and serve as
 4698 inputs to the explainability layer described in Section 6.4.5.

4699 6.4.2 Resource Normalization and Conflict Resolution

4700 The direct assembly of Section 6.4.1 guarantees spatial completeness but not
 4701 physical feasibility. The aggregate intervention intensity requested across all regions
 4702 may exceed the total available suppression capacity C_k defined by the resource data
 4703 layer $U_{Res,k}$. A normalization pass is therefore applied following assembly.

4704 Let D_k denote the integrated intervention intensity over the domain (103):

$$D_k = \sum_{(x,y) \in \mathcal{G}} U_{DSS,k}(x,y) \quad (103)$$

4705 If $D_k \leq C_k$, the assembled field is feasible and is passed to the simulation unchanged.

4706 If $D_k > C_k$, the field is scaled uniformly by the ratio $\frac{C_k}{D_k}$ (104):

$$U_{DSS,k}(x,y) \leftarrow U_{DSS,k}(x,y) \cdot \frac{C_k}{D_k} \quad (104)$$

4707 This uniform scaling preserves the relative spatial distribution of intervention
 4708 intensity (i.e., cells that were assigned higher priority by the local agents continue to
 4709 receive proportionally more intervention capacity after normalization) while
 4710 ensuring that the total allocated resources remain within the physically available
 4711 budget. The normalized field continues to satisfy $U_{DSS,k}(x,y) \in [0,1]$ since both
 4712 $U_{DSS,k}(x,y) \leq 1$ and $\frac{C_k}{D_k} \leq 1$, when normalization is triggered.

4713 Severity-based prioritization is embedded upstream in the local fuzzy inference of
 4714 Section 6.3, where the candidate intervention $u_{k,m}^i(x,y)$ is already monotonic in the
 4715 severity concepts $c_{k,1}$, $c_{k,3}$, and $c_{k,4}$. Uniform capacity scaling therefore preserves,
 4716 rather than overrides, the severity-weighted prioritization produced by the regional
 4717 agents; hard priority cutoffs that forcibly zero low-severity cells under capacity stress

4718 are outside the scope of the non-inferential coordination layer and are accessible
4719 through the operator override interface of Section 6.5.3.

4720 The uniform scaling of Equation (104) ensures aggregate feasibility but does not
4721 distinguish between intervention types that may have independent resource pools or
4722 heterogeneous capacity constraints. When the operational scenario involves distinct
4723 resource categories (e.g., aerial assets for direct suppression versus ground crews for
4724 fuel management versus evacuation support units), a per-action-type conflict
4725 resolution step provides finer-grained capacity enforcement.

4726 For each intervention type $m \in \{1, \dots, M\}$, define the type-specific integrated
4727 demand over the active domain:

$$D_{k,m} = \sum_{(x,y) \in \mathcal{G}} U_{DSS,k}^m(x,y) \quad (105)$$

4728 and the type-specific available capacity, allocated from the total capacity C_k
4729 proportionally to the priority weights α_m of Equation (71):

$$C_{k,m} = C_k \cdot \bar{\alpha}_m, \bar{\alpha}_m = \frac{\alpha_m}{\sum_{r=1}^M \alpha_r} \quad (106)$$

4730 The conflict ratio for each intervention type is (107) where $\epsilon > 0$ is a small
4731 numerical stabilizer.

$$\chi_{k,m} = \min\left(1, \frac{C_{k,m}}{D_{k,m} + \epsilon}\right) \quad (107)$$

4732 The conflict-aware per-type field is (108).

$$\hat{u}_{k,m}(x,y) = \chi_{k,m} \cdot U_{DSS,k}^m(x,y) \quad (108)$$

4733 When $\chi_{k,m} = 1$, the demand for action type m is within capacity and no scaling is
4734 applied. When $\chi_{k,m} < 1$, all cells requesting that action type are proportionally
4735 reduced, preserving the relative spatial distribution of the intervention within each
4736 type while ensuring type-level feasibility.

4737 **Remark 6.6 (Relation to Scalar Normalization).** If a single shared capacity pool
 4738 is used (i.e., all intervention types draw from the same resource), the per-type
 4739 formulation reduces to the scalar normalization of Equation (104) when the priority
 4740 weights are uniform. The per-type extension is therefore a strict generalization that
 4741 collapses to the scalar case under homogeneous resource assumptions.

4742 **Remark 6.7 (Well-Posedness and Structural Properties of $\mathcal{A}_{DSS}$).** The global
 4743 coordination operator $\mathcal{A}_{DSS}$ defined in Definition 6.2 satisfies three structural
 4744 properties that ensure its well-posedness as a single-step spatial assembly
 4745 mechanism.

4746 *Existence and uniqueness.* For any collection of regional fields $\{U_{DSS,k}^i(x,y)\}_{i=1}^N$
 4747 with $U_{DSS,k}^i \in [0,1]$ for all i and all $(x,y) \in \Omega_i$, the output $U_{DSS,k}(x,y)$ exists and is
 4748 unique for every $(x,y) \in \mathcal{G}$. Existence follows from the completeness of the partition
 4749 $\mathcal{G} = \bigcup_{i=1}^N \Omega_i$ (every cell is assigned). Uniqueness follows from the disjointness $\Omega_i \cap$
 4750 $\Omega_j = \emptyset$ for $i \neq j$ (no cell receives conflicting assignments). Together, these
 4751 guarantee that $\mathcal{A}_{DSS}^{sp}$ is a well-defined function, not a relation.

4752 *Input-output stability.* The operator $\mathcal{A}_{DSS}$ is non-expansive with respect to the L^∞
 4753 norm over the spatial domain:

$$\| U_{DSS,k} \|_\infty \leq \max_i \| U_{DSS,k}^i \|_\infty \leq 1 \quad (109)$$

4754 The assembly step is a direct copy (non-expansive by construction), and the
 4755 normalization step multiplies by a factor $\frac{C_k}{D_k} \leq 1$ when triggered, which can only
 4756 reduce the supremum. Small perturbations in any regional field $U_{DSS,k}^i$ propagate to
 4757 the global field with unit gain (assembly) and at most unit gain (normalization),
 4758 ensuring that the operator does not amplify input uncertainty.

4759 *Degenerate case behavior.* When all regional fields are identically zero
 4760 ($U_{DSS,k}^i(x,y) = 0$ for all i and all (x,y)), the global field is zero and the
 4761 normalization step is not triggered ($D_k = 0 \leq C_k$), producing the well-defined no-
 4762 intervention output. When a single region Ω_i produces nonzero output while all

4763 others are zero, the global field inherits this regional output without distortion. When
 4764 the aggregate demand D_k exceeds capacity C_k by an arbitrarily large factor, the
 4765 normalization ratio C_k/D_k approaches zero smoothly, producing a uniformly
 4766 attenuated field rather than a discontinuous rejection.

4767 It is important to note that $\mathcal{A}_{DSS}$ is a single-step mapping executed once per time
 4768 step k , not an iterative procedure. Consequently, convergence in the classical fixed-
 4769 point sense does not apply. The closed-loop dynamics of the overall system, in which
 4770 the DSS output influences the next state S_{k+1} and thereby the next observation O_{k+1} ,
 4771 are characterized by the Lipschitz continuity of π_{DSS}^i (Proposition 5.4) and the
 4772 boundedness of the state transition operator Φ established in Chapter 4, which
 4773 together ensure that the feedback loop does not exhibit unbounded amplification.

4774 6.4.3 Feedback Injection and Loop Closure

4775 The normalized global intervention field $U_{DSS,k}(x, y)$ is injected into the external
 4776 input set $\mathcal{F}_{in,k}$ of the Simulation Core, as described in Section 4.2.1. Through the
 4777 suppression function f_{supp} , the field is translated into physical intervention
 4778 parameters (i.e., suppression capacity, fuel reduction rate, and resource reallocation)
 4779 that influence the state transition $S_{k+1}(x, y) = \Phi(S_k(x, y), \mathcal{F}_{in,k})$ at the next step.

4780 This injection closes the feedback loop introduced in Section 6.1.5: the wildfire state
 4781 evolves in response to the DSS-generated intervention, the updated state produces
 4782 new observations through the Observation Layer, and the cycle repeats. The global
 4783 field thereby constitutes the sole interface between the cognitive DSS layer and the
 4784 physical simulation layer, ensuring that all intervention recommendations
 4785 (regardless of which local agent generated them) reach the simulation through a
 4786 single, normalized, and globally consistent channel.

6.4.4 Decision Quality Evaluation

Following the injection of the global intervention field into the Simulation Core, the decision quality evaluation function f_{eval} computes the scalar quality score $Q_{DSS,k} \in [0,1]$ introduced in Chapter 2. This score measures how well the generated intervention $U_{DSS,k}(x, y)$ satisfies the four operational criteria identified in Section 5.1.4 and determines whether the satisfaction condition $Q_{DSS,k} \geq \eta_k$ of (1) is met.

Definition 6.3 (Decision Quality Evaluation Function). *The multi-criteria quality score is defined as the weighted average of four partial criterion scores (110) where $w_j > 0$, $\sum_{j=1}^4 w_j = 1$, each partial score $q_{j,k} \in [0,1]$ measures local satisfaction over the active domain $\mathcal{G}_k^+ = \{(x, y) \in \mathcal{G} : B_k(x, y) = 1 \text{ or } c_{k,1}^i(x, y) > 0\}$.*

$$Q_{DSS,k} = f_{eval}(O_k(x, y), \mathcal{F}_{DSS,k}, U_{DSS,k}(x, y)) = \sum_{j=1}^4 w_j \cdot q_{j,k} \quad (110)$$

Each partial score is defined as the spatially averaged local satisfaction between the intervention field and the corresponding concept value (111)

$$q_{j,k} = \frac{1}{|\mathcal{G}_k^+|} \sum_{(x,y) \in \mathcal{G}_k^+} (1 - |u_{k,m_j}^i(x, y) - c_{k,h_j}^i(x, y)|) \quad (111)$$

The mapping from criterion j to concept index h_j and the default weights are specified in Table 5.18. The table specifies, for each criterion j , both the concept index h_j and the intervention-type index $m_j \in \{1, \dots, M\}$ against which satisfaction is measured. The scalar $q_{j,k}$ therefore compares per-type intervention components to the corresponding concept field, preserving type-consistency. The local satisfaction term $1 - |U_{DSS,k}(x, y) - c_{k,h_j}^i(x, y)|$ reaches its maximum value of 1 when the intervention intensity exactly matches the concept activation level and decreases linearly as the two diverge. This formulation is consistent with the satisfaction-based decision paradigm of Chapter 2: a decision is fully satisfactory at a cell when the

4808 deployed intervention degree equals the situational need encoded in the
4809 corresponding concept.

4810 **Remark 6.8 (Boundedness of $Q_{DSS,k}$).** Since $U_{DSS,k}(x, y) \in [0,1]$ and $c_{k,h_j}^i(x, y) \in$
4811 $[0,1]$ by construction, the absolute difference $|U_{DSS,k} - c_{k,h_j}^i| \in [0,1]$, hence each
4812 local satisfaction term lies in $[0,1]$. Each $q_{j,k}$ is an average of values in $[0,1]$, so
4813 $q_{j,k} \in [0,1]$. The weighted average $Q_{DSS,k}$ hence satisfies $Q_{DSS,k} \in [0,1]$ for all k .

4814 **Definition 6.4 (Adaptive Satisfaction Threshold).** *The time-varying threshold η_k*
4815 *is designed to become stricter when the operational environment permits higher-*
4816 *quality decisions and to relax when conditions are degraded or the incident demands*
4817 *rapid response under uncertainty. Define three global context indicators:*

4818 **(i) Mean observation confidence:** the spatial average of the confidence coefficient
4819 over the active domain:

$$\bar{\rho}_k = \frac{1}{|\mathcal{G}_k^+|} \sum_{(x,y) \in \mathcal{G}_k^+} conf_k^i(x, y) \quad (112)$$

4820 **(ii) Mean resource sufficiency:** the ratio of available capacity to aggregate demand,
4821 clamped to $[0,1]$:

$$\bar{s}_k^* = \min\left(1, \frac{C_k}{D_k + \epsilon}\right) \quad (113)$$

4822 **(iii) Mean operational severity:** the spatial average of the threat, exposure, and
4823 urgency concepts:

$$\bar{c}_{sev,k} = \frac{1}{|\mathcal{G}_k^+|} \sum_{(x,y) \in \mathcal{G}_k^+} \frac{\bar{c}_{k,1}^i + \bar{c}_{k,3}^i + \bar{c}_{k,4}^i}{3} \quad (114)$$

4824 The extended adaptive threshold is then defined as:

$$\eta_k = \eta_{min} + (\eta_{max} - \eta_{min}) \cdot [\gamma_1 \cdot \bar{\rho}_k + \gamma_2 \cdot \bar{s}_k^* + \gamma_3 \cdot (1 - \bar{c}_{sev,k})] \quad (115)$$

subject to

$$\begin{aligned}\gamma_r &\geq 0 \\ \sum_{r=1}^3 \gamma_r &= 1 \\ 0 &< \eta_{min} < \eta_{max} < 1\end{aligned}$$

4825 When mean observation confidence $\bar{\rho}_k$ is high, the DSS has access to reliable
 4826 information and the threshold rises, demanding higher decision quality. When mean
 4827 resource sufficiency $\bar{s}_k^*$ is high, operational slack is available and the threshold rises.
 4828 When mean operational severity $\bar{c}_{sev,k}$ is high, the $(1 - \bar{c}_{sev,k})$ term decreases,
 4829 lowering the threshold to permit the acceptance of imperfect but urgent decisions
 4830 during crisis phases.

4831 **Remark 6.9 (Reduction to Urgency-Driven Threshold).** Setting $\gamma_1 = \gamma_2 = 0$,
 4832 $\gamma_3 = 1$, $\eta_{min} = 0$, and $\eta_{max} = 1$, and substituting the urgency-based severity
 4833 with the pure urgency average, the extended threshold reduces to the urgency-driven
 4834 threshold of Equation (115). The extended formulation is therefore a strict
 4835 generalization. The threshold satisfies $\eta_k \in [\eta_{min}, \eta_{max}]$ for all k , ensuring that the
 4836 satisfaction condition $Q_{DSS,k} \geq \eta_k$ remains well-defined and bounded.

4837 6.4.5 Decision Acceptance Rule and Fail-Safe Policy

4838 The satisfaction condition $Q_{DSS,k} \geq \eta_k$ is evaluated through Definition 6.3
 4839 determines whether the generated intervention field $U_{DSS,k}(x, y)$ is deemed
 4840 operationally adequate. When the condition is met, the intervention field is injected
 4841 into the Simulation Core without modification. When the condition is not met, the
 4842 system must determine how to proceed: a full rejection of the intervention field
 4843 would create a dangerous operational gap, while unconditional acceptance of a low-
 4844 quality decision undermines the purpose of the evaluation layer. To resolve this, a
 4845 soft fail-safe policy is defined that attenuates the generated intervention toward a
 4846 conservative baseline in proportion to the degree of quality shortfall.

4847 **Definition 6.5 (Decision Acceptance and Fail-Safe).** *The final intervention field*
 4848 *injected into $\mathcal{F}_{in,k}$ is defined as (116) and (117) where the attenuation coefficient*
 4849 *is (118) and $U_{safe,k}(x, y)$ is a conservative baseline intervention defined for the*
 4850 *three-action configuration ($M = 3$) as (119).*

$$U_{DSS,k}^{final}(x, y) = U_{DSS,k}(x, y), \text{ if } Q_{DSS,k} \geq \eta_k \quad (116)$$

$$U_{DSS,k}^{final}(x, y) = \zeta_k \cdot U_{DSS,k}(x, y) + (1 - \zeta_k) \cdot U_{safe,k}(x, y) \quad (117)$$

if $Q_{DSS,k} < \eta_k$

$$\zeta_k = \frac{Q_{DSS,k}}{\eta_k + \epsilon} \in [0,1] \quad (118)$$

$$U_{safe,k}(x, y) = [\bar{c}_{k,4}^i(x, y), 0, \bar{c}_{k,3}^i(x, y)]^T \quad (119)$$

4851 The three components correspond to: ($m = 1$) direct suppression, set to the urgency
 4852 concept $\bar{c}_{k,4}^i$ to concentrate intervention effort where urgency is highest; ($m = 2$)
 4853 preventive fuel reduction, set to zero under fail-safe conditions to conserve
 4854 resources; and ($m = 3$) asset protection, set to the asset exposure risk concept $\bar{c}_{k,3}^i$
 4855 to ensure that the most exposed assets receive protective attention regardless of
 4856 overall decision quality.

4857 **Remark 6.10 (Graduated Response).** When $Q_{DSS,k}$ is close to η_k , the attenuation
 4858 coefficient ζ_k is close to 1 and the intervention field is only marginally modified.
 4859 When $Q_{DSS,k}$ is near zero, ζ_k approaches zero and the system falls back almost
 4860 entirely to the conservative baseline. This response avoids both the operational
 4861 paralysis of binary rejection and the risk of executing a severely deficient decision.

4862 **Remark 6.11 (Boundedness).** Since $\zeta_k \in [0,1]$, $U_{DSS,k}(x, y) \in [0,1]$, and
 4863 $U_{safe,k}(x, y) \in [0,1]$ by the boundedness of the concept values, the final
 4864 intervention field satisfies $U_{DSS,k}^{final}(x, y) \in [0,1]$ for all (x, y) and all k .

4865 6.5 Explainability and AI Governance Layer

4866 Explainability and governance mechanisms are not embedded directly within the
4867 core decision-generation pipeline. Instead, they are treated as an auxiliary
4868 supervisory layer that can be integrated on top of the DSS architecture without
4869 modifying the underlying reasoning structure. This design choice reflects the
4870 principle of separation of concerns: the reasoning mechanisms of Sections 6.2
4871 through 6.4 operate independently of any governance logic, ensuring that
4872 explainability features can be added, modified, or disabled without affecting the
4873 functional correctness of the decision output.

4874 The layer operates by inspecting the intermediate outputs produced at each stage of
4875 the pipeline (i.e., the concept vector $\mathbf{c}_k^i(x, y)$ from Section 6.2, the candidate
4876 intervention degrees $\{u_{k,m}^i\}$ from Section 6.3, and the normalized global field
4877 $U_{DSS,k}(x, y)$ from Section 6.4) and generating human-readable justifications for the
4878 intervention recommendations issued at each time step. Because every computation
4879 in the pipeline is expressed in the same $[0, 1]$ space with linguistic variables, the
4880 provenance of any output can be traced back to the specific feature values and rule
4881 activations that produced it.

4882 **Remark 6.12 (Scope of the Explainability Guarantee).** The explainability
4883 guarantee provided by this layer is rule-level per decision. For any generated
4884 intervention at cell (x, y) and time step k , the activated rules and concept values are
4885 recoverable through the traceability mechanism of Section 6.5.1. It is not a systemic
4886 claim that the full 351-rule base is holistically comprehensible by a single human
4887 operator at a single point in time. Contribution C4 therefore refers to per-decision
4888 traceability and auditability, not to a cognitive claim about rule-base-wide human
4889 comprehensibility.

4890 The governance layer addresses three distinct requirements identified in Chapter 1
4891 (C4):

Traceability ensures that every intervention recommendation can be traced back through the rule activations, concept values, and feature observations that produced it. Given the fuzzy inference structure of Section 6.3, this traceability is intrinsic: the activation strength α_R of each rule R and its singleton output v_R are individually inspectable at every time step and grid cell.

Auditability provides post-hoc access to the decision record across the full simulation timeline. The governance layer maintains a structured log of concept vectors, rule activations, and intervention outputs, enabling incident commanders and analysts to review how and why particular intervention recommendations were generated at any point during a wildfire event.

Override and constraint mechanisms allow human operators to impose hard constraints or soft preferences on the intervention field prior to injection into $\mathcal{F}_{in,k}$. These mechanisms operate downstream of the global coordination layer and do not alter the DSS reasoning process; they act as a final filter between the system's autonomous recommendation and the physical simulation.

The specific implementation choices for each of these components, which includes the log structure, the natural-language generation method for traceability reports, and the interface design for operator overrides, are subject to ongoing discussion and will be elaborated in a subsequent revision of this section following advisory consultation.

6.5.1 Traceability Mechanism

The traceability mechanism provides a cell-level causal chain from the final intervention output back through the intermediate computational stages that produced it. At each time step k and for each grid cell (x,y) , the mechanism identifies the dominant rule that contributed most strongly to each intervention type.

4917 **Definition 6.6 (Dominant Explanation Rule).** For each intervention type $m \in$
 4918 $\{1, \dots, M\}$ and each grid cell (x, y) , the dominant explanation rule is the rule with
 4919 the highest activation strength in the corresponding rule base:

$$R_m^*(x, y) = \operatorname{argmax}_{R \in \pi_{i,m}} \alpha_R(x, y) \quad (120)$$

4920 where $\alpha_R(x, y)$ is the activation strength of rule R computed by Equation (81). In
 4921 the case of ties, the rule with the more extreme consequent value v_R (i.e., the rule
 4922 whose output is furthest from the neutral midpoint 0.5) is selected, as it has a larger
 4923 influence on the aggregated candidate degree.

4924 For each dominant rule $R_m^*(x, y)$, the traceability report includes: (a) the four
 4925 antecedent linguistic terms of the rule and their corresponding concept membership
 4926 values, identifying which concept conditions triggered the rule; (b) the consequent
 4927 value v_R , identifying the recommended intervention intensity; and (c) the two
 4928 highest-contributing concept-term pairs by membership magnitude, providing a
 4929 concise natural-language explanation of the form: “**Rule activated primarily**
 4930 **because [Concept A] is [Term] and [Concept B] is [Term].**”

4931 This structure enables field operators to inspect any grid cell in the intervention map
 4932 and obtain a human-readable explanation of why that particular intervention level
 4933 was recommended, grounded in the specific feature observations, concept values,
 4934 and rule activations at that cell.

4935 **Example 6.1 (Traceability Report for a High-Threat Scenario).** To illustrate the
 4936 traceability mechanism concretely, consider a grid cell (x, y) at time step k with the
 4937 following observed concept vector: $c_1 = 0.82$ (Fire Threat Level: High), $c_2 = 0.25$
 4938 (Suppression Feasibility: Low), $c_3 = 0.71$ (Asset Exposure Risk: High), $c_4 = 0.68$
 4939 (Intervention Urgency: Medium–High). The traceability chain for intervention type
 4940 $m = 1$ (Direct Suppression) proceeds as follows.

4941 **Step 1: Concept-level fuzzification.** Each concept value is fuzzified over the same
 4942 three-term design (Equation 53). For $c_1 = 0.82$ with parameters (0.20, 0.50, 0.80):
 4943 $\mu_L = 0, \mu_M = 0, \mu_H = 1.0$. For $c_2 = 0.25$ with parameters (0.30, 0.60, 0.85):

4944 $\mu_L = 1.0, \mu_M = 0, \mu_H = 0$. For $c_3 = 0.71$ with parameters (0.15, 0.45, 0.75):
 4945 $\mu_L = 0, \mu_M \approx 0.13, \mu_H \approx 0.87$. For $c_4 = 0.68$ with parameters (0.30, 0.60, 0.85):
 4946 $\mu_L = 0, \mu_M \approx 0.68, \mu_H \approx 0.32$.

4947 **Step 2: Dominant rule identification.** The rule with antecedent ($c_1 = H, c_2 = L, c_3$
 4948 $= H, c_4 = M$) fires with activation strength $\alpha_R = \min(1.0, 1.0, 0.87, 0.68) = 0.68$.
 4949 Per Definition 6.6, this is identified as R_1^* with consequent value $v_R = 0.25$ (low
 4950 intervention recommended), reflecting the rule logic that high fire threat combined
 4951 with low suppression feasibility warrants *reduced* direct suppression to avoid
 4952 committing resources to infeasible positions.

4953 **Step 3: Top-2 contributing concepts.** The two highest-contributing concept-term
 4954 pairs by membership magnitude are (c_1, H) with $\mu = 1.0$ and (c_2, L) with $\mu = 1.0$.
 4955 These two concepts jointly determine the dominant rule's activation.

4956 **Step 4: Natural-language explanation.** The traceability mechanism assembles the
 4957 following explanation string from the dominant rule's structure:

4958 *"Direct Suppression at cell (x, y) is recommended at intensity 0.25*
 4959 *(low). Rule activated primarily because Fire Threat Level is High (μ*
 4960 *$= 1.00$) and Suppression Feasibility is Low ($\mu = 1.00$). Despite severe*
 4961 *fire conditions, resource constraints preclude effective direct*
 4962 *intervention at this location."*

4963 This explanation is operationally meaningful: it not only states what the DSS
 4964 recommends but explains *why* through the concept values that drove the decision. A
 4965 field commander reading this output can immediately verify whether the system's
 4966 assessment of fire threat and suppression feasibility matches ground-truth
 4967 conditions, and can exercise the operator override mechanism (Section 6.5.3) if the
 4968 assessment is judged to be inconsistent with field observations.

4969 **Definition 6.7 (Explanation Function).** For each intervention type m and grid cell
 4970 (x, y) , the explanation function E_m maps the dominant rule and its associated concept
 4971 activations to a structured explanation record where R_m^* is the dominant rule from

4972 *Definition 6.6, v_R is its consequent value, α_R is its activation strength, and $P_m^{(1)}, P_m^{(2)}$*
 4973 *are the two highest-contributing (concept, term, membership) triples sorted by*
 4974 *membership magnitude.*

$$E_m(x, y) = (m, R_m^*, v_R, \alpha_R, P_m^{(1)}, P_m^{(2)}) \quad (121)$$

4975 The explanation function is deterministic, stateless, and computed in constant time
 4976 per cell per intervention type (one pass over four concept memberships per rule). It
 4977 requires no post-hoc approximation, no sampling, and no model-external attribution
 4978 technique. This intrinsic explainability distinguishes the DisasterAware architecture
 4979 from systems that require post-hoc methods such as SHAP or LIME to generate
 4980 explanations, which provide statistical attributions rather than causal rule-level
 4981 traceability (Ribeiro, Singh, & Guestrin, 2016).

4982 6.5.2 Audit Log Structure

4983 The audit log maintains a complete computational record of the DSS pipeline at
 4984 every time step and every grid cell. This record enables post-hoc analysis of decision
 4985 sequences, comparative evaluation of alternative parameter configurations, and
 4986 external review of the system's behavior over the course of a simulated event.

4987 **Definition 6.8 (Audit Record).** *For each time step k and each grid cell (x, y) , the*
 4988 *audit record is the ordered tuple:*

$$\mathcal{L}_k(x, y) = (\mathbf{z}_k(x, y), \boldsymbol{\mu}_k(x, y), \bar{\mathbf{c}}_k(x, y), \mathbf{u}_{DSS,k}^i(x, y), U_{DSS,k}(x, y), Q_{DSS,k}, \eta_k, R_1^*, \dots, R_M^*) \quad (122)$$

4989 where $\mathbf{z}_k(x, y)$ is the six-dimensional feature vector from Section 6.2.1; $\boldsymbol{\mu}_k(x, y)$ is
 4990 the 3×6 membership matrix from Section 6.2.2; $\bar{\mathbf{c}}_k(x, y)$ is the confidence-gated
 4991 concept vector from Section 6.2.4; $\mathbf{u}_{DSS,k}^i(x, y)$ is the local multi-action vector from
 4992 Section 6.3; $U_{DSS,k}(x, y)$ is the globally coordinated intervention vector from Section
 4993 6.4; $Q_{DSS,k}$ is the decision quality score from Definition 6.3; η_k is the adaptive

4994 threshold from Definition 6.4; and $R_1^*, \dots, R_M^*$ are the dominant explanation rules
4995 from Definition 6.6.

4996 The audit log over a full simulation run is the indexed collection $\{\mathcal{L}_k(x, y) : (x, y) \in$
4997 $\mathcal{G}, k = 0, \dots, K\}$. It provides a complete, reproducible trace of the DSS computation
4998 that is independent of the visualization or reporting layer and can be serialized for
4999 external analysis without requiring access to the runtime system.

5000 6.5.3 Operator Override Interface

5001 The operator override interface provides human-in-the-loop control downstream of
5002 the global coordination layer. Override actions modify the final intervention field
5003 $U_{DSS,k}^{final}(x, y)$ after it has been produced by the decision acceptance mechanism of
5004 Section 6.4.5 but before it is injected into $\mathcal{F}_{in,k}$. Three override modes are supported.

5005 **(i) Cell-level hard constraint.** The operator specifies a set of grid cells $\Omega_{lock} \subseteq \mathcal{G}$
5006 and a fixed intervention value $u_{lock} \in [0, 1]$ for those cells. The override replaces
5007 the DSS output at the specified cells: $U_{DSS,k}^{final}(x, y) = u_{lock}$ for all $(x, y) \in \Omega_{lock}$.
5008 This mode supports operational scenarios where certain locations must receive
5009 mandatory intervention (e.g., critical infrastructure) or where intervention must be
5010 withheld (e.g., areas declared unsafe for crew deployment).

5011 **(ii) Region-level soft preference.** The operator assigns a weight $w_{op,i} \in (0, 2]$ to
5012 one or more regions Ω_i . The override scales the regional contribution:
5013 $U_{DSS,k}^{final}(x, y) \leftarrow \min(1, w_{op,i} \cdot U_{DSS,k}^{final}(x, y))$ for all $(x, y) \in \Omega_i$. This mode
5014 allows field commanders to increase or decrease the relative priority of specific
5015 regions based on tactical judgment without overriding individual cell decisions.

5016 **(iii) Global attenuation.** The operator specifies a global scale factor $\lambda_{op} \in [0, 1]$
5017 that is applied uniformly to the entire intervention field: $U_{DSS,k}^{final}(x, y) \leftarrow \lambda_{op} \cdot$
5018 $U_{DSS,k}^{final}(x, y)$. This mode supports precautionary reduction of DSS authority when

5019 the operator has reason to distrust the system’s outputs, for example during transition
5020 phases or when field conditions are known to deviate from the simulation model.

5021 All override actions are recorded in the audit log with a timestamp, the affected cells
5022 or regions, and the operator identifier, ensuring that post-hoc analysis can distinguish
5023 between DSS-generated decisions and human-modified decisions at every point in
5024 the simulation timeline. Override actions do not modify the DSS reasoning pipeline;
5025 they act exclusively on the output layer, preserving the separation of concerns
5026 between autonomous reasoning and human supervisory control.

5027 **6.6 Summary of the DisasterAware Decision Support System**

5028 This chapter presents the complete architectural realization of the DisasterAware
5029 Decision Support System as a layered, distributed, and concept-driven reasoning
5030 architecture. The chapter addresses the central design challenge of this thesis:
5031 generating spatially distributed, operationally meaningful, and uncertainty-robust
5032 intervention guidance at every time step of a dynamic wildfire simulation, without
5033 direct access to the underlying system state and without requiring assumptions of
5034 precise or complete observability.

5035 The architectural decomposition proceeds through five functional layers, each with
5036 a clearly defined input, output, and formal guarantee. The Observation Layer
5037 (Section 6.1.2) derives graded, uncertainty-affected observations $O_k^i(x, y)$ from the
5038 Grid State Manager for each agent region. The Concept Mapping and Abstraction
5039 Layer (Section 6.2) transforms these observations through feature extraction,
5040 fuzzification, and concept construction into a four-dimensional concept vector
5041 $\mathbf{c}_k^i(x, y) \in [0,1]^4$, reducing the 18-dimensional membership representation by a
5042 factor of 4.5 while preserving its epistemic structure through the Lipschitz-
5043 continuous membership encoding. The Local DSS Model (Section 6.3) evaluates the
5044 81-rule intervention rule base per intervention type against this concept vector,
5045 producing candidate degrees for three intervention types that are consolidated by the

5046 priority-weighted aggregation function f_{agg} into a per-cell regional output
5047 $U_{DSS,k}^i(x, y)$. The Global Coordination Layer (Section 6.4) assembles the N regional
5048 fields into a spatially complete global intervention field $U_{DSS,k}(x, y)$, normalizes it
5049 against the available resource capacity C_k , and injects it into the Simulation Core
5050 through the external input set $\mathcal{F}_{in,k}$. Finally, the Explainability and AI Governance
5051 Layer (Section 6.5) provides traceability, auditability, and operator override
5052 mechanisms as a non-invasive supervisory overlay.

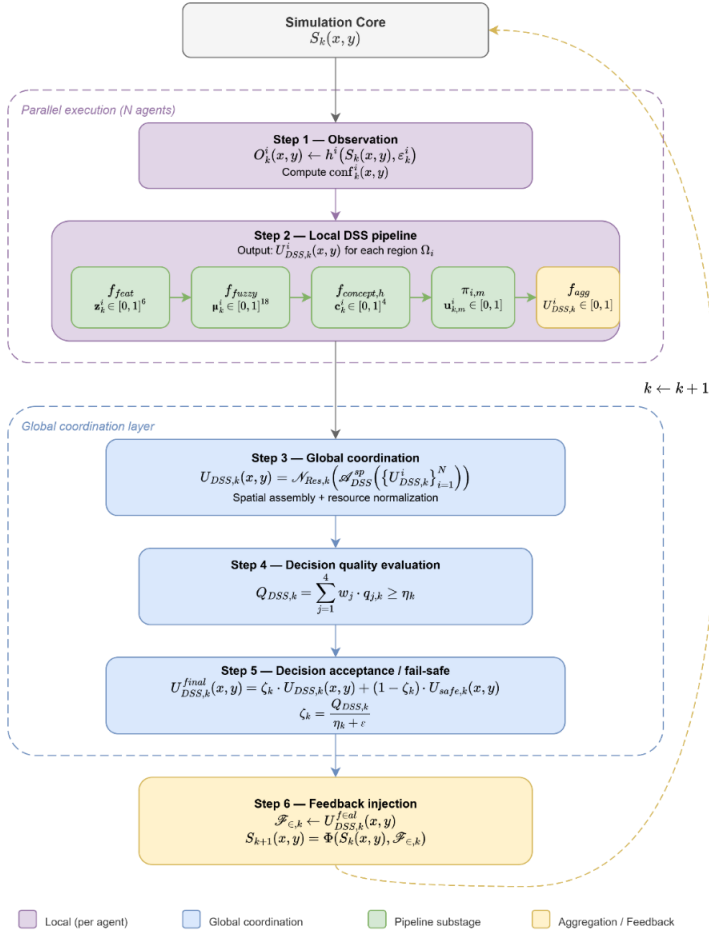
5053 Three formal properties hold throughout the pipeline. First, all intermediate and final
5054 outputs are bounded in $[0, 1]$, enabling concept and intervention spaces to share the
5055 same representational framework without inter-layer normalization. Second, the
5056 partition-of-unity property ensures that every observation is fully represented across
5057 the three linguistic terms with no evidential weight lost. Third, Lipschitz continuity
5058 of the concept formation mapping guarantees that small perturbations in observable
5059 information produce proportionally small changes in the intervention output.
5060 Together, these properties ensure that the DSS generates decisions that are feasible,
5061 consistent, and robust by construction.

5062 The closed-loop structure operationalizes the decision formulation of Chapter 2: the
5063 intervention field $U_{DSS,k}(x, y)$ influences the next state $S_{k+1}(x, y)$ through the
5064 Simulation Core, whose updated output feeds back into the DSS as new observations.
5065 This realizes the dynamic decision formulation and addresses the structural gaps
5066 identified in Chapter 1. The architecture does not converge to a fixed policy but
5067 generates context-sensitive guidance that continuously tracks the evolving wildfire
5068 state under partial observability.

5069 **6.7 Algorithmic Summary of DisasterAware Execution Cycle**

5070 Figure 6.7 illustrates the complete execution cycle of the DisasterAware DSS within
5071 a single time step k . The cycle consists of six steps organized in two phases. Phase I

5072 (Steps 1–2) executes in parallel across N regional agents. Phase II (Steps 3–6)
 5073 operates globally.



5074
 5075 Figure 6.7 Overall algorithm of DisasterAware DSS
 5076 **Phase I: Parallel Local Execution**

5077 **Step 1 Observation.** Each agent i derives its region-specific observation $O_k^i(x,y)$
5078 from the Simulation Core through the observation model (84)–(91). The
5079 confidence coefficient $conf_k^i(x,y)$ quantifies the reliability of each observation.

5080 **Step 2 Local DSS Pipeline.** Each agent passes $O_k^i(x,y)$ through the five-stage
5081 pipeline of Definition 5.1:

5082 (a) **Feature extraction.** f_{feat} maps the observation to a six-dimensional feature
5083 vector $z_k^i(x,y) \in [0,1]^6$ (77).

5084 (b) **Fuzzification.** f_{fuzzy} transforms each feature into three membership values
5085 (Low, Medium, High), producing an 18-dimensional representation (78).

5086 (c) **Concept construction.** $f_{concept,h}$ aggregates the membership values into four
5087 concept values $c_k^i(x,y) \in [0,1]^4$ (79).

5088 (d) **Intervention inference.** Each rule base π_m^i evaluates 81 rules against the concept
5089 vector and produces a candidate degree $u_{k,m}^i(x,y)$ per intervention type (80)–(81).

5090 (e) **Action aggregation.** f_{agg} consolidates the three candidate degrees into a single
5091 regional output $U_{DSS,k}^i(x,y)$ via priority-weighted averaging (82).

5092 **Phase II: Global Coordination**

5093 **Step 3 Spatial assembly and normalization.** The global coordination operator
5094 $\mathcal{A}_{DSS}$ assembles the N regional fields into a spatially complete intervention field
5095 $U_{DSS,k}(x,y)$ (102). The resource normalization operator scales this field to respect
5096 the available capacity C_k (103)–(108).

5097 **Step 4 Decision quality evaluation.** The evaluation function f_{eval} computes the
5098 quality score $Q_{DSS,k}$ as the weighted average of four partial criteria (110). The
5099 adaptive threshold η_k adjusts based on observation confidence, resource sufficiency,
5100 and operational severity (115).

5101 **Step 5 Decision acceptance.** If $Q_{DSS,k} \geq \eta_k$, the intervention field is accepted
5102 without modification (116). The fail-safe policy blends the DSS output with the
5103 conservative baseline $U_{safe,k}(x,y)$ using the coefficient ζ_k (117)–(119).

5104 **Step 6 Feedback injection.** The final intervention field $U_{DSS,k}^{final}(x,y)$ is injected into
5105 the external input set $\mathcal{F}_{in,k}$. The Simulation Core computes the next state:
5106 $S_{k+1}(x,y) = \Phi(S_k(x,y), \mathcal{F}_{in,k})$.

5107 This closes the feedback loop. The updated state $S_{k+1}(x,y)$ feeds back into Step 1 at
5108 the next time step, as indicated by the dashed return arrow in Figure 6.7. The cycle
5109 repeats at every discrete time step throughout the simulation.

5110

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5173		

5174 **7.6.3 Priority Weights ($\alpha_1, \alpha_2, \alpha_3$) and Concept Weights**

5175

5176 **7.6.4 Decision Quality Weights (w_1, w_2, w_3, w_4)**

5177

5178 **7.6.5 Adaptive Threshold Behavior (η_k)**

5179

5180 **7.7 Discussion**

5181

5182 **7.7.1 Strengths and Limitations of the Experimental Results**

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5184 **7.7.2 Implications for Operational Deployment**

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5188 **7.8 Summary of Simulation and Performance Results**

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5191

5192

5193

CHAPTER 8

5194

CONCLUSION AND FUTURE WORKS

5195

5196 **8.1 Conclusions**

5197

5198 **8.2 Future Works**

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APPENDICES

A. Simulation Mathematics for Evolution of Burning Status

This appendix provides the mathematical formulations underlying the wildfire simulation model, ensuring consistency with the state transition framework defined in Chapter 4.

The rate of spread $R_{spread,k}(x,y)$ represents the forward propagation potential of wildfire at grid cell (x,y) and time step k . It is not an externally supplied parameter; rather, it is computed internally within the hybrid fire spread model using meteorological forcing, terrain properties, fuel characteristics, and fuel moisture. The spread formulation is not intended to reproduce physical wildfire propagation in metric units. It provides a grid-consistent, DSS-oriented abstraction inspired by semi-empirical wildfire behavior models.

The formulation adopted in this work is a semi-empirical, Rothermel-like mapping designed to preserve structural interpretability (i.e., each multiplicative term maps to a physically meaningful effect (wind, slope, fuel) even though the output $R_{spread,k}$ is expressed in normalized cells-per-step rather than in metric units) while remaining computationally stable and easily calibratable for DSS simulation purposes (Rothermel, 1972). The formulation follows a semi-empirical wildfire spread abstraction inspired by Rothermel-type models rather than attempting to reproduce detailed physical fire dynamics. The rate of spread $R_{spread,k}(x,y)$ is calculated by f_{ros} (47). Each multiplicative component of the rate-of-spread function f_{ros} corresponds directly to a physically interpretable effect and is tied to an already-defined state or input variable (123). All terms are evaluated at (x,y) and time k .

$$R_{spread,k}(x,y) = r_{base} \left(F_{type}(x,y) \right) \times g_{moist}(x,y) \times g_{wind}(x,y) \times g_{slope}(x,y) \times g_{aspect}(x,y) \quad (123)$$

5434 Fuel-Type Base Spread Rate $r_{base}(\cdot)$ defines the baseline spread potential of each
 5435 fuel class under nominal environmental conditions. It is calibrated in units of cell per
 5436 simulation step.

5437 Moisture damping factor g_{moist} is (124). If fuel moisture equals the extinction
 5438 threshold, spread is suppressed (125). The damping structure is selected for stability
 5439 and interpretability; it prevents negative spread rates and ensures bounded behavior.

$$g_{moist}(x, y) = \max\left(0, 1 - \frac{F_{moist,k}(x, y)}{m_{ext}(F_{type}(x, y))}\right) \quad (124)$$

$F_{moist,k}(x, y)$: surface fuel moisture fraction

$m_{ext}(F_{type})$: extinction moisture threshold for the given fuel type

$$F_{moist,k} \geq m_{ext} \Rightarrow g_{moist} = 0 \quad (125)$$

5440 Wind amplification factor g_{wind} is (126). This factor models the acceleration of
 5441 fire spread due to wind speed. The hyperbolic tangent ensures that spread increases
 5442 with wind speed but does not grow unbounded for extreme winds. The use of
 5443 $\tanh(\cdot)$ guarantees numerical stability. The wind scaling constant w_0 controls the
 5444 saturation behavior of wind amplification. A value of 10 m/s is selected to reflect
 5445 moderate-to-strong wind regimes while preserving numerical stability.

$$g_{wind}(x, y) = 1 + a_w(F_{type}(x, y)) \cdot \tanh\left(\frac{W_{ws,k}(x, y)}{w_0}\right)$$

$W_{ws,k}(x, y)$ is the wind speed. (126)

$a_w(F_{type})$ is a wind sensitivity coefficient.

w_0 is a scaling constant.

5446 Slope acceleration term g_{slope} is (127). Spread accelerates in uphill directions. Fire
 5447 spreads faster uphill due to flame preheating of fuels. The tangent function reflects
 5448 nonlinear growth with increasing slope angle.

$$g_{slope}(x, y) = 1 + a_s(F_{type}(x, y)) \cdot \tan(G_{slope}(x, y)) \quad (127)$$

$G_{slope}(x, y)$: terrain slope (radians)

$a_s(F_{type})$: slope sensitivity coefficient

Aspect–wind orientation term g_{aspect} is (128). This term captures interaction between terrain orientation and wind direction. If wind direction aligns with terrain orientation, $\cos(\cdot) \approx 1$ and spread increases. If wind opposes terrain orientation, $\cos(\cdot) < 0$ and spread decreases.

$$g_{aspect}(x, y) = 1 + a_{asp}(F_{type}(x, y)) \cdot \cos(G_{aspect}(x, y) - W_{wd,k}(x, y))$$

(128)

$G_{aspect}(x, y)$: terrain orientation angle

$W_{wd,k}(x, y)$: wind direction field

$a_{asp}(F_{type})$: aspect sensitivity coefficient

Different fuel types exhibit distinct intrinsic combustion and propagation characteristics due to variations in fuel structure, packing ratio, moisture retention capacity, and heat-release behavior. These differences are parameterized in Table A.1 through a fuel-dependent base spread rate and sensitivity coefficients that modulate environmental amplification effects. The base spread parameter r_{base} is expressed in units of grid cells per simulation step and therefore must be interpreted relative to the spatial grid resolution Δx and the discrete simulation time step Δt .

In contrast, the extinction moisture threshold m_{ext} is a dimensionless mass fraction, representing the critical fuel moisture level above which sustained combustion is suppressed. The environmental sensitivity coefficients a_w , a_s , and a_{asp} are also dimensionless parameters, governing the relative amplification of spread rate due to wind speed, terrain slope, and wind–aspect alignment, respectively. Together, these parameters provide a resolution-consistent, yet physically interpretable abstraction of fuel-dependent wildfire spread dynamics suitable for discrete-time grid-based simulation.

Table A.1 Spread rate parameters for each fuel type

Fuel Type	r_{base}	m_{ext}	a_w	a_s	a_{asp}
Grass	1.20	0.25	2.0	0.8	0.30
Shrub	0.80	0.35	1.5	1.0	0.25

5469

Pine litter	0.60	0.45	1.2	1.2	0.20
Hardwood	0.40	0.50	0.8	0.6	0.15

B. Simulation Mathematics for Fuel Mass Evolution

This section provides detailed mathematical formulation and explanation for both the fractional burn rate calculation and the suppression of fuel reduction maps.

Fractional burn rate calculation

The discrete combustion consumption ratio $F_{burn,k}(x, y)$ represents the fraction of available fuel mass consumed at grid cell (x, y) during a single simulation time step Δt under active burning conditions. Unlike fuel load $F_{load,k}$, which is a dynamic state variable, $F_{burn,k}$ is not stored as part of the state vector. Instead, it is computed algebraically at each simulation step as a bounded function of fuel type and surface fuel moisture (129).

$$F_{burn,k}(x, y) = \text{sat}_{[0,1]} \left(b_{base} (F_{type}(x, y)) \cdot (1 - F_{moist,k}(x, y)) \right)$$

$b_{base}(F_{type}) \in (0,1)$: the baseline combustion coefficient associated with the fuel class,

$F_{moist,k}(x, y) \in [0,1]$: the surface fuel moisture fraction, (129)

$\text{sat}_{[0,1]}(\cdot)$: the saturation operator ensuring boundedness.

The term $(1 - F_{moist,k})$: a linear moisture damping factor, reducing combustion efficiency as moisture increases.

The coefficient b_{base} is expressed as a *fraction of fuel consumed per simulation time step* and therefore depends on the chosen time resolution Δt . It is not a universal physical constant but a calibrated simulation parameter. Different fuel classes exhibit distinct combustion rates due to structural density, packing ratio, and heat-release characteristics. These differences are abstracted through the fuel-dependent baseline coefficient b_{base} . Surface fuel moisture inhibits combustion efficiency. When $F_{moist,k} \rightarrow 1$, the burn fraction approaches zero. When $F_{moist,k} \rightarrow 0$, combustion proceeds at the baseline rate.

5488 The baseline combustion coefficients used in this study are summarized in Table
 5489 B.1. Values are selected to produce realistic burn residence times under nominal dry
 5490 conditions for the adopted grid resolution and simulation time step.

5491 Table B.1 Baseline combustion coefficients per fuel type

Fuel Type	b_{base} (fraction per step)	Typical Burn Duration (qualitative)
Grass	0.25	Short, fast consumption
Shrub	0.15	Moderate consumption
Pine litter	0.10	Sustained surface burn
Hardwood	0.05	Slow combustion

5492 These coefficients are calibrated by such that, in dry conditions and absent
 5493 suppression, fuel mass decreases significantly over several time steps without
 5494 causing numerical instability.

5495 **Suppression of fuel reduction map**

5496 The suppression function $f_{supp}(\cdot)$ produces a **step-normalized fuel reduction**
 5497 **fraction** applied at grid cell (x, y) during simulation step k . Unlike continuous-time
 5498 rate formulations, the present model operates in discrete simulation steps. Therefore,
 5499 the suppression term directly represents the fraction of fuel reduced within a single
 5500 time step, consistent with the combustion fraction $F_{burn,k}$. The suppression-induced
 5501 fuel reduction term is defined as (130). This multiplicative structure guarantees
 5502 bounded and interpretable mitigation dynamics.

5503 The suppression formulation is defined per simulation step and therefore represents
 5504 the maximum achievable mitigation effect within a single discrete simulation
 5505 interval. This design ensures compatibility with the discrete-time wildfire state
 5506 transition model defined in Chapter 4.

$$F_{red,k}(x, y) = \alpha_s \cdot \eta_{cap,k} \cdot \eta_{avail,k} \cdot \eta_{reach,k}(x, y) \cdot \eta_{eff,k}(x, y)$$

$\alpha_s \in [0,1]$: global suppression scaling coefficient (130)
 $\eta_{cap,k}$: operational capacity factor

$\eta_{avail,k}$: operational availability factor

$\eta_{reach,k}(x, y)$: reachability factor

$\eta_{eff,k}(x, y)$: local suppression effectiveness factor

5507 Global scaling term $\alpha_s \in [0,1]$ is a global suppression scaling coefficient
 5508 representing overall doctrine or scenario-level mitigation strength per simulation
 5509 step. In practical terms, α_s absorbs the modeling choice of “how strong one step of
 5510 suppression can be” under nominal conditions.

5511 Suppression cannot exceed the operational capability that is actually present. This is
 5512 captured by separating *capacity magnitude* $\eta_{cap,k}$ (131) from *availability gating*
 5513 $\eta_{avail,k}$ (132).

$$\eta_{cap,k} = \frac{R_{cap,k}}{R_{cap,max}}$$

$R_{cap,k}$: the available suppression capacity (e.g., water volume, (131)
 manpower equivalent, sortie capability) at step k

$R_{cap,max}$: the maximum reference capacity used for normalization

5514 Thus $\eta_{cap,k} \in [0,1]$ expresses the fraction of maximal capability currently available.

$$\eta_{avail,k} = R_{avail,k} \quad (132)$$

$R_{avail,k} \in \{0,1\}$: availability of the suppression resources

5515 This factor encodes a strict necessary condition. If resources are not deployable,
 5516 suppression impact is zero regardless of other terms.

5517 Even if resources exist, mitigation effectiveness must decay with increasing travel
 5518 time and poor accessibility. This is represented by a smooth reachability term:

$$\eta_{reach,k}(x, y) = \exp(-\beta_t R_{time,k}(x, y)) \cdot G_{access}(x, y)$$

$R_{time,k}(x, y)$: the estimated travel time to reach (x, y) at step k (133)

$G_{access}(x, y) \in [0,1]$: the static accessibility index of the terrain

$\beta_t > 0$: the decay rate with travel delay

5519 The exponential structure ensures boundedness and numerical stability.
 5520 Operationally, it implements the principle that interventions arriving late cannot
 5521 mitigate near-term fire growth at the same step, and inaccessible terrain reduces the
 5522 usable effectiveness of resources.

5523 Suppression success is not independent of local fire severity. High intensity fires
 5524 typically resist suppression, reducing the effective mitigation that can be achieved
 5525 within a single step. This is modeled by:

$$\eta_{eff,k}(x,y) = \frac{R_{eff,k}(x,y)}{1 + \gamma_I I_k(x,y)}$$

$R_{eff,k}(x,y) \in [0,1]$: the nominal suppression efficiency at (x,y) (134)

$I_k(x,y) \in [0,1]$: the local fire intensity proxy

$\gamma_I \geq 0$: intensity resistance

5526 As I_k increases, the denominator increases and $\eta_{eff,k}$ decreases. This captures the
 5527 intuitive and operationally observed phenomenon that extreme fire behavior reduces
 5528 the marginal impact of suppression. Suppression effectiveness is computed using the
 5529 current-state intensity $I_k(x,y)$, ensuring that mitigation decisions are based on
 5530 observable fire conditions at time step k , prior to state transition.

5531 The final multiplicative formulation of f_{supp} in (50) is explicitly written as (135).
 5532 It encodes the following operational semantics in a compact and interpretable form:

- 5533 • If resources are unavailable, mitigation becomes zero
- 5534 • If the cell is unreachable, mitigation decays smoothly toward zero
- 5535 • If accessibility is poor, mitigation is proportionally reduced
- 5536 • If fire intensity is high, mitigation effectiveness decreases
- 5537 • The overall effect remains bounded and suitable for discrete-time simulation

$$F_{red,k}^{raw}(x,y) = \alpha_s \cdot \left(\frac{R_{cap,k}}{R_{cap,max}} \right) \cdot R_{avail,k} \cdot \exp \left(-\beta_t R_{time,k}(x,y) \right) \cdot G_{access}(x,y) \cdot \frac{R_{eff,k}(x,y)}{1 + \gamma_I I_k(x,y)} \quad (135)$$

5538 The calibration parameters summarized in Table B.2 regulate the strength, spatial
 5539 decay, and operational sensitivity of the suppression mapping. These parameters are
 5540 not arbitrary numerical coefficients; rather, they encode interpretable characteristics
 5541 of wildfire response systems under discrete-time grid simulation.

5542 To preserve physical realism, suppression-driven fuel reduction cannot exceed
 5543 available fuel mass $0 \leq F_{red,k}(x,y) \leq F_{load,k}(x,y)$. Implementation uses the
 5544 saturation rule $F_{red,k}(x,y) = \min\{F_{load,k}(x,y), F_{red,k}^{raw}(x,y)\}$. This prevents
 5545 negative fuel states and ensures compatibility with the fuel evolution equation.

5546 The global suppression gain α_s represents the maximum achievable fuel reduction
 5547 fraction within a single simulation step and is therefore dimensionless (fraction per
 5548 step). The travel-time decay parameter β_t has units of inverse time (e.g., 1/min or
 5549 1/hour depending on the chosen simulation time scale) and governs the exponential
 5550 attenuation of suppression effectiveness as arrival time increases. The intensity
 5551 resistance coefficient γ_I is dimensionless and models the nonlinear reduction in
 5552 suppression efficiency under increasing fire intensity.

5553 All parameters are normalized to ensure compatibility with the discrete-time
 5554 formulation of the model. Calibration may be performed using historical wildfire
 5555 response data, operational doctrine constraints, or structured sensitivity analysis
 5556 across scenario ensembles.

5557 Table B.2 Suppression mapping calibration parameters

Parameter	Symbol	Typical Range	Effect When Increased
Global suppression gain	α_s	0.01 – 0.30	Stronger overall mitigation per simulation step

Travel-time decay coefficient	β_t	0.05 – 0.50	Faster decay of suppression effectiveness with increasing delay
Intensity resistance factor	γ_I	0.5 – 5.0	High-intensity fires become progressively harder to suppress

5558 This appendix provides a calibrated, decision-aware abstraction of suppression that
5559 influences wildfire evolution through physically interpretable state variables,
5560 without directly overwriting the burning state.

5561 C. Simulation Mathematics for Fire Intensity Evolution

5562 The fire intensity proxy $I_k(x, y)$ is a bounded indicator representing local
 5563 combustion strength. It does not model flame temperature or heat flux directly but
 5564 provides a normalized scalar quantity suitable for suppression resistance modeling;
 5565 risk prioritization; decision-layer feature extraction; and visualization. To ensure
 5566 numerical stability, scale invariance, and resolution consistency, intensity is
 5567 constructed using normalized inputs (136).

$$\begin{aligned}\tilde{F}_k(x, y) &= \frac{F_{load,k}(x, y)}{F_{load,max}} \\ \tilde{W}_k(x, y) &= \frac{W_{ws,k}(x, y)}{W_{ws,max}} \\ \tilde{S}(x, y) &= \frac{\tan(G_{slope}(x, y))}{\tan(G_{slope,max})}\end{aligned}\quad (136)$$

$F_{load,max}$: the maximum fuel load,

$W_{ws,max}$: a reference maximum wind speed (e.g., 20 m/s),

$G_{slope,max}$: the maximum slope angle allowed in the terrain model.

Thus: $0 \leq \tilde{F}, \tilde{W}, \tilde{S} \leq 1$

5568 The local fire intensity proxy is defined as (137). The intensity proxy is normalized
 5569 and bounded within the interval [0,1] in order to ensure numerical stability and
 5570 consistent interpretation across grid cells.

$$I_{k+1}(x, y) = B_{k+1}(x, y) \cdot \tanh\left(\beta \left(\tilde{F}_k(x, y) + \gamma_W \tilde{W}_k(x, y) + \gamma_S \tilde{S}(x, y)\right)\right)$$

$\beta > 0$: the global intensity gain parameter,

$\gamma_W \in [0,1]$: the wind weighting coefficient,

$\gamma_S \in [0,1]$: the slope weighting coefficient.

B_{k+1} ensures that non-burning cells have zero intensity.

5571 The normalized structure ensures that fuel load remains the dominant contributor;
5572 wind and slope act as amplifying modifiers; intensity remains bounded in [0,1]. The
5573 fire intensity evolution parameters are listed in Table C.1.

5574 Table C.1 Fire intensity evolution parameters

Parameter	Range	Suggested Value	Interpretation
β	$1 \leq \beta \leq 3$	2.0	Moderate global intensity amplification
γ_w	$0 \leq \gamma_w \leq 0.7$	0.5	Wind moderately increases intensity
γ_s	$0 \leq \gamma_s \leq 0.5$	0.3	Slope has secondary influence

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CURRICULUM VITAE

Commented [ÇA13]: Patent, Article Update.

5577

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2007-2007	TUBITAK	Part Time Engineer

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5680 **HOBBIES**

5681 Playing piano, playing chess, playing table tennis, and swimming